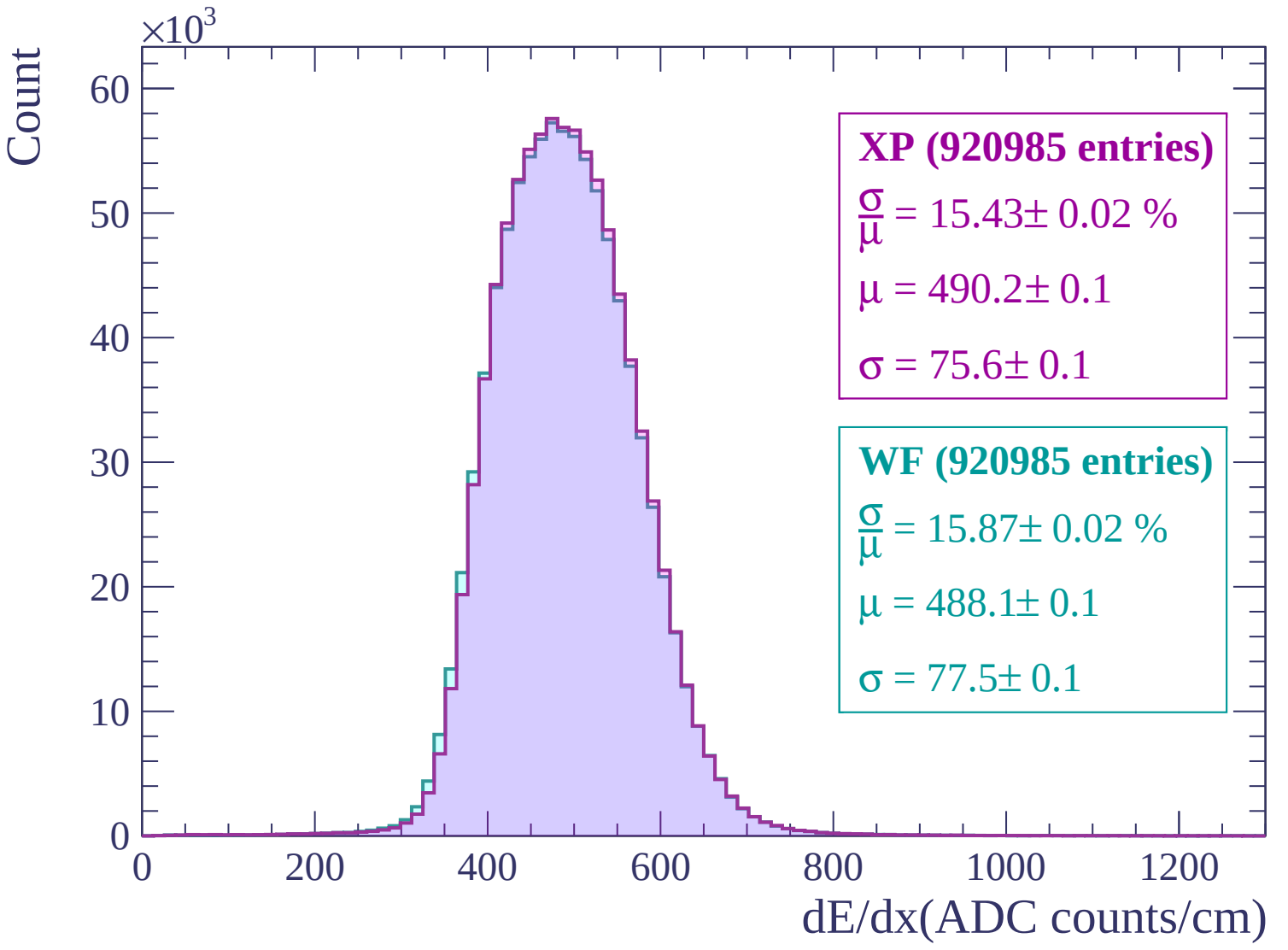
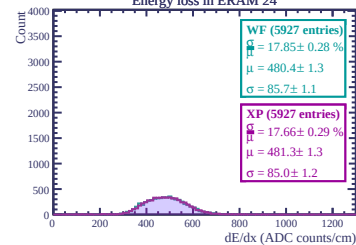
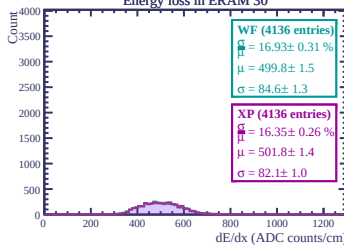




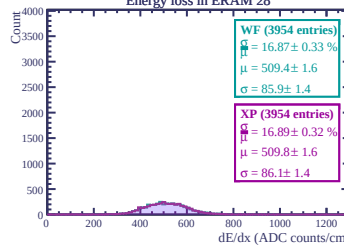
Energy loss in ERAM 14



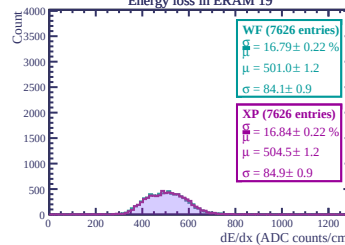
Energy loss in ERAM 30



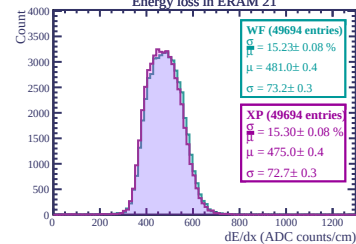
Energy loss in ERAM 28



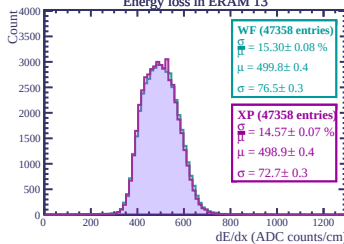
Energy loss in ERAM 10



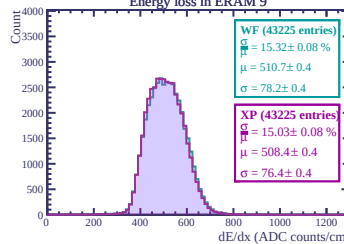
Energy loss in ERAM 21



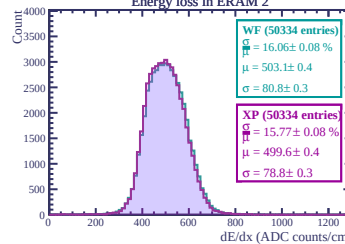
Energy loss in ERAM 13



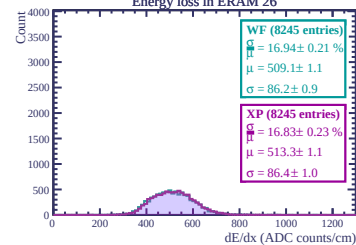
Energy loss in ERAM 9



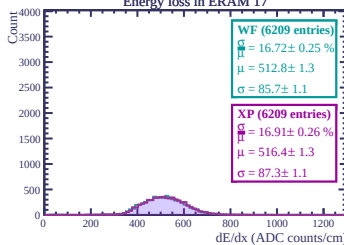
Energy loss in ERAM 2



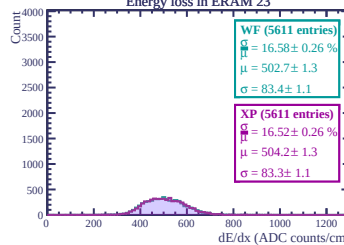
Energy loss in ERAM 26



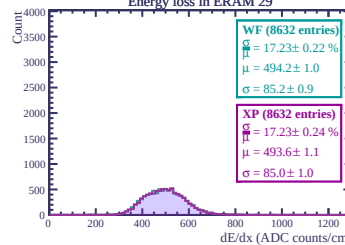
Energy loss in ERAM 17



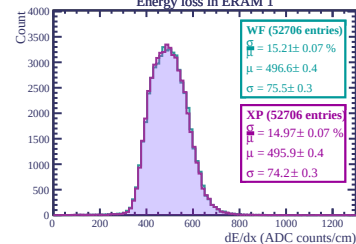
Energy loss in ERAM 23



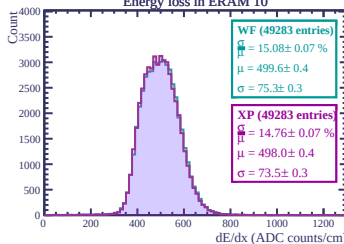
Energy loss in ERAM 29



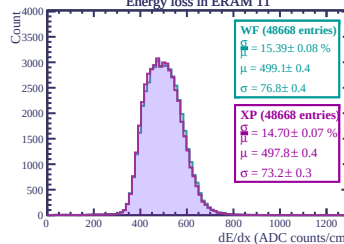
Energy loss in ERAM 1



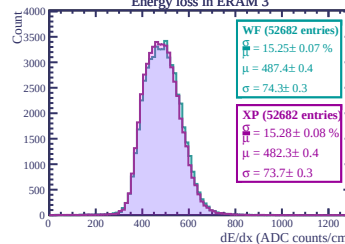
Energy loss in ERAM 10

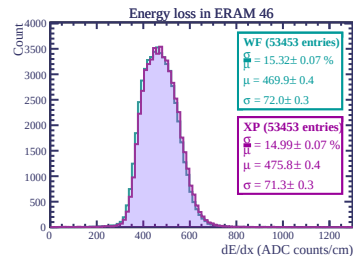
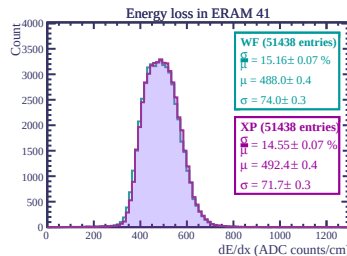
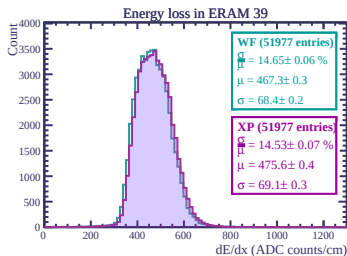
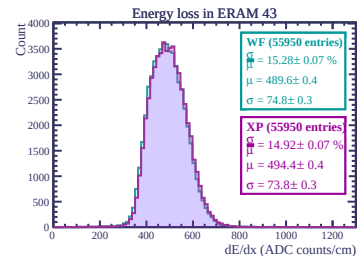
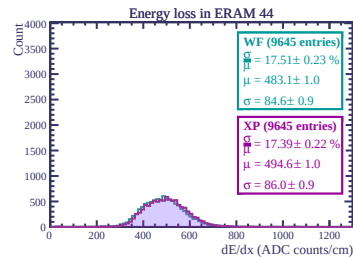
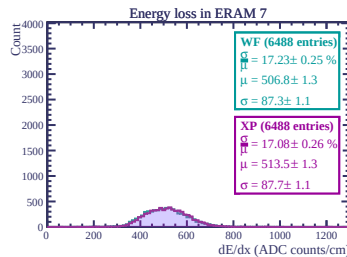
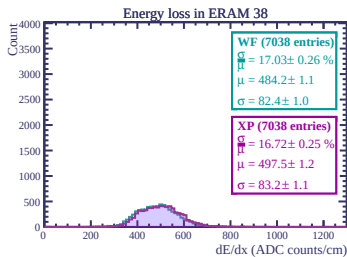
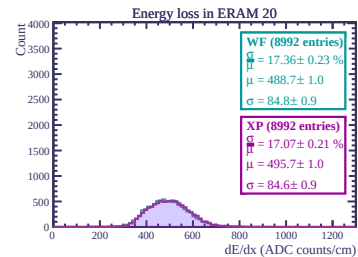
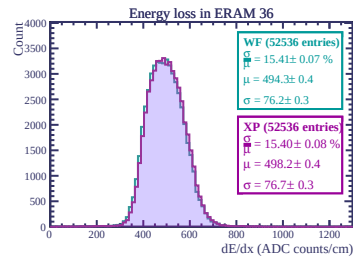
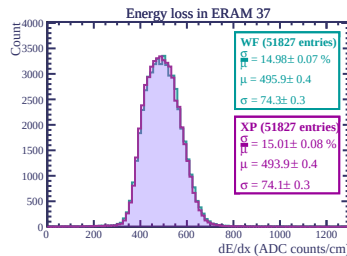
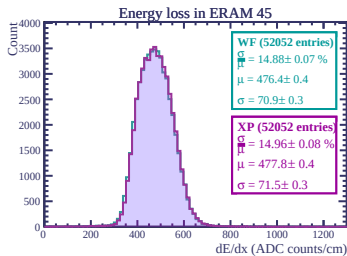
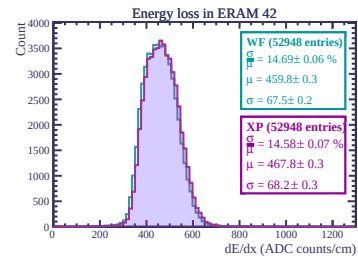
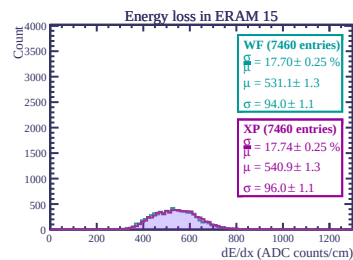
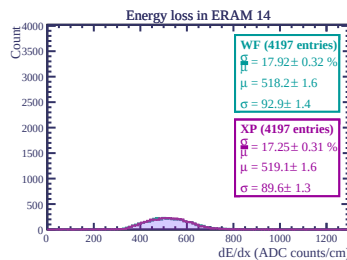
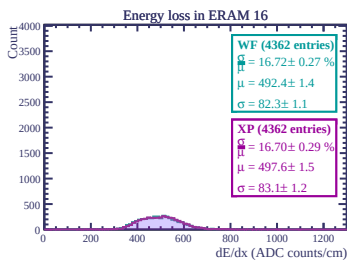
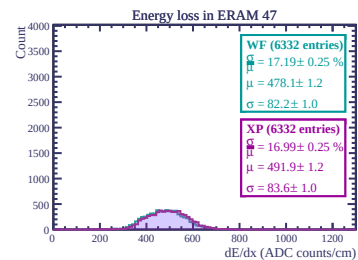


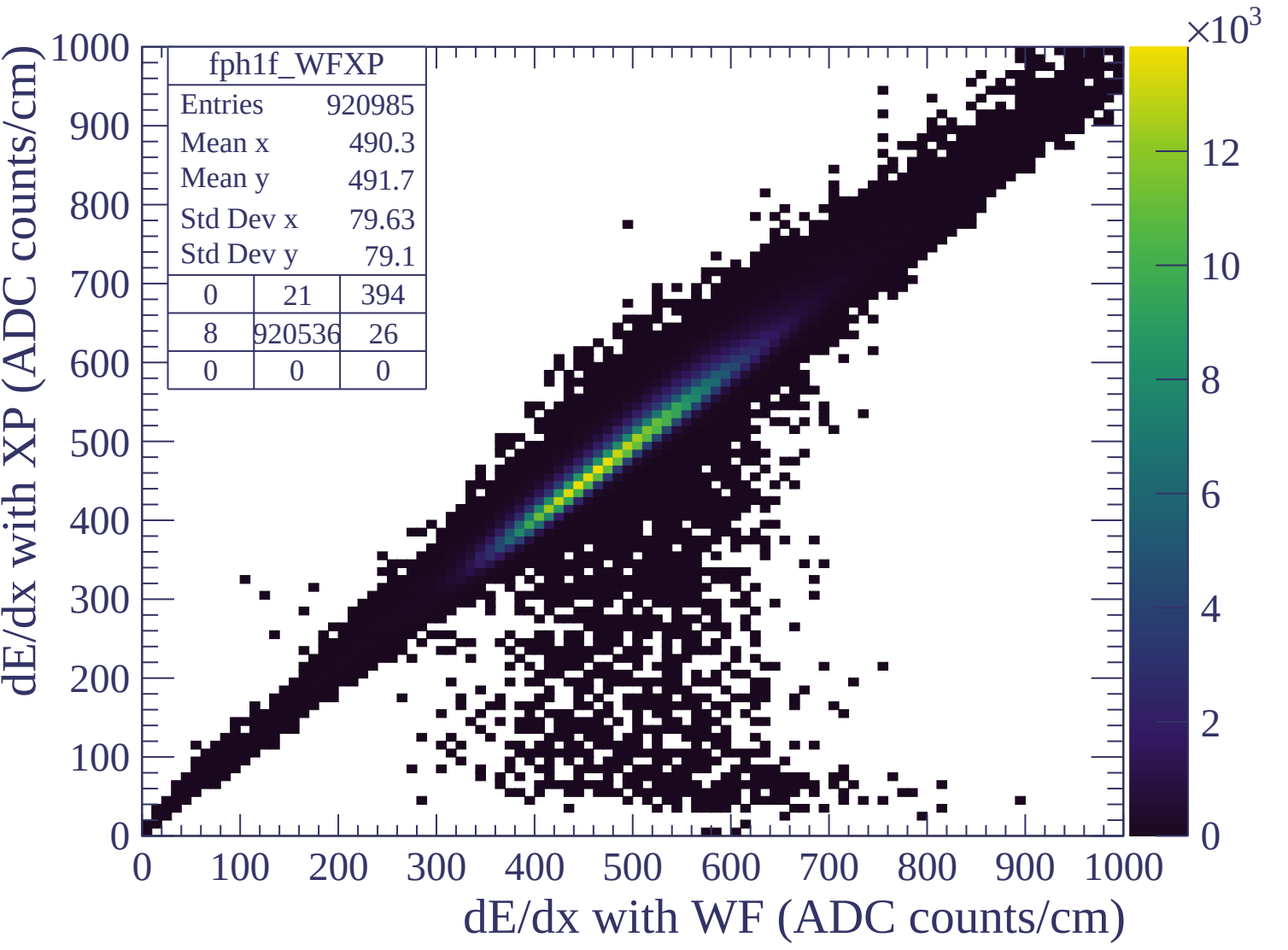
Energy loss in ERAM 11



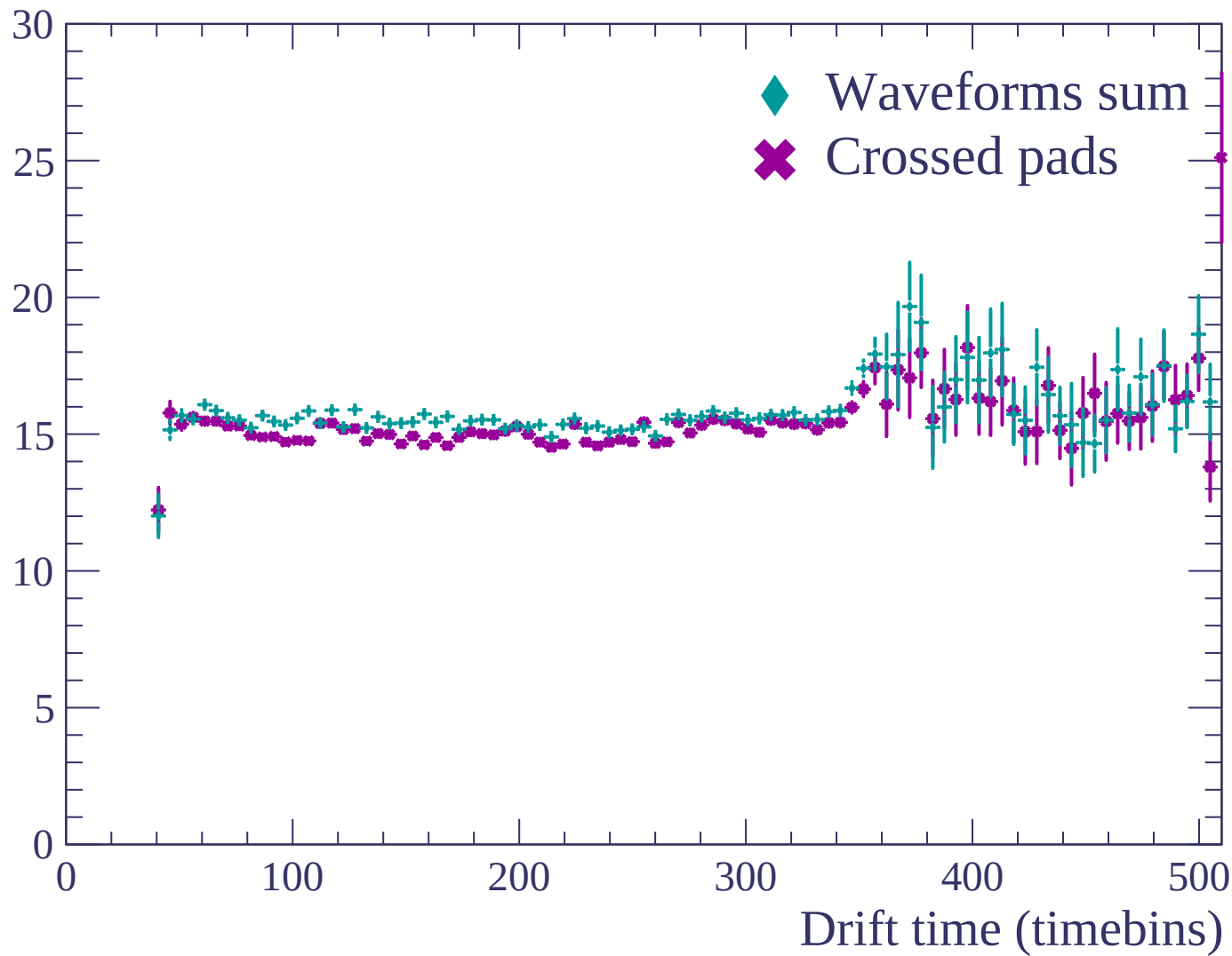
Energy loss in ERAM 3

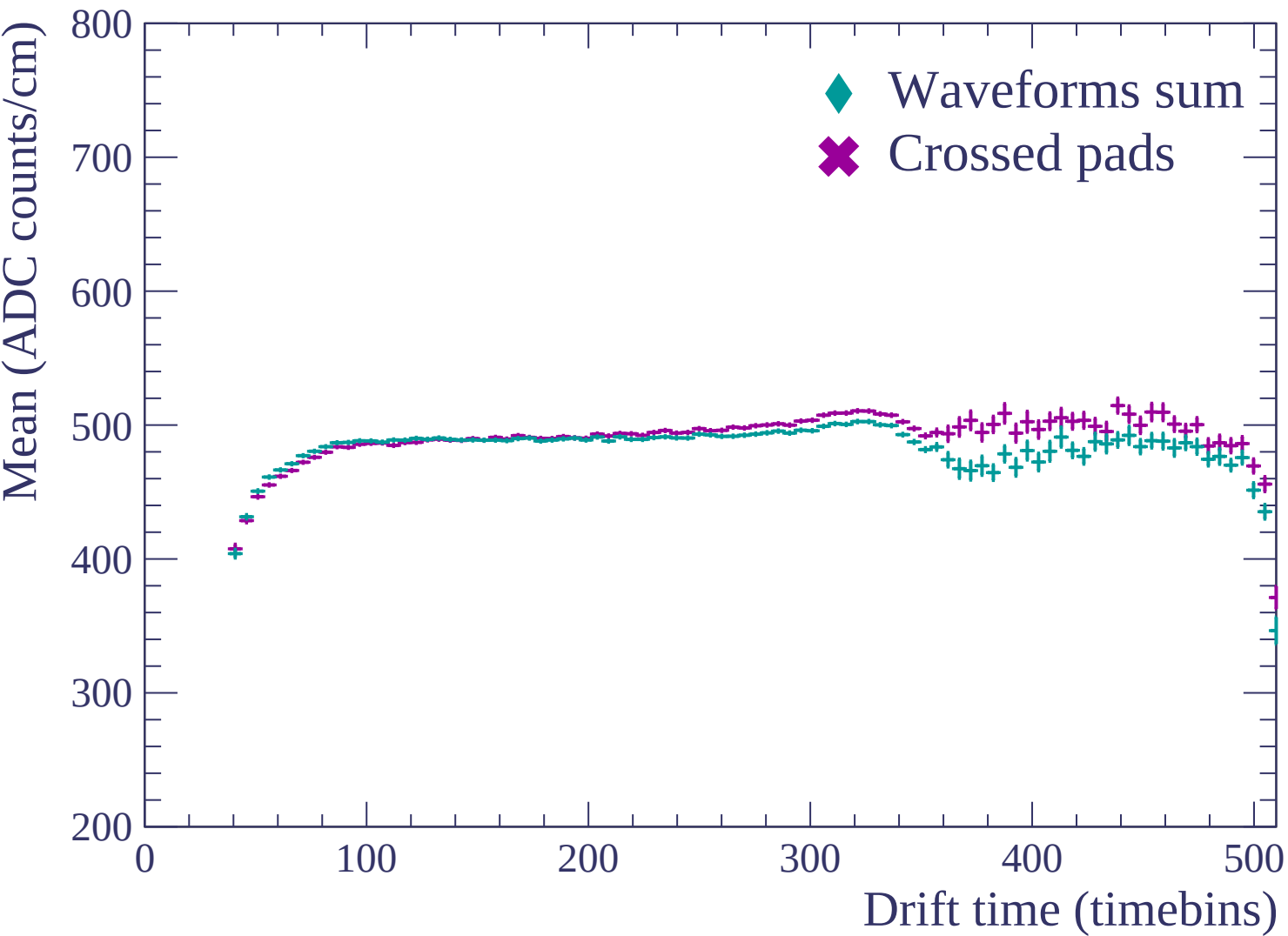


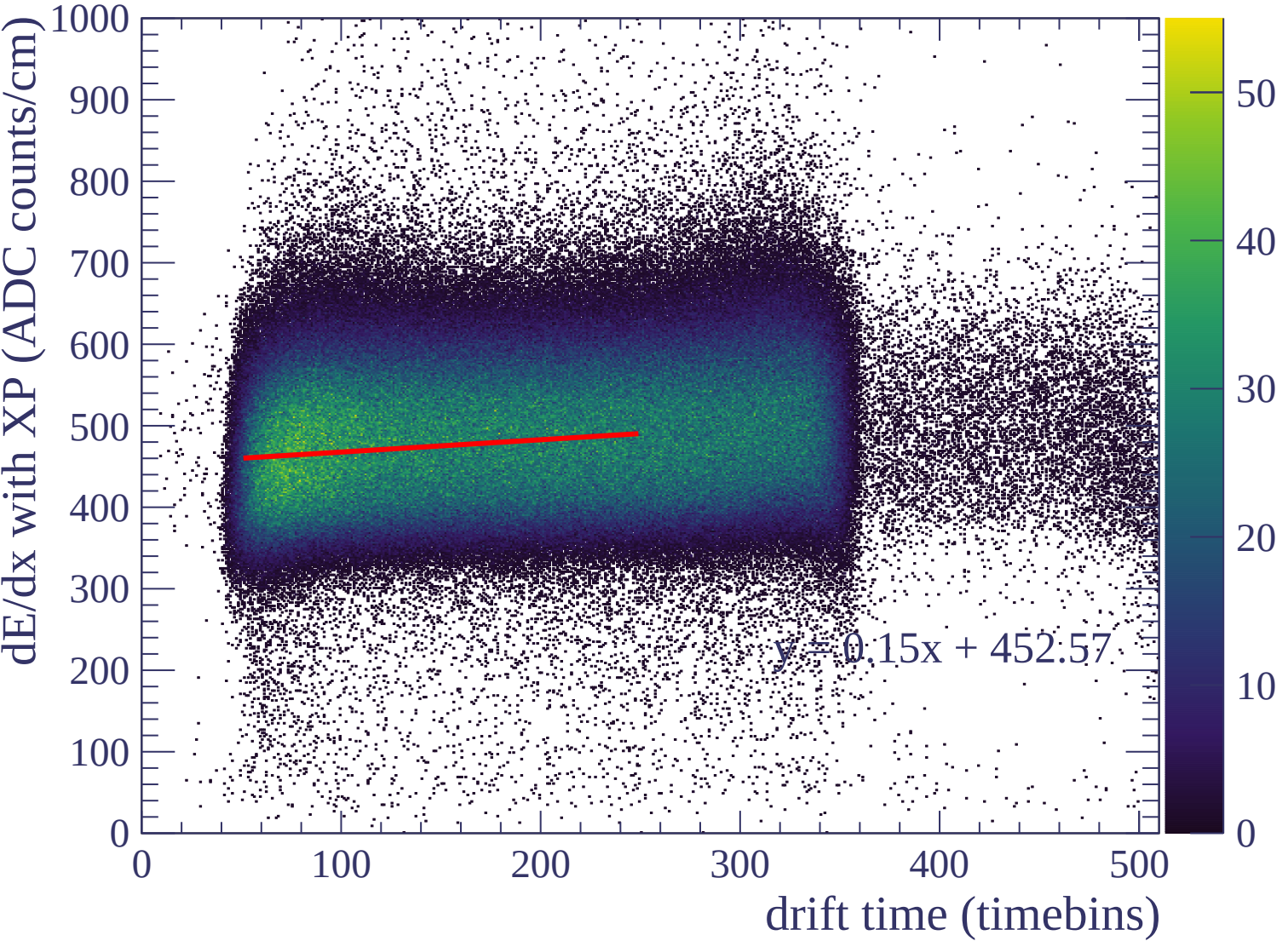


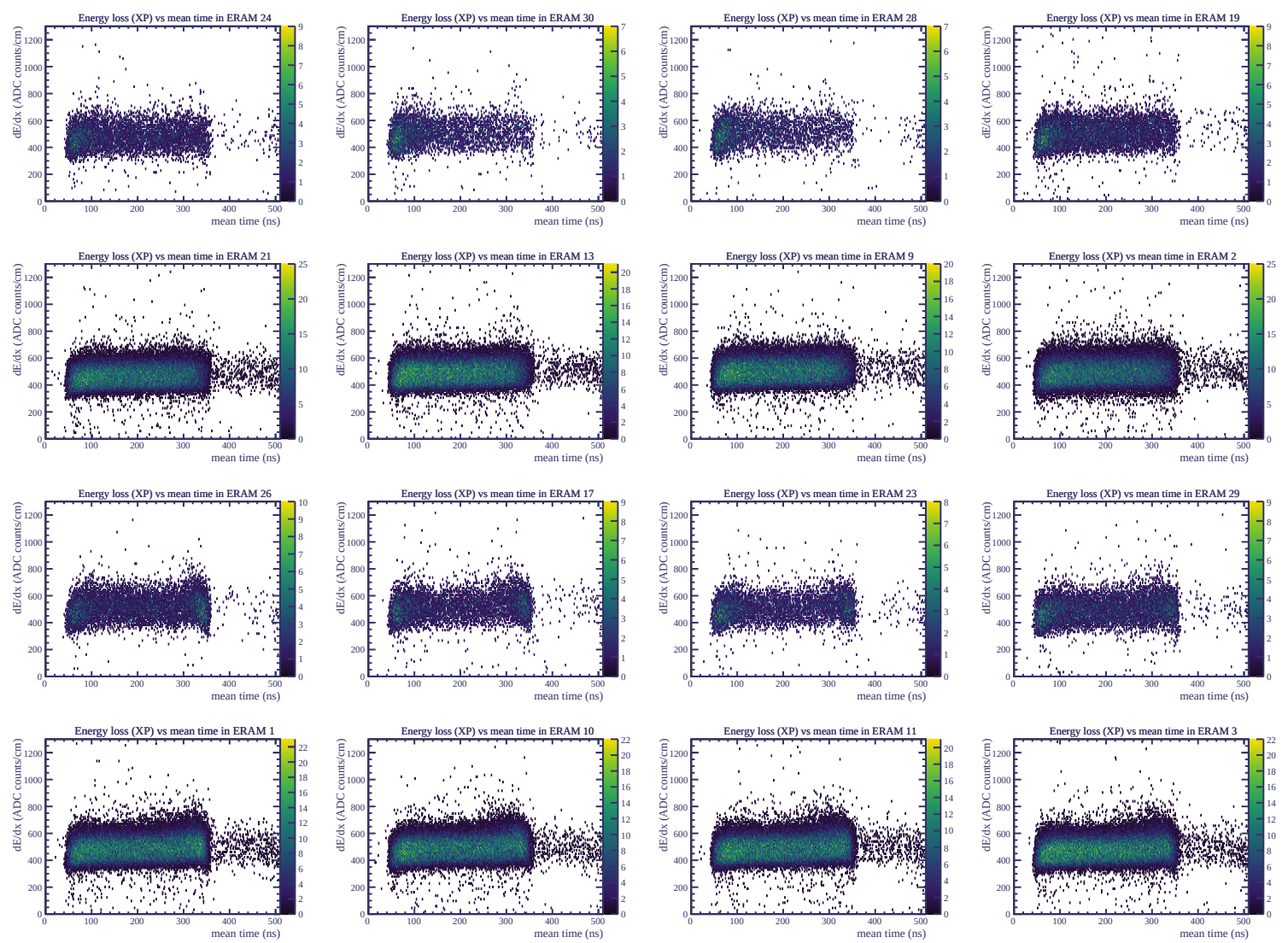


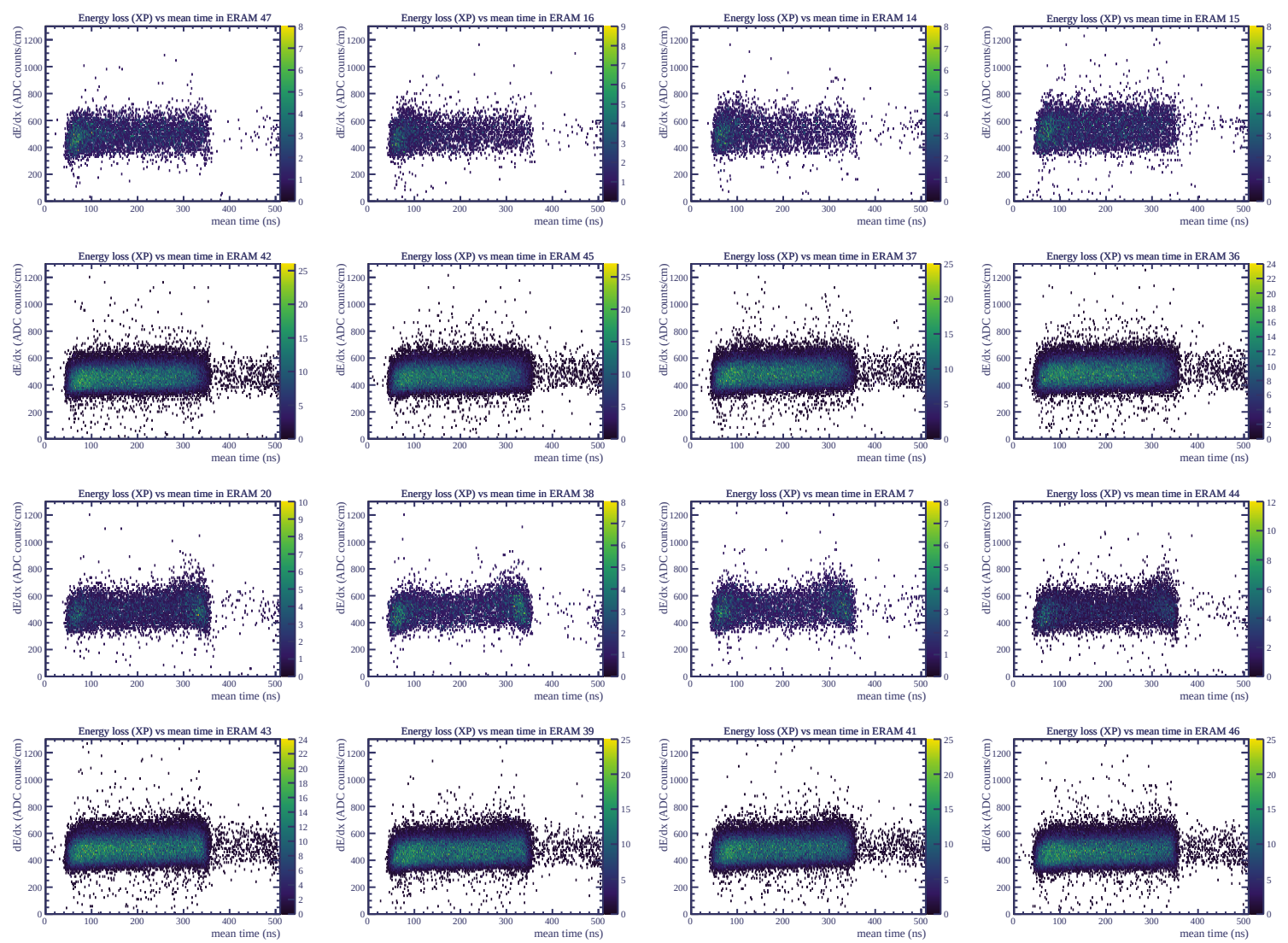
Resolution (%)

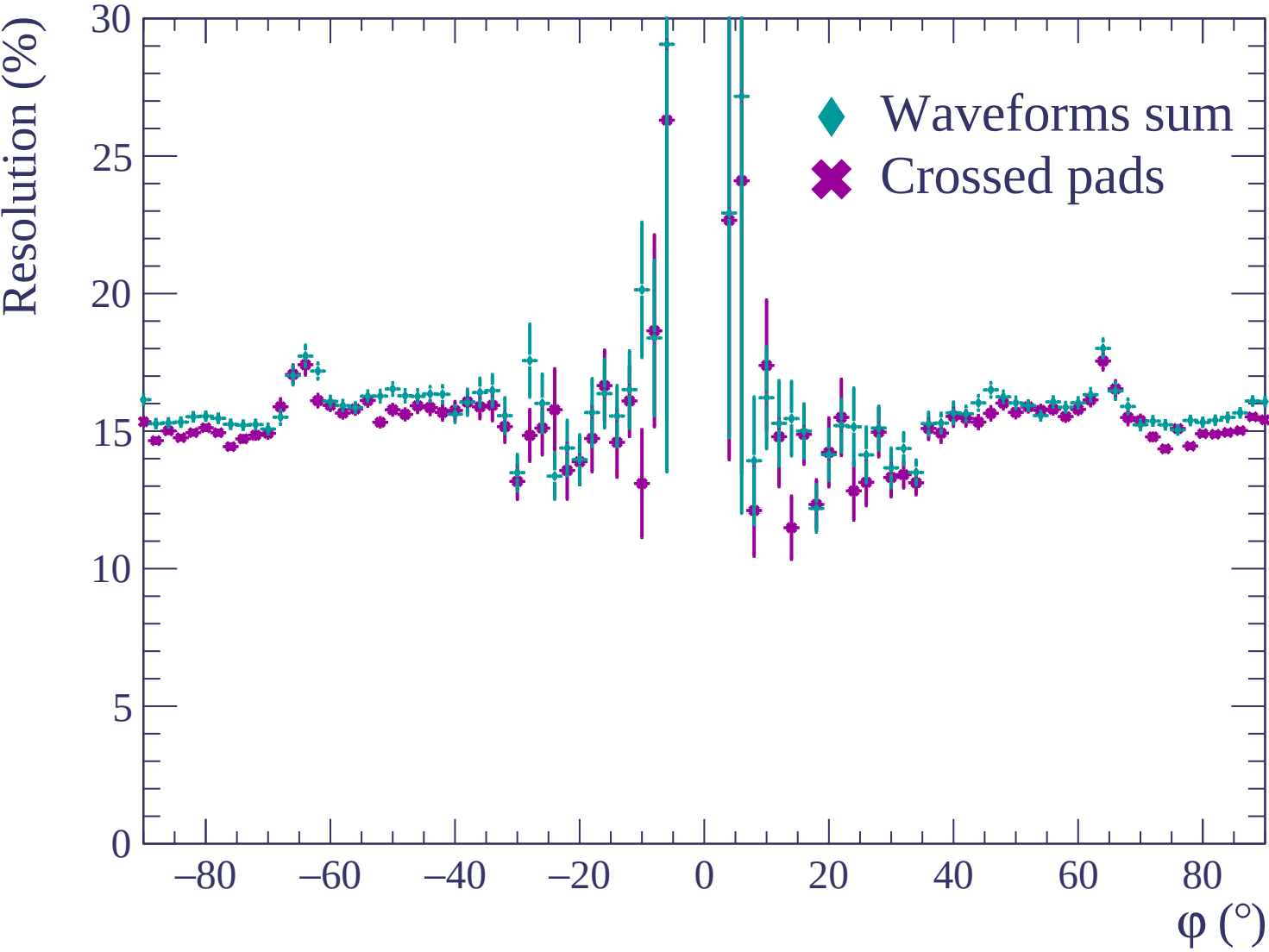


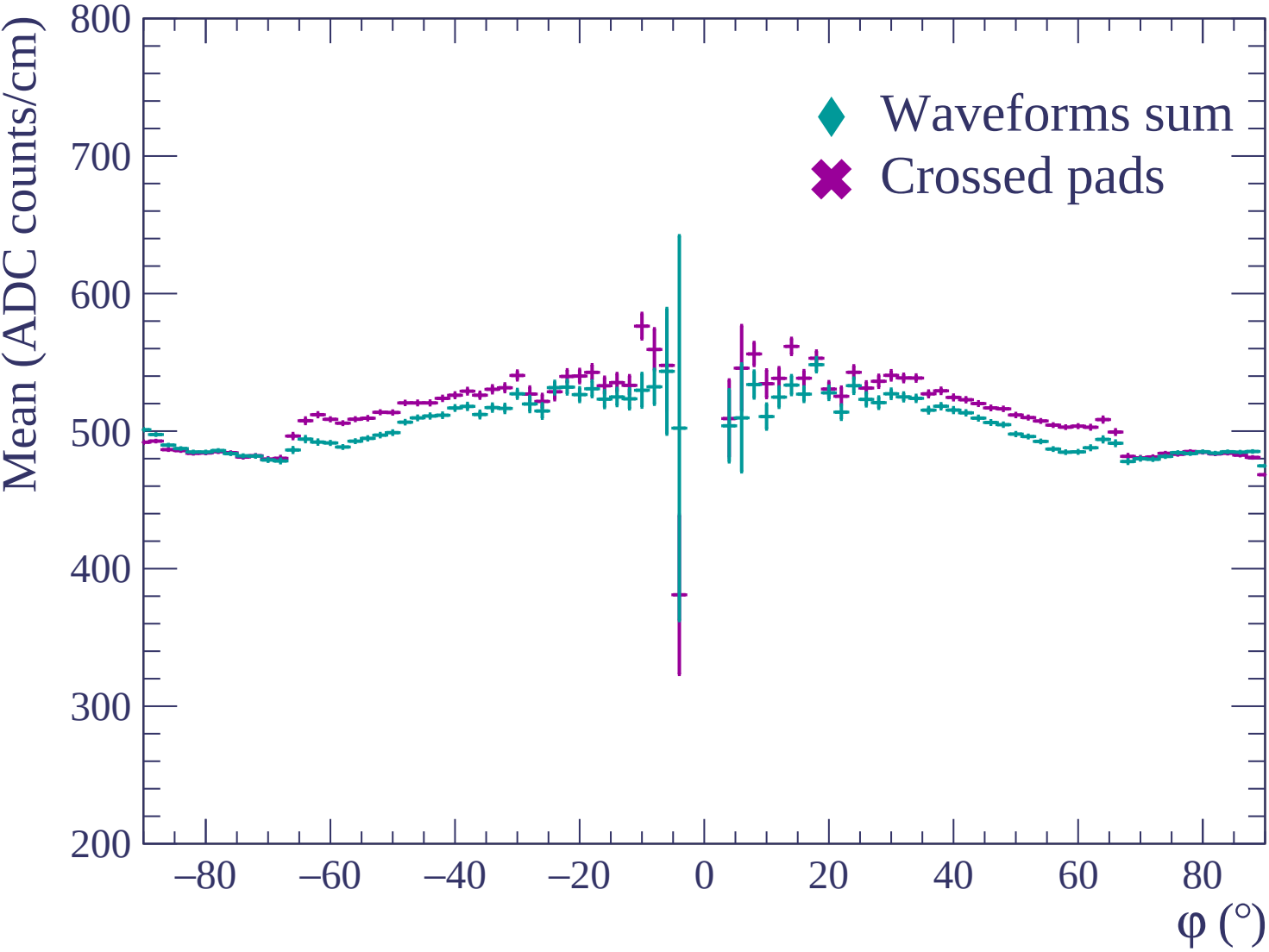


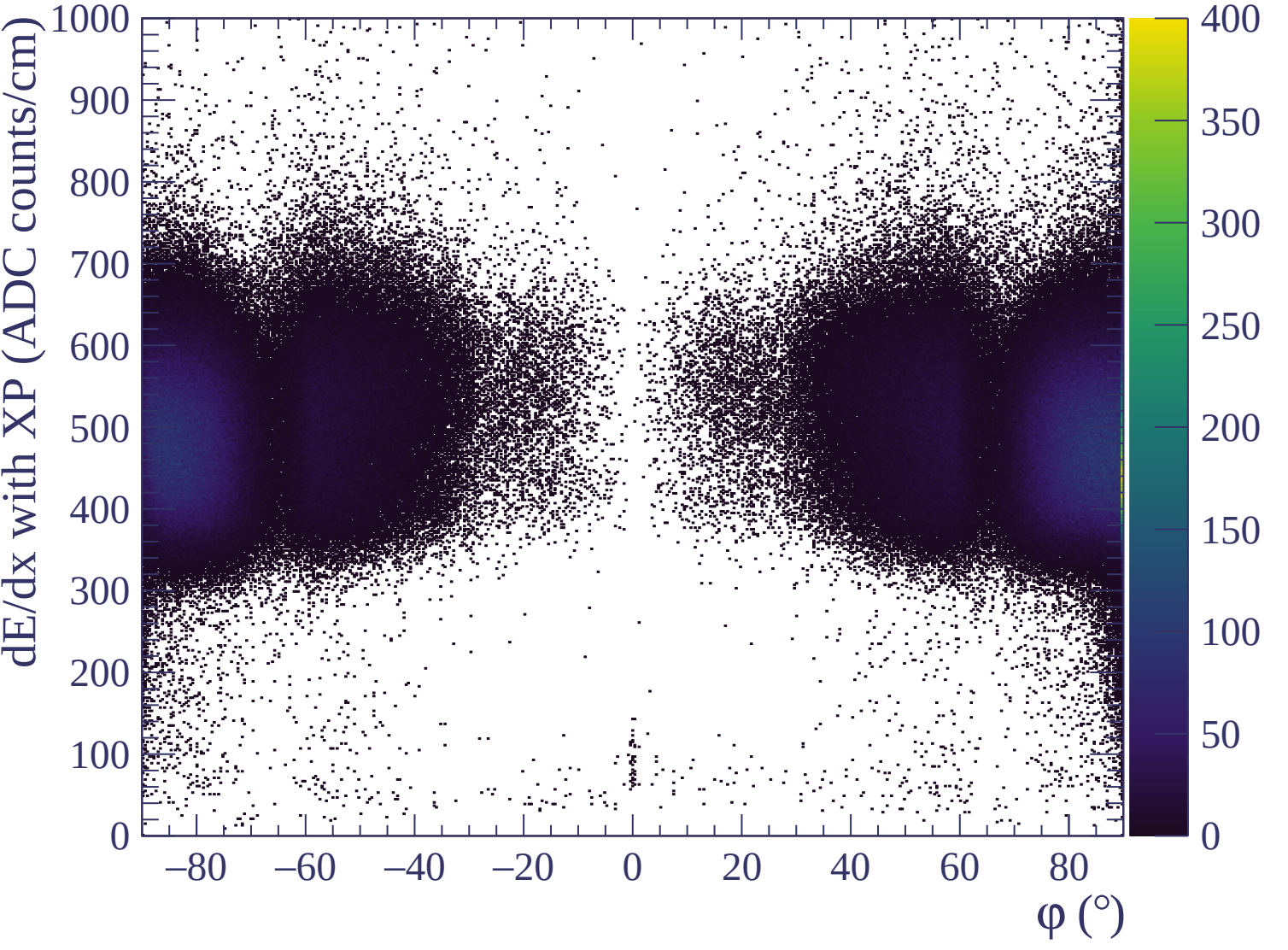


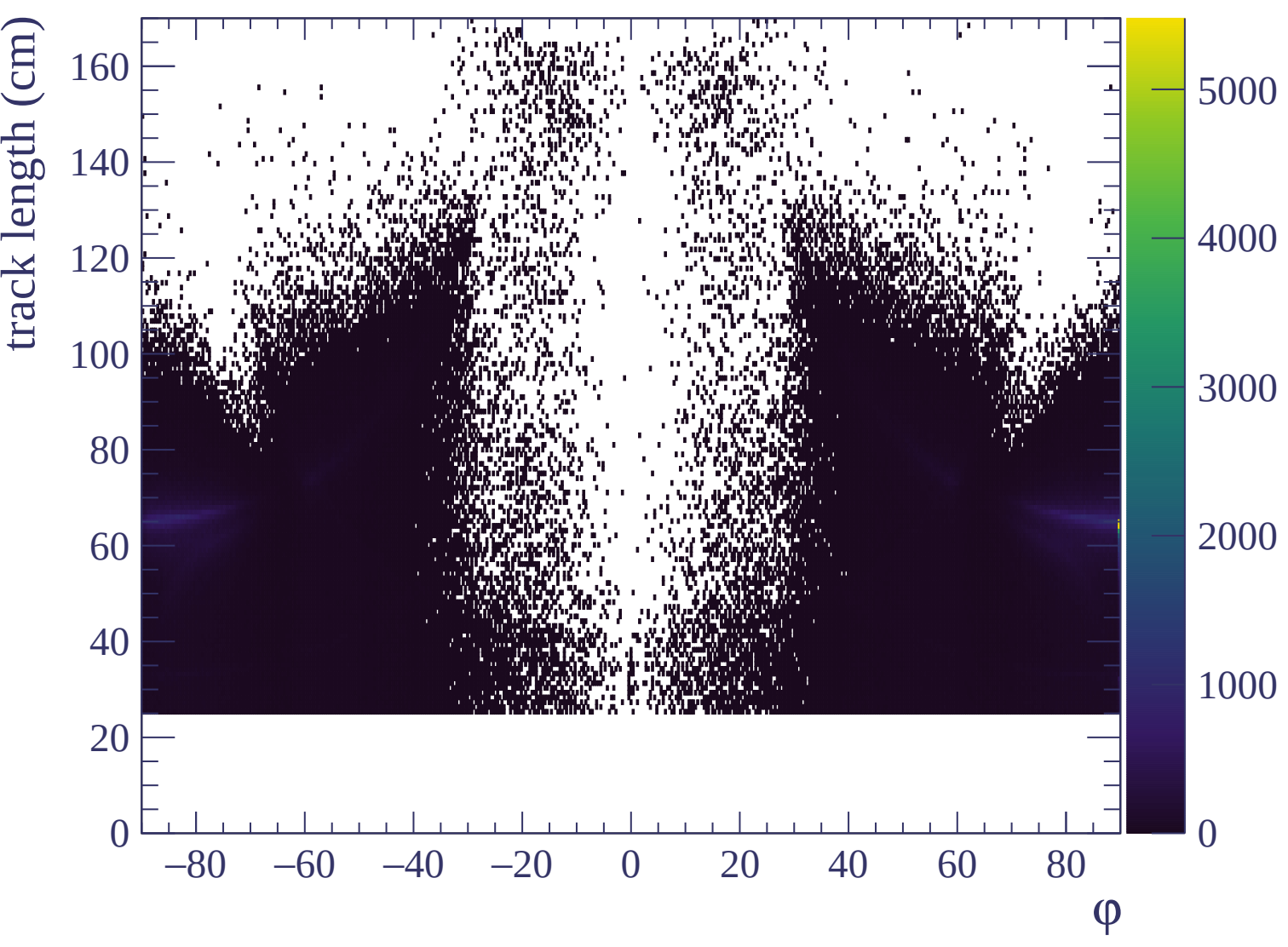


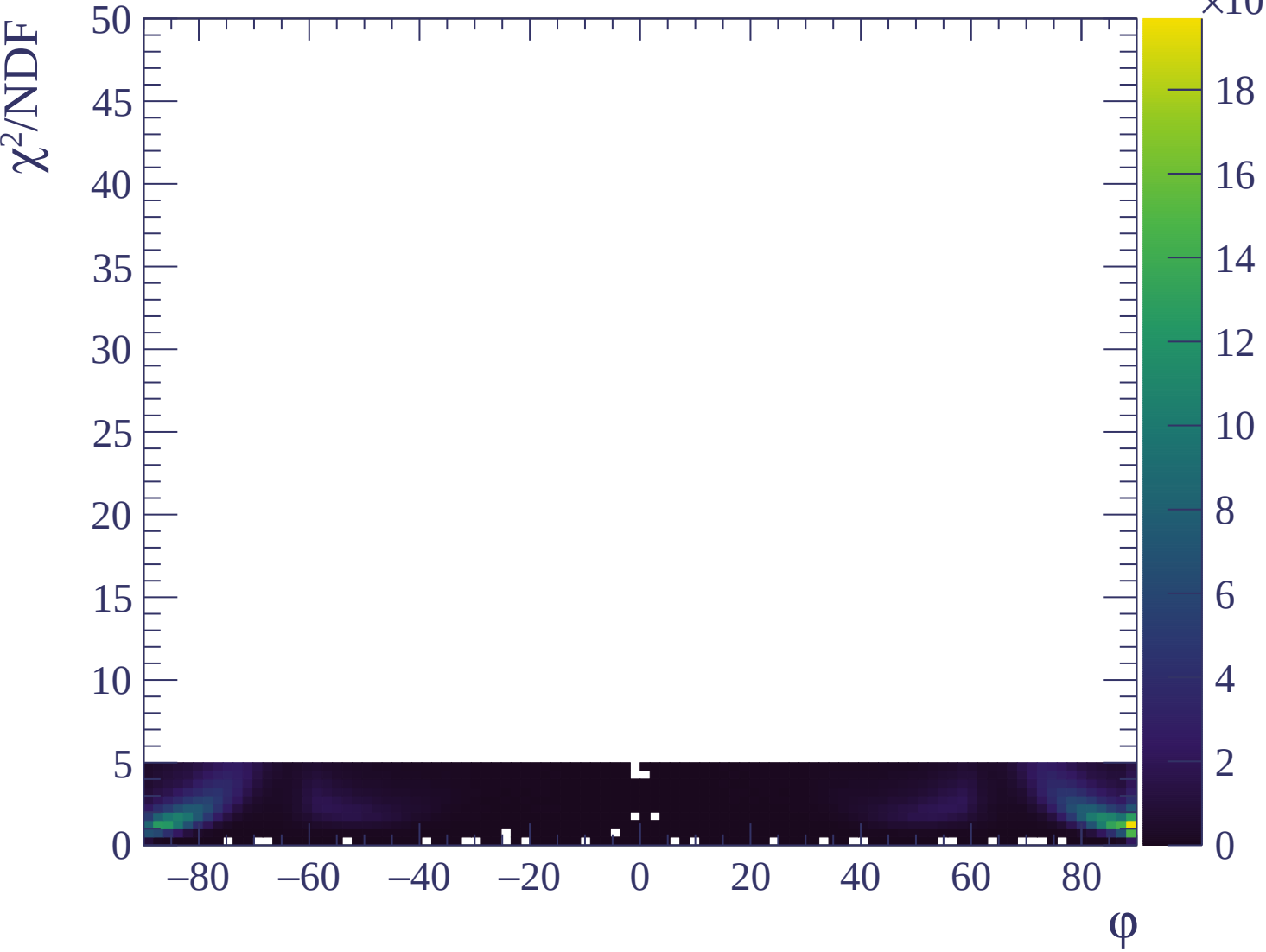


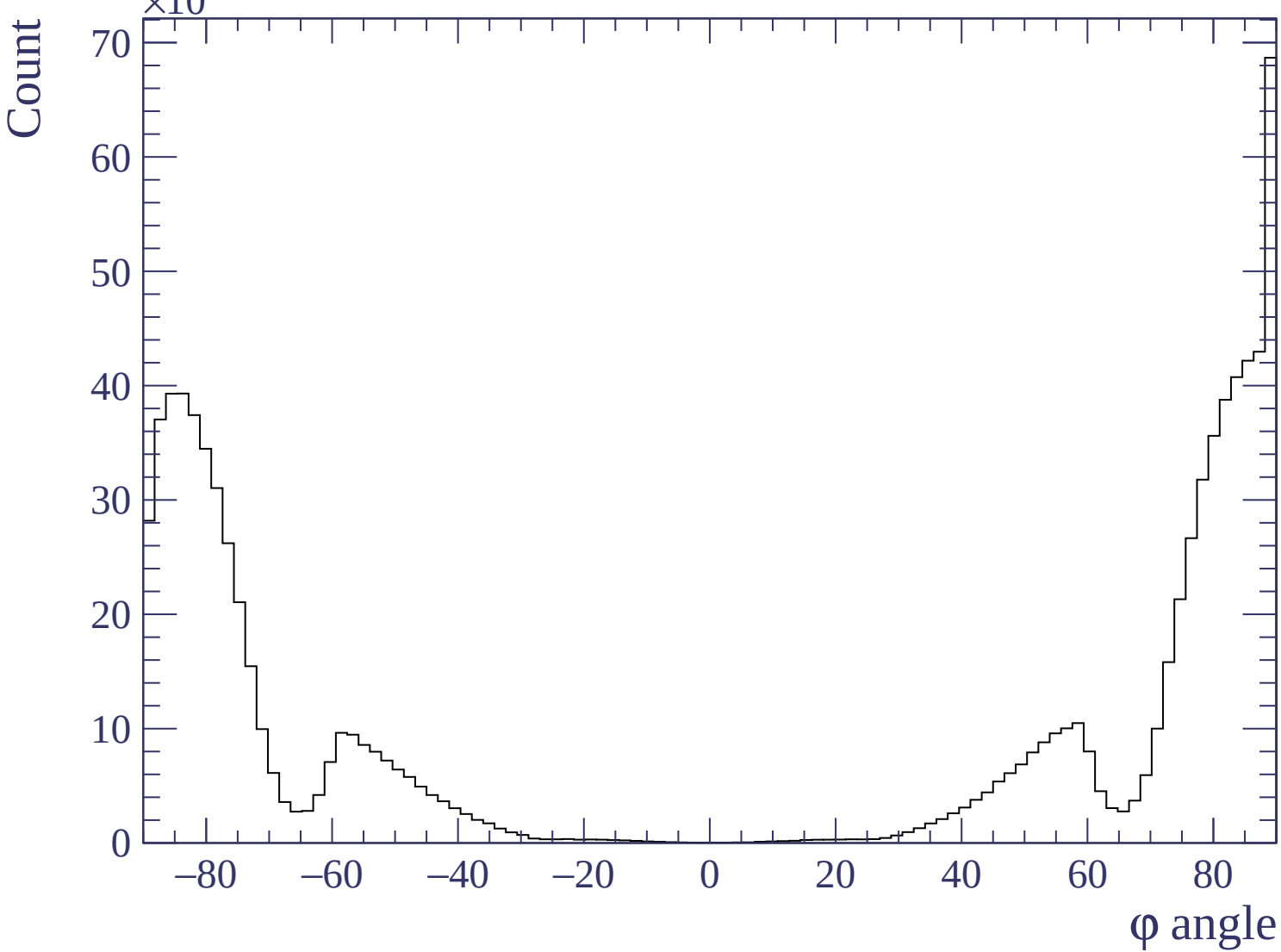


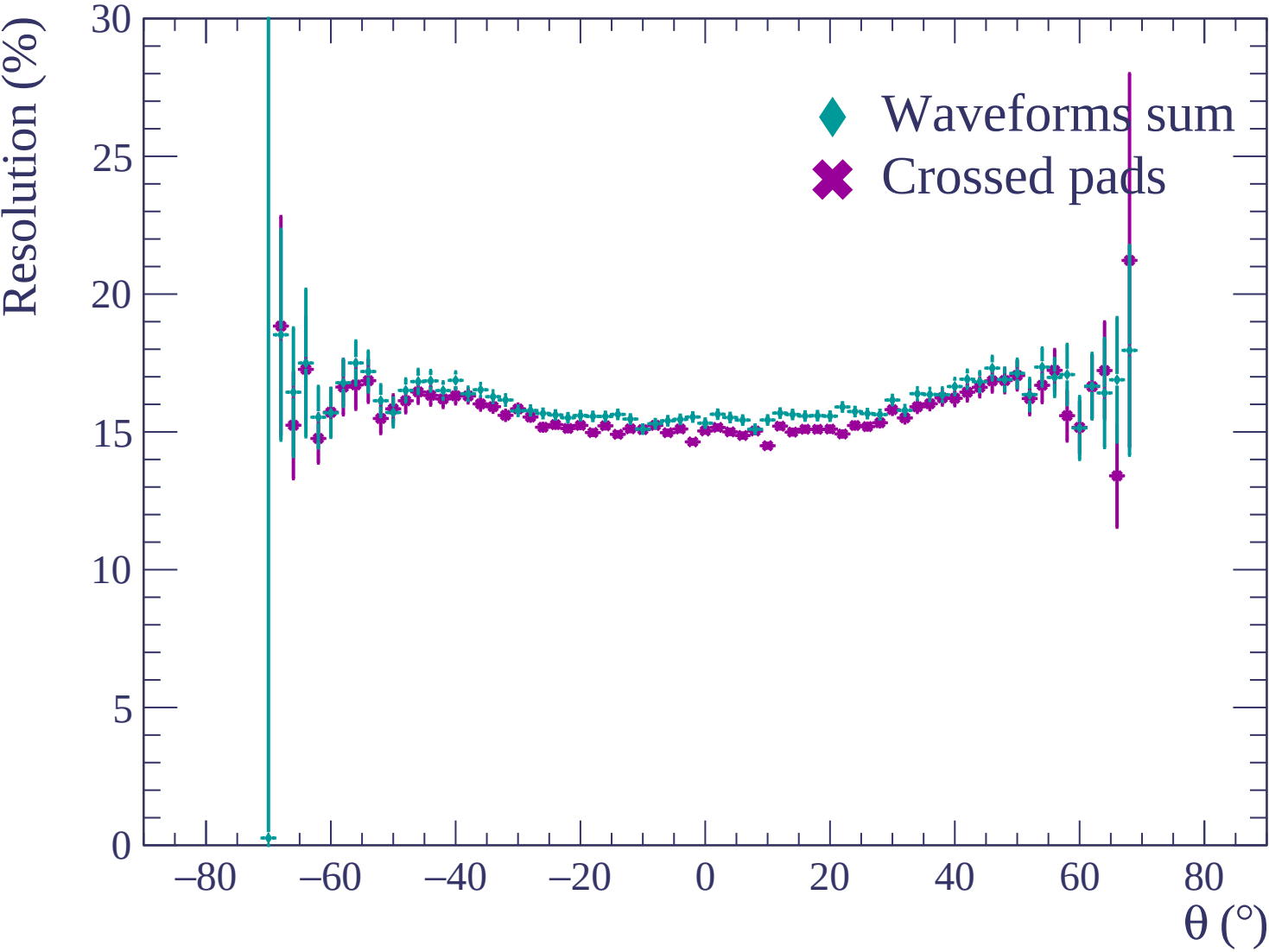


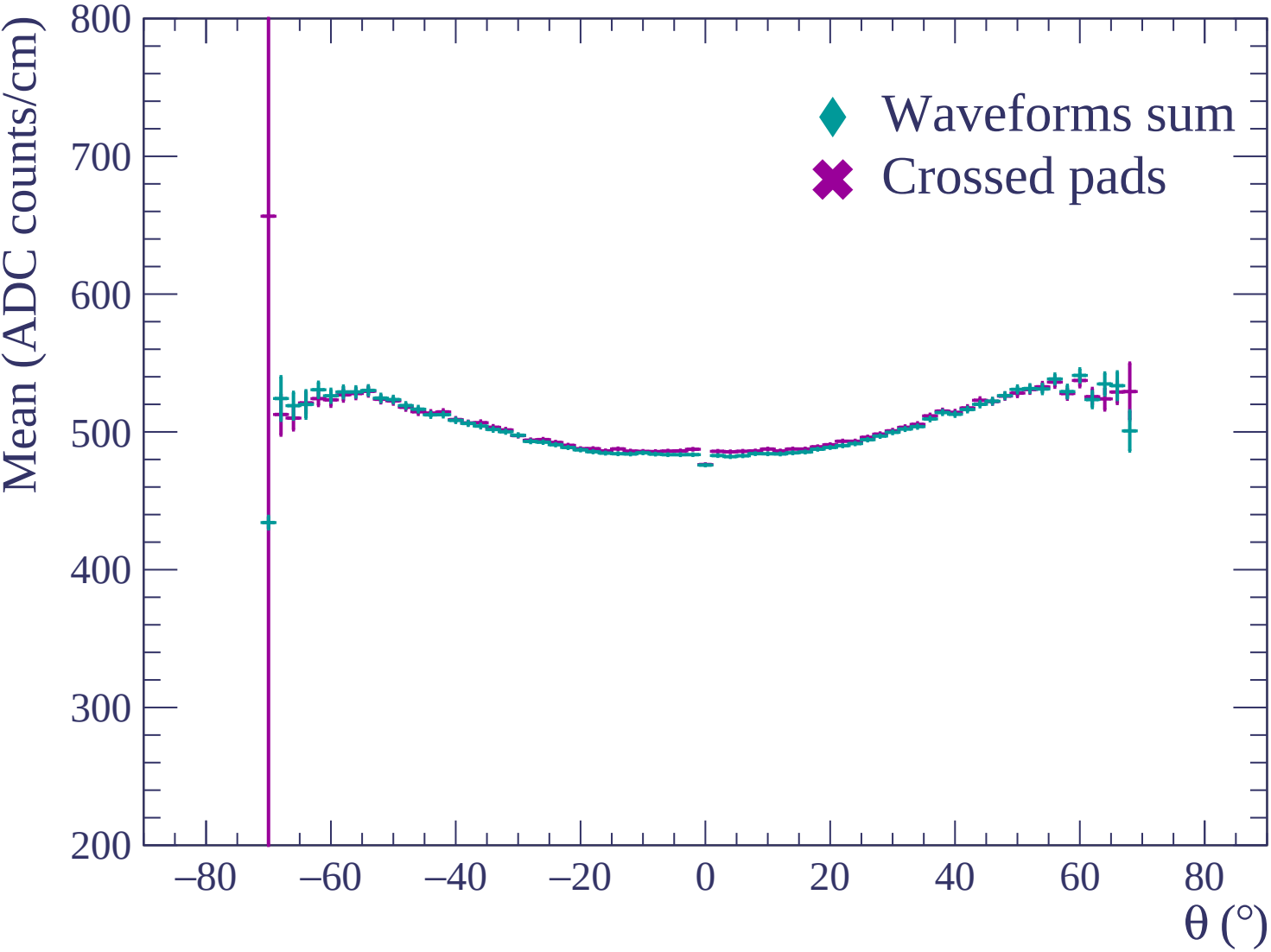


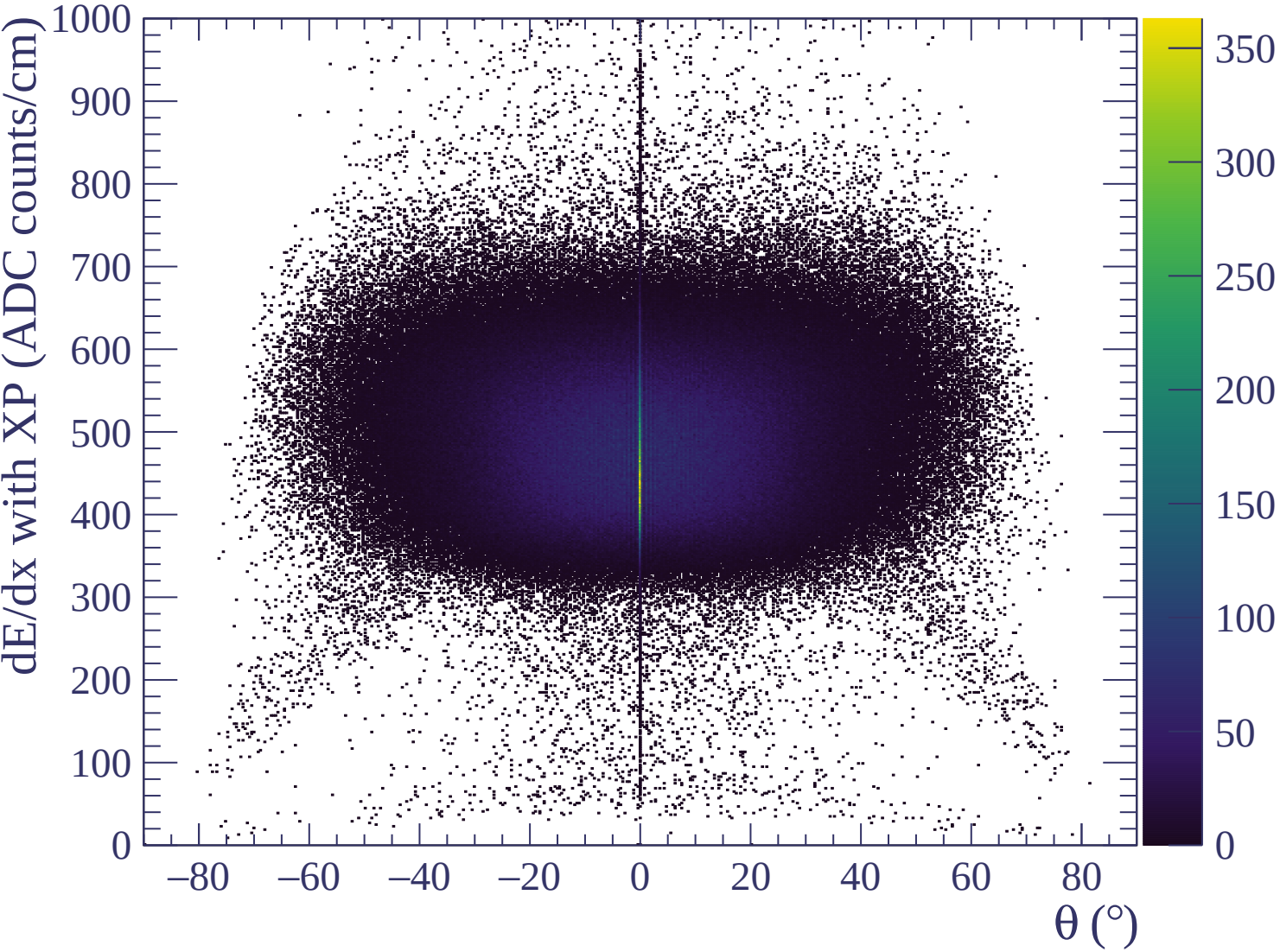


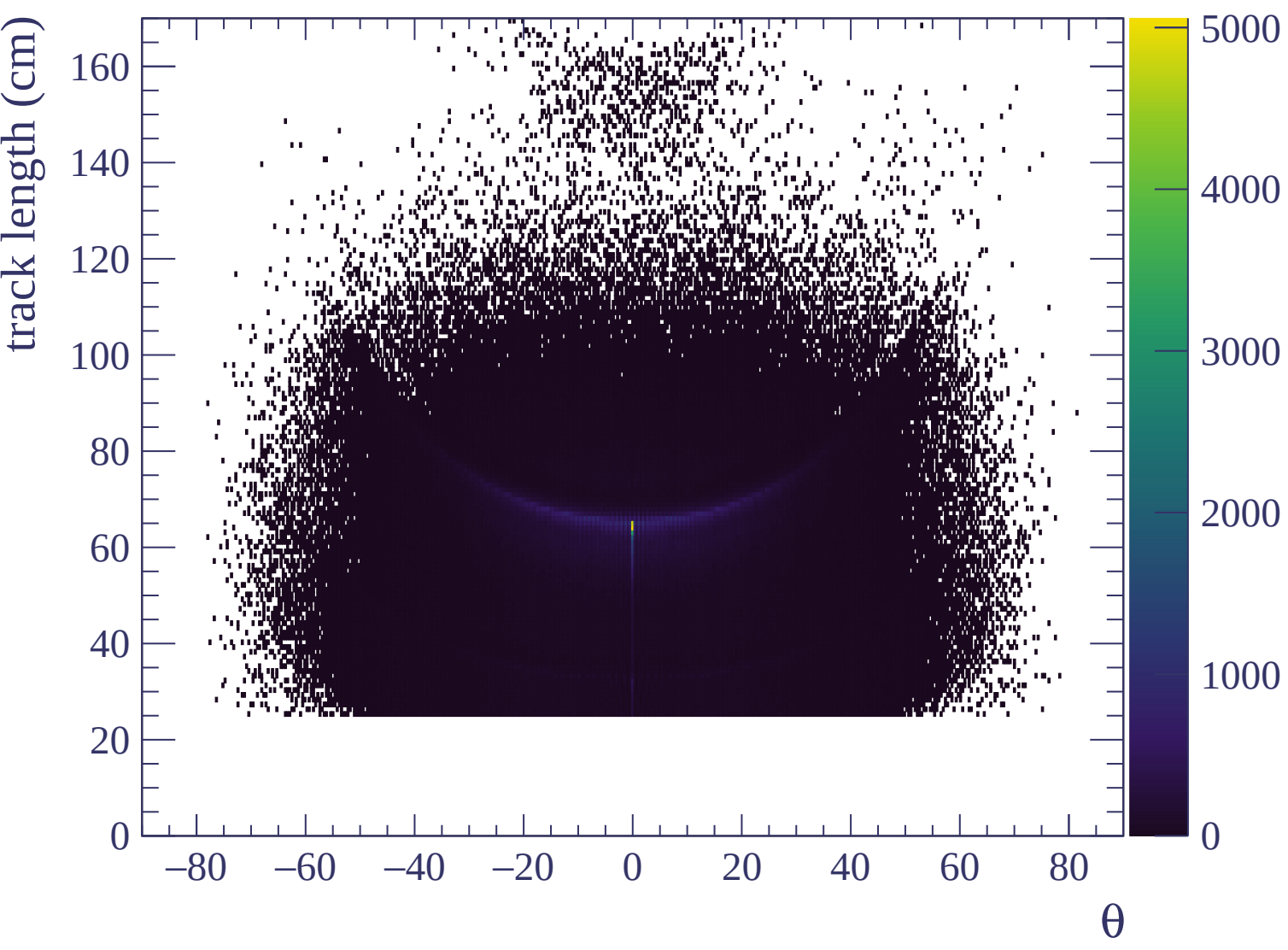


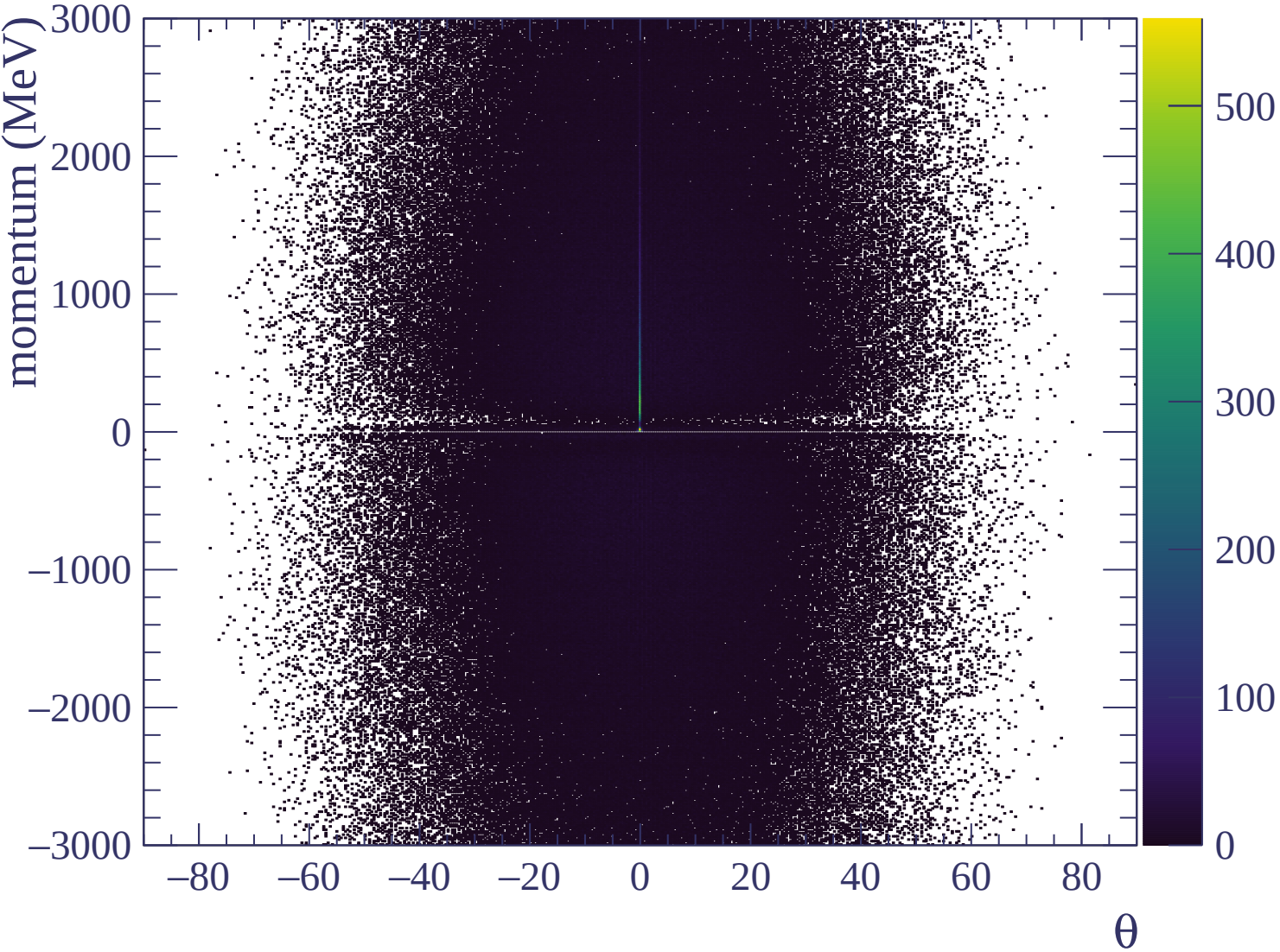


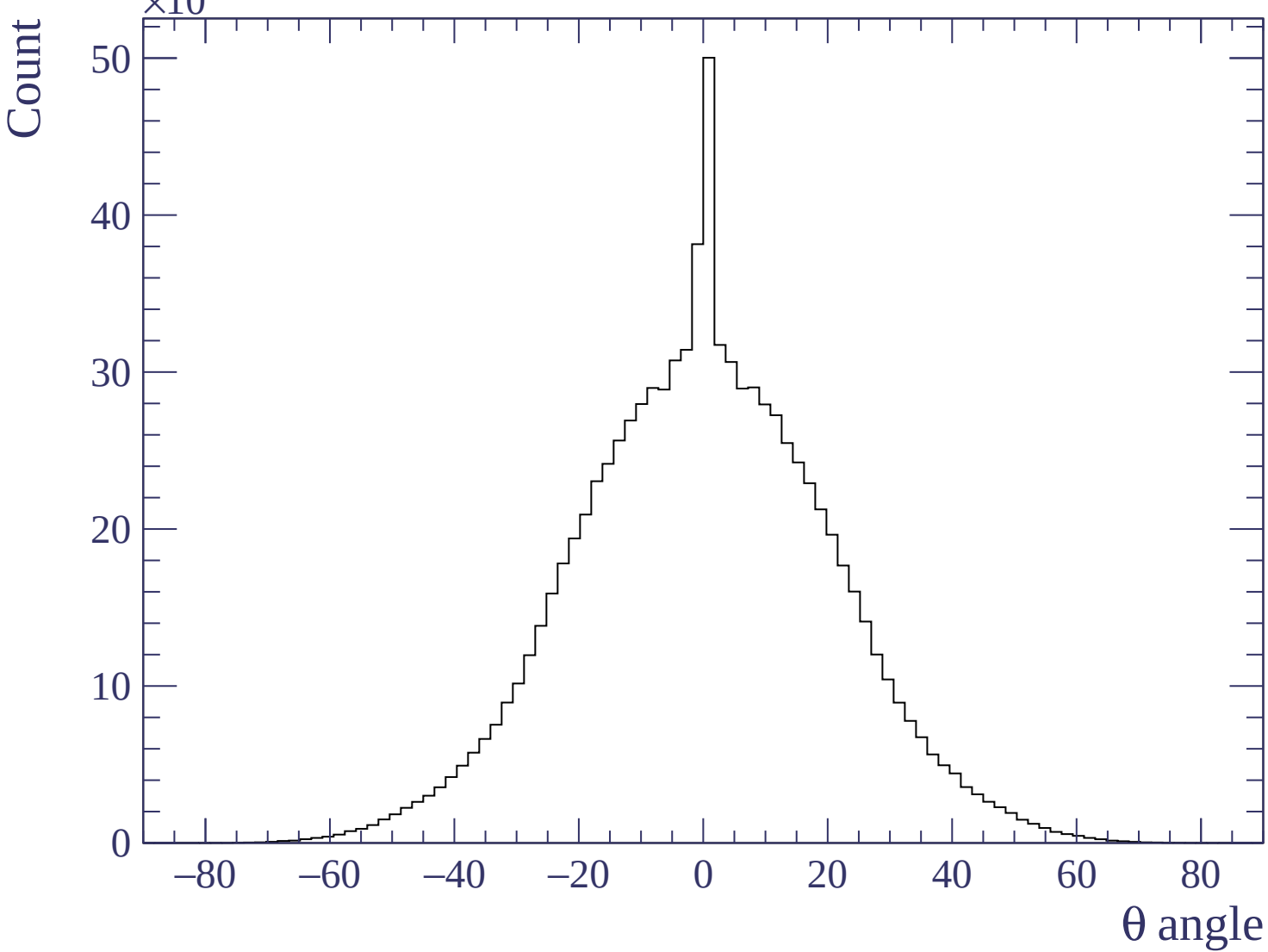


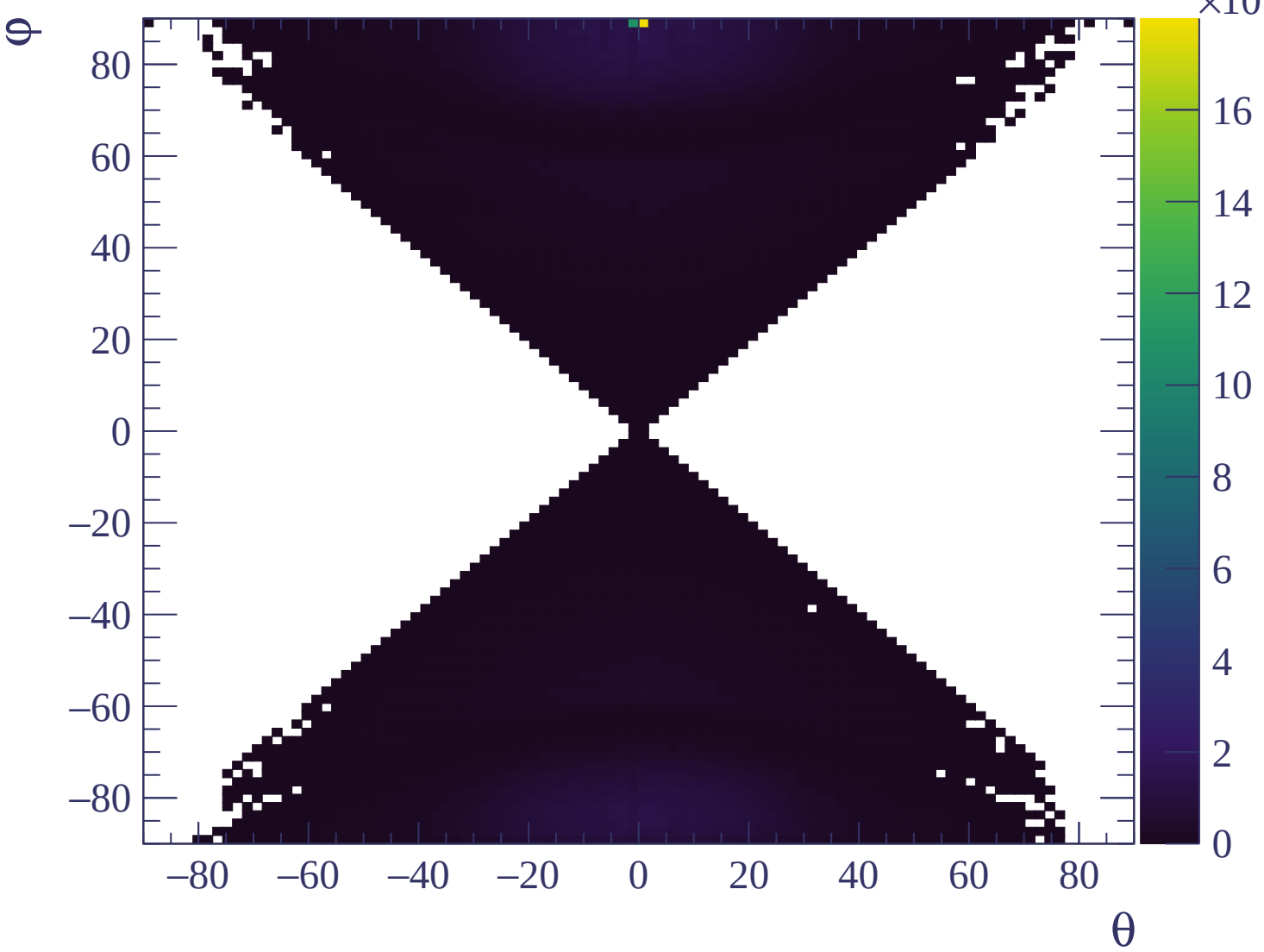


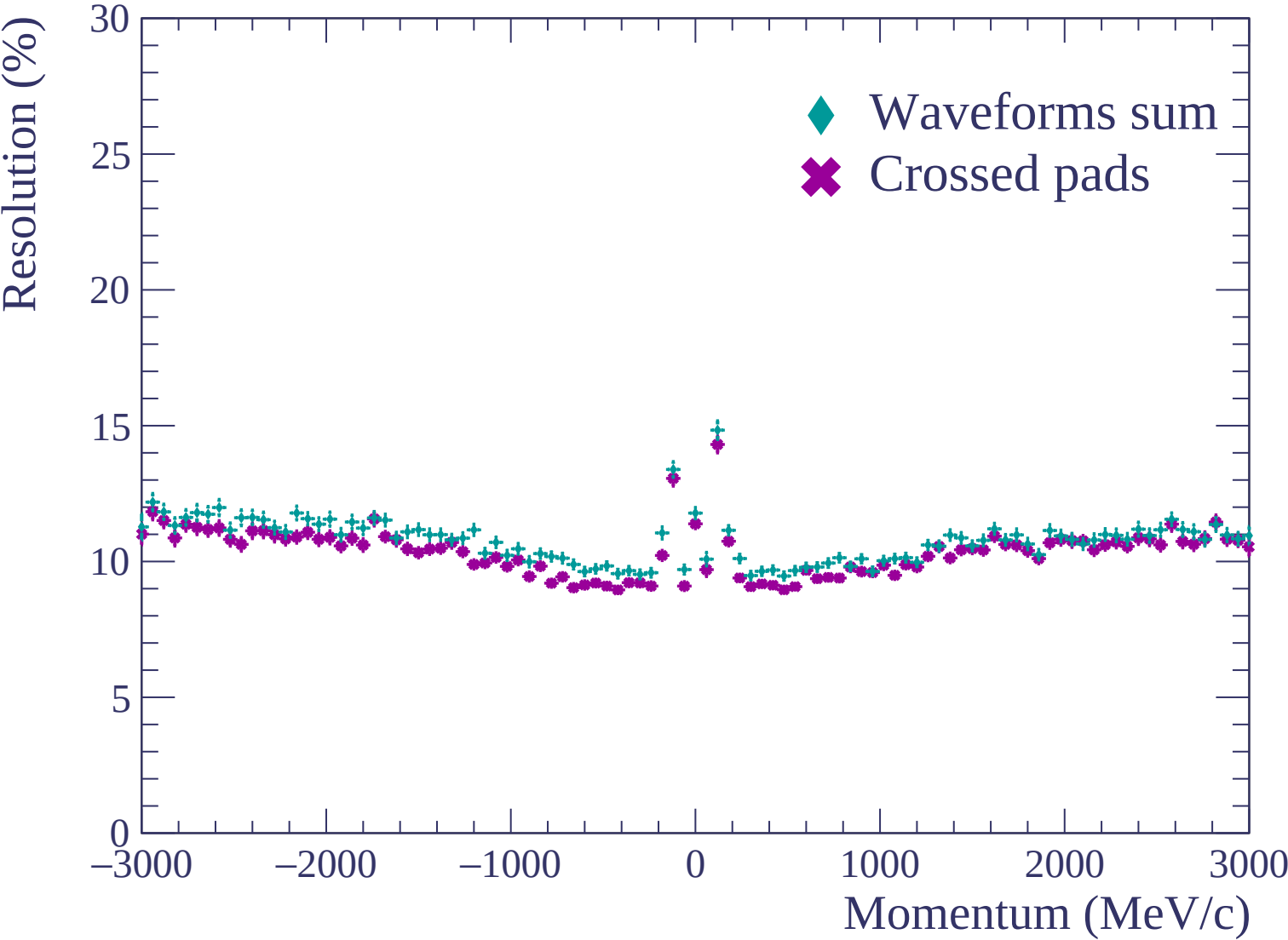


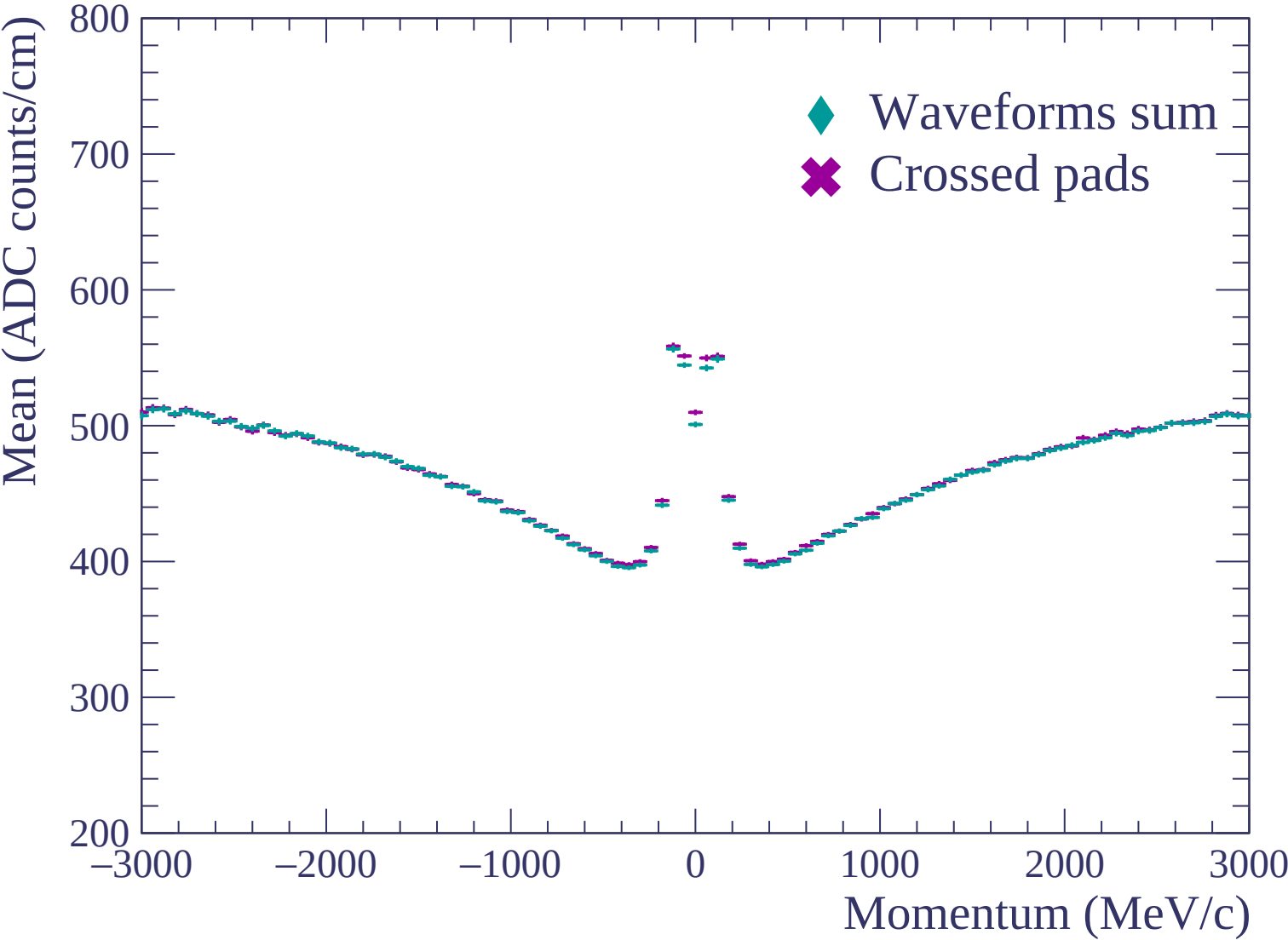


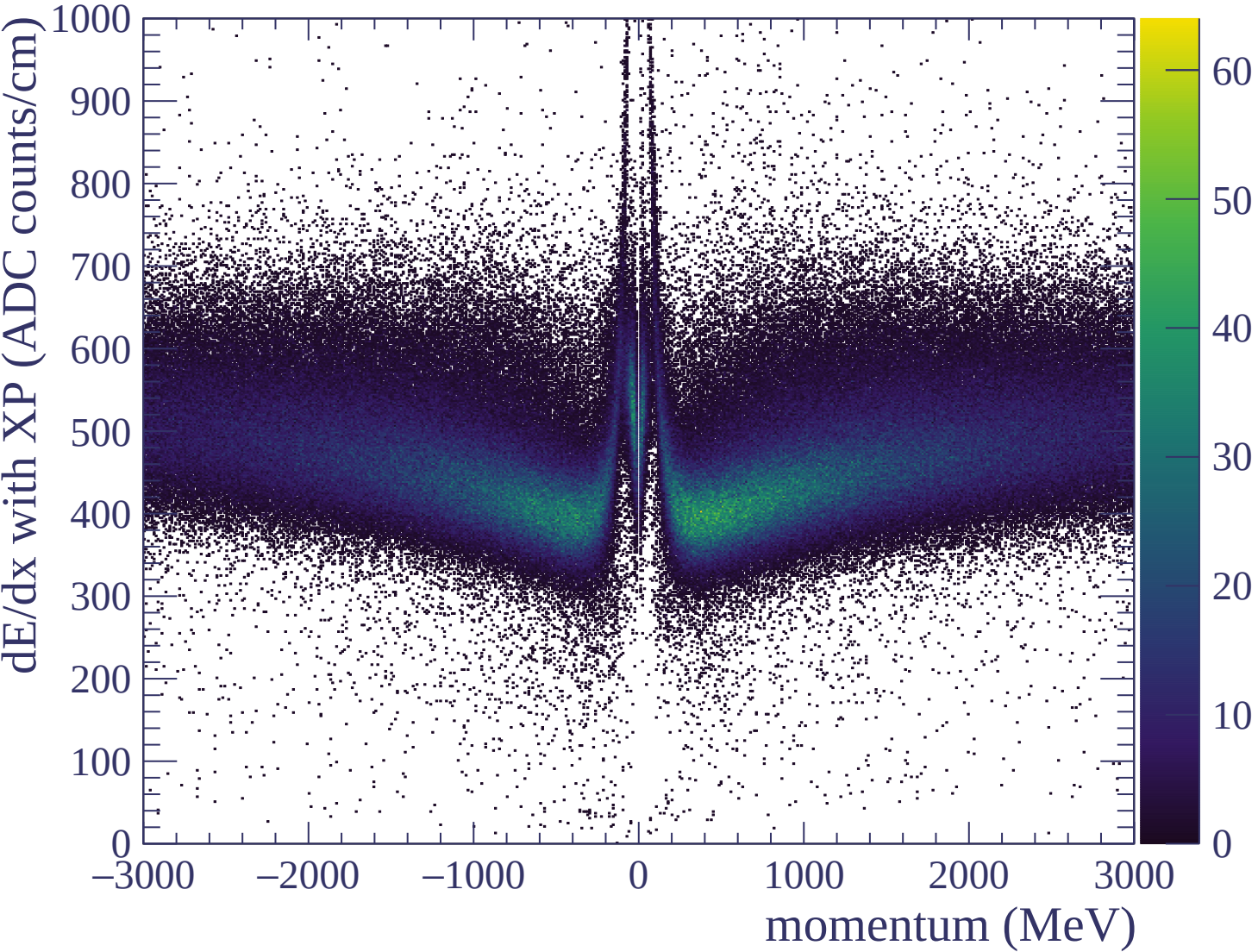


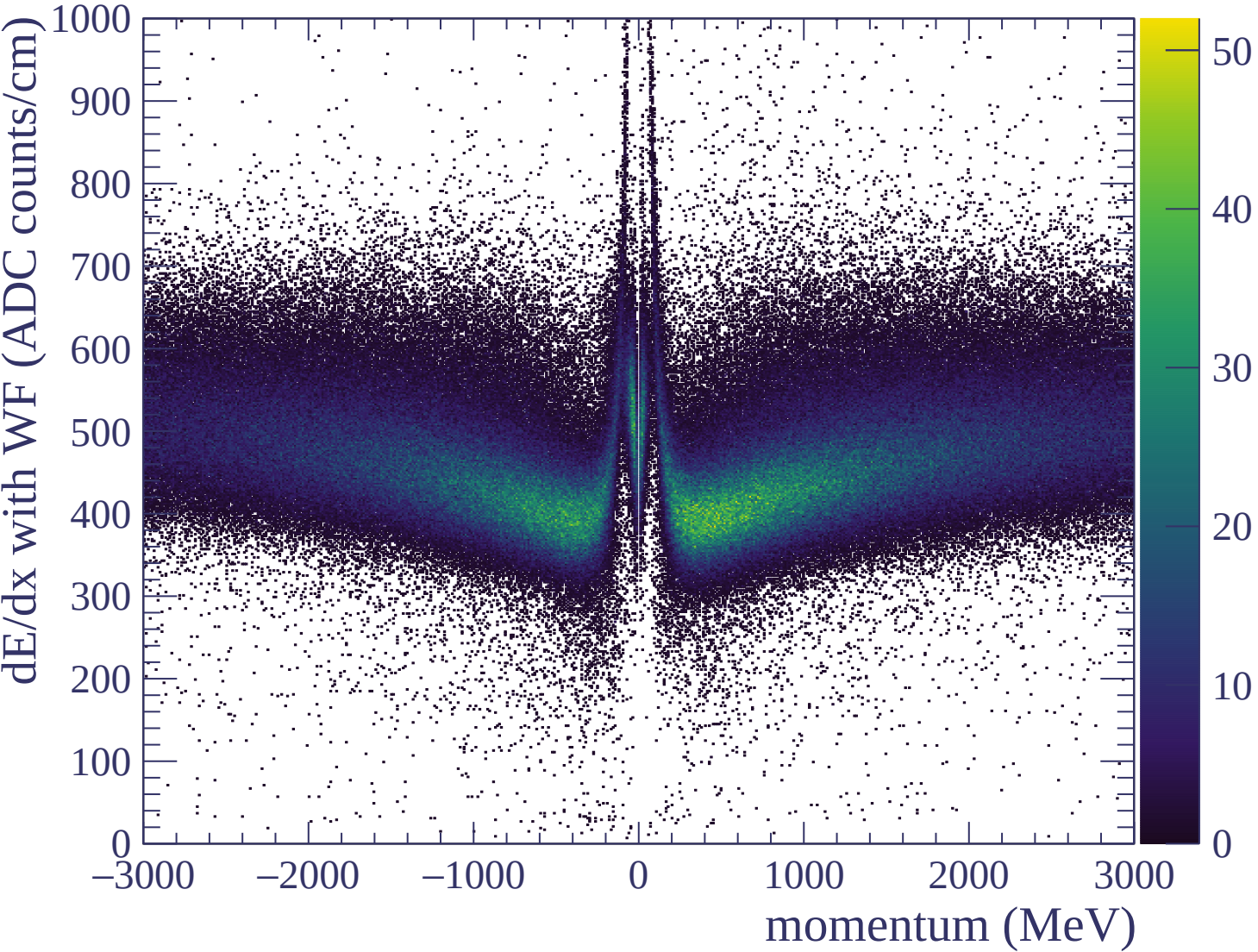


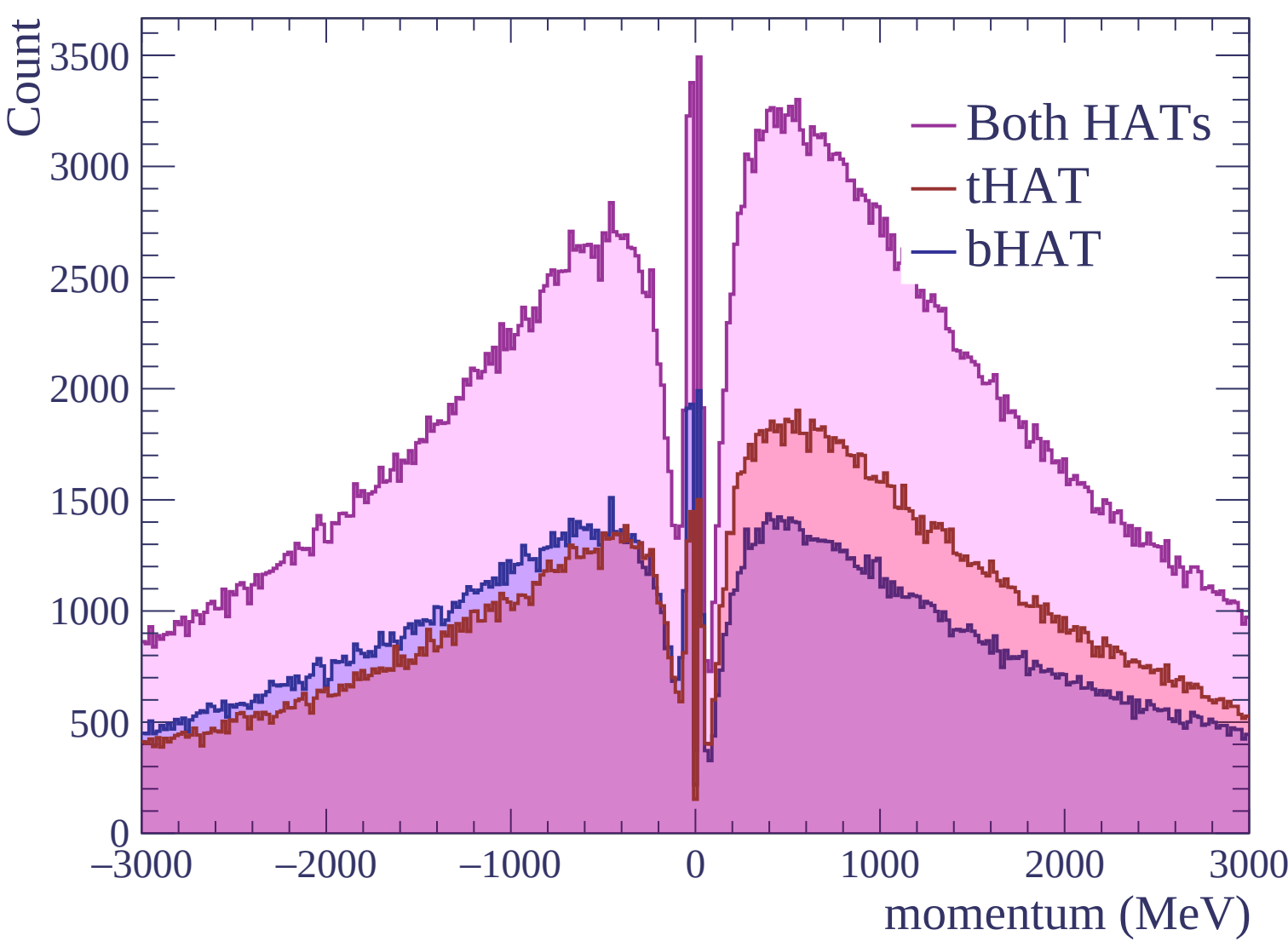


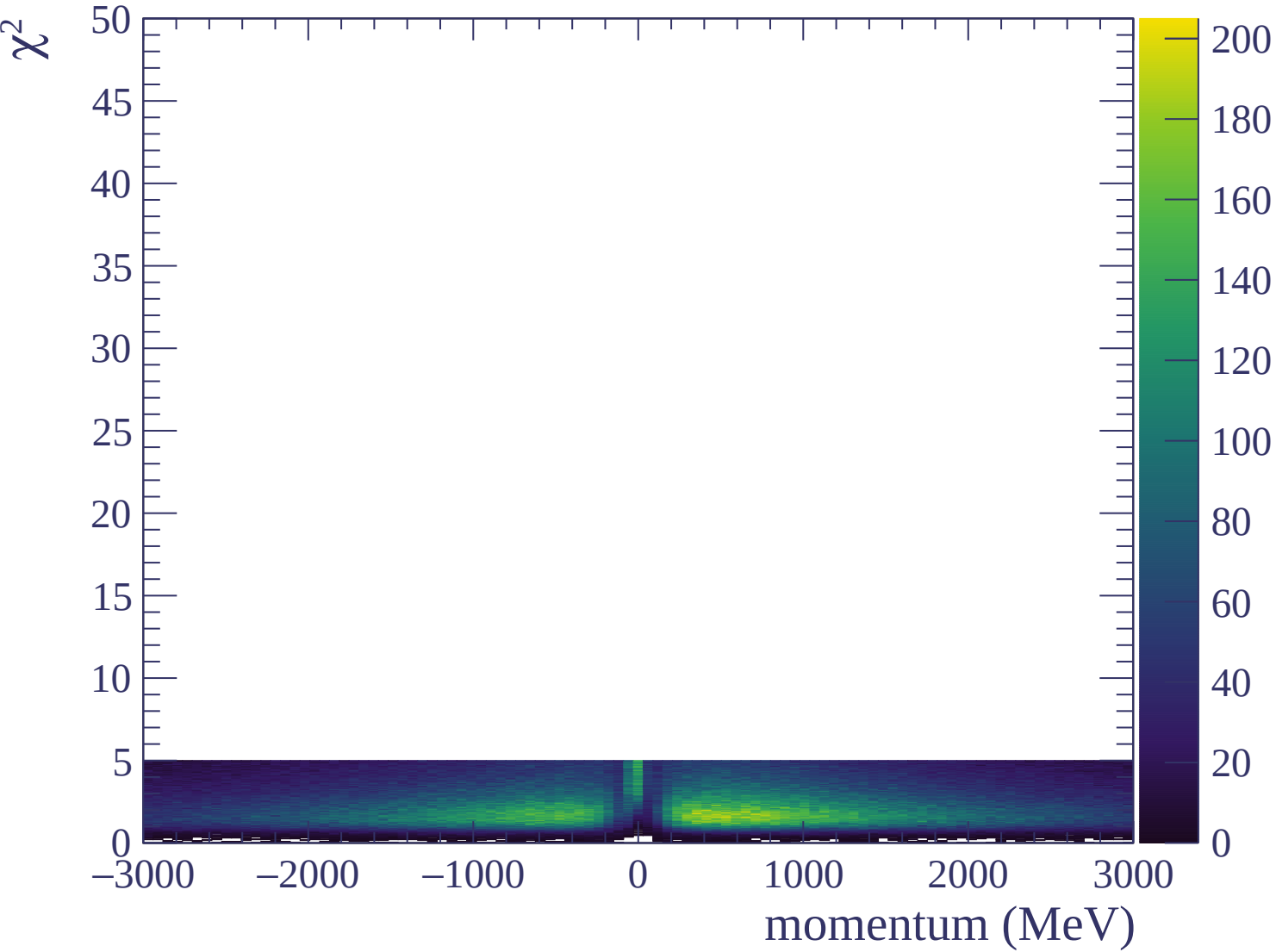


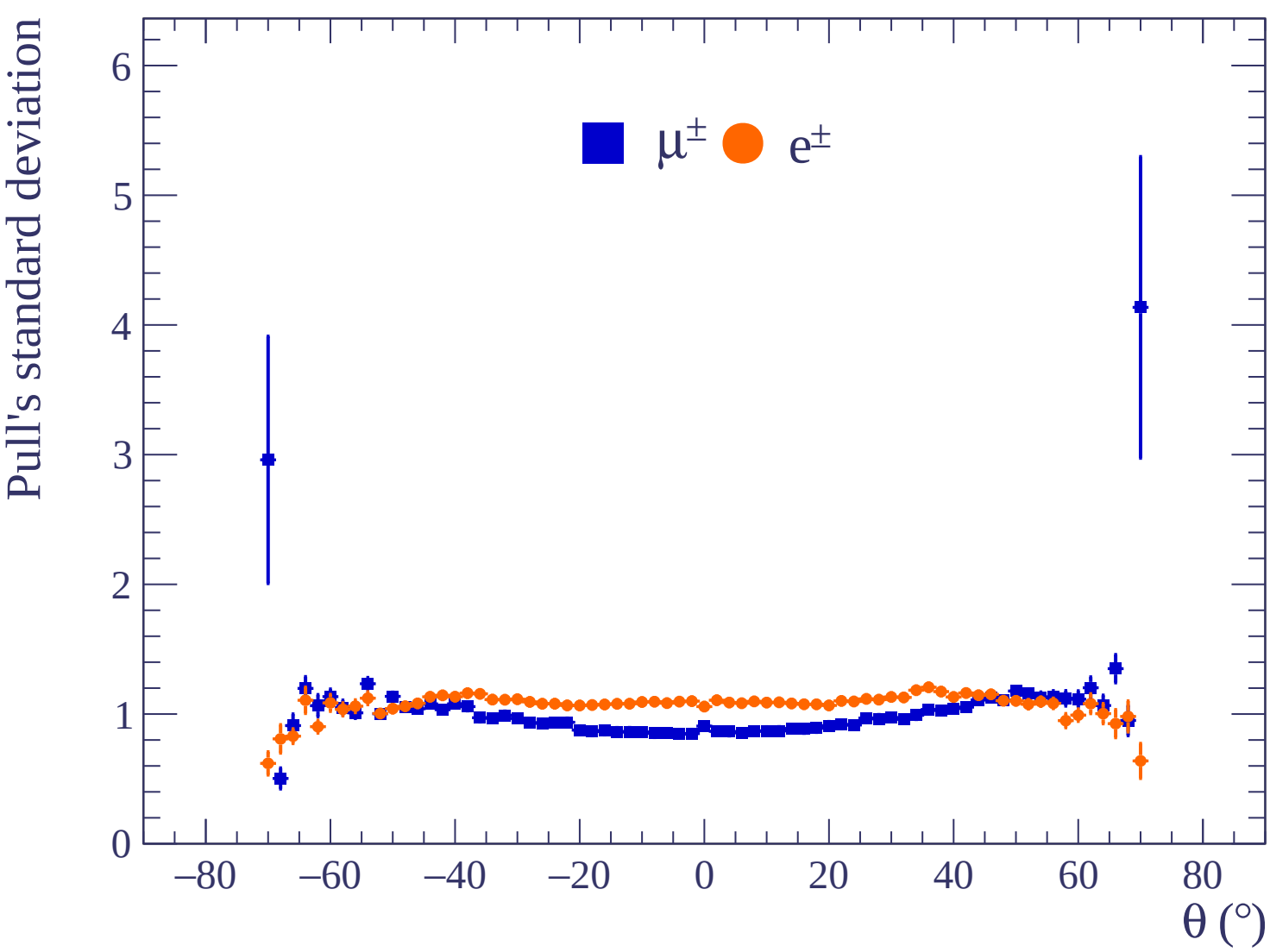


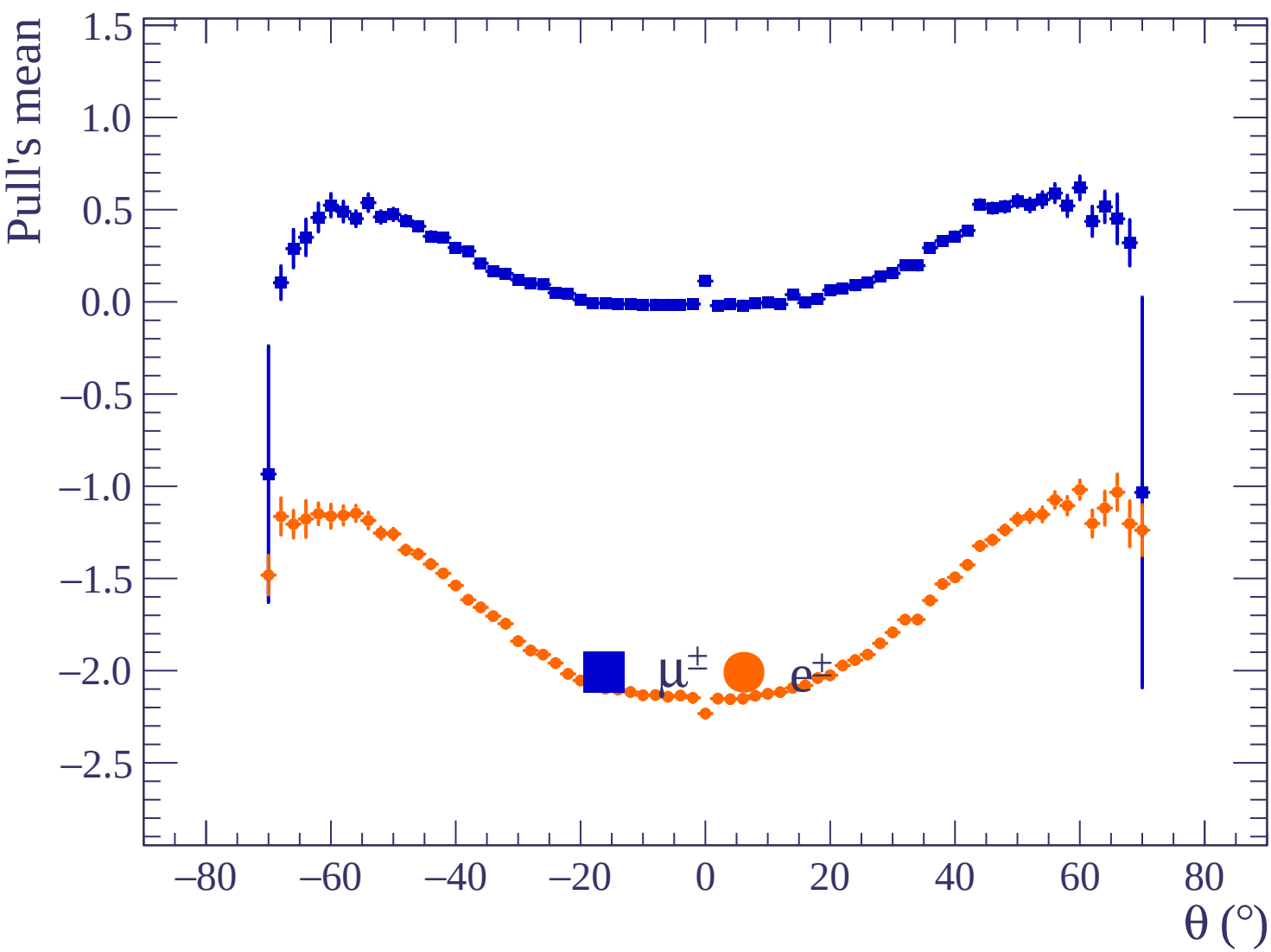


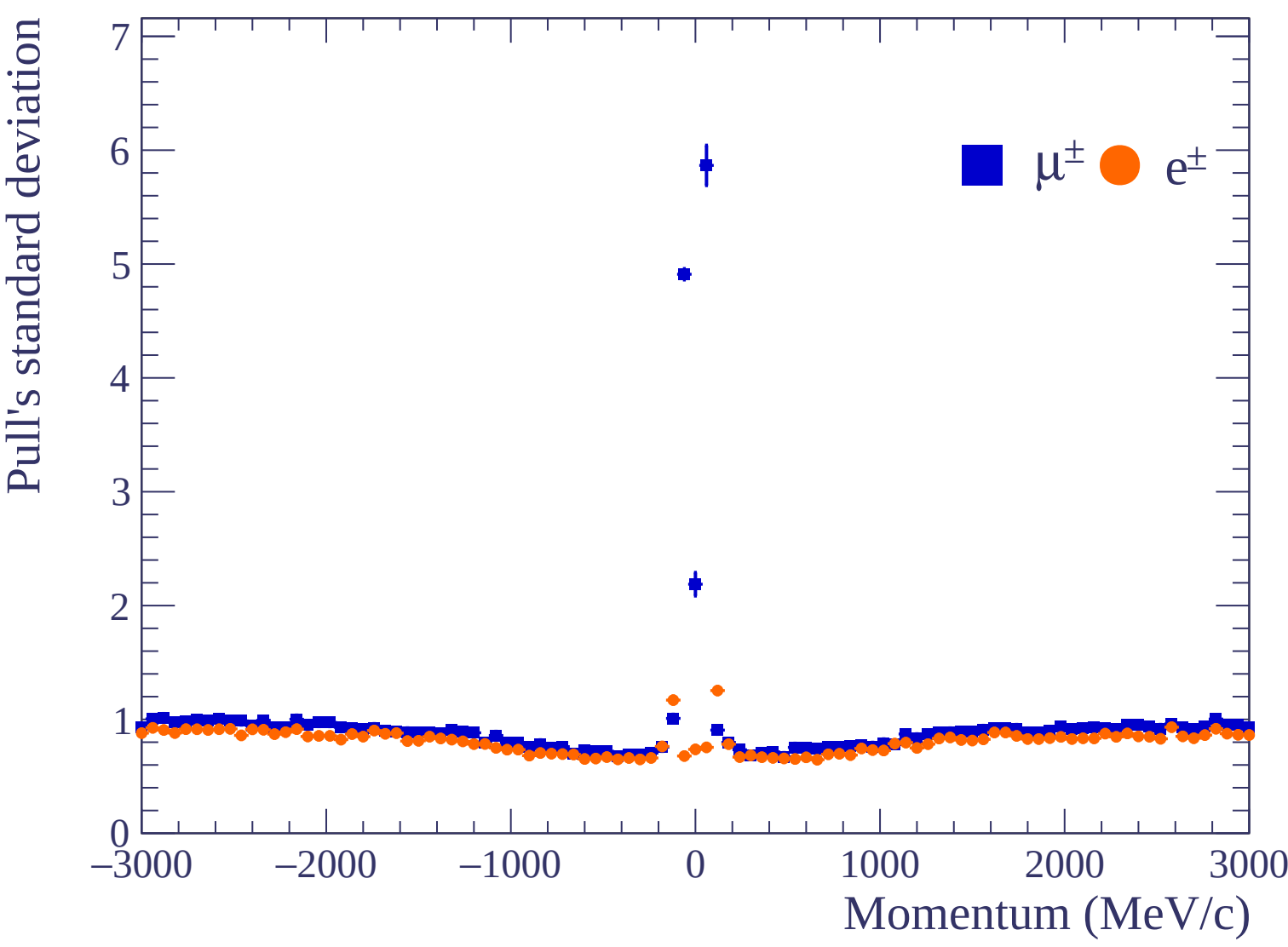


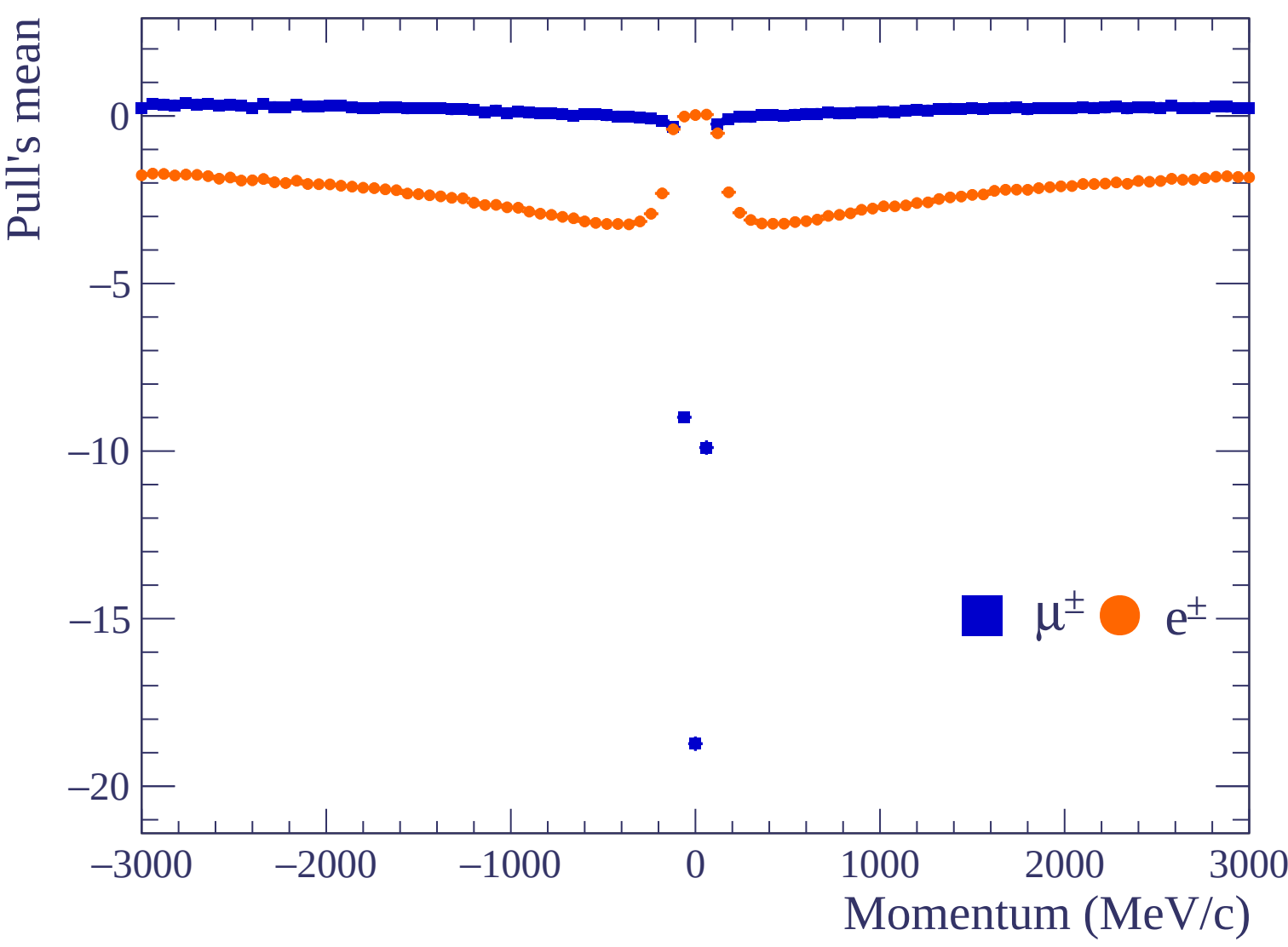


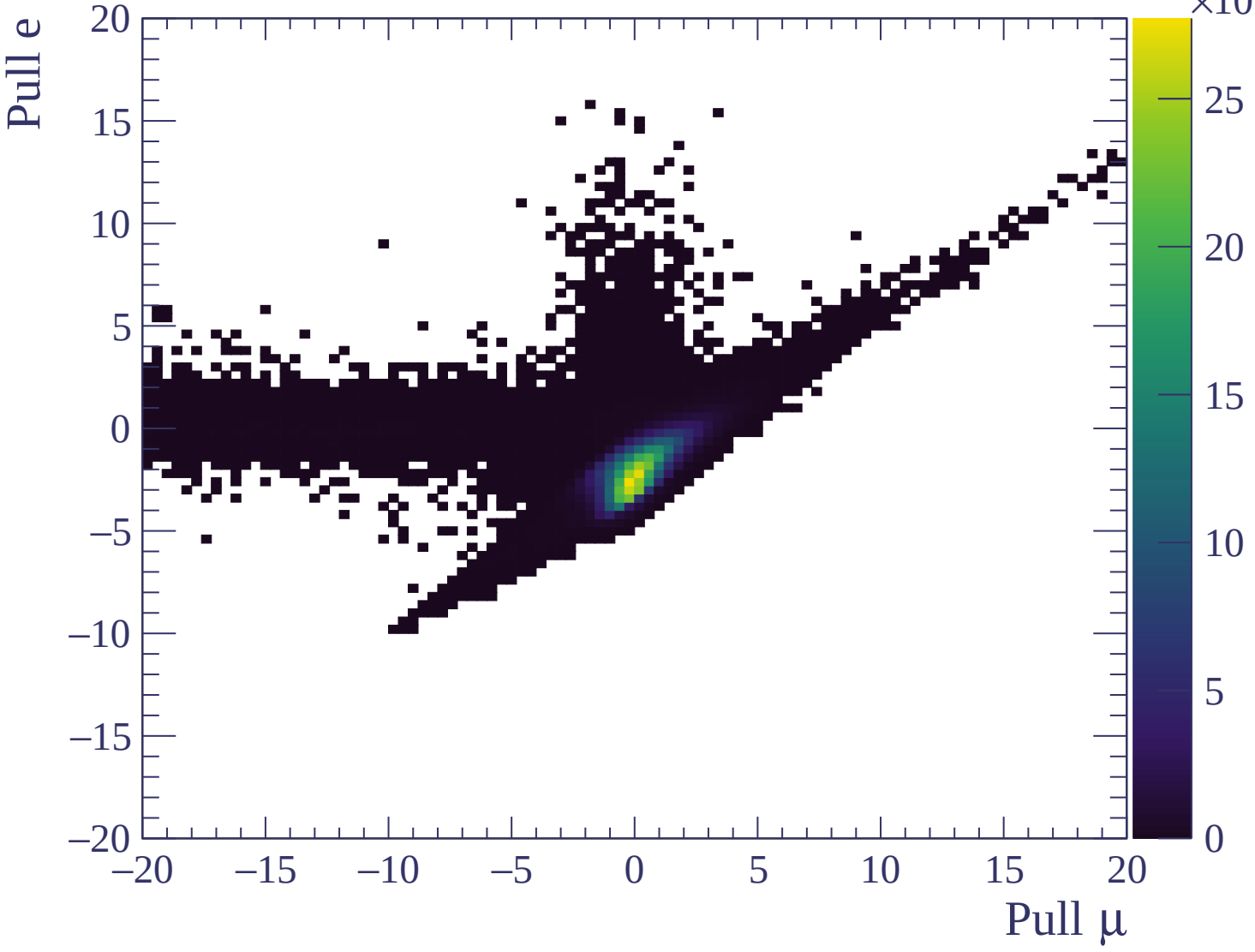


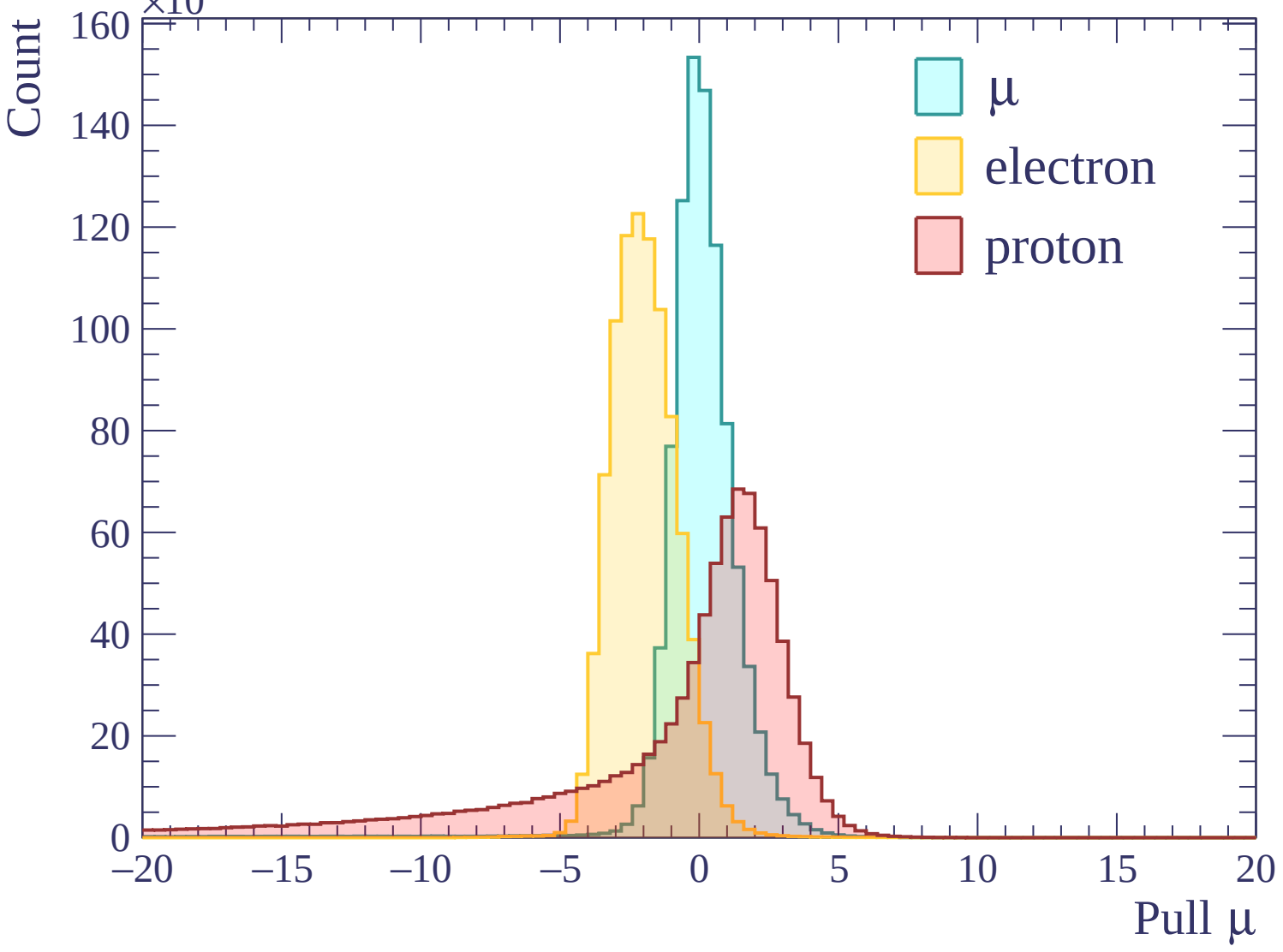


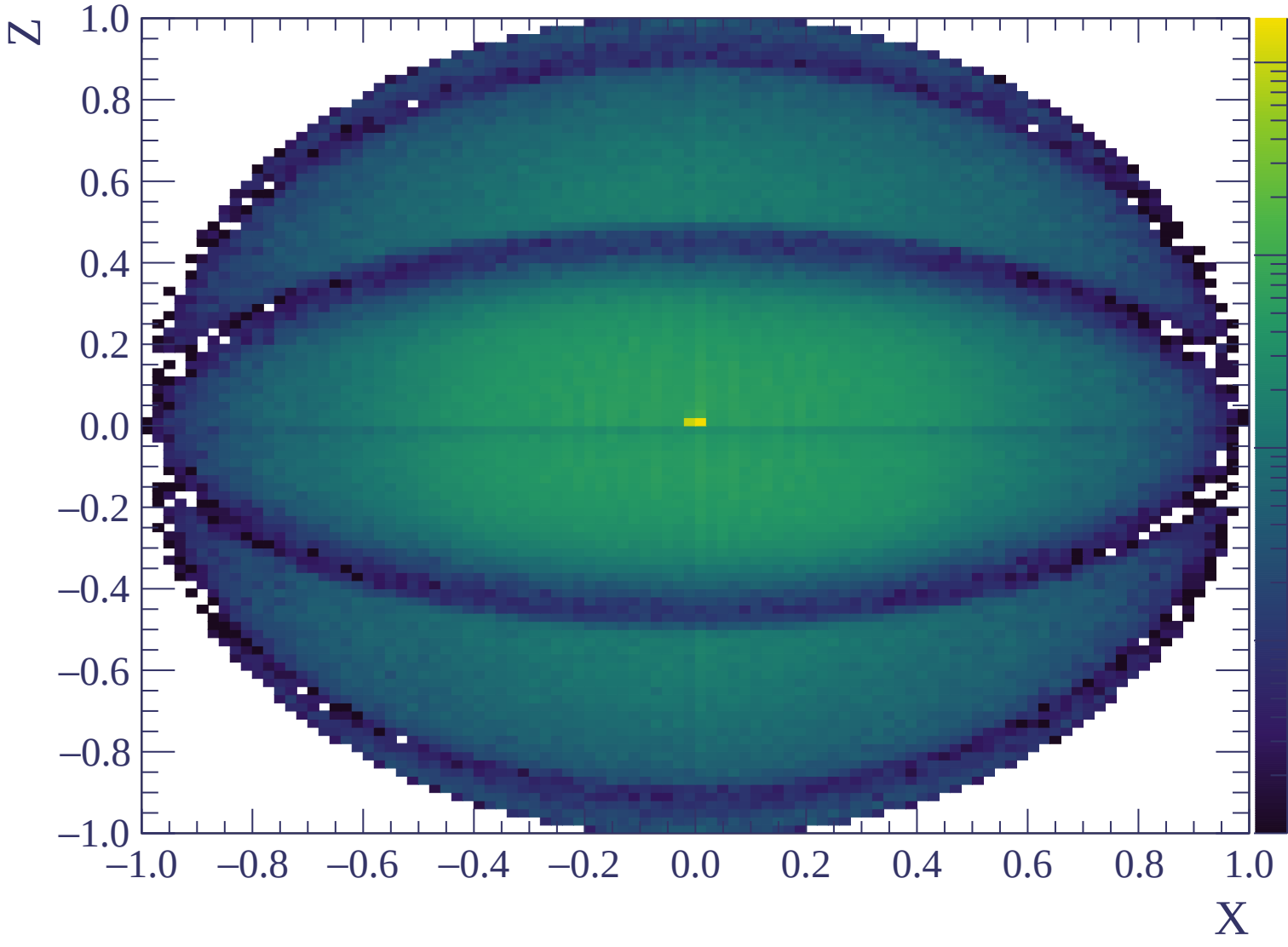


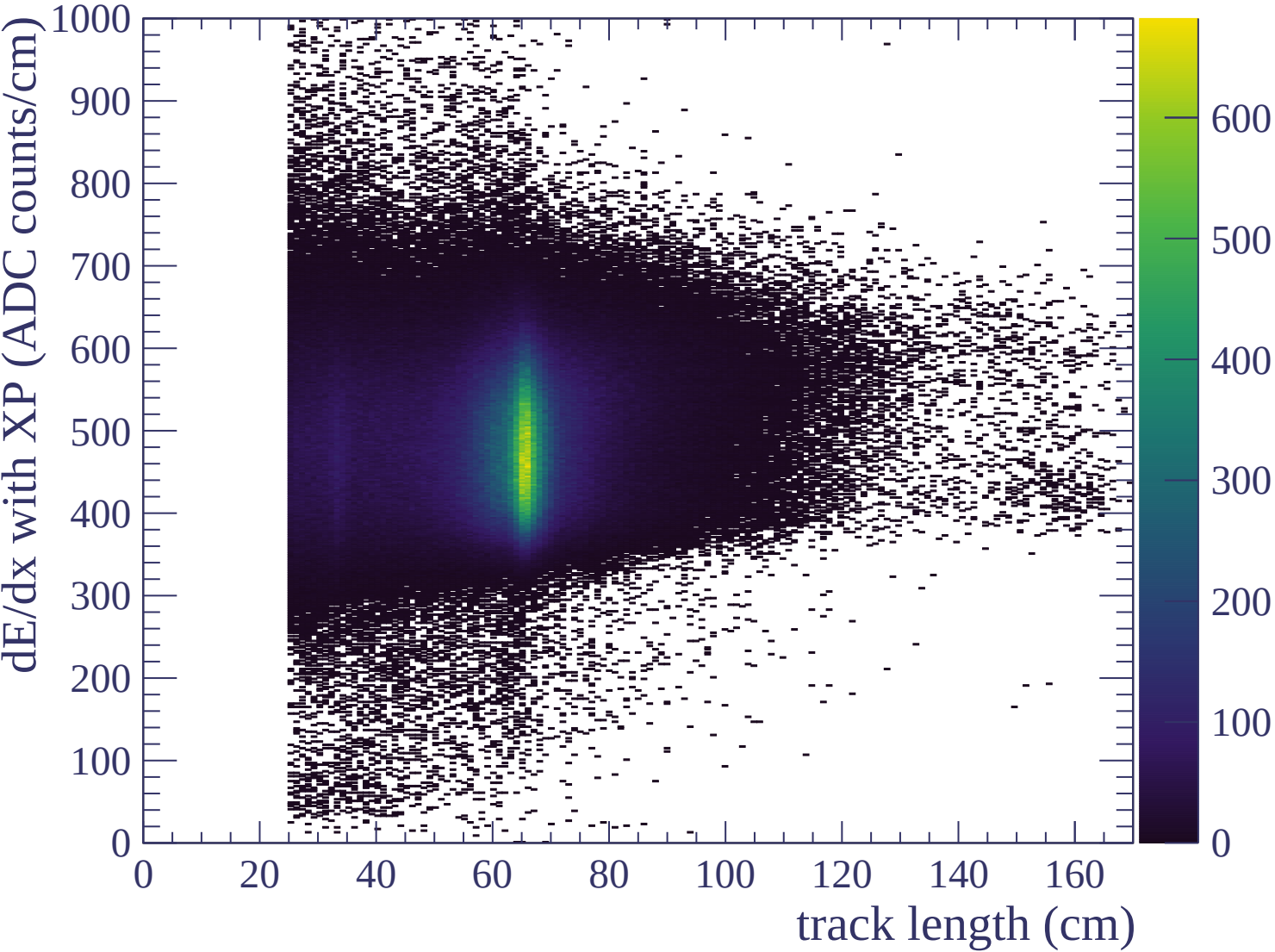


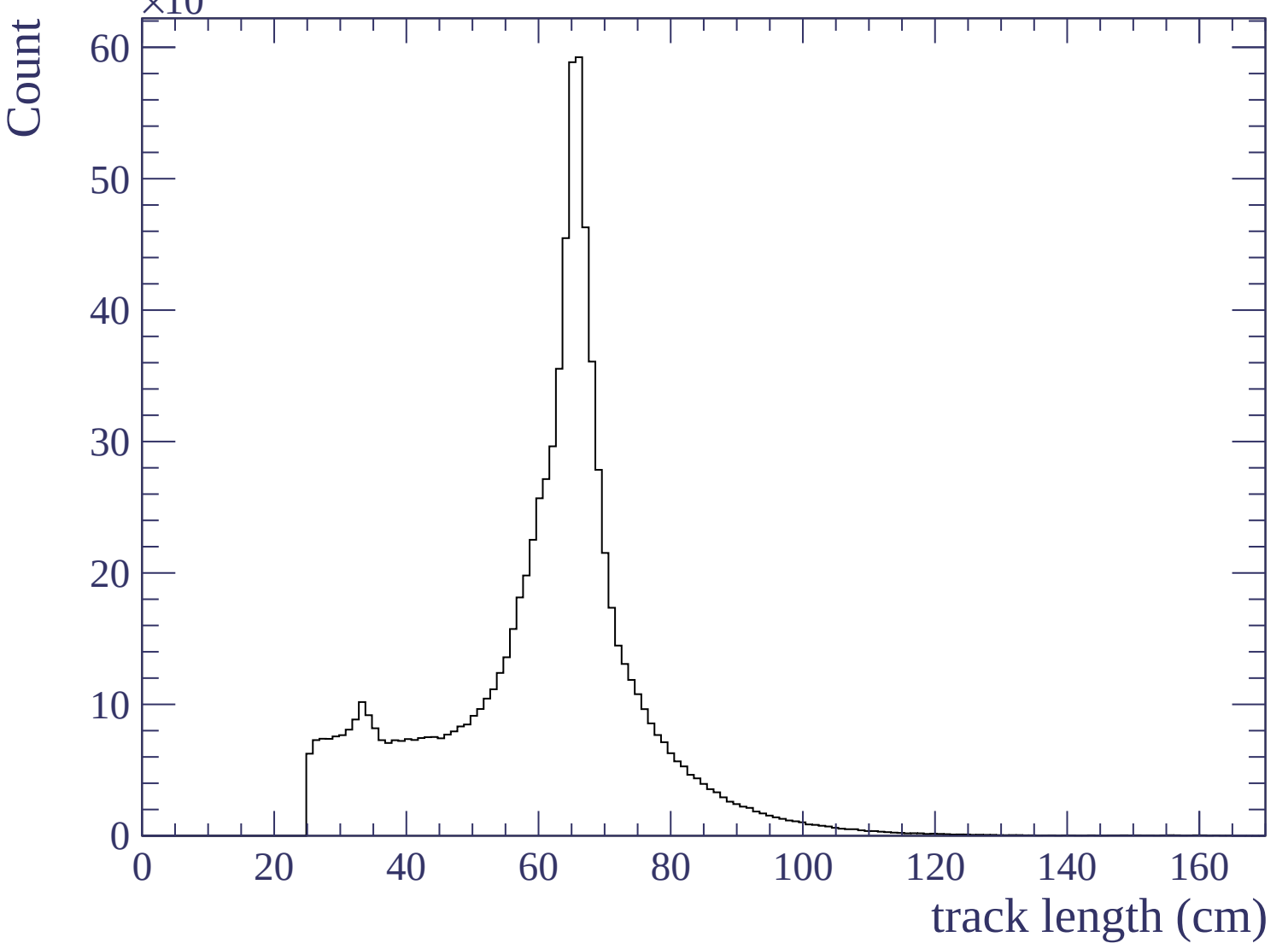


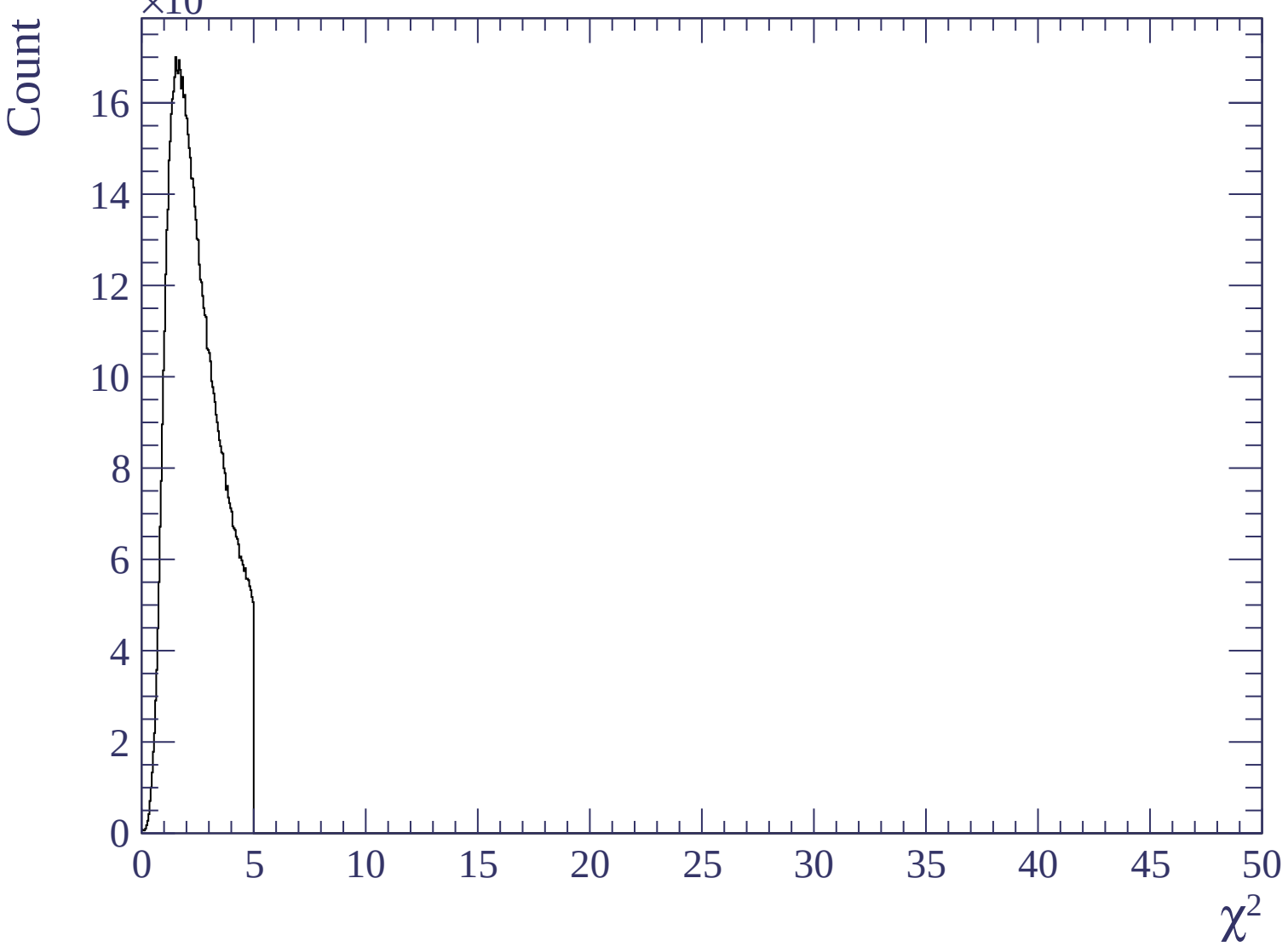


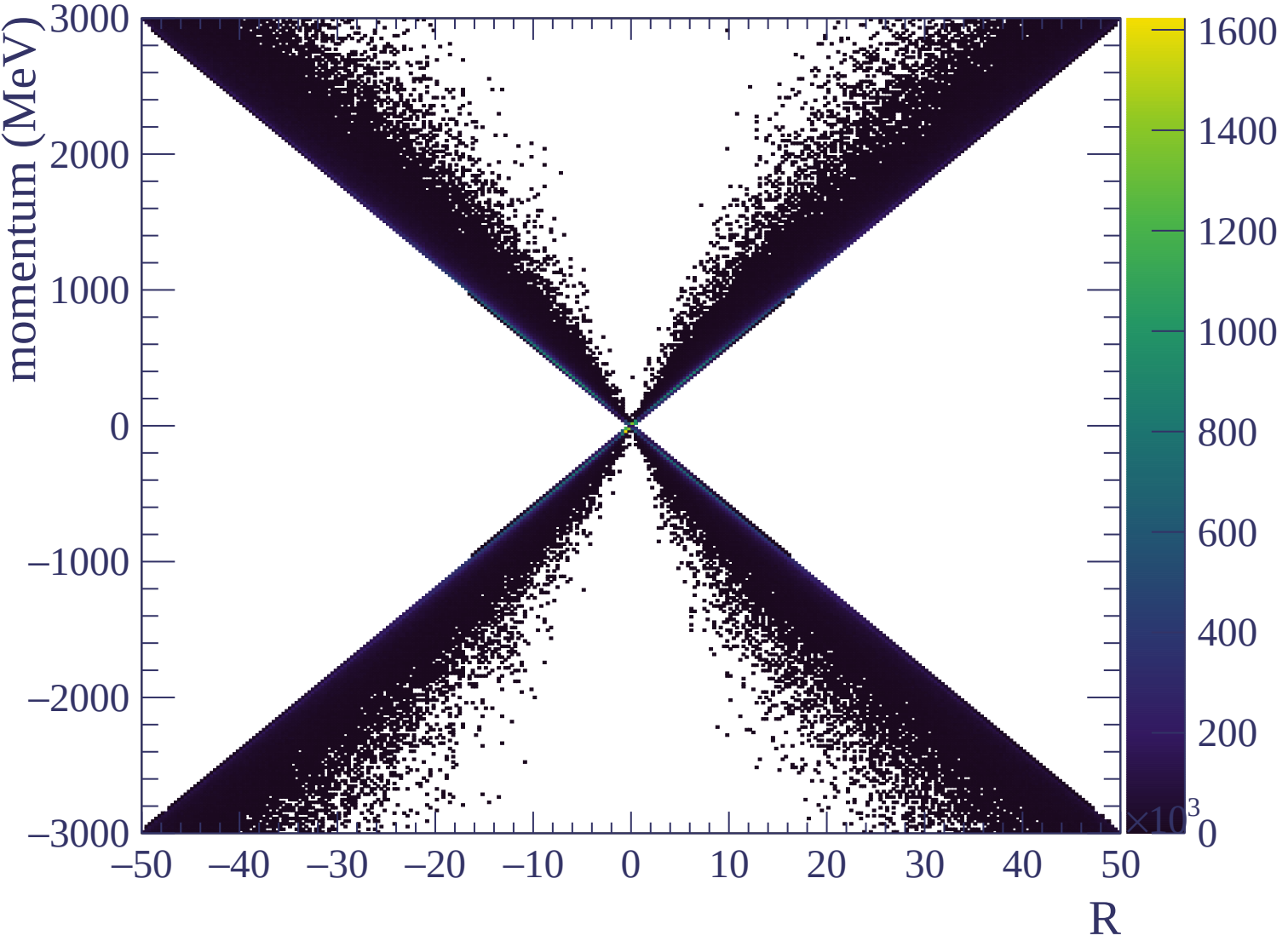


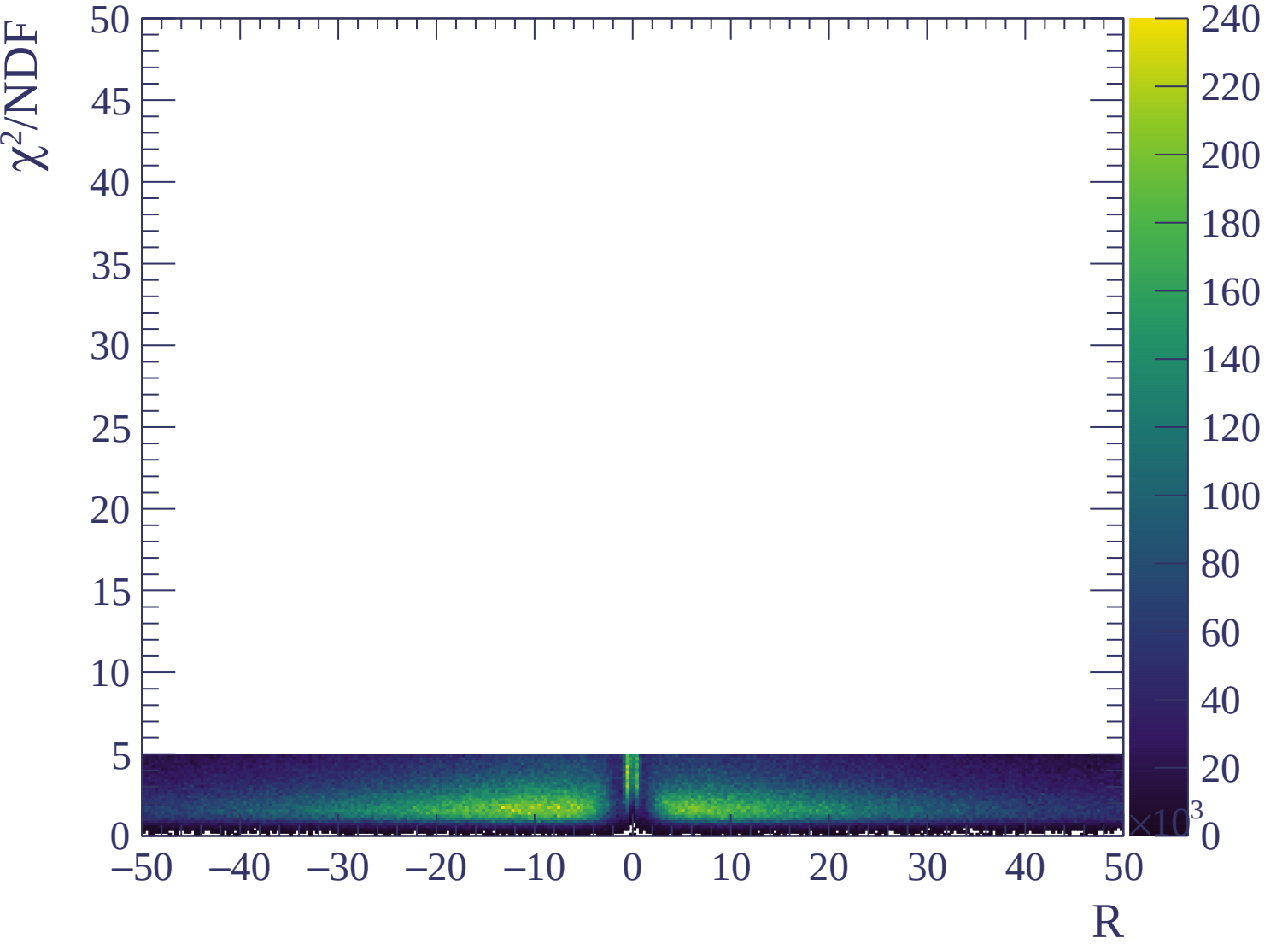






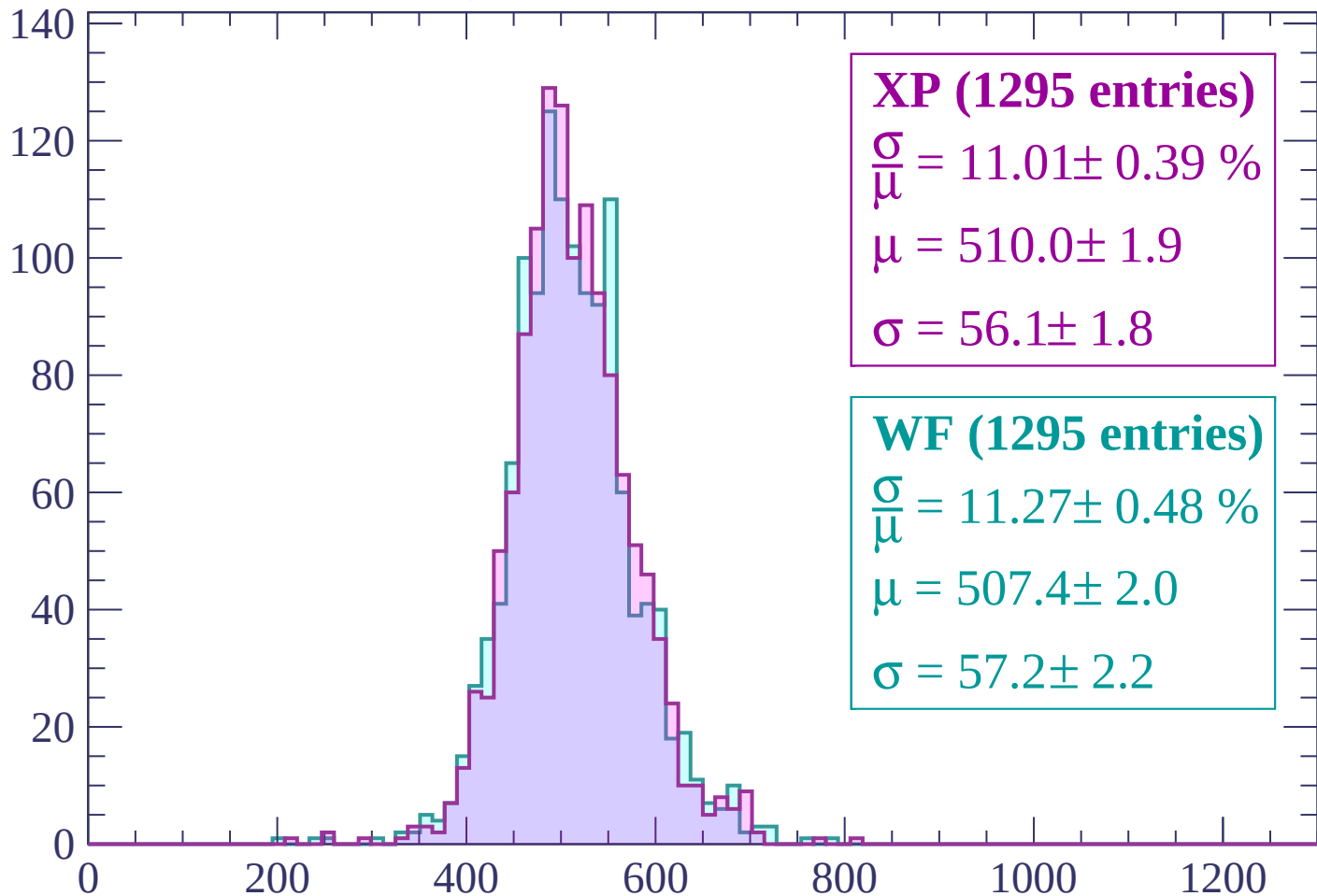






Energy loss | $-3000 < p < -2940$

Count



XP (1295 entries)

$$\frac{\sigma}{\mu} = 11.01 \pm 0.39 \%$$

$$\mu = 510.0 \pm 1.9$$

$$\sigma = 56.1 \pm 1.8$$

WF (1295 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.48 \%$$

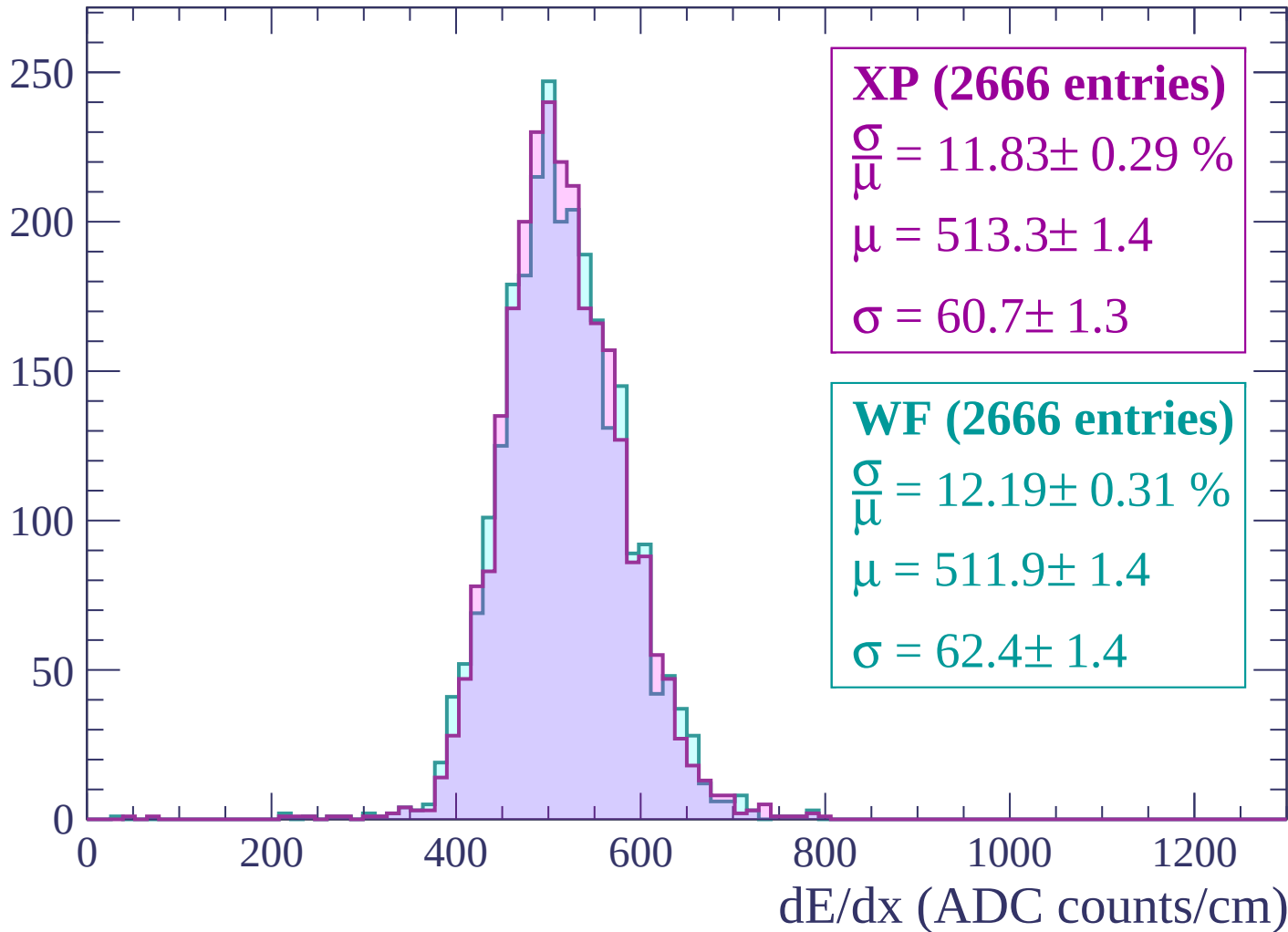
$$\mu = 507.4 \pm 2.0$$

$$\sigma = 57.2 \pm 2.2$$

dE/dx (ADC counts/cm)

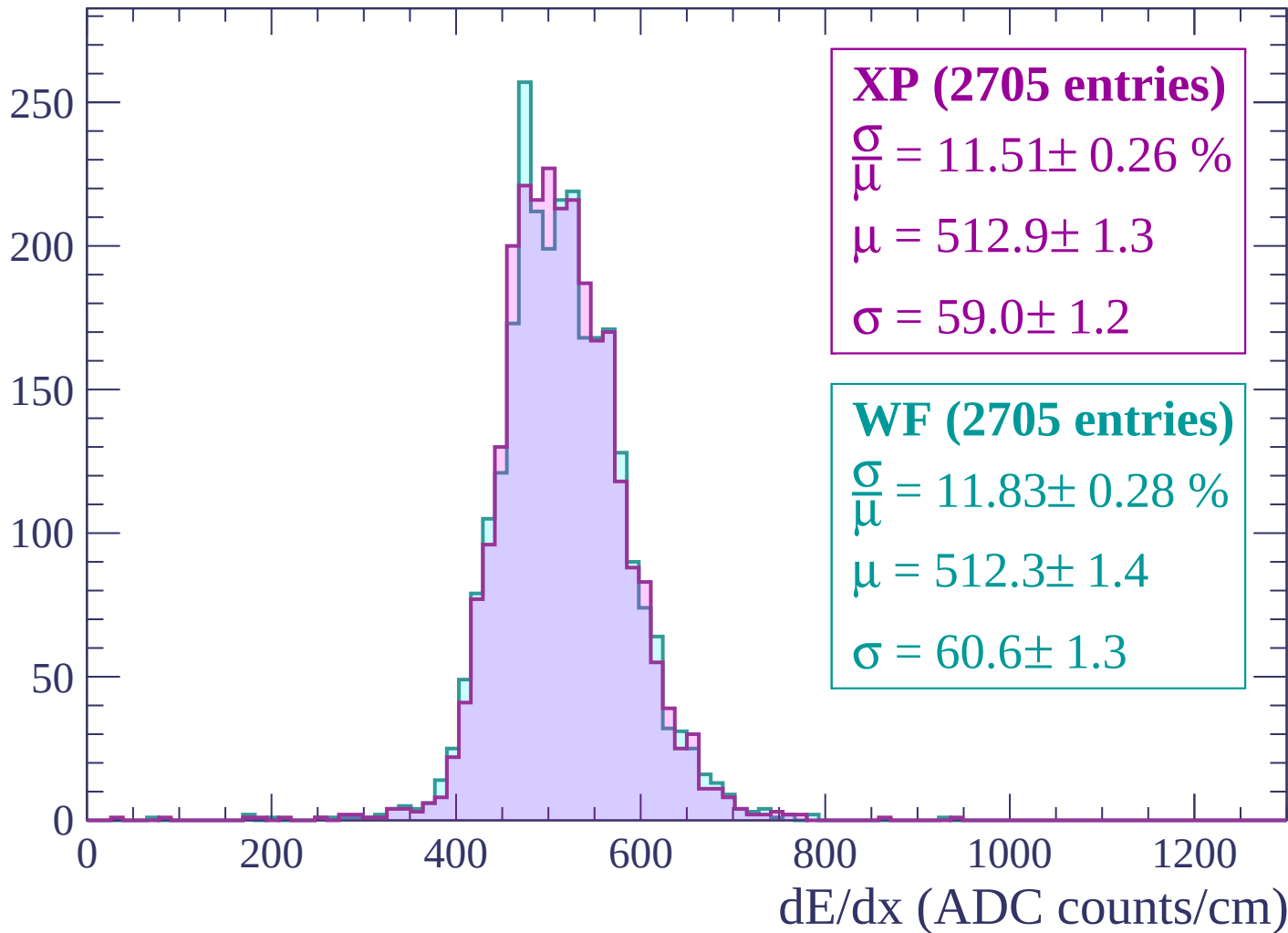
Energy loss | $-2940 < p < -2880$

Count



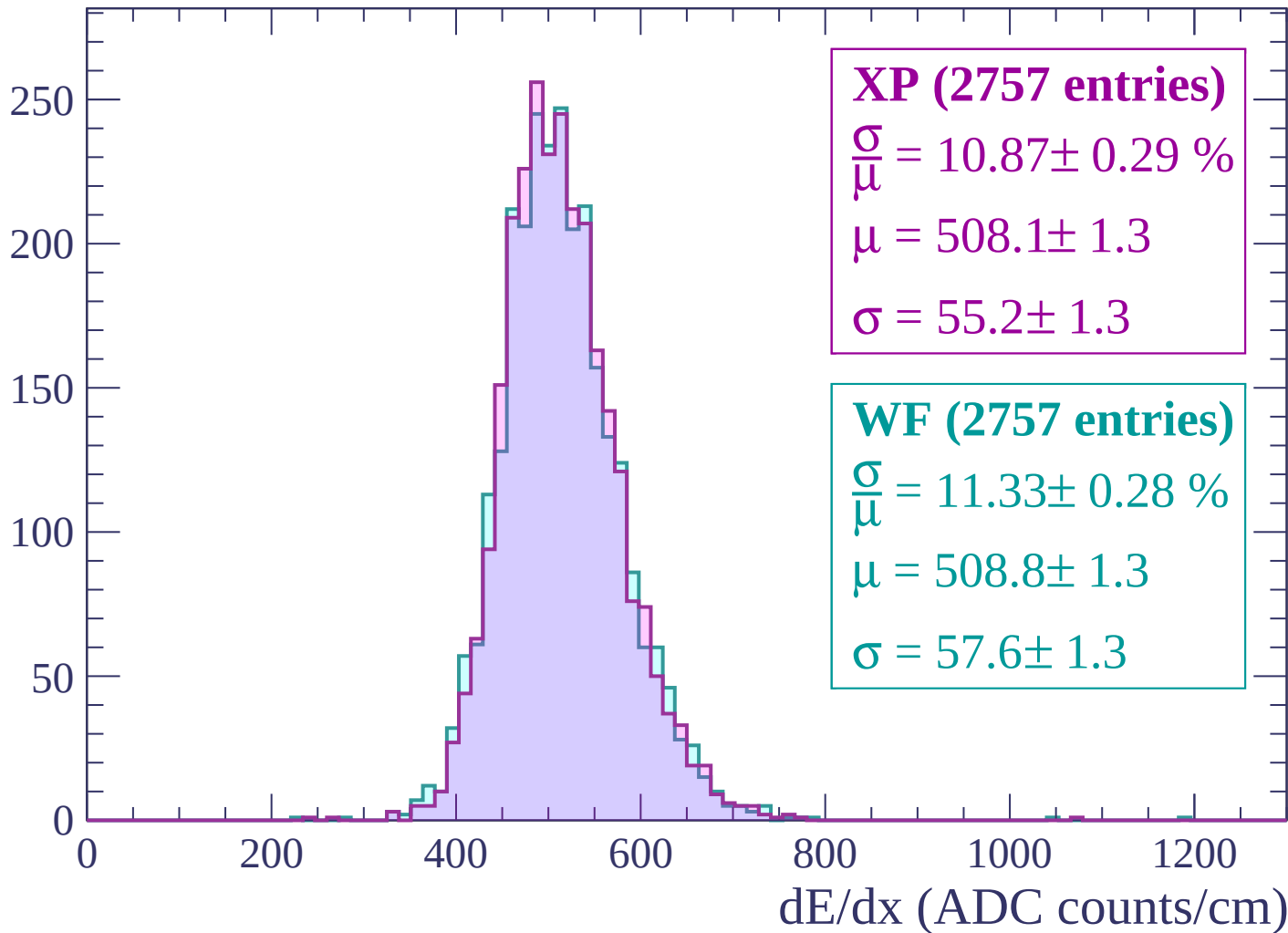
Energy loss | $-2880 < p < -2820$

Count



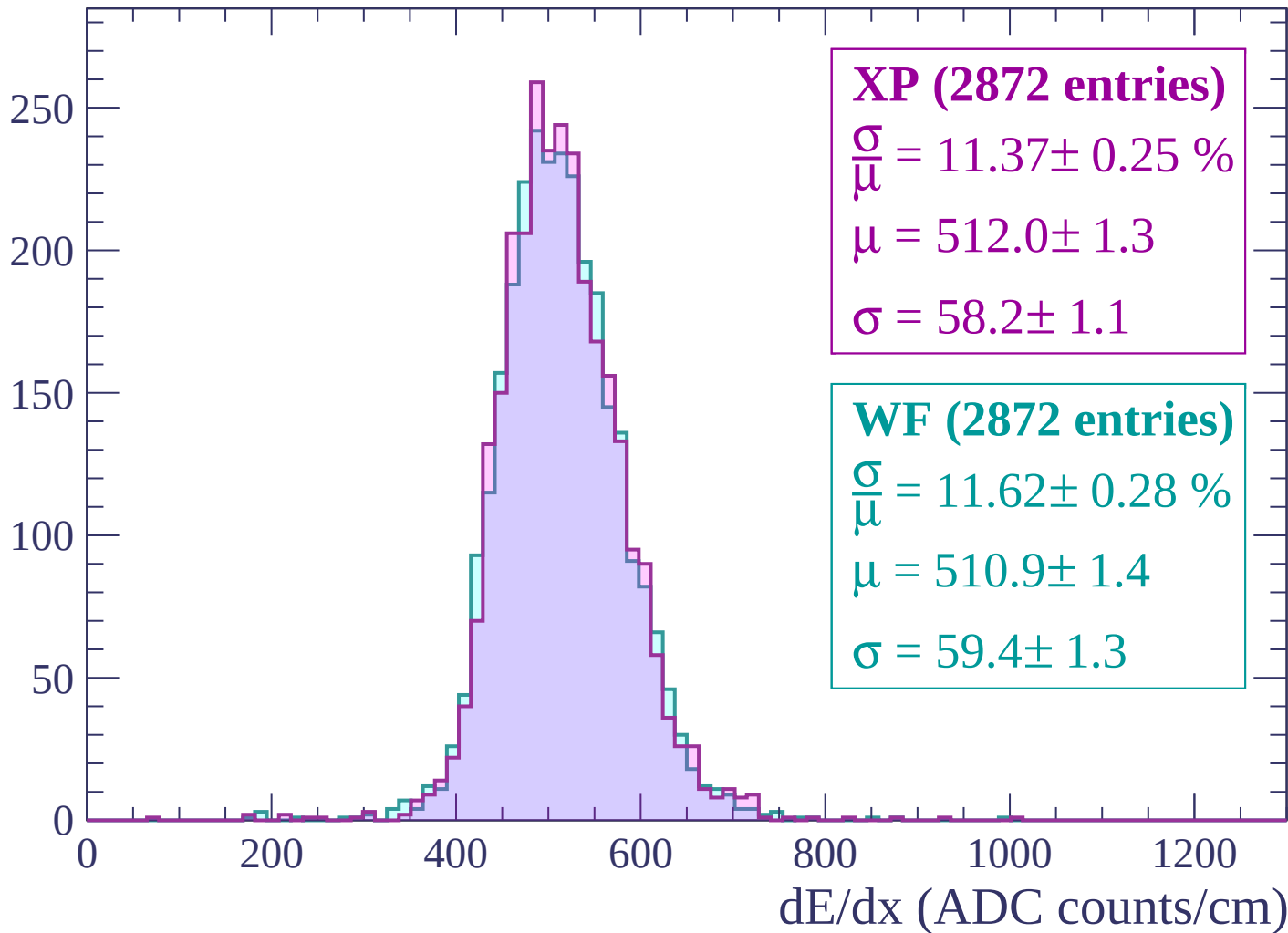
Energy loss | $-2820 < p < -2760$

Count



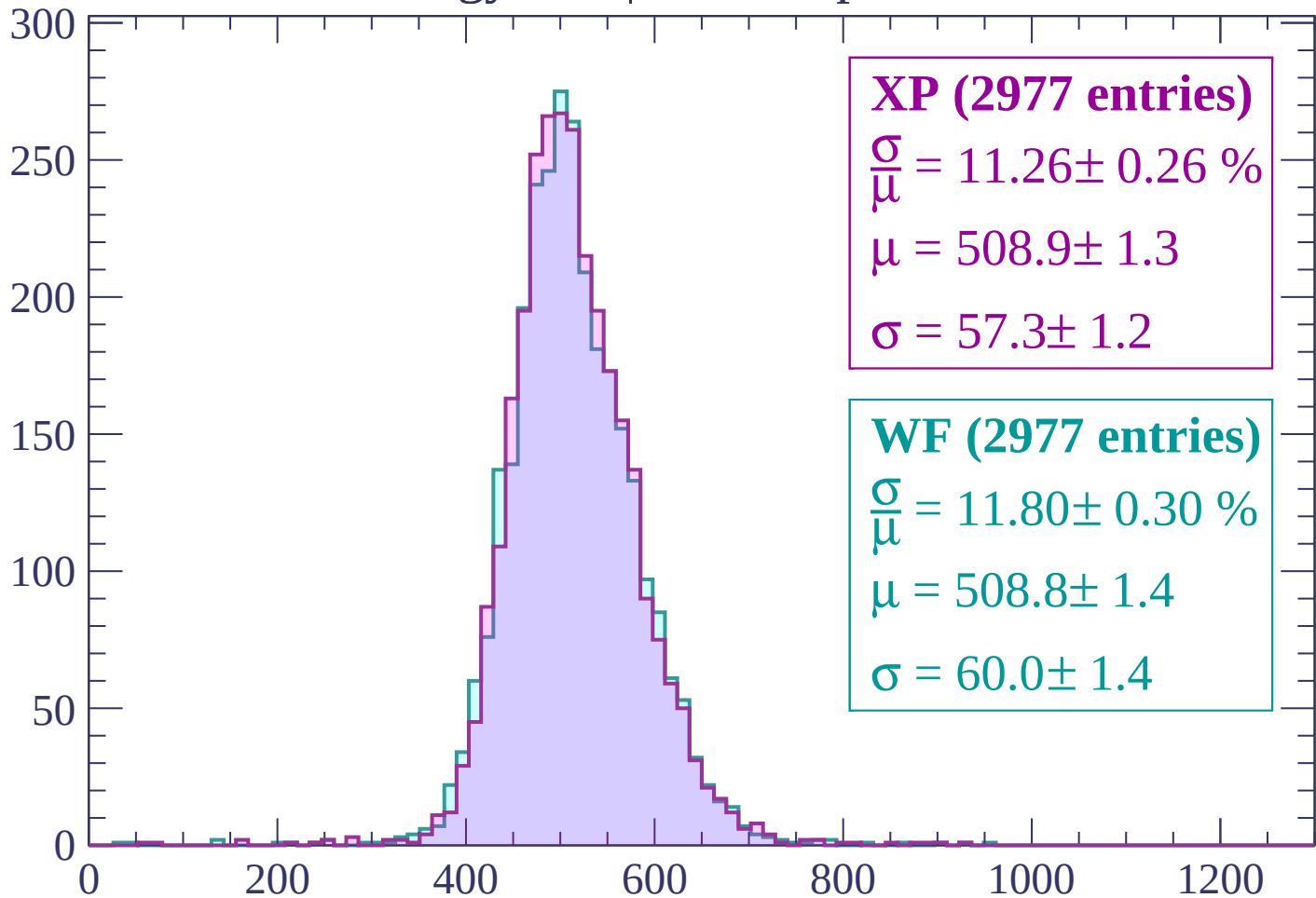
Energy loss | $-2760 < p < -2700$

Count



Energy loss | -2700 < p < -2640

Count



XP (2977 entries)

$$\frac{\sigma}{\mu} = 11.26 \pm 0.26 \%$$

$$\mu = 508.9 \pm 1.3$$

$$\sigma = 57.3 \pm 1.2$$

WF (2977 entries)

$$\frac{\sigma}{\mu} = 11.80 \pm 0.30 \%$$

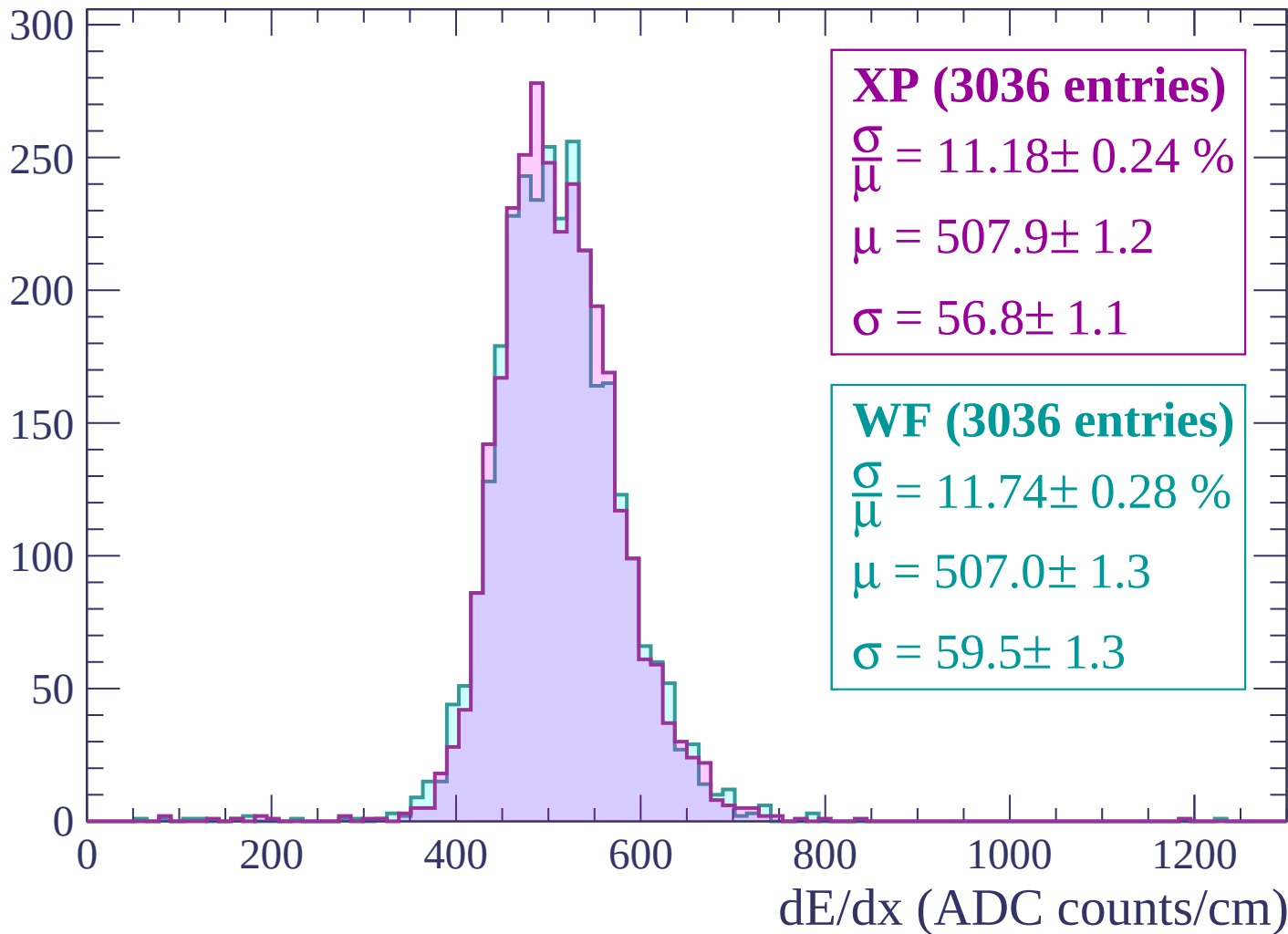
$$\mu = 508.8 \pm 1.4$$

$$\sigma = 60.0 \pm 1.4$$

dE/dx (ADC counts/cm)

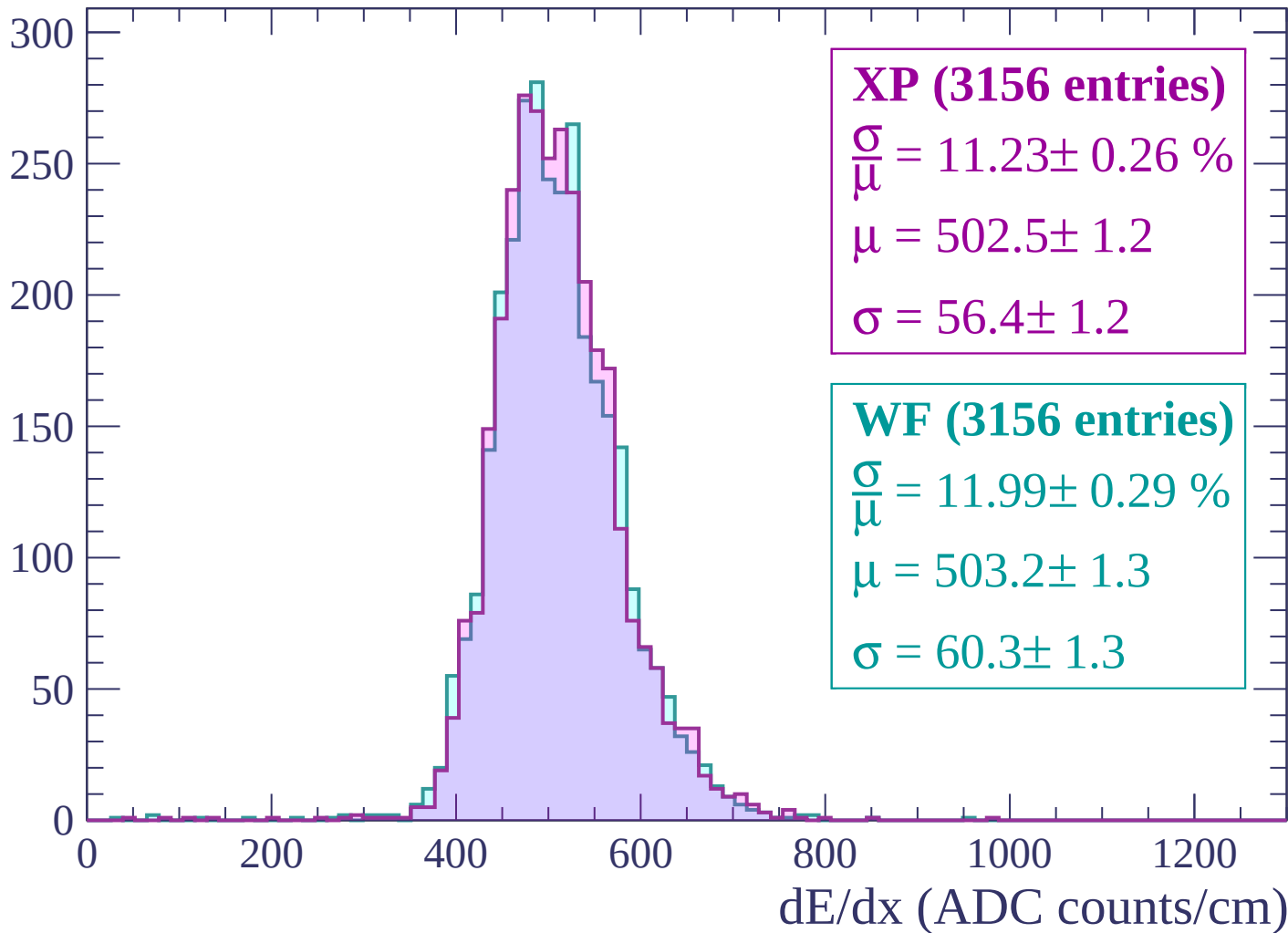
Energy loss | $-2640 < p < -2580$

Count



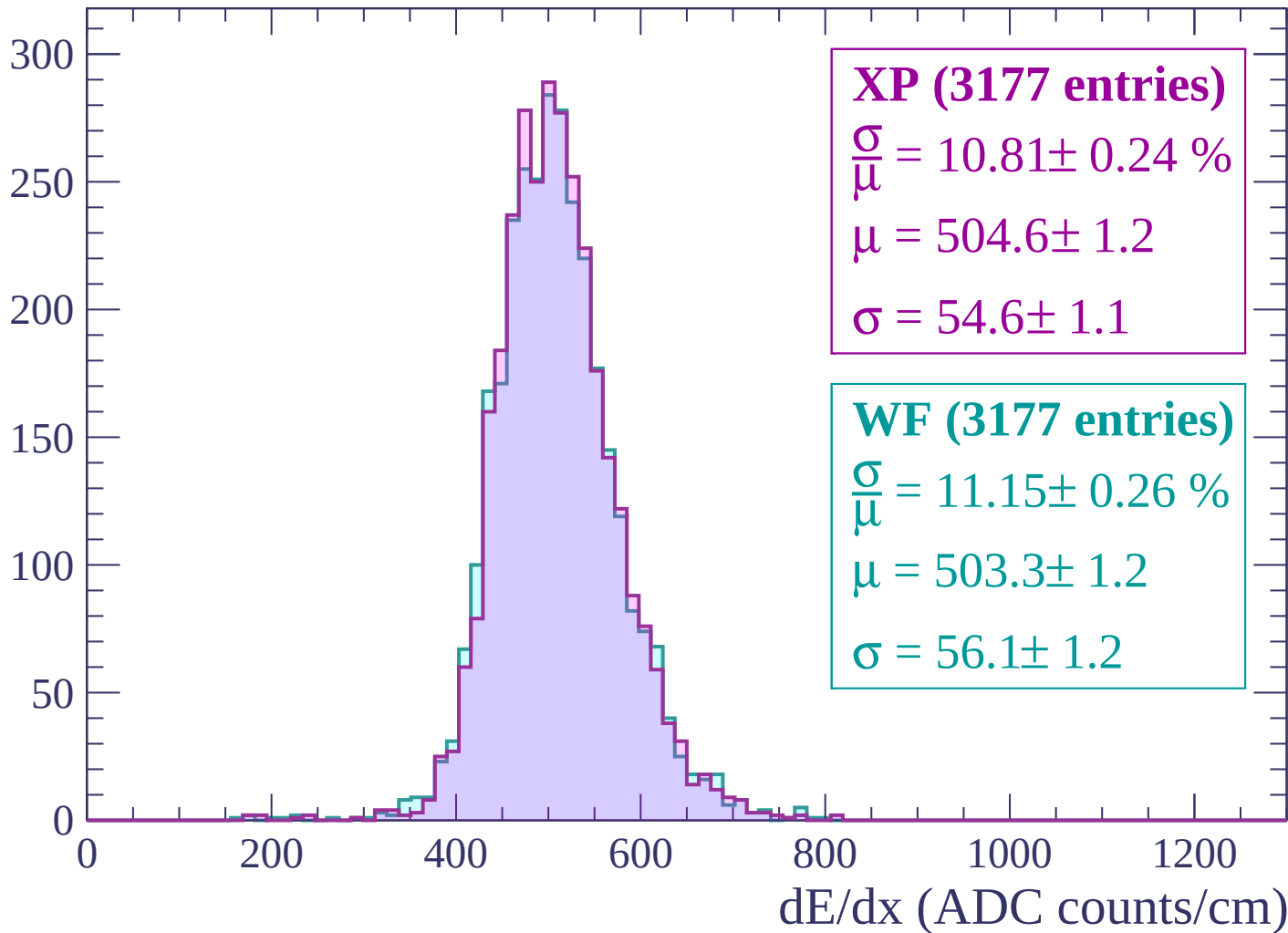
Energy loss | -2580 < p < -2520

Count



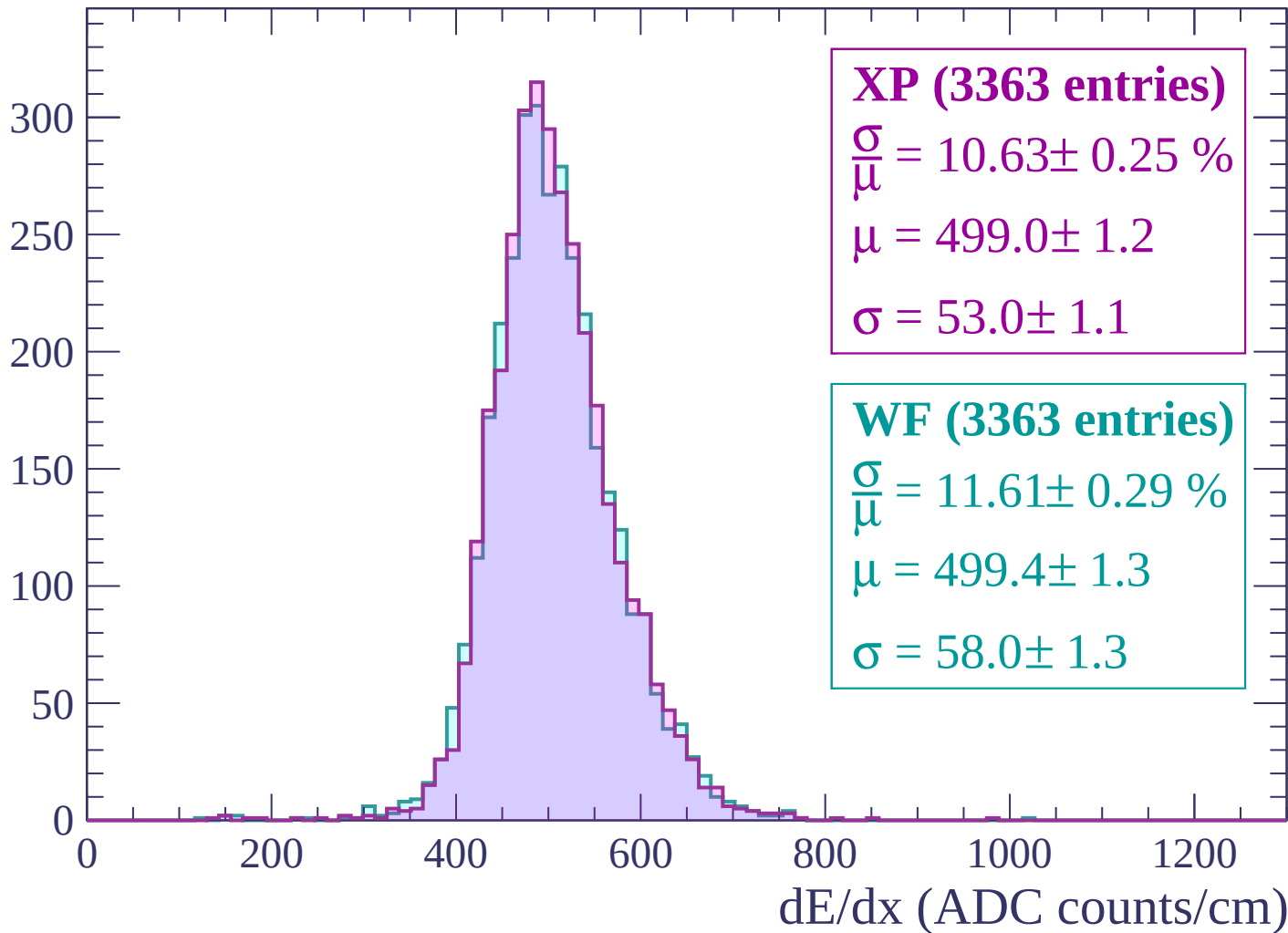
Energy loss | -2520 < p < -2460

Count



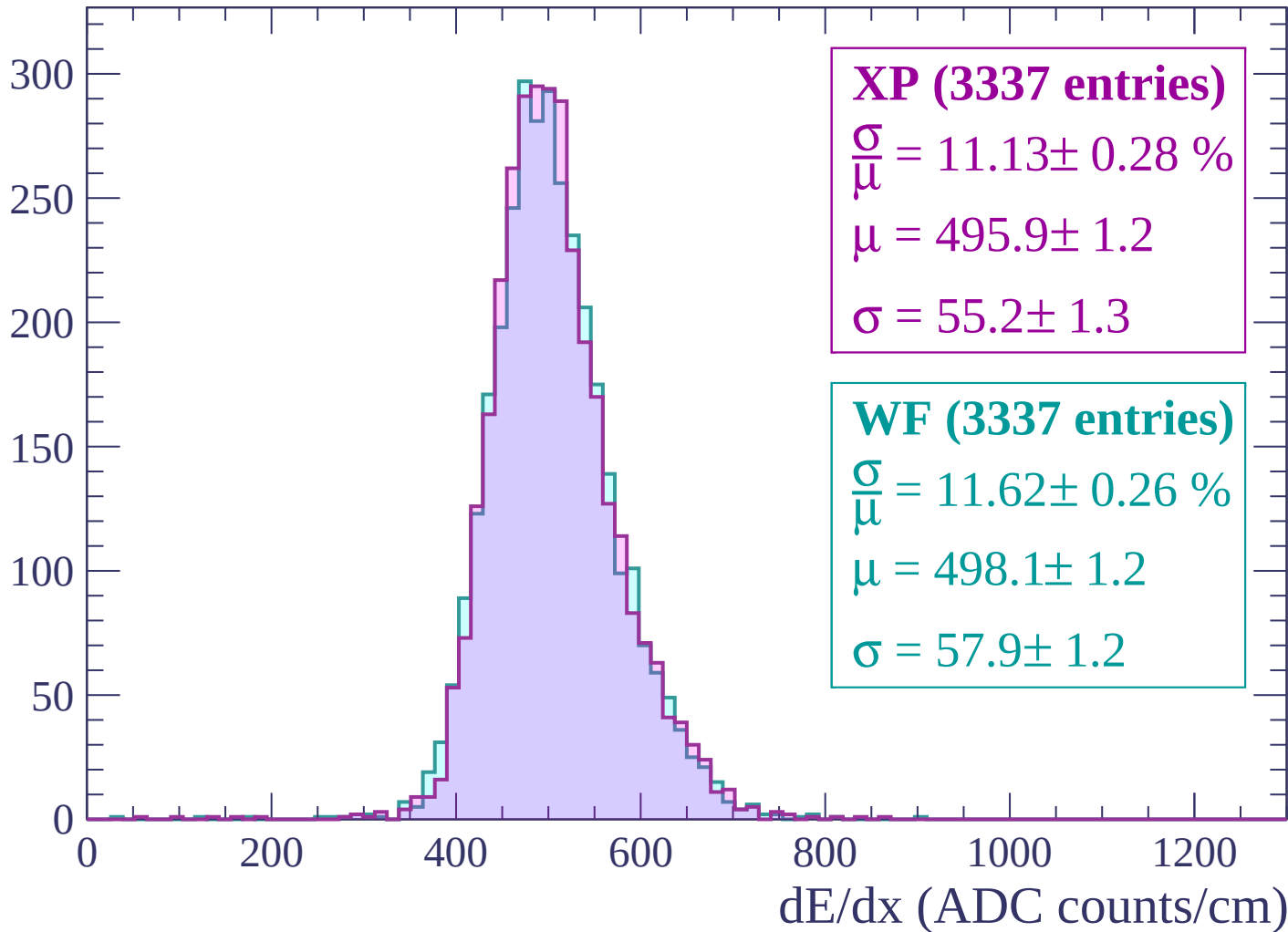
Energy loss | $-2460 < p < -2400$

Count



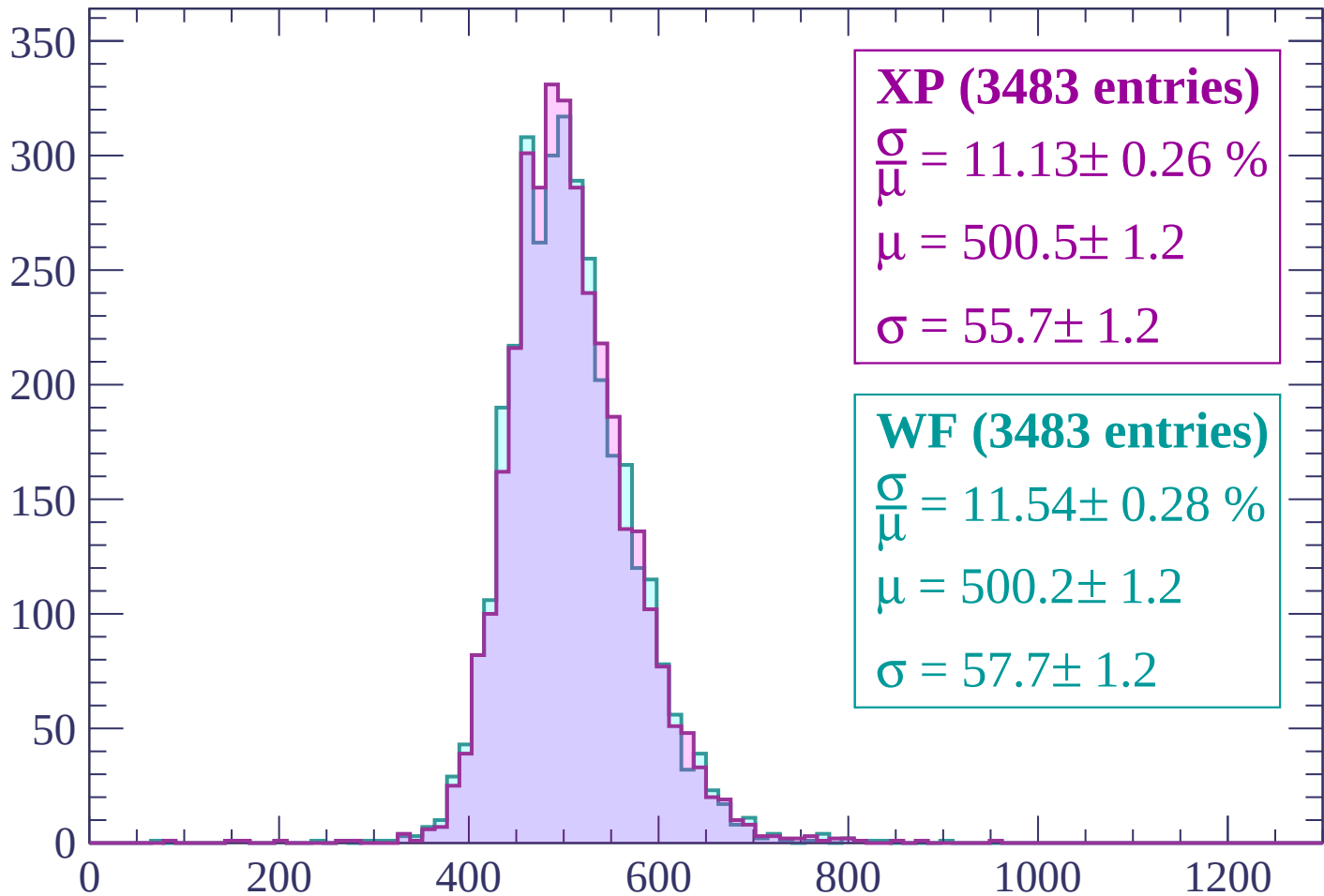
Energy loss | -2400 < p < -2340

Count



Energy loss | $-2340 < p < -2280$

Count



XP (3483 entries)

$$\frac{\sigma}{\mu} = 11.13 \pm 0.26 \%$$

$$\mu = 500.5 \pm 1.2$$

$$\sigma = 55.7 \pm 1.2$$

WF (3483 entries)

$$\frac{\sigma}{\mu} = 11.54 \pm 0.28 \%$$

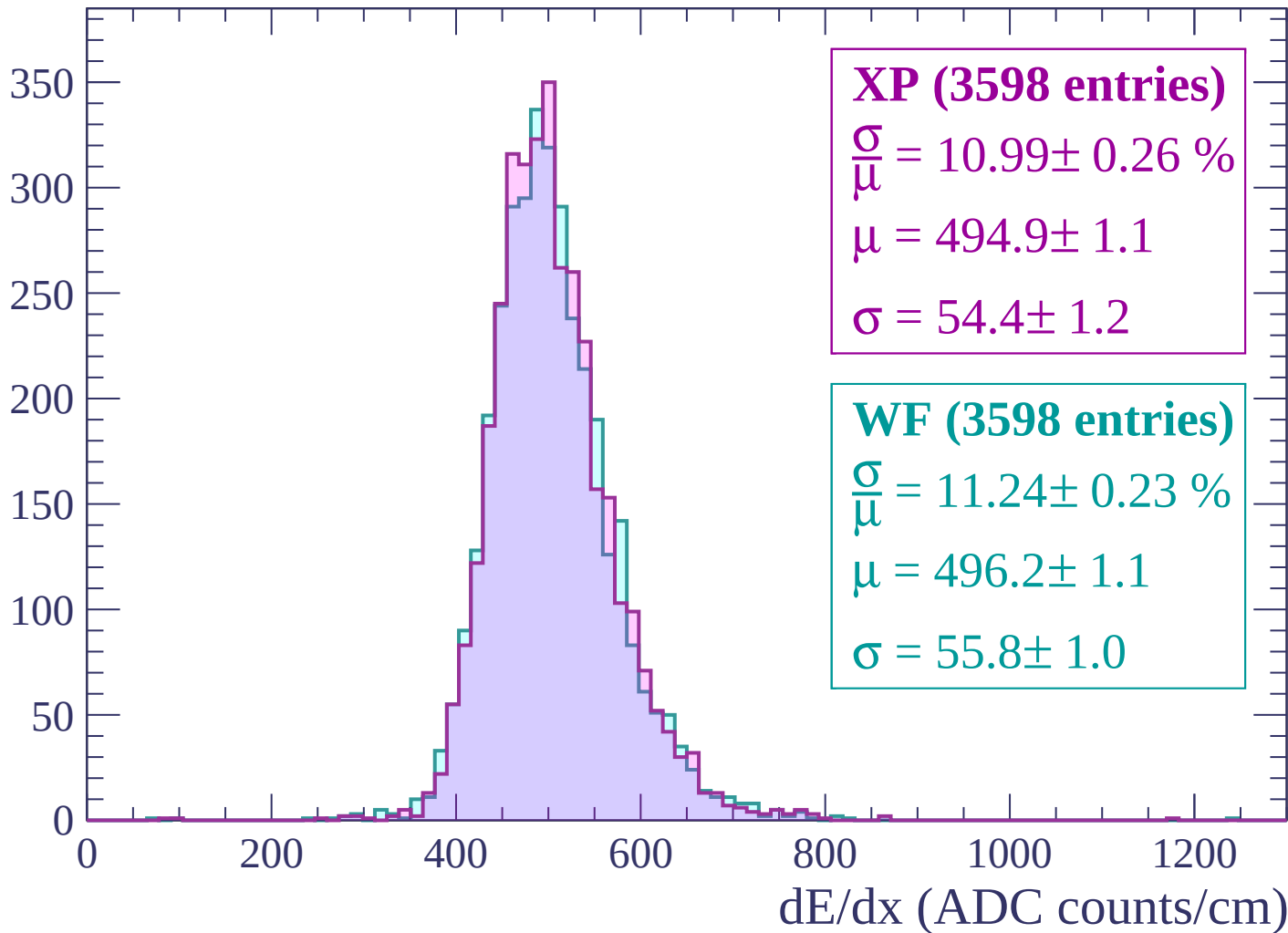
$$\mu = 500.2 \pm 1.2$$

$$\sigma = 57.7 \pm 1.2$$

dE/dx (ADC counts/cm)

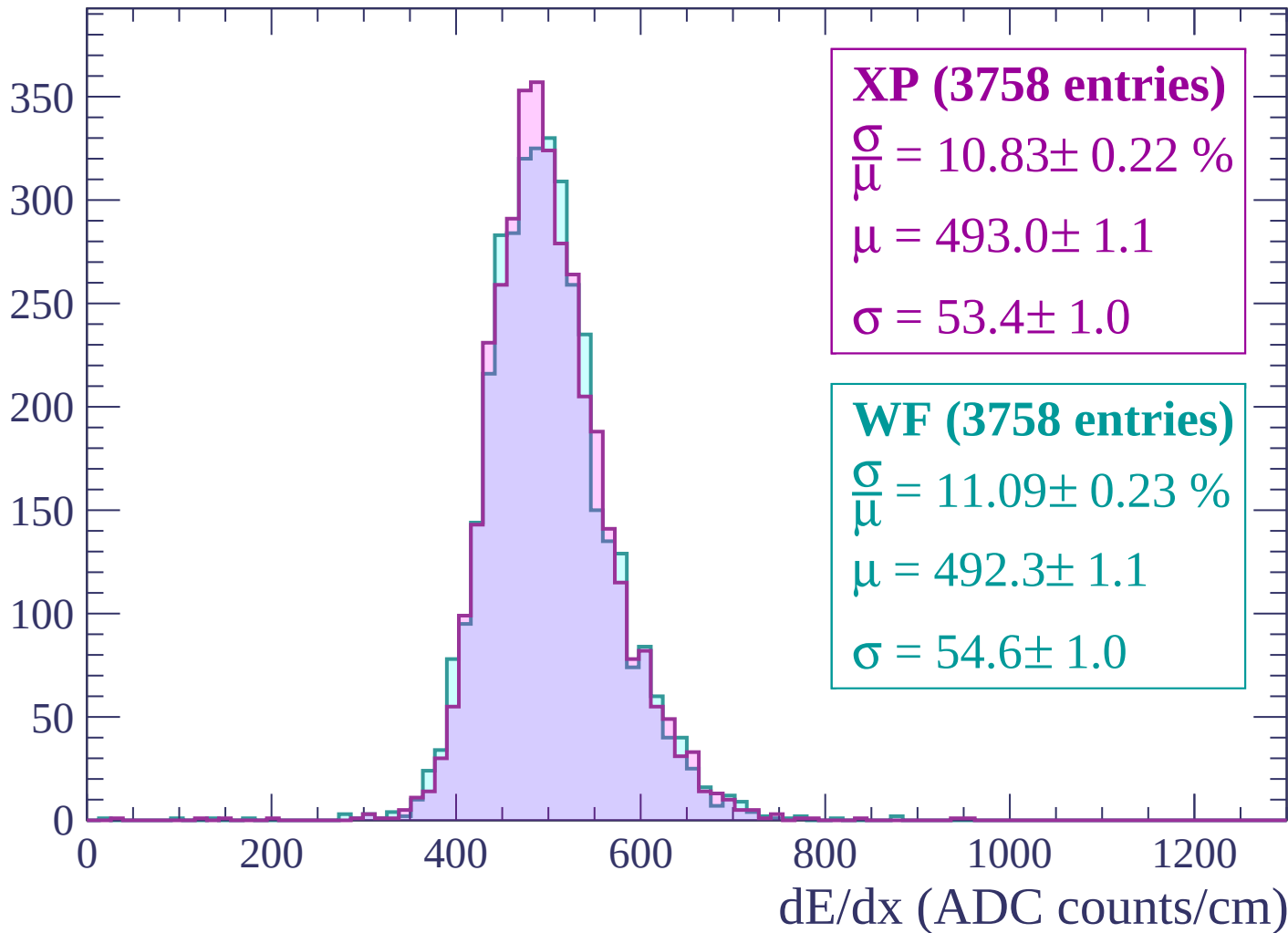
Energy loss | $-2280 < p < -2220$

Count



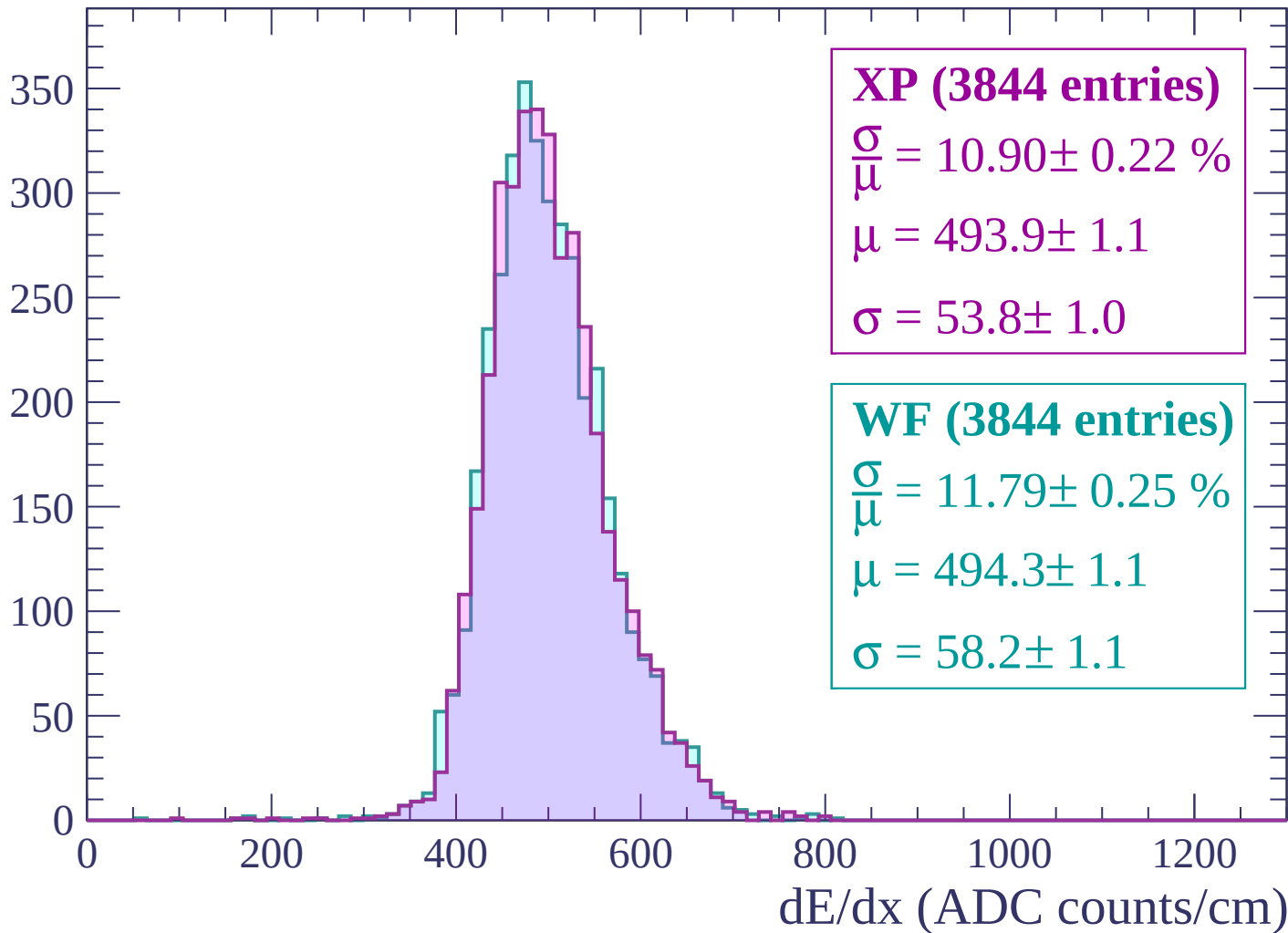
Energy loss | -2220 < p < -2160

Count



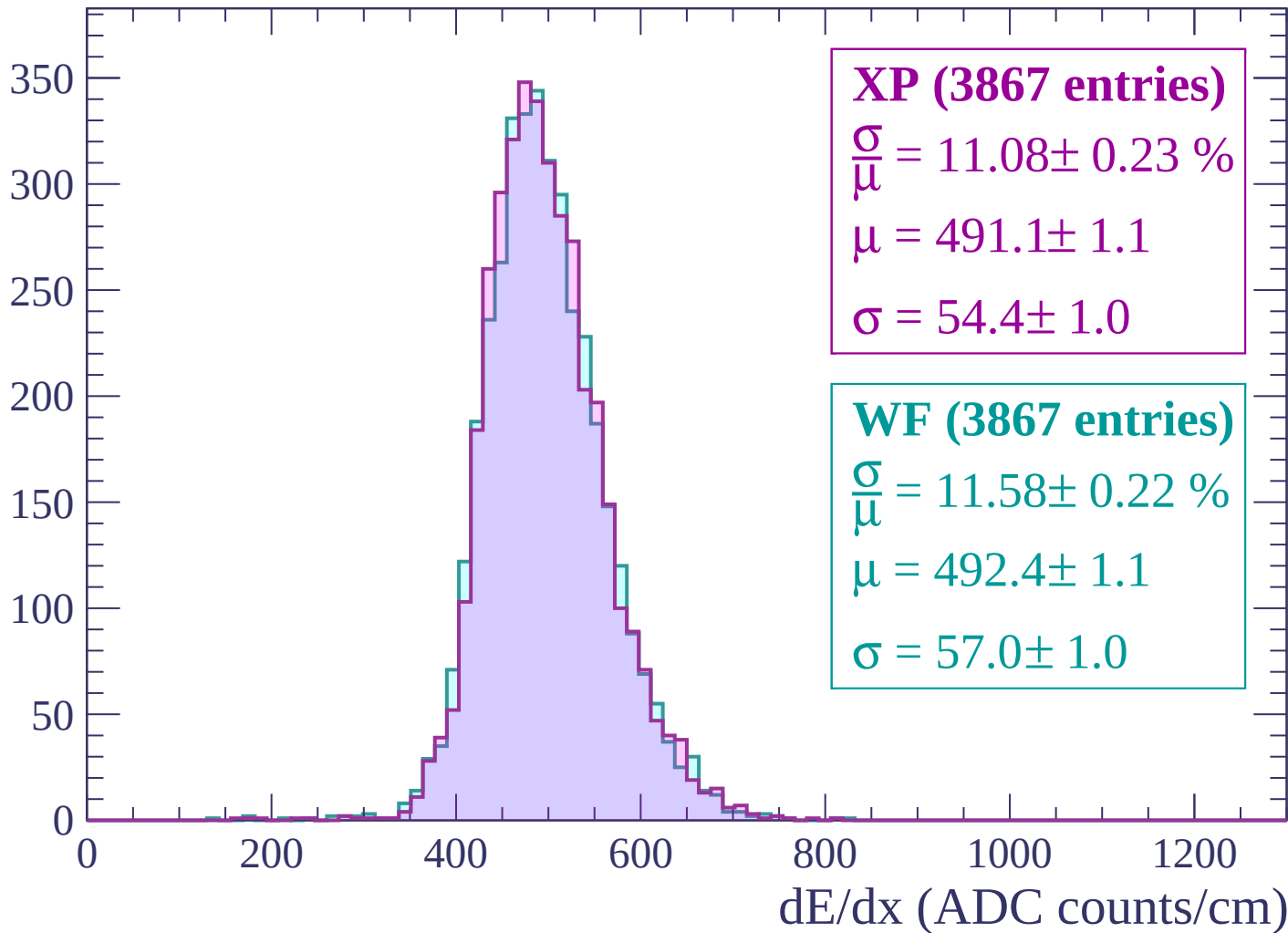
Energy loss | -2160 < p < -2100

Count



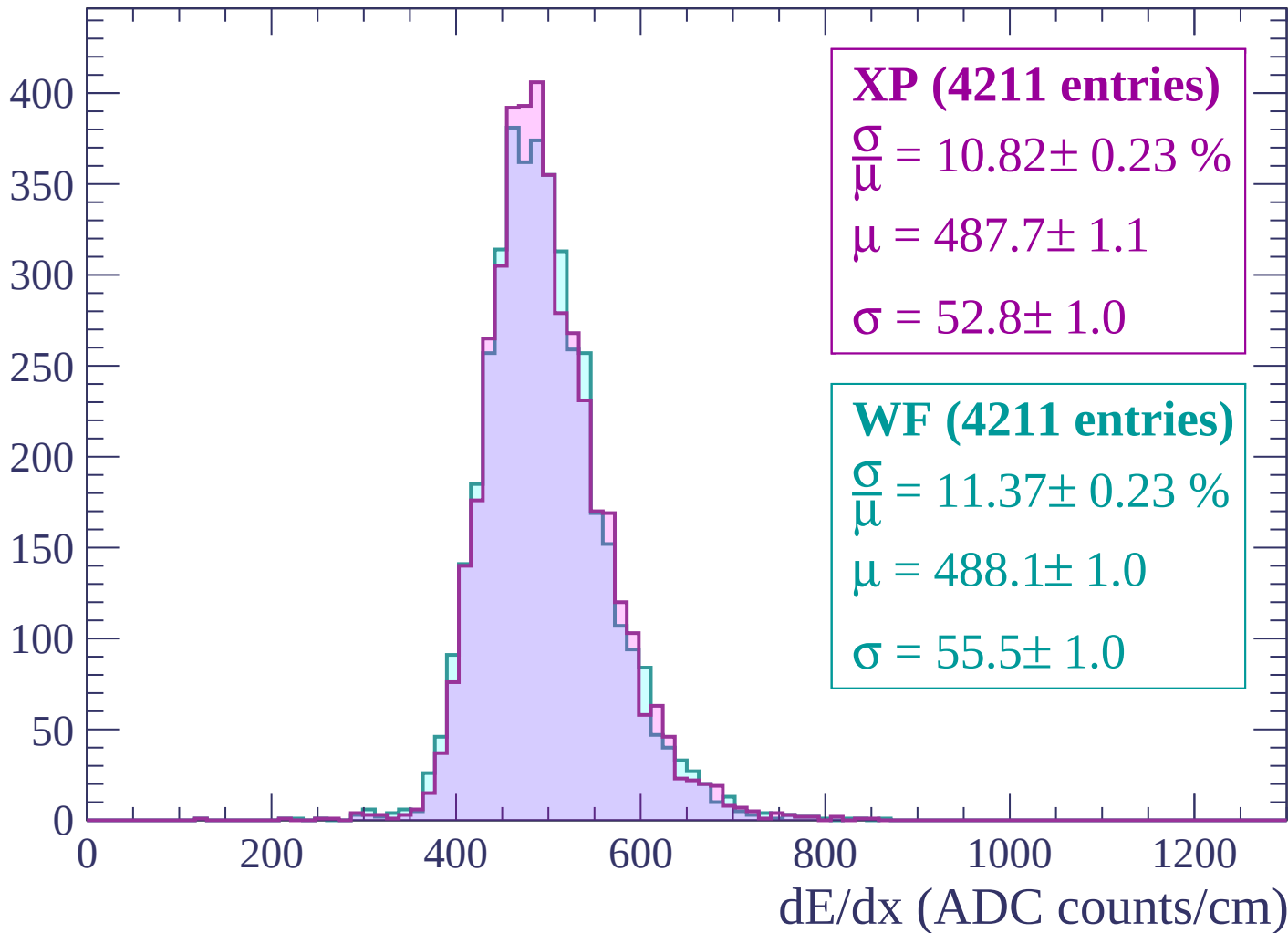
Energy loss | -2100 < p < -2040

Count



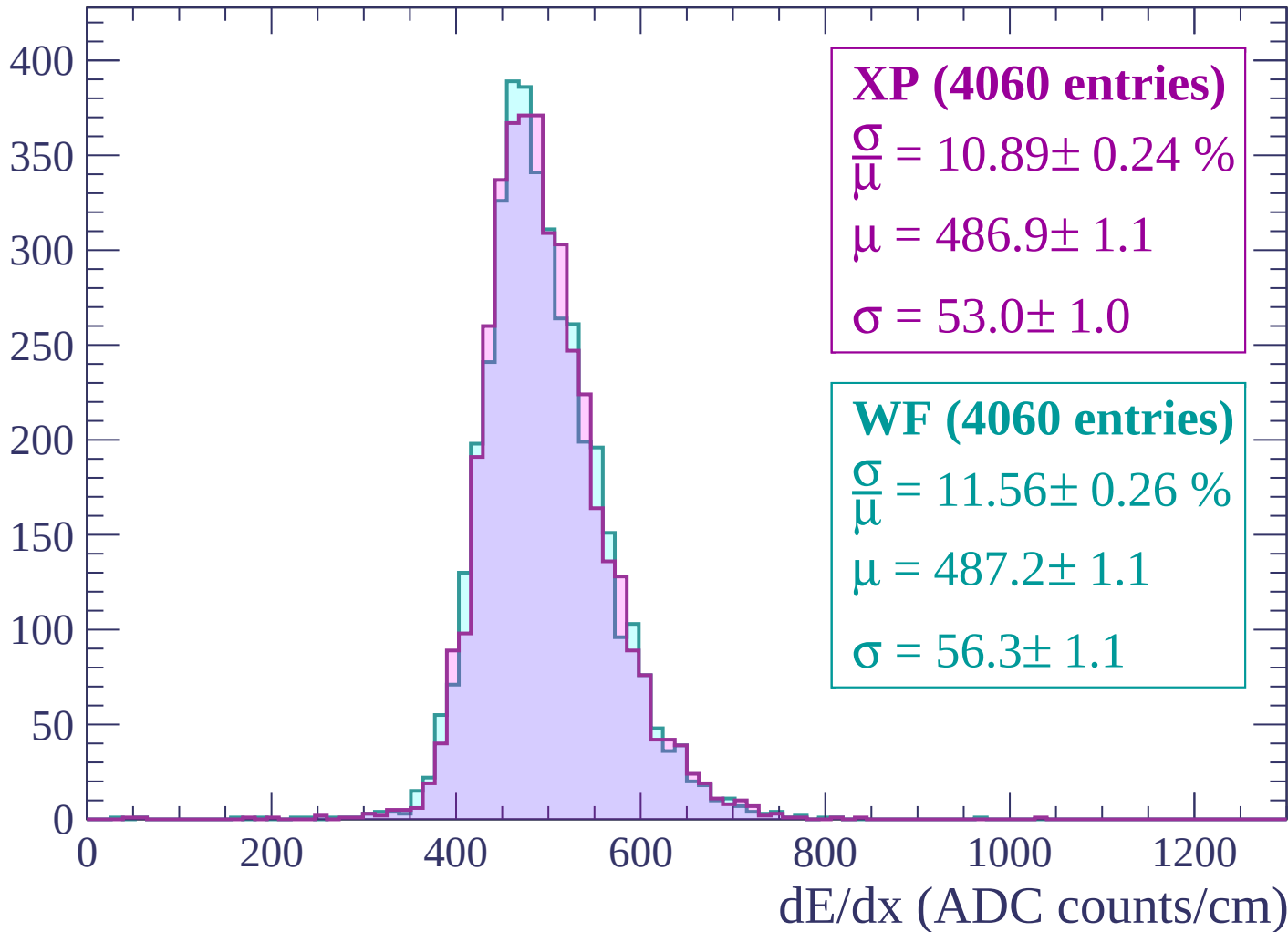
Energy loss | -2040 < p < -1980

Count



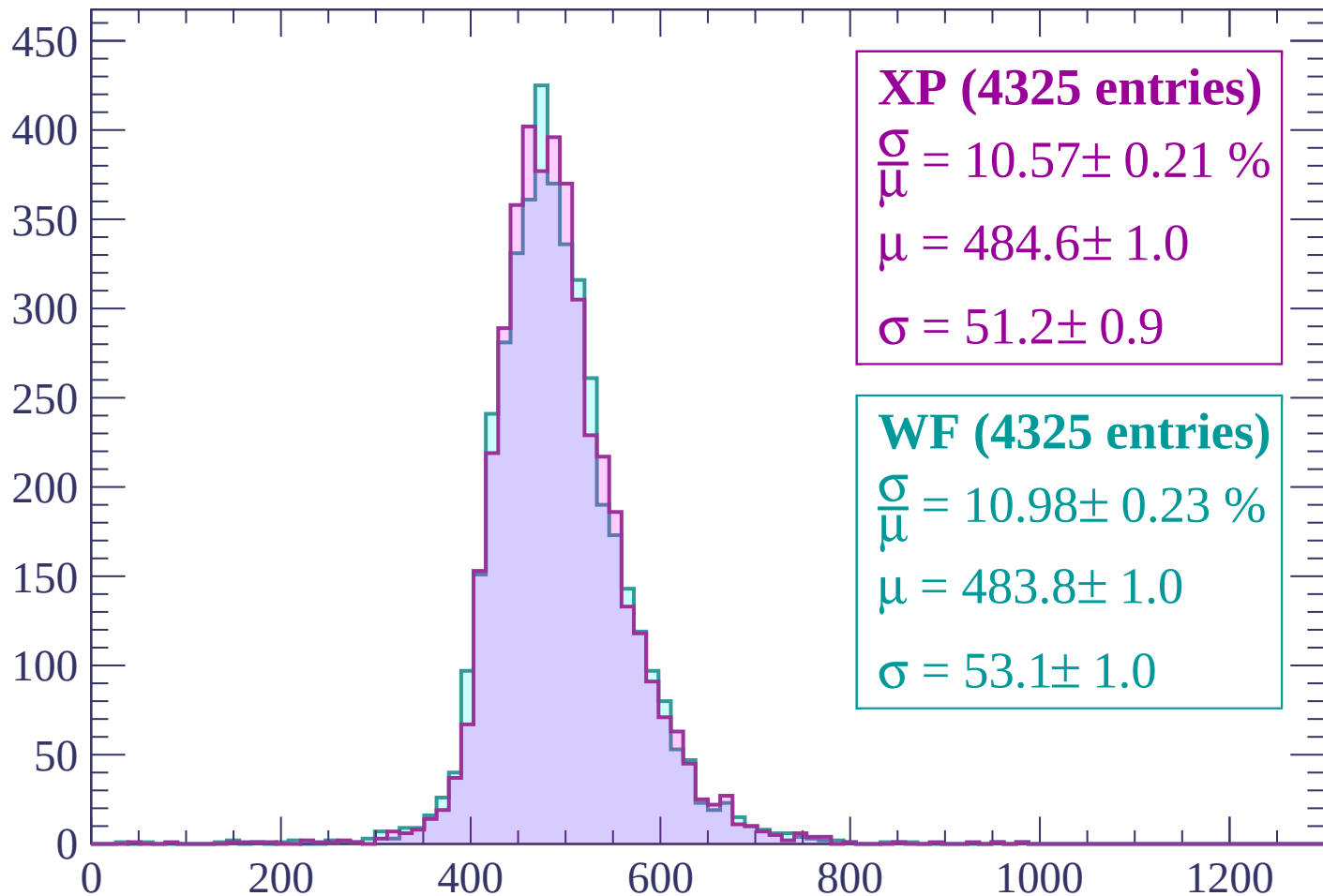
Energy loss | -1980 < p < -1920

Count



Energy loss | -1920 < p < -1860

Count



XP (4325 entries)

$\frac{\sigma}{\mu} = 10.57 \pm 0.21 \%$

$\mu = 484.6 \pm 1.0$

$\sigma = 51.2 \pm 0.9$

WF (4325 entries)

$\frac{\sigma}{\mu} = 10.98 \pm 0.23 \%$

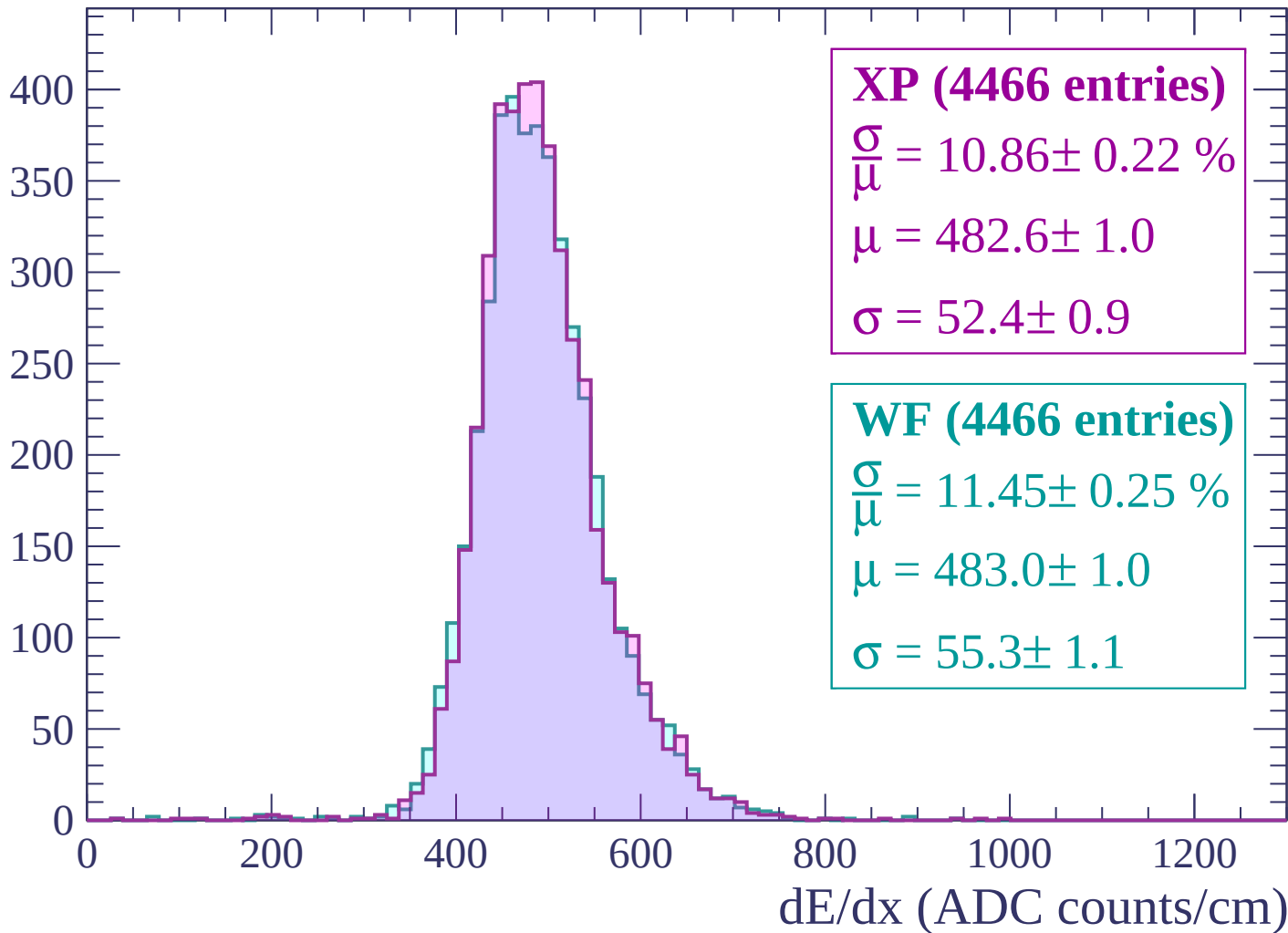
$\mu = 483.8 \pm 1.0$

$\sigma = 53.1 \pm 1.0$

dE/dx (ADC counts/cm)

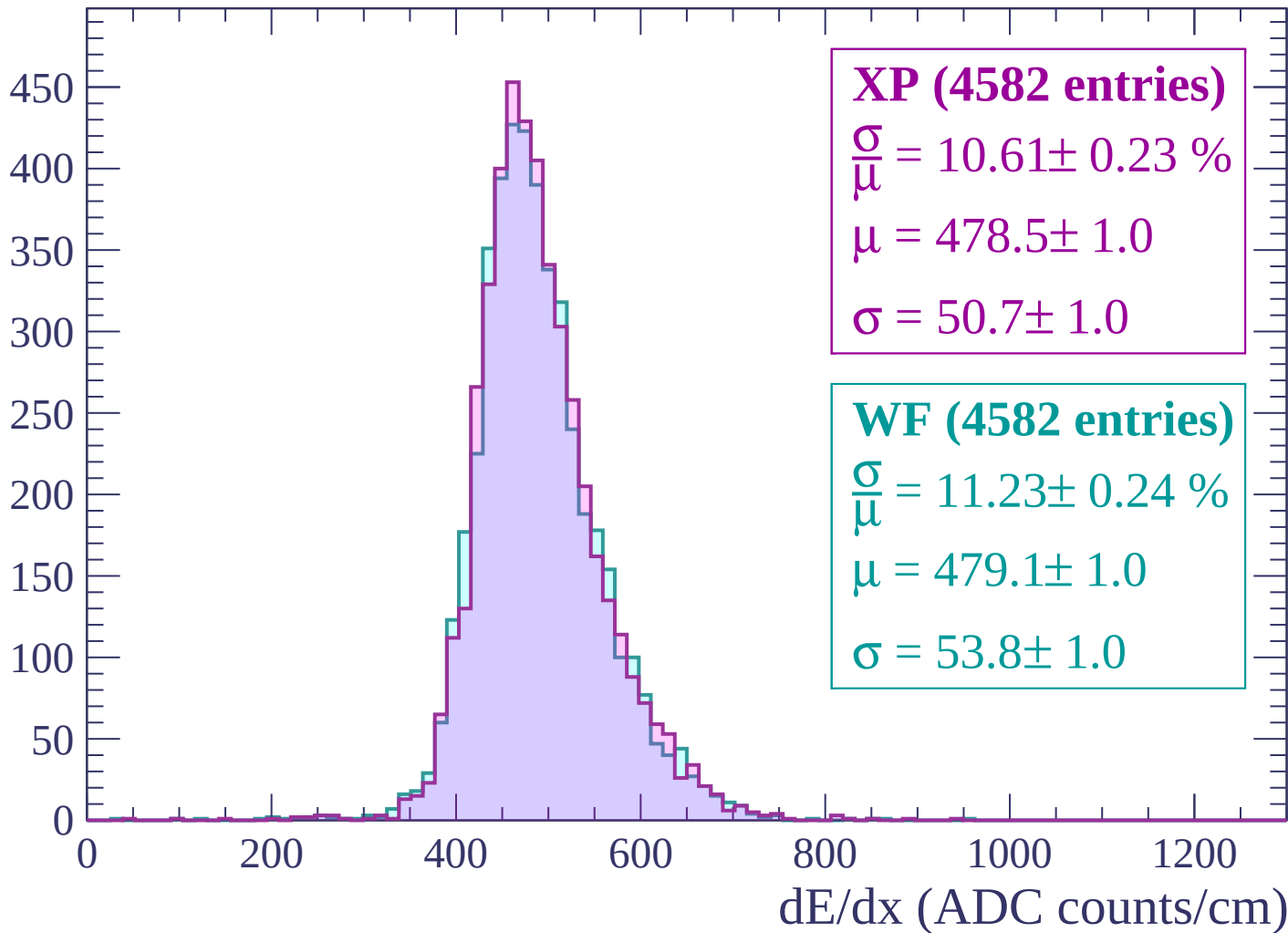
Energy loss | -1860 < p < -1800

Count



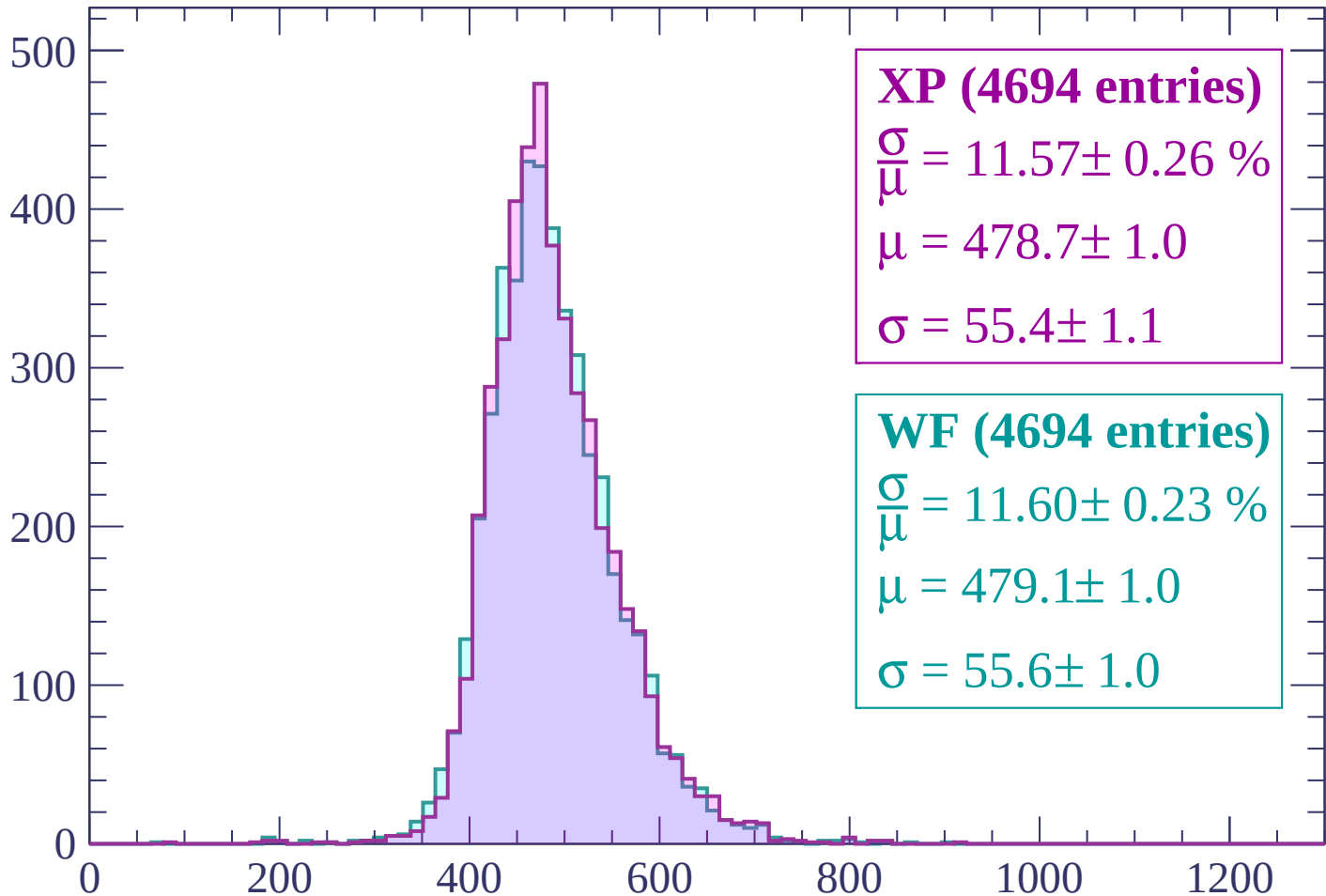
Energy loss | $-1800 < p < -1740$

Count



Energy loss | $-1740 < p < -1680$

Count



XP (4694 entries)

$\frac{\sigma}{\mu} = 11.57 \pm 0.26 \%$

$\mu = 478.7 \pm 1.0$

$\sigma = 55.4 \pm 1.1$

WF (4694 entries)

$\frac{\sigma}{\mu} = 11.60 \pm 0.23 \%$

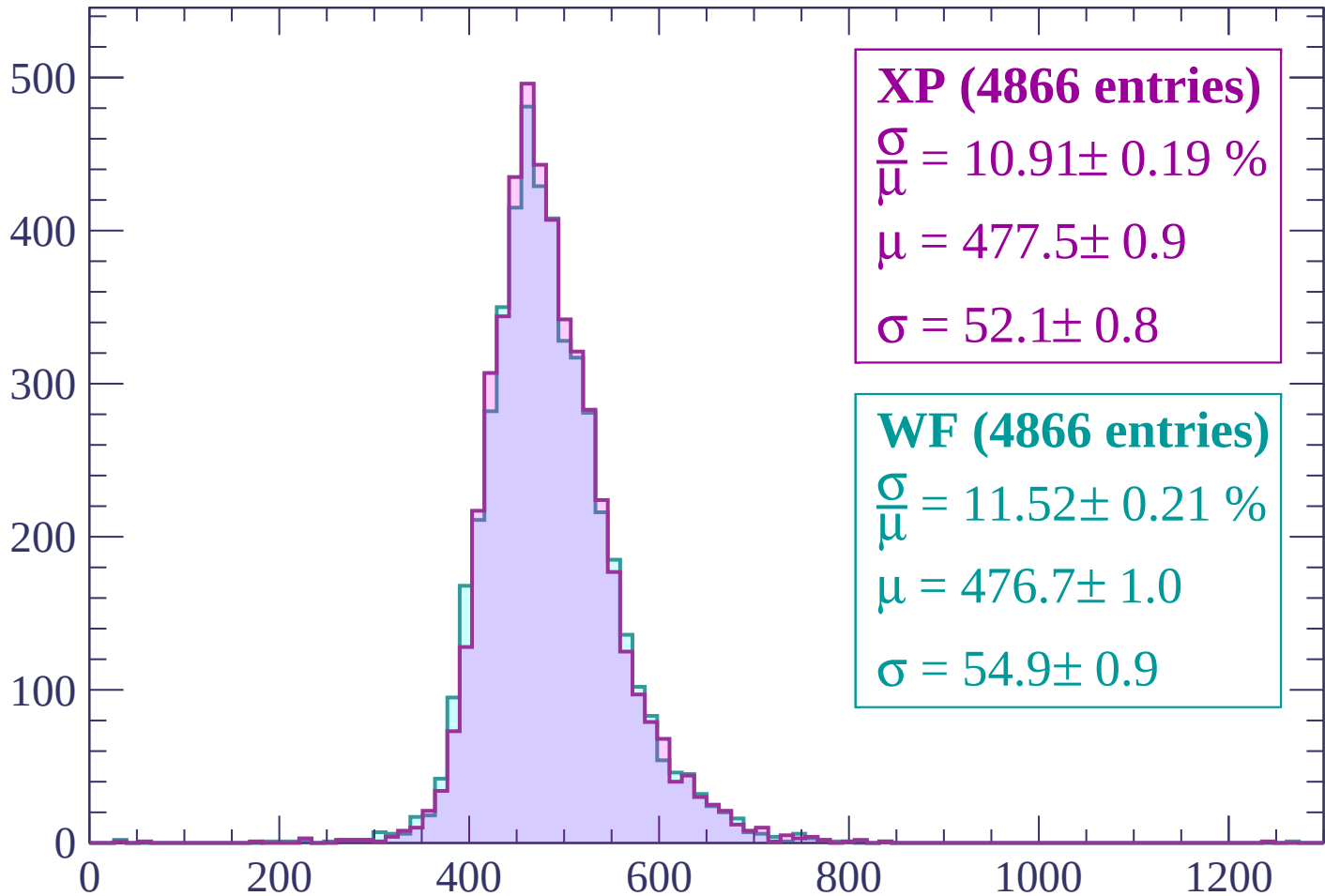
$\mu = 479.1 \pm 1.0$

$\sigma = 55.6 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$

Count



XP (4866 entries)

$\frac{\sigma}{\mu} = 10.91 \pm 0.19 \%$

$\mu = 477.5 \pm 0.9$

$\sigma = 52.1 \pm 0.8$

WF (4866 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.21 \%$

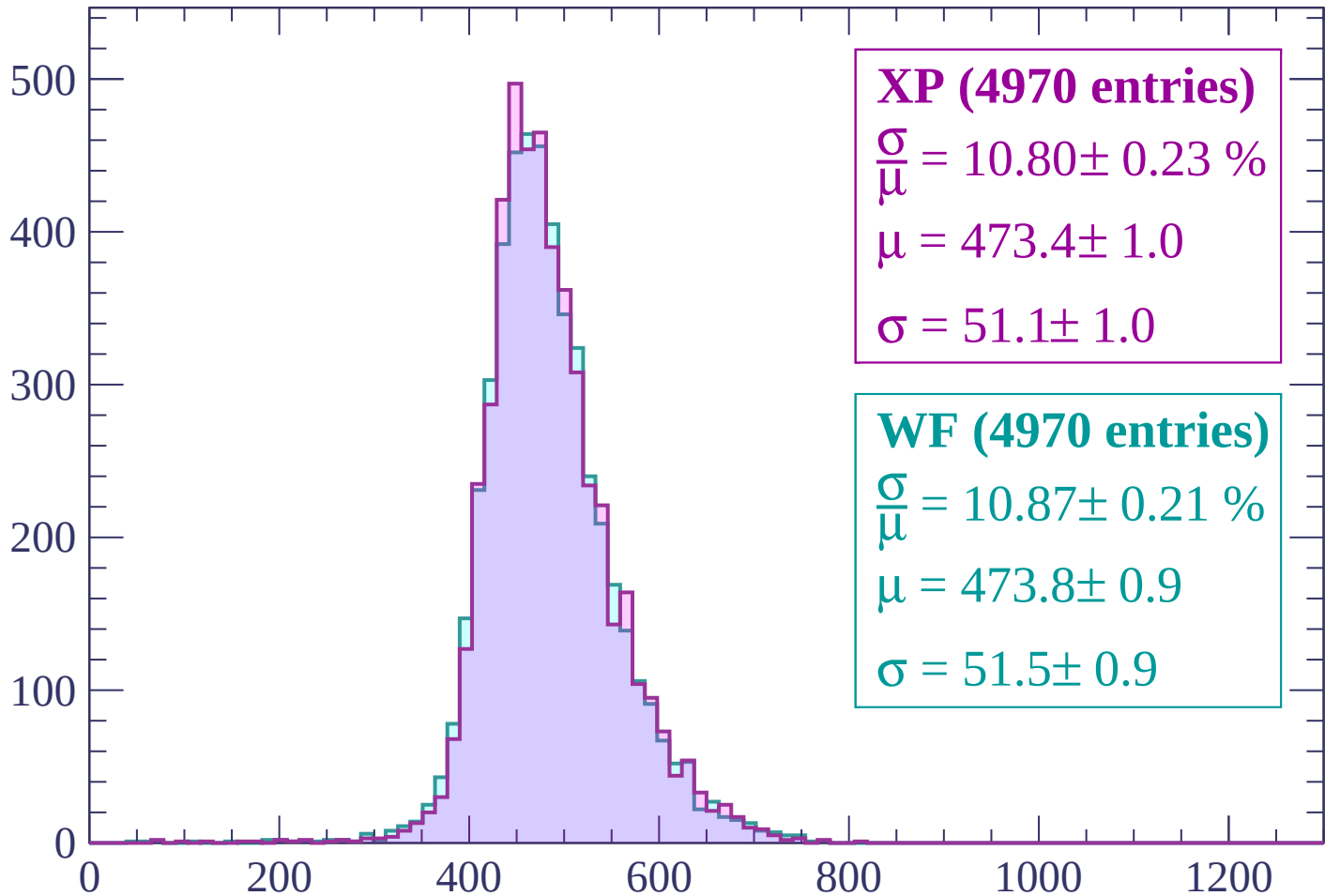
$\mu = 476.7 \pm 1.0$

$\sigma = 54.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-1620 < p < -1560$

Count



XP (4970 entries)

$$\frac{\sigma}{\mu} = 10.80 \pm 0.23 \%$$

$$\mu = 473.4 \pm 1.0$$

$$\sigma = 51.1 \pm 1.0$$

WF (4970 entries)

$$\frac{\sigma}{\mu} = 10.87 \pm 0.21 \%$$

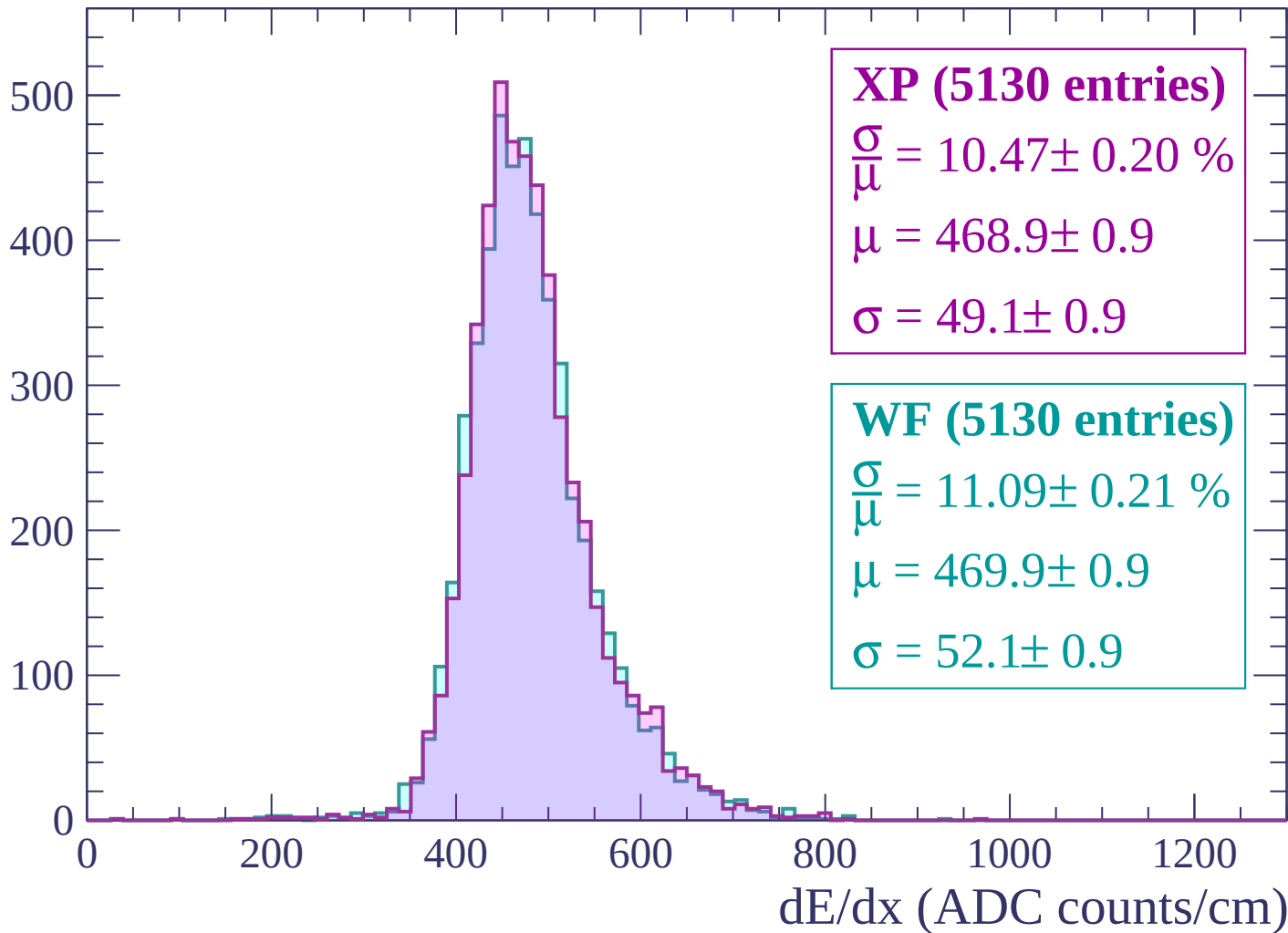
$$\mu = 473.8 \pm 0.9$$

$$\sigma = 51.5 \pm 0.9$$

dE/dx (ADC counts/cm)

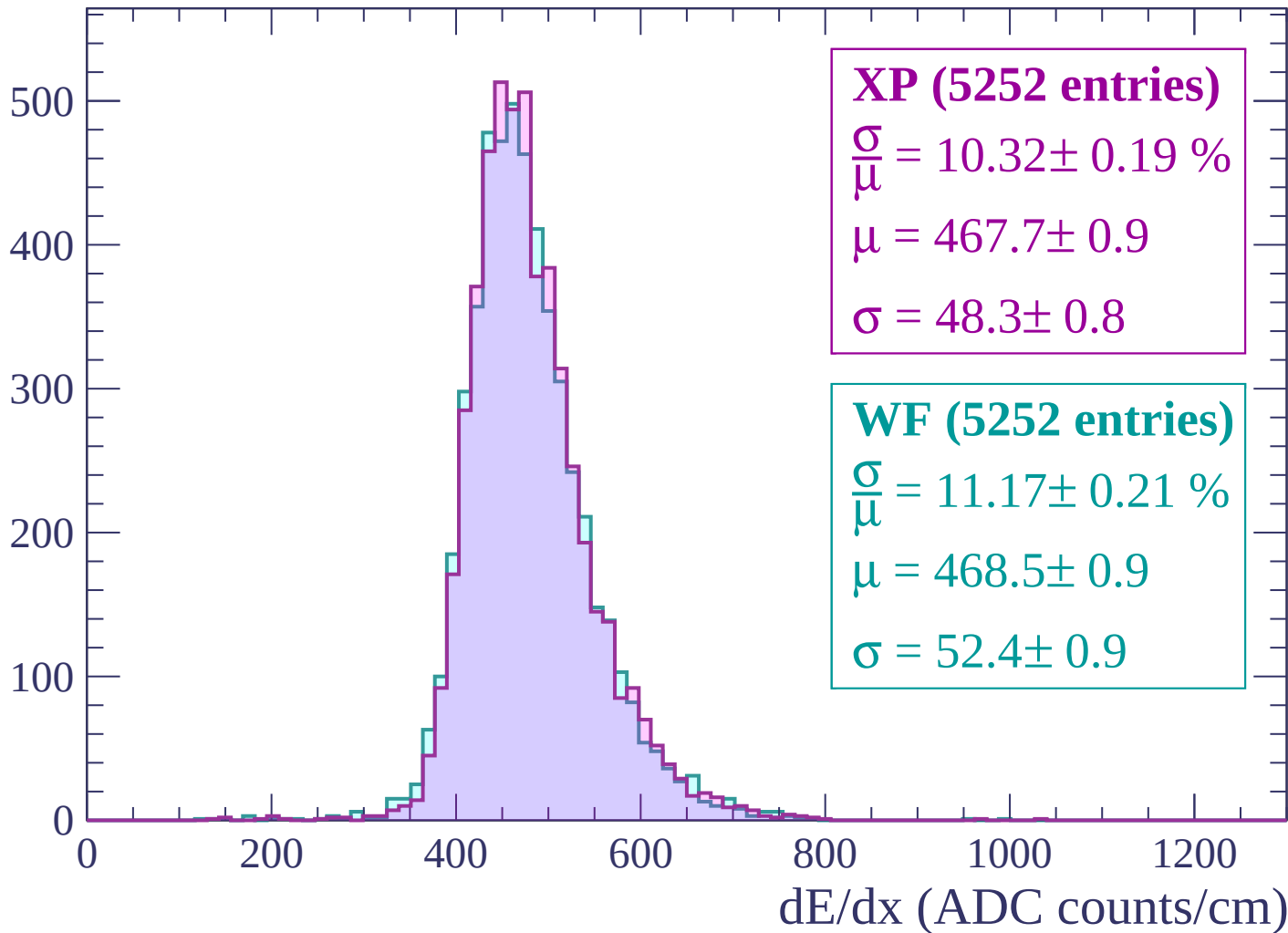
Energy loss | $-1560 < p < -1500$

Count



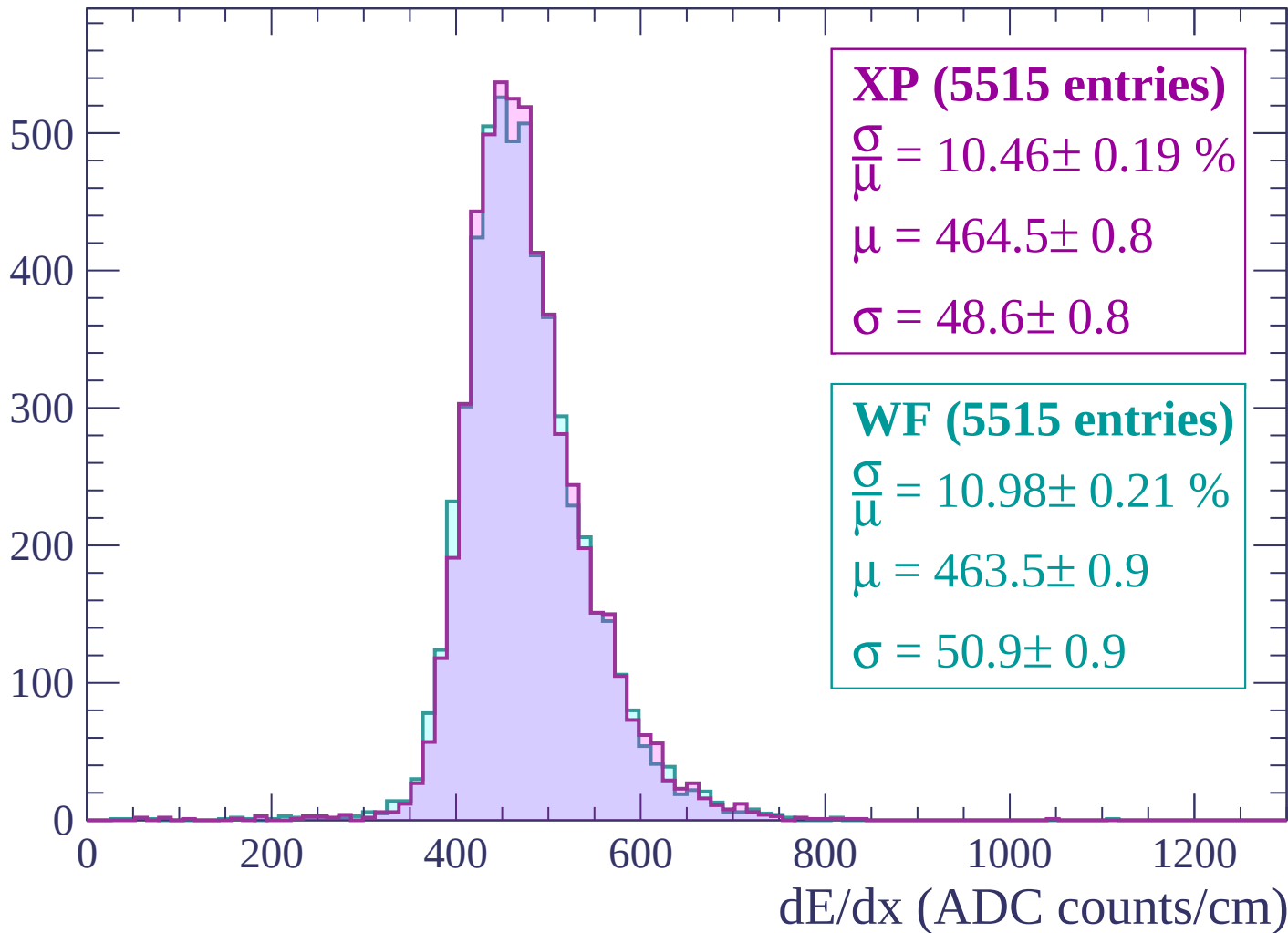
Energy loss | $-1500 < p < -1440$

Count

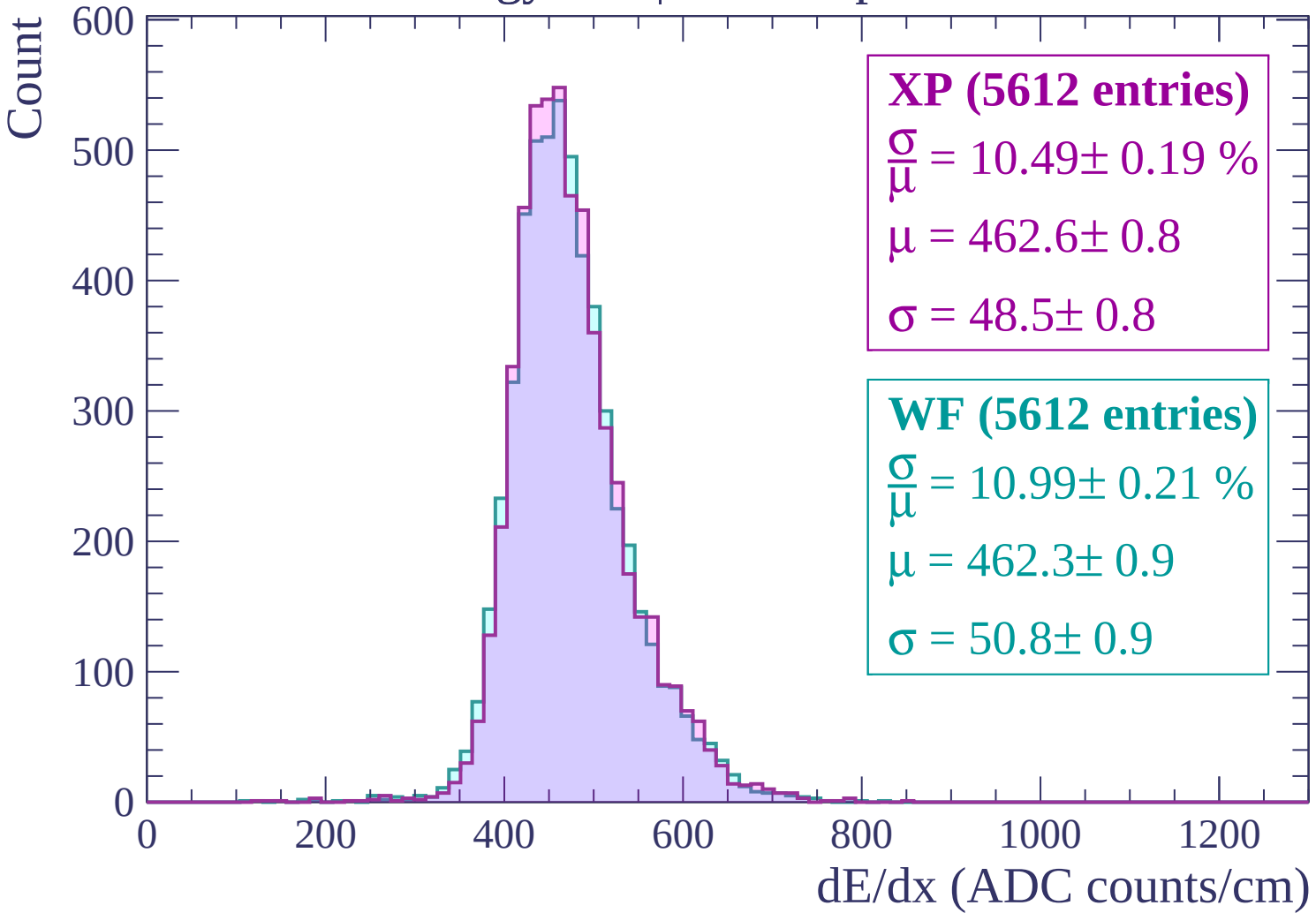


Energy loss | $-1440 < p < -1380$

Count

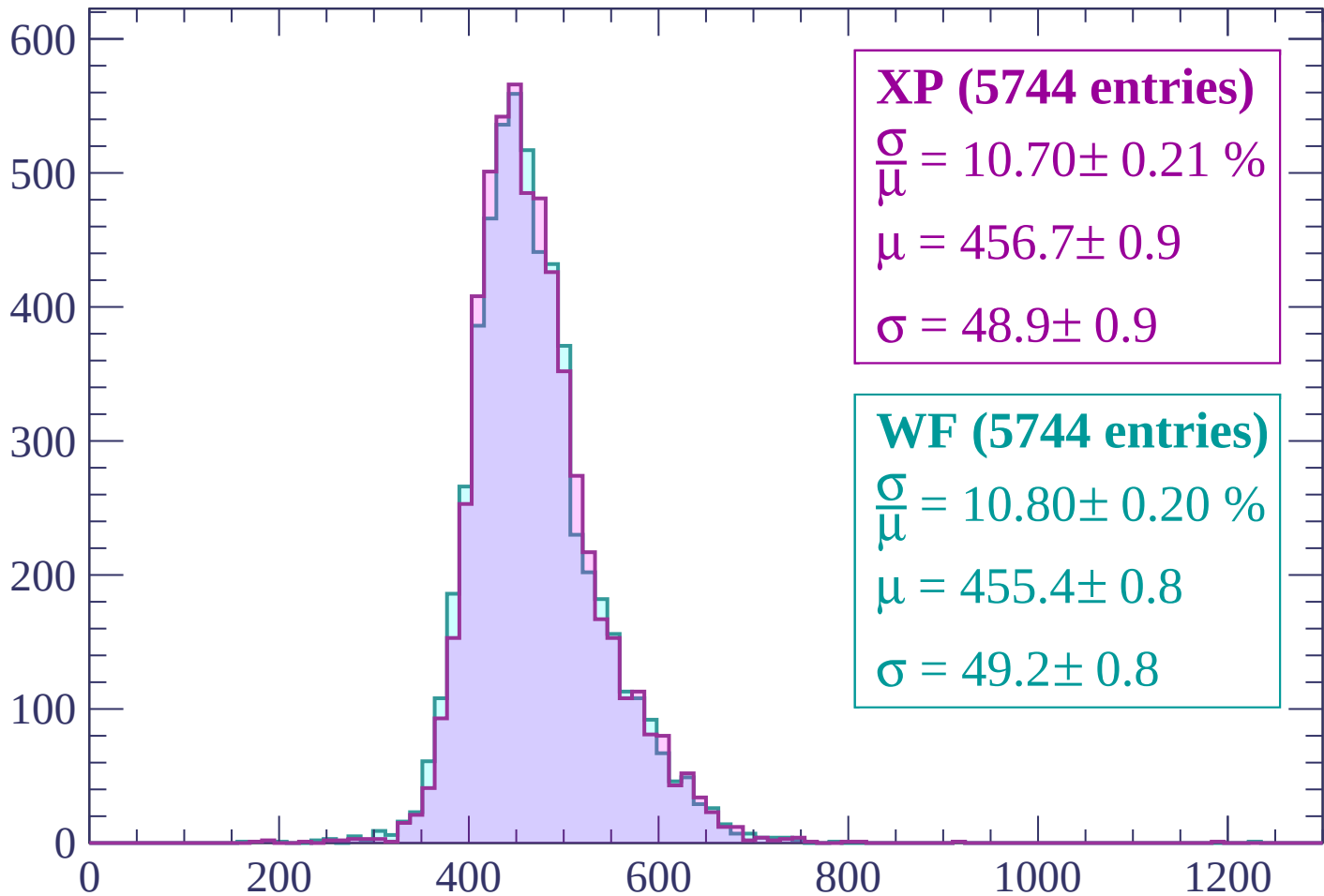


Energy loss | -1380 < p < -1320



Energy loss | -1320 < p < -1260

Count



XP (5744 entries)

$\frac{\sigma}{\mu} = 10.70 \pm 0.21 \%$

$\mu = 456.7 \pm 0.9$

$\sigma = 48.9 \pm 0.9$

WF (5744 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.20 \%$

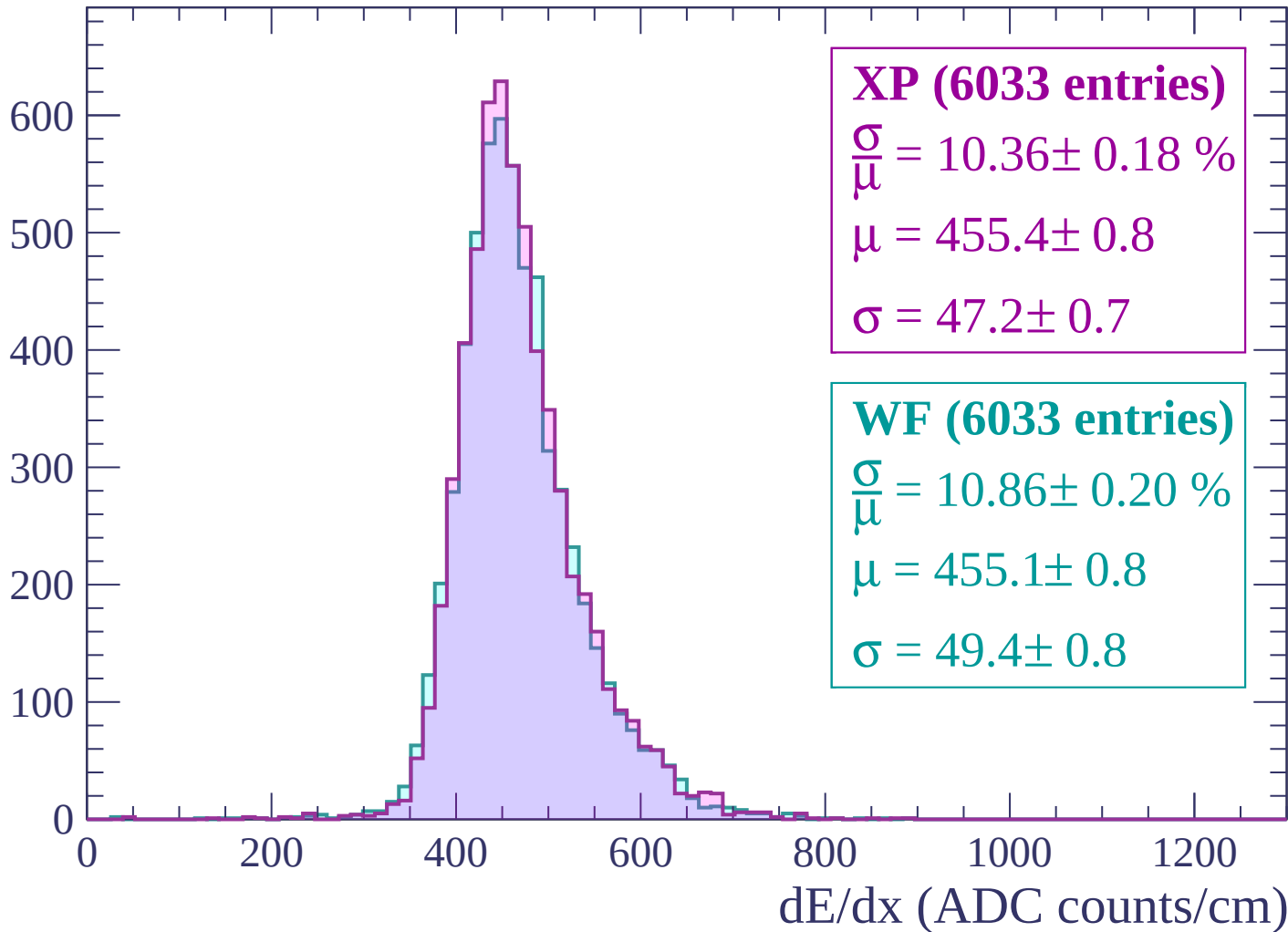
$\mu = 455.4 \pm 0.8$

$\sigma = 49.2 \pm 0.8$

dE/dx (ADC counts/cm)

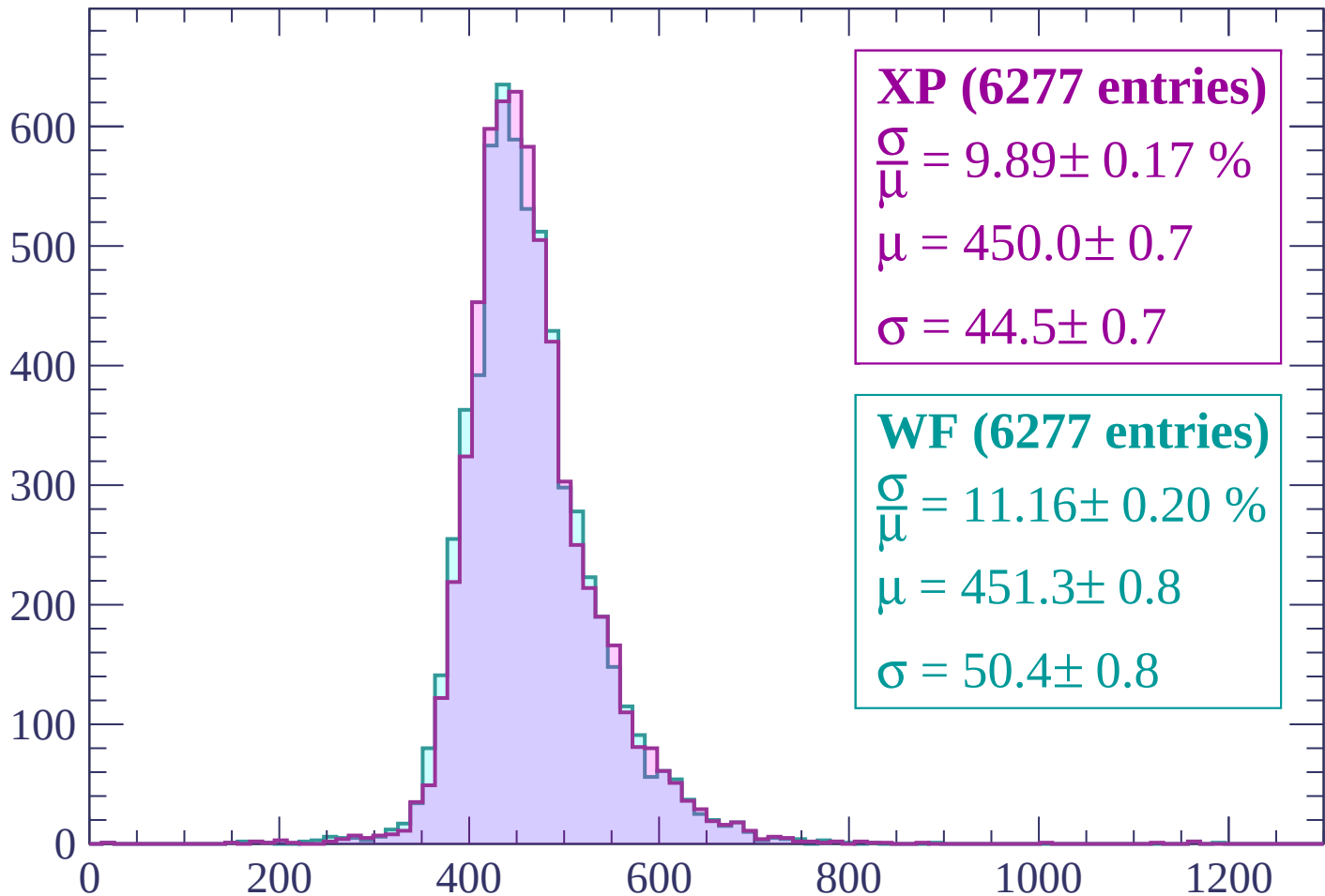
Energy loss | $-1260 < p < -1200$

Count



Energy loss | -1200 < p < -1140

Count



XP (6277 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.17 \%$$

$$\mu = 450.0 \pm 0.7$$

$$\sigma = 44.5 \pm 0.7$$

WF (6277 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.20 \%$$

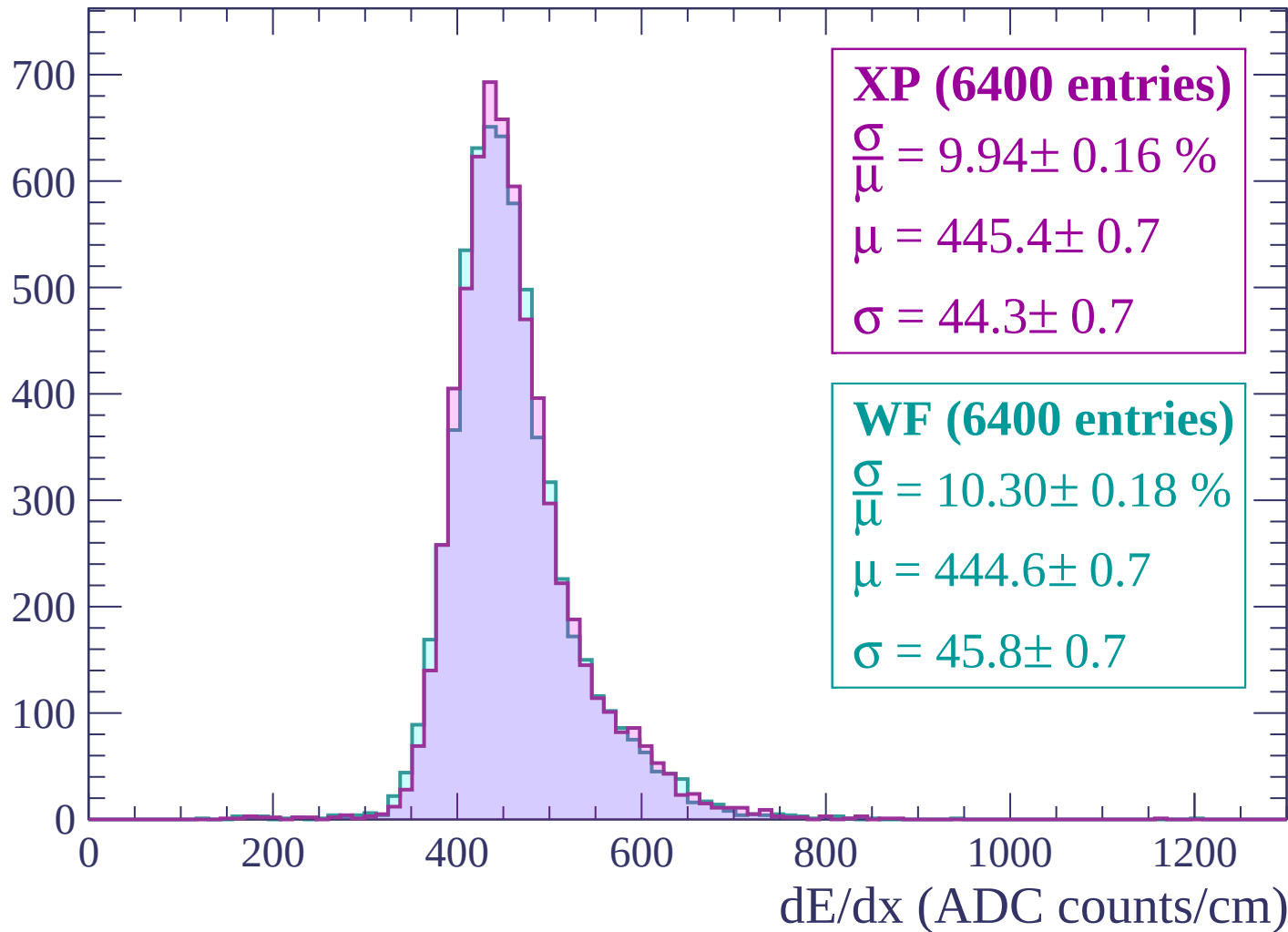
$$\mu = 451.3 \pm 0.8$$

$$\sigma = 50.4 \pm 0.8$$

dE/dx (ADC counts/cm)

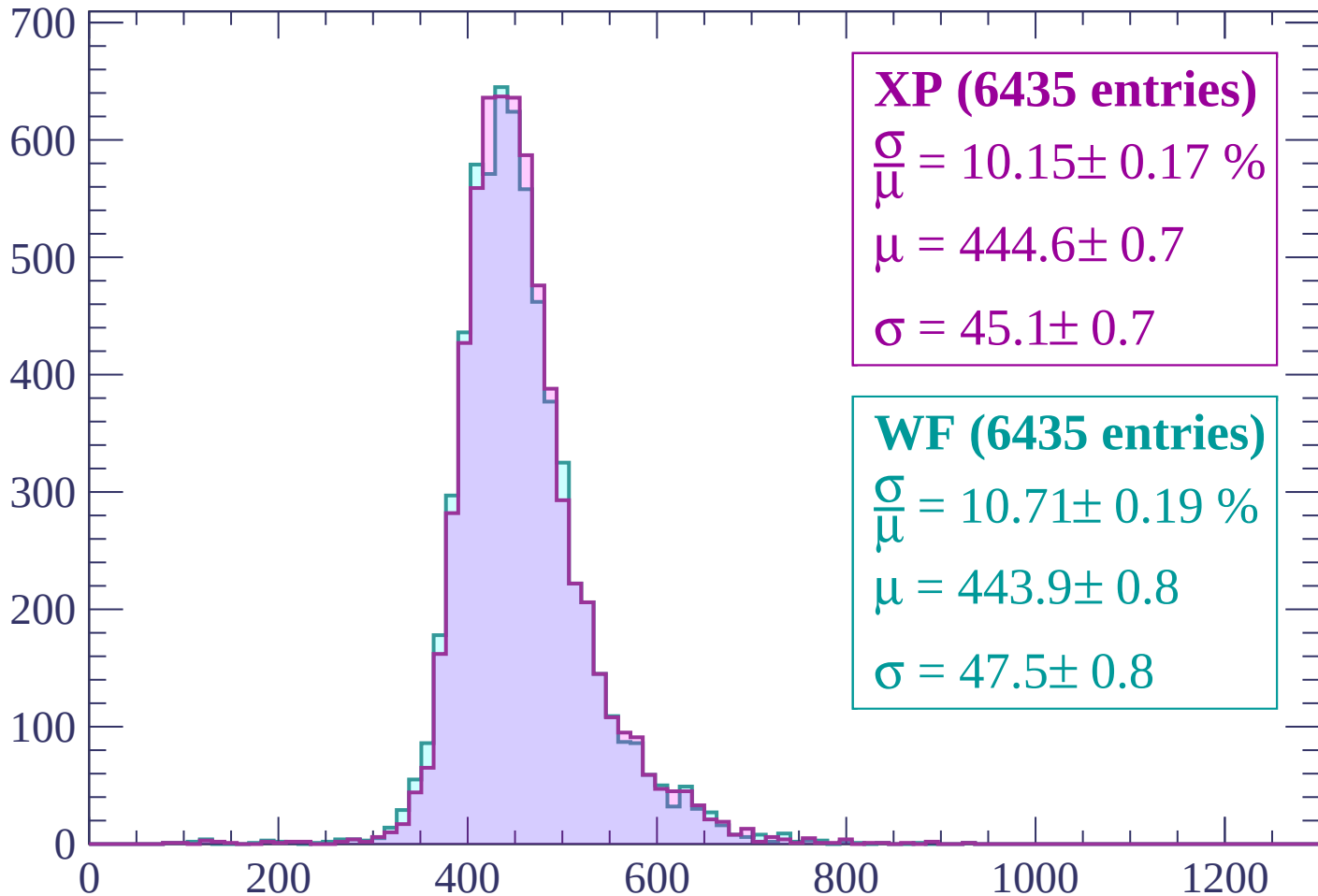
Energy loss | $-1140 < p < -1080$

Count



Energy loss | -1080 < p < -1020

Count



XP (6435 entries)

$$\frac{\sigma}{\mu} = 10.15 \pm 0.17 \%$$

$$\mu = 444.6 \pm 0.7$$

$$\sigma = 45.1 \pm 0.7$$

WF (6435 entries)

$$\frac{\sigma}{\mu} = 10.71 \pm 0.19 \%$$

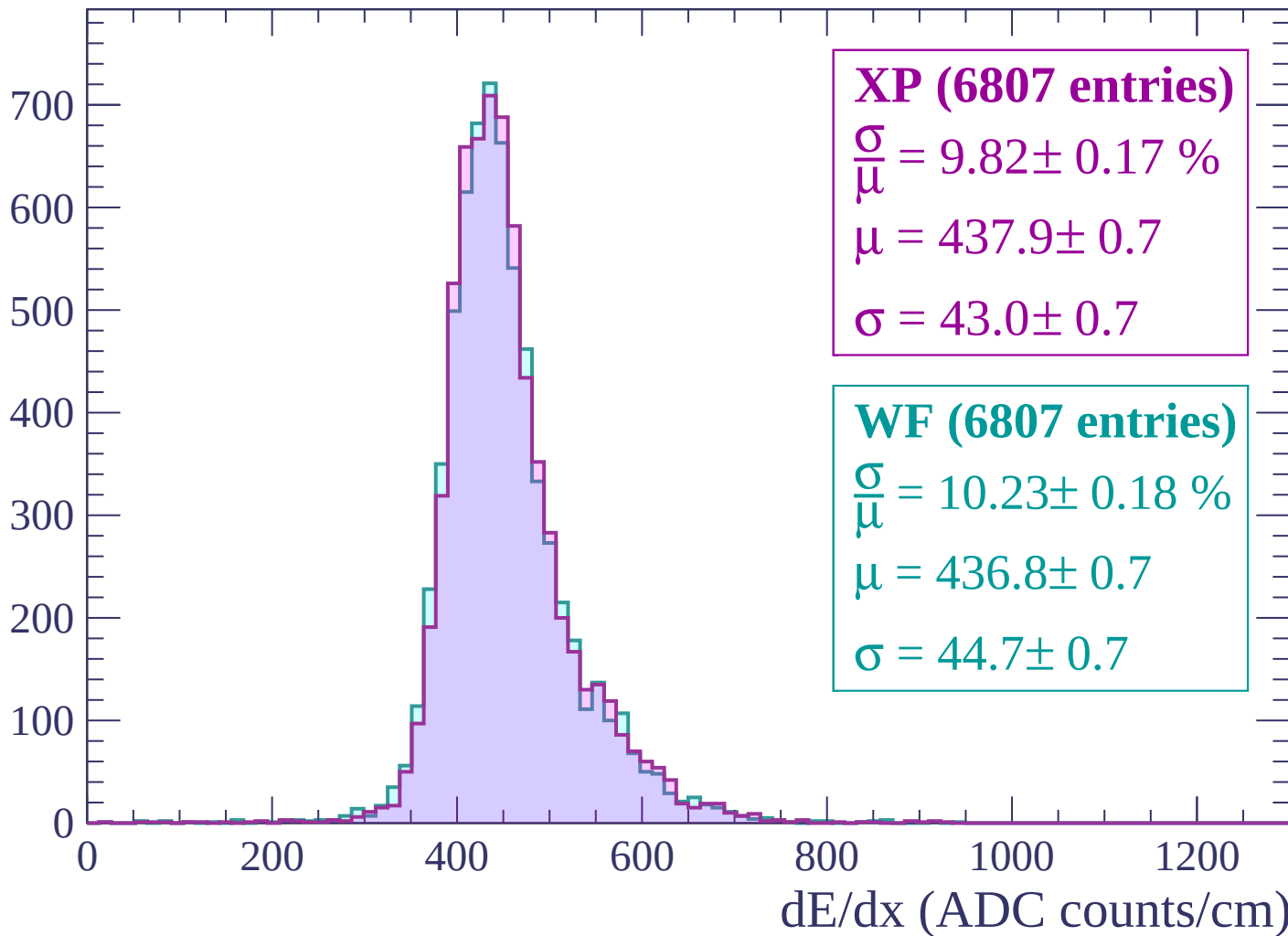
$$\mu = 443.9 \pm 0.8$$

$$\sigma = 47.5 \pm 0.8$$

dE/dx (ADC counts/cm)

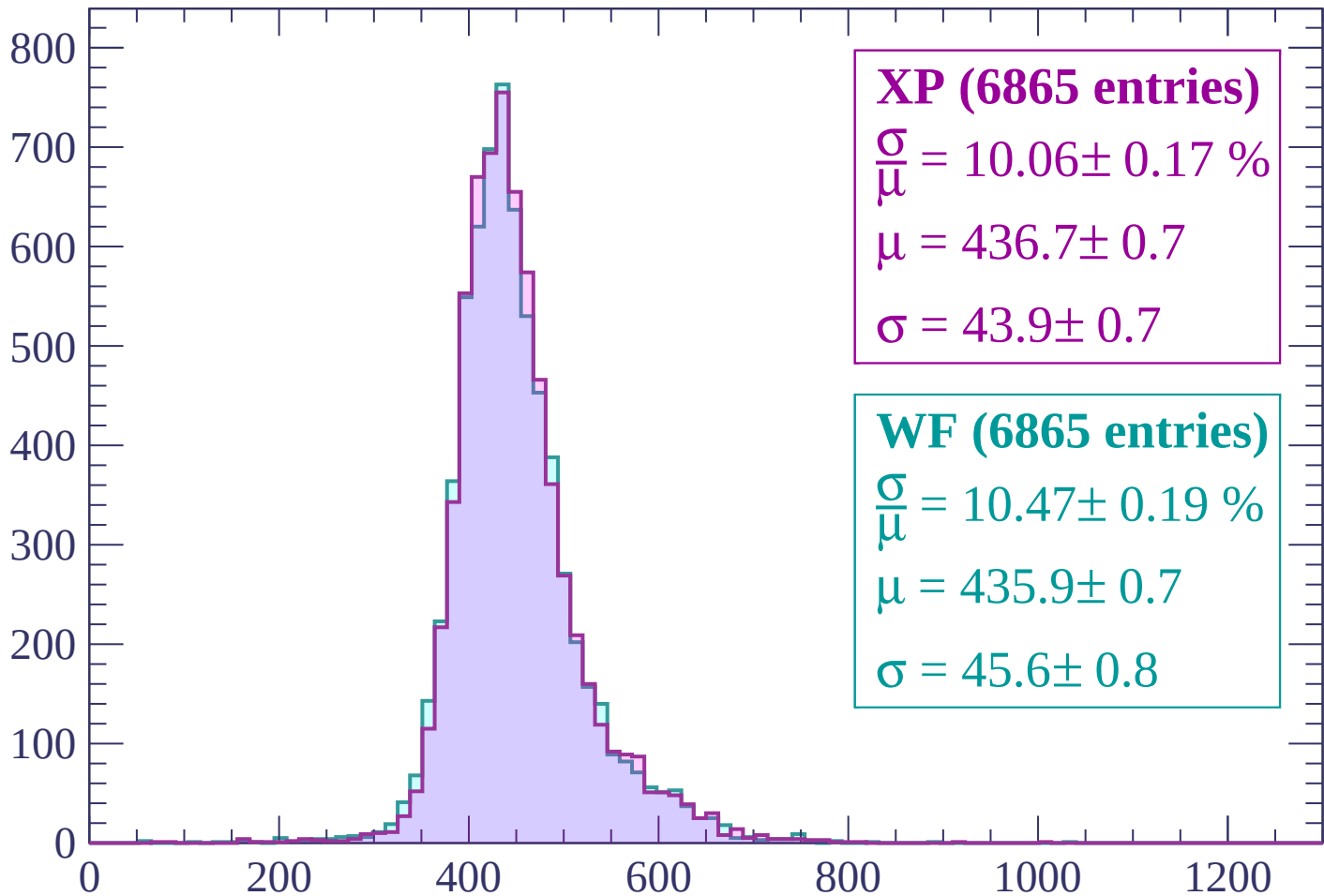
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (6865 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.17 \%$$

$$\mu = 436.7 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

WF (6865 entries)

$$\frac{\sigma}{\mu} = 10.47 \pm 0.19 \%$$

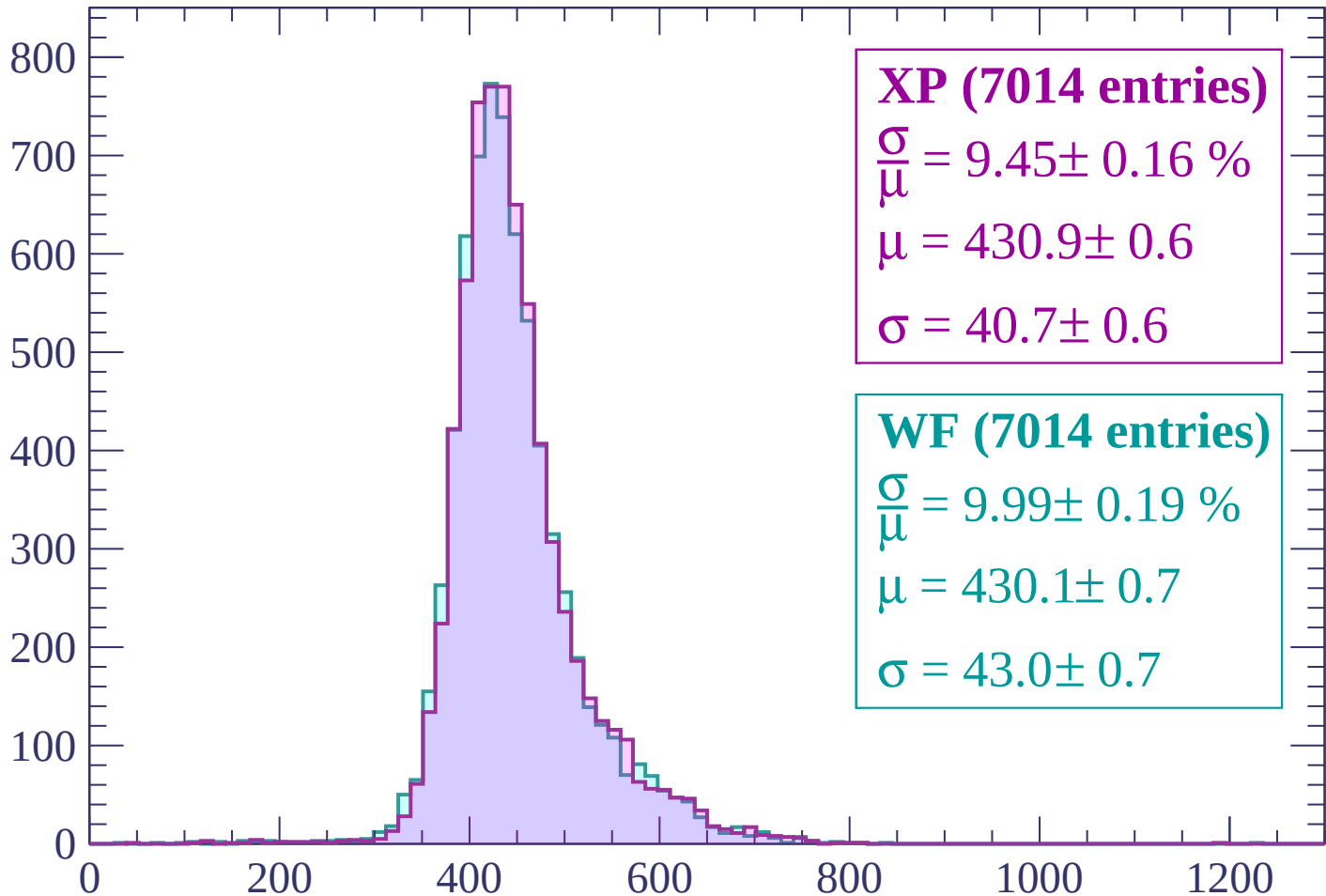
$$\mu = 435.9 \pm 0.7$$

$$\sigma = 45.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (7014 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.16 \%$$

$$\mu = 430.9 \pm 0.6$$

$$\sigma = 40.7 \pm 0.6$$

WF (7014 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.19 \%$$

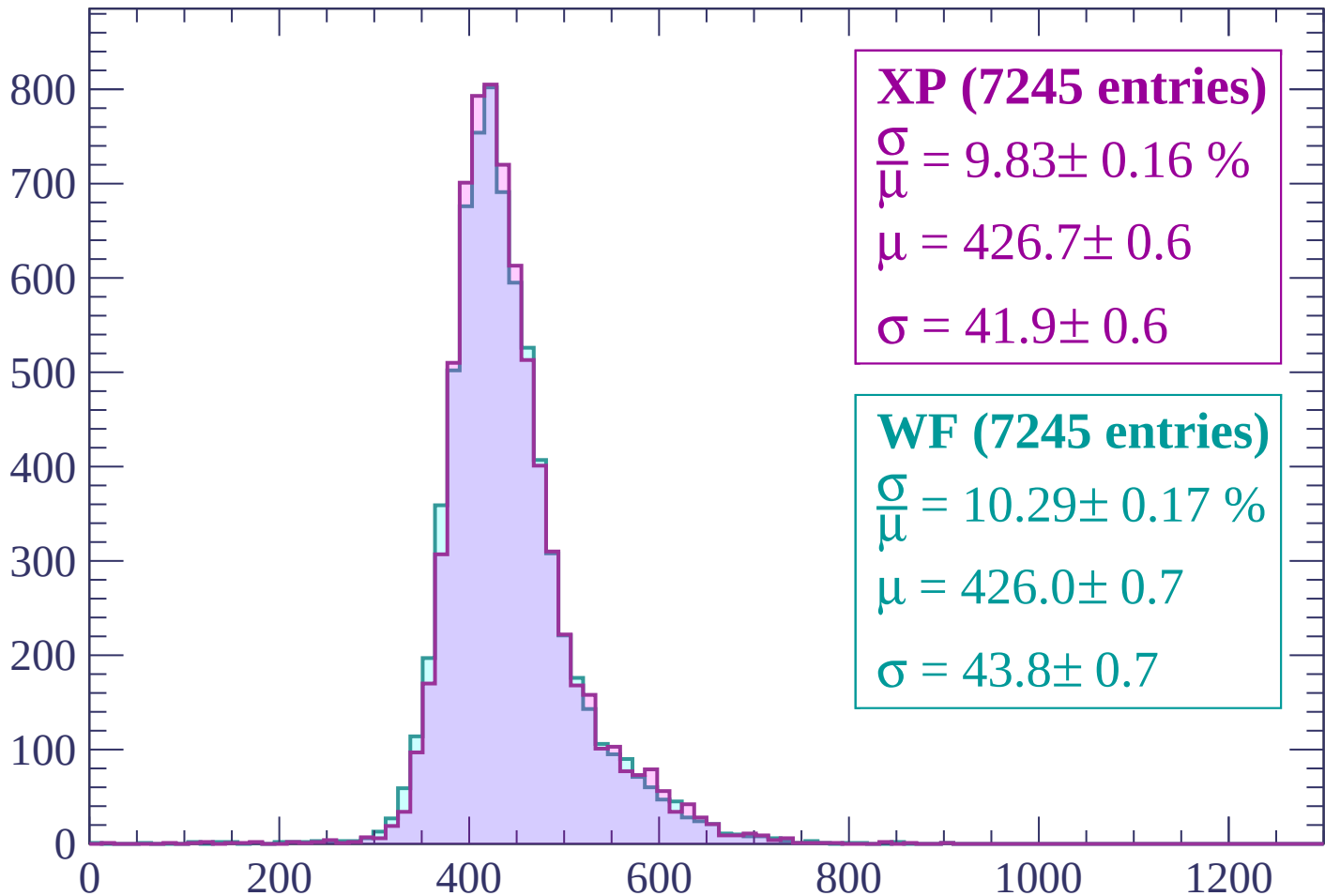
$$\mu = 430.1 \pm 0.7$$

$$\sigma = 43.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (7245 entries)

$$\frac{\sigma}{\mu} = 9.83 \pm 0.16 \%$$

$$\mu = 426.7 \pm 0.6$$

$$\sigma = 41.9 \pm 0.6$$

WF (7245 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.17 \%$$

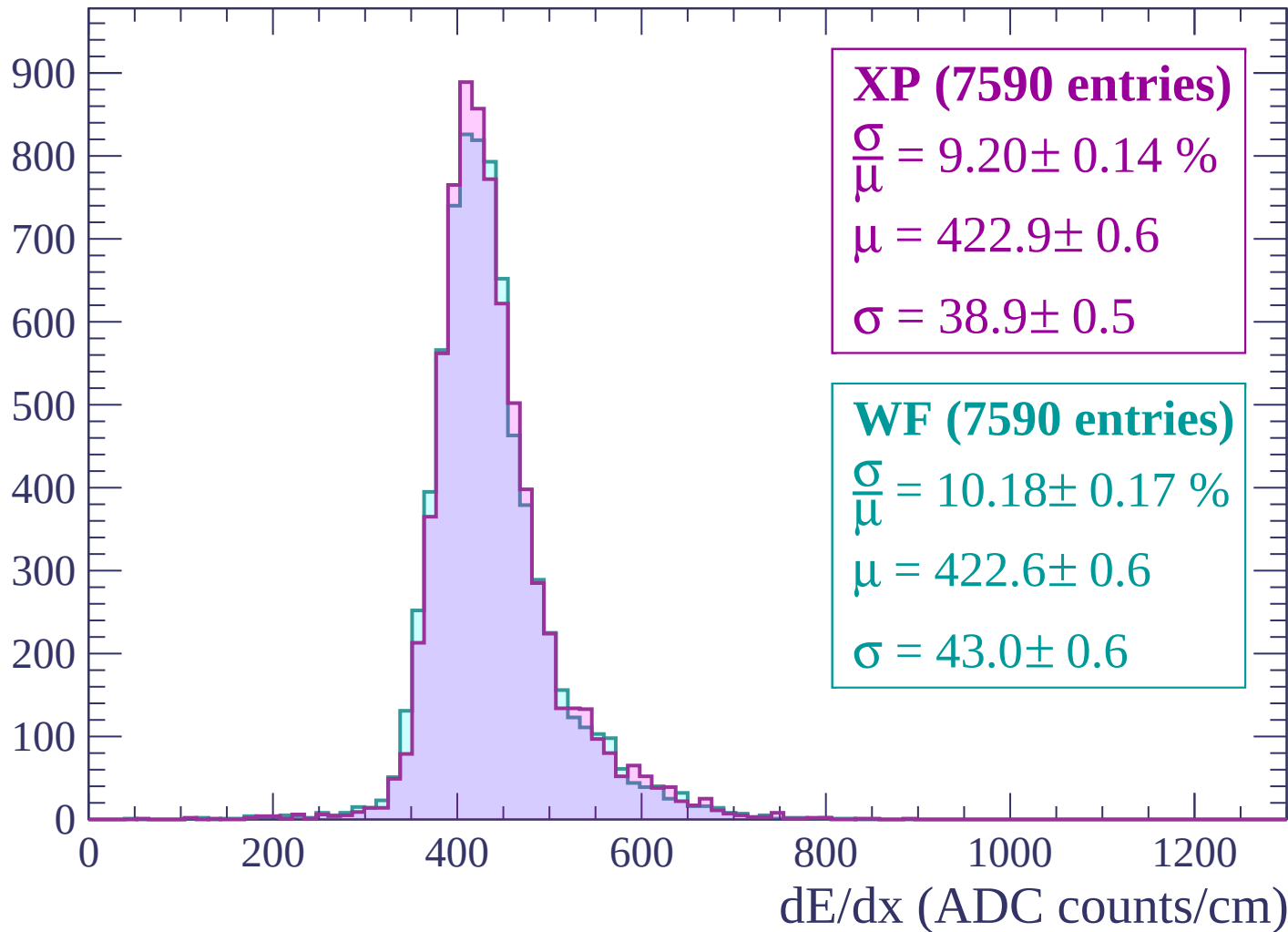
$$\mu = 426.0 \pm 0.7$$

$$\sigma = 43.8 \pm 0.7$$

dE/dx (ADC counts/cm)

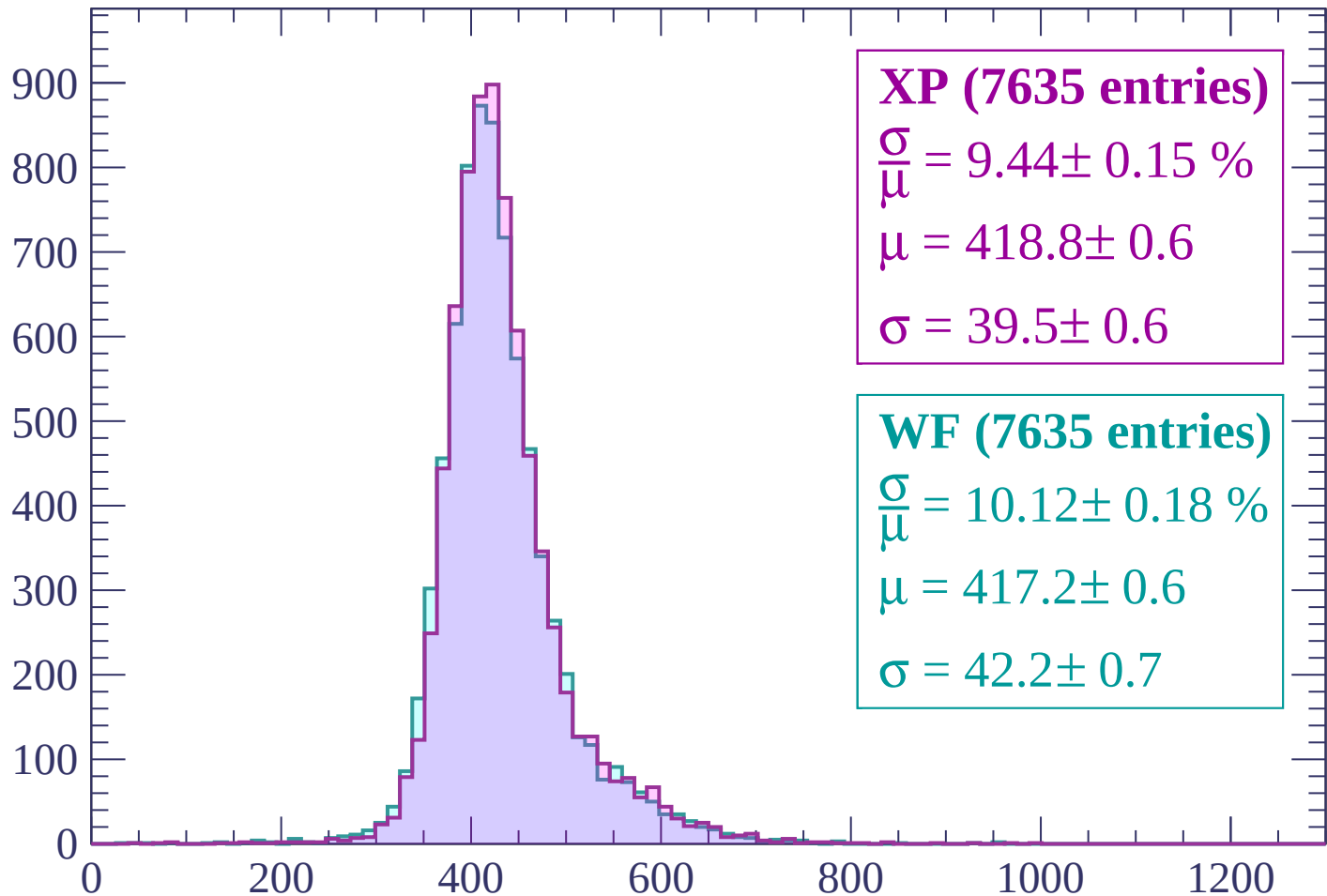
Energy loss | $-780 < p < -720$

Count



Energy loss | $-720 < p < -660$

Count



XP (7635 entries)

$$\frac{\sigma}{\mu} = 9.44 \pm 0.15 \%$$

$$\mu = 418.8 \pm 0.6$$

$$\sigma = 39.5 \pm 0.6$$

WF (7635 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.18 \%$$

$$\mu = 417.2 \pm 0.6$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7994 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.15 \%$

$\mu = 413.1 \pm 0.5$

$\sigma = 37.3 \pm 0.6$

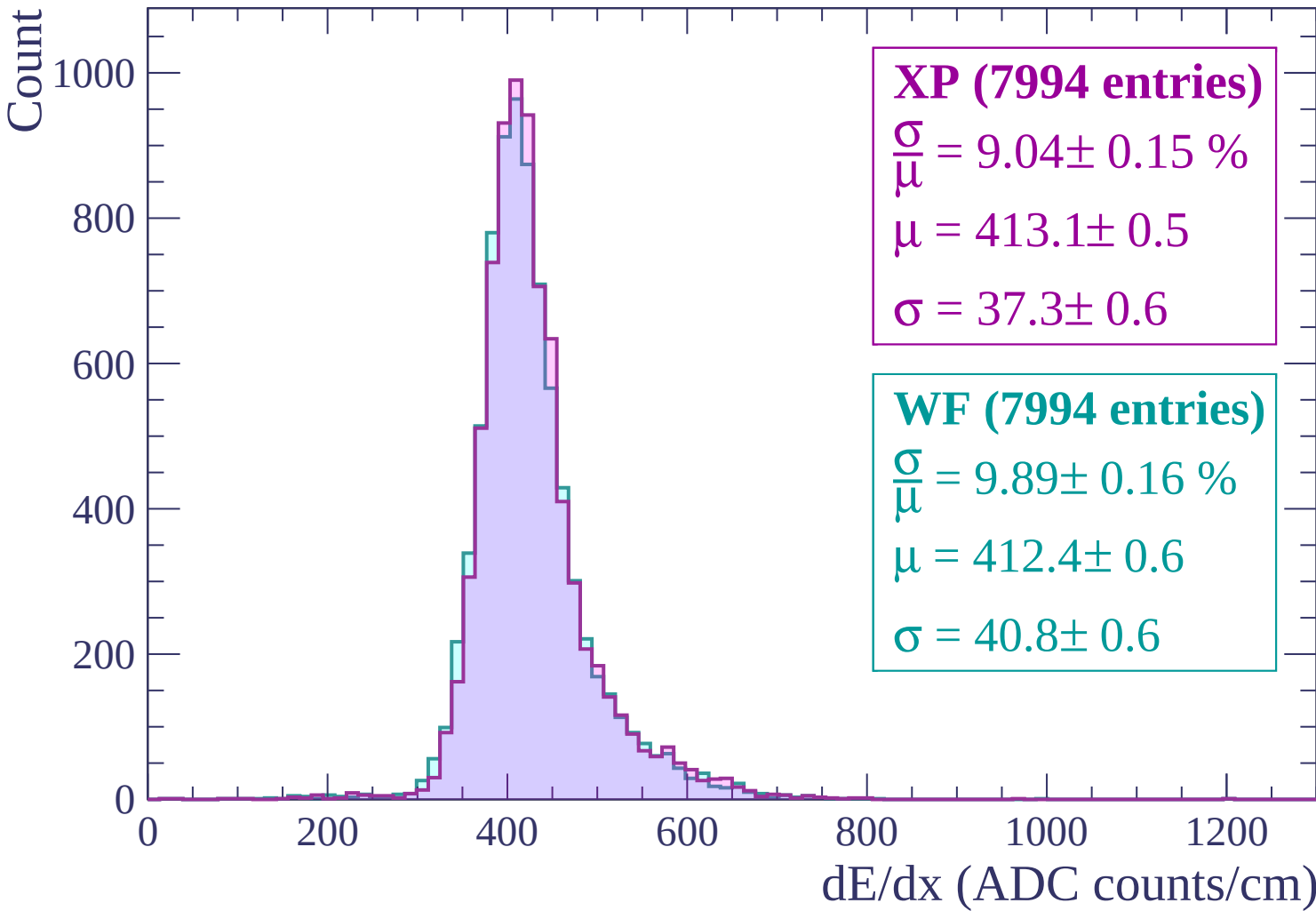
WF (7994 entries)

$\frac{\sigma}{\mu} = 9.89 \pm 0.16 \%$

$\mu = 412.4 \pm 0.6$

$\sigma = 40.8 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $-600 < p < -540$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8001 entries)

$\frac{\sigma}{\mu} = 9.14 \pm 0.15 \%$

$\mu = 409.4 \pm 0.5$

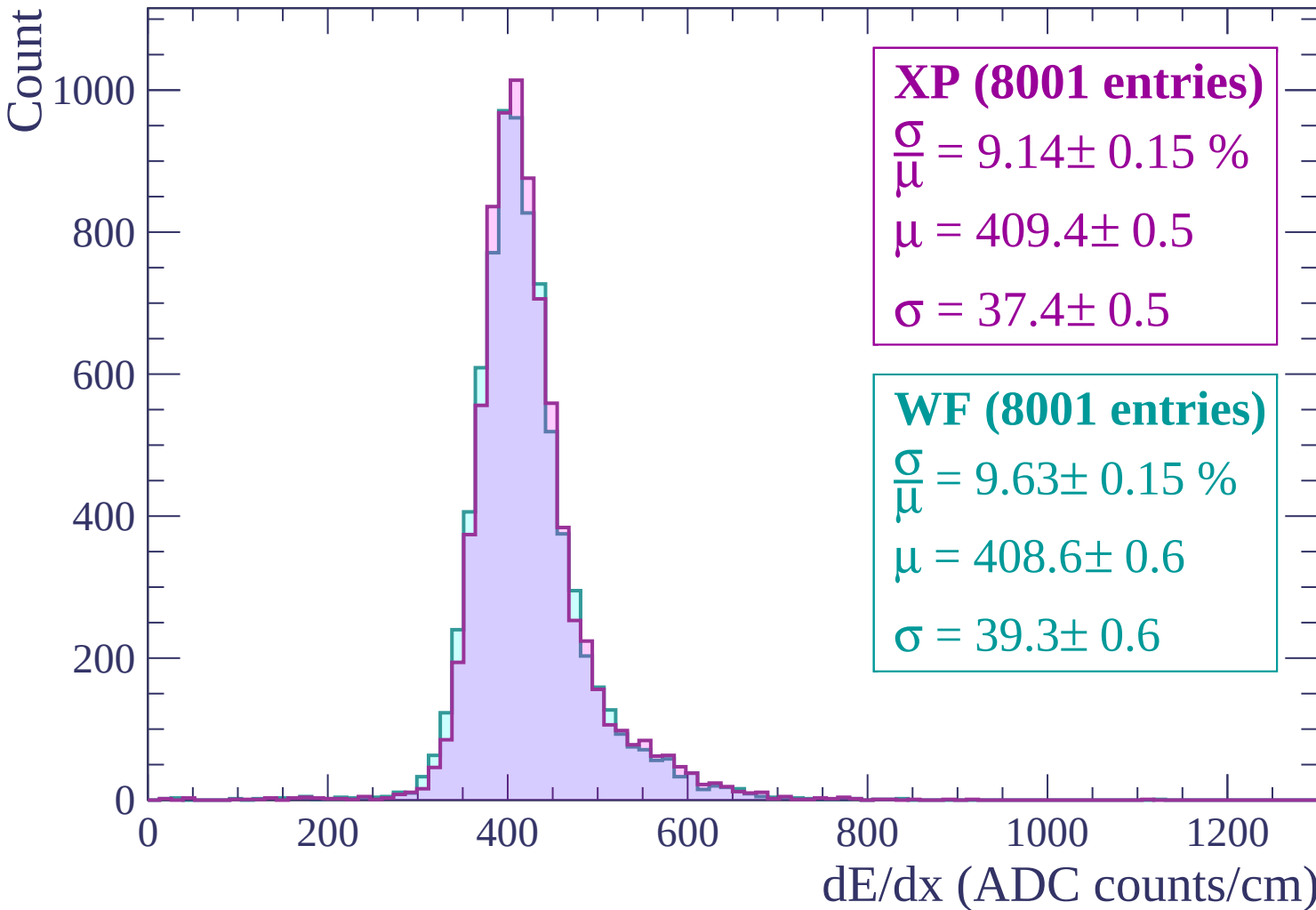
$\sigma = 37.4 \pm 0.5$

WF (8001 entries)

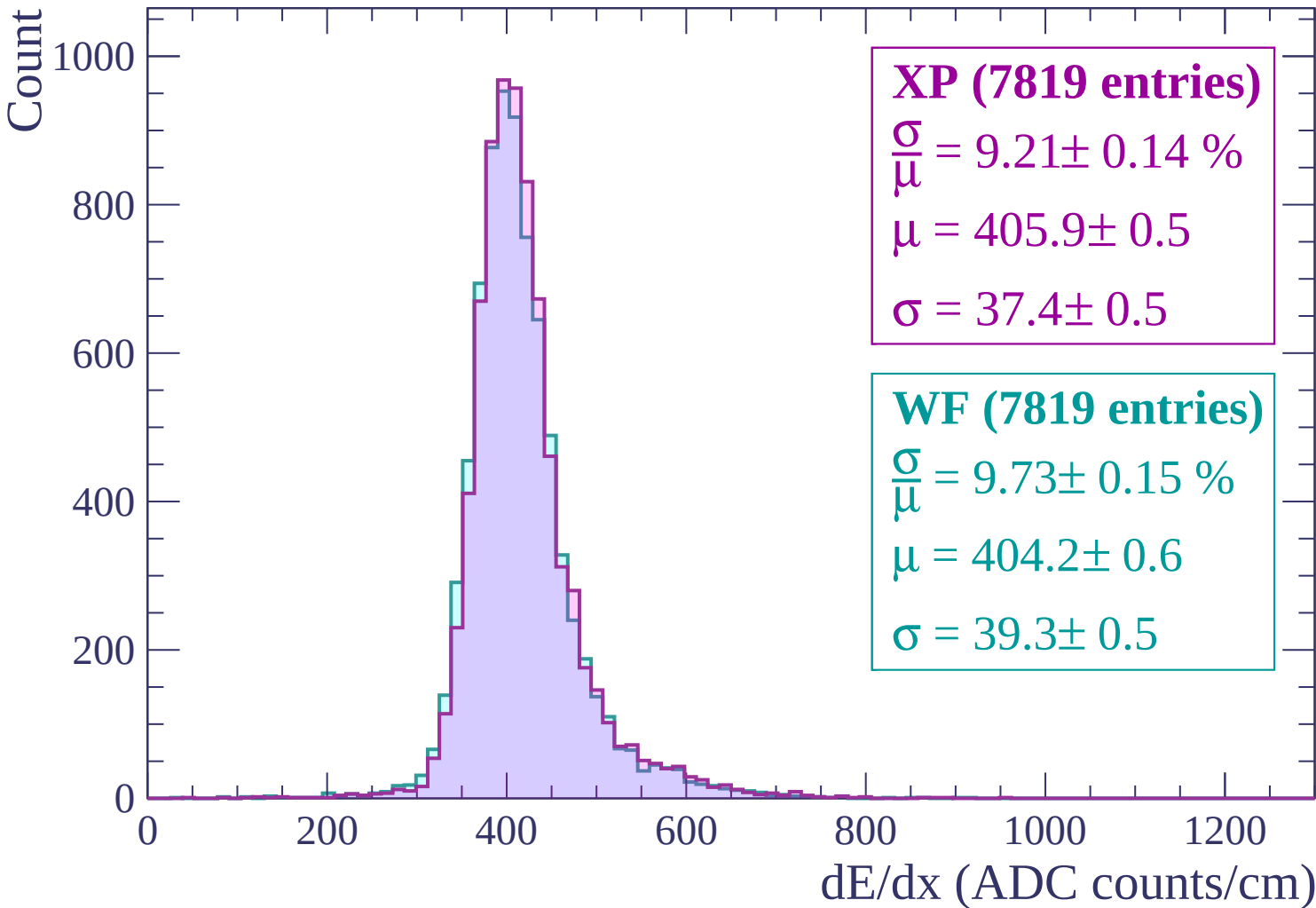
$\frac{\sigma}{\mu} = 9.63 \pm 0.15 \%$

$\mu = 408.6 \pm 0.6$

$\sigma = 39.3 \pm 0.6$



Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$

Count

1000

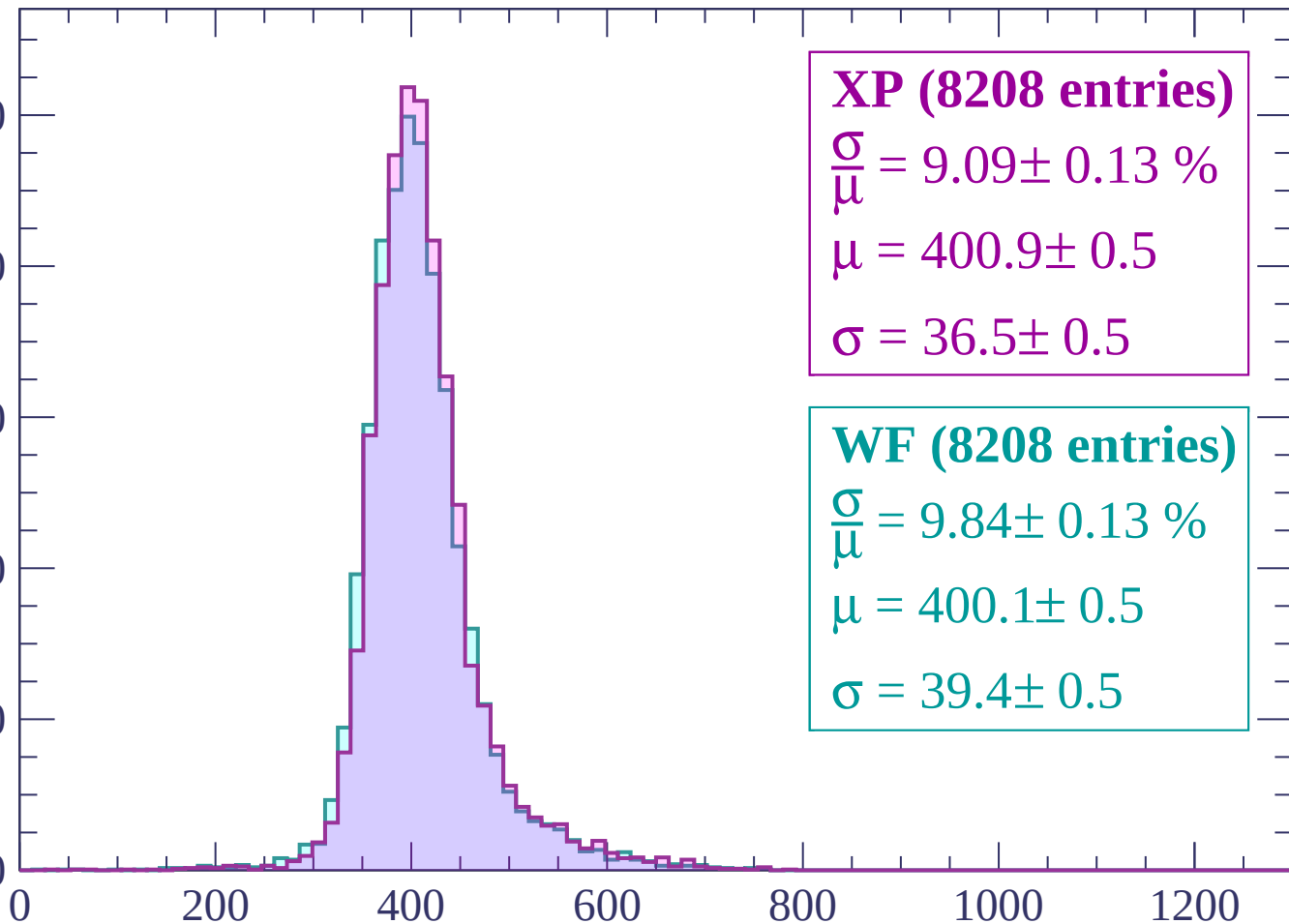
800

600

400

200

0



XP (8208 entries)

$\frac{\sigma}{\mu} = 9.09 \pm 0.13 \%$

$\mu = 400.9 \pm 0.5$

$\sigma = 36.5 \pm 0.5$

WF (8208 entries)

$\frac{\sigma}{\mu} = 9.84 \pm 0.13 \%$

$\mu = 400.1 \pm 0.5$

$\sigma = 39.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8195 entries)

$$\frac{\sigma}{\mu} = 8.95 \pm 0.13 \%$$

$$\mu = 398.8 \pm 0.5$$

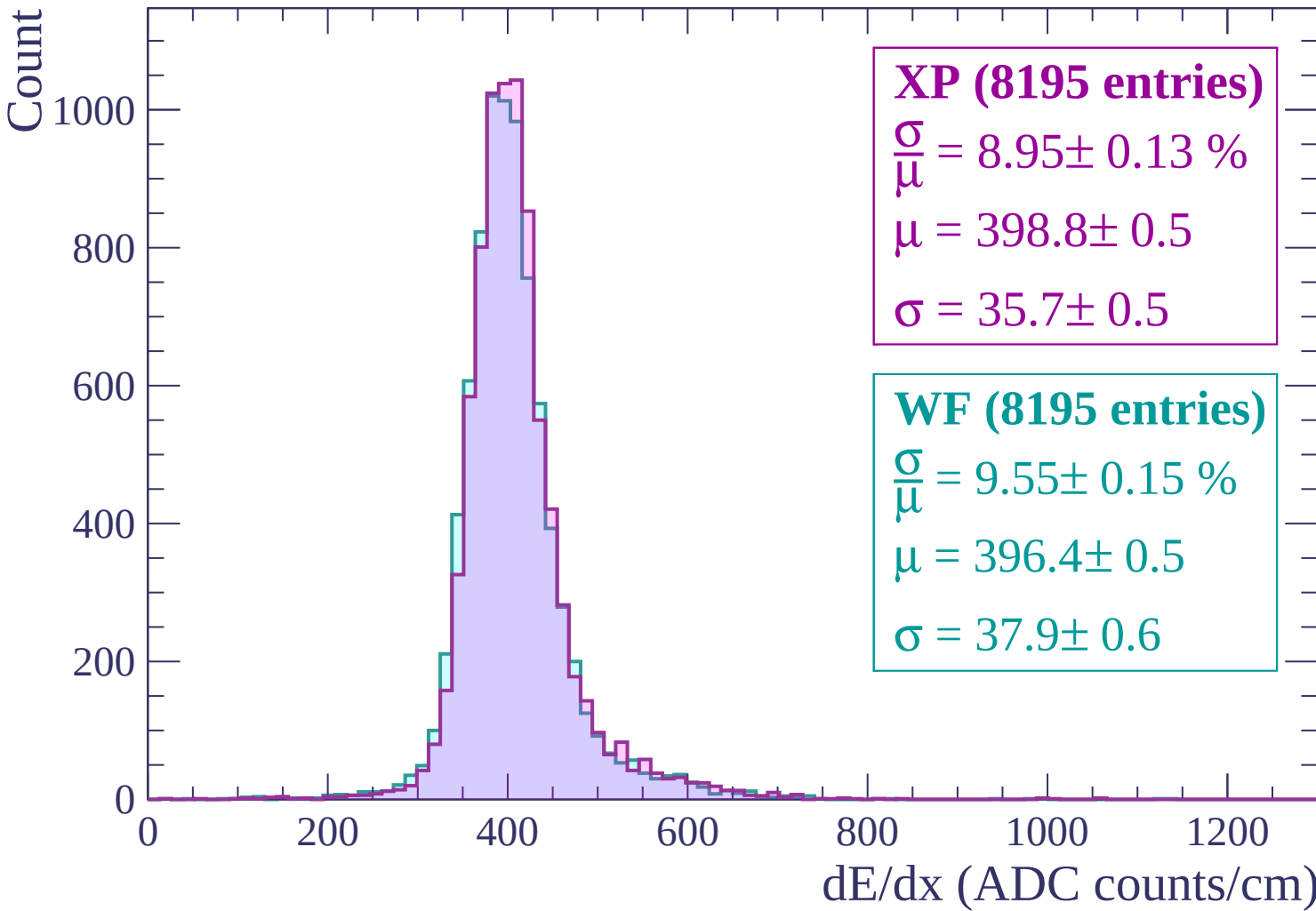
$$\sigma = 35.7 \pm 0.5$$

WF (8195 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.15 \%$$

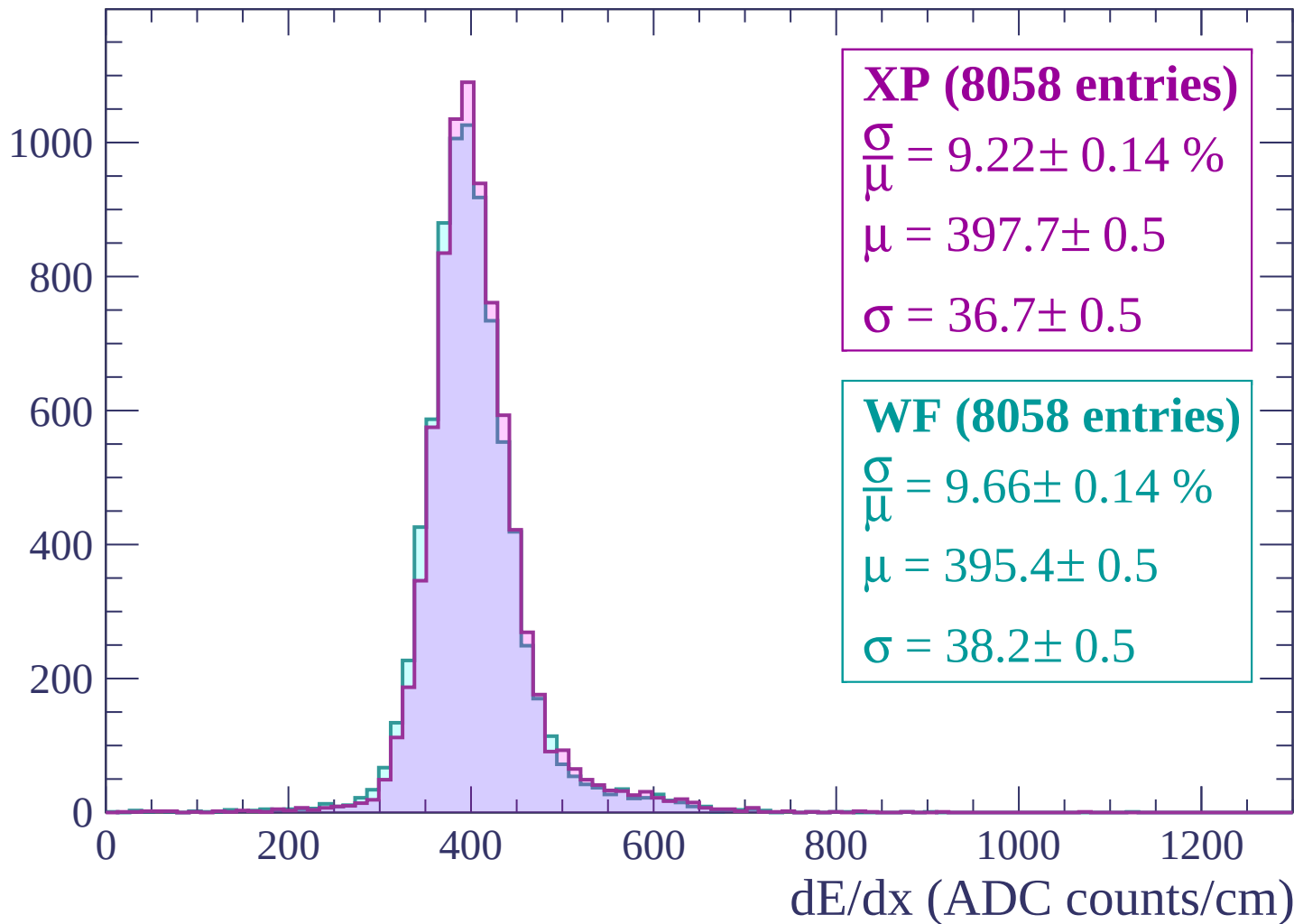
$$\mu = 396.4 \pm 0.5$$

$$\sigma = 37.9 \pm 0.6$$



Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count

1000

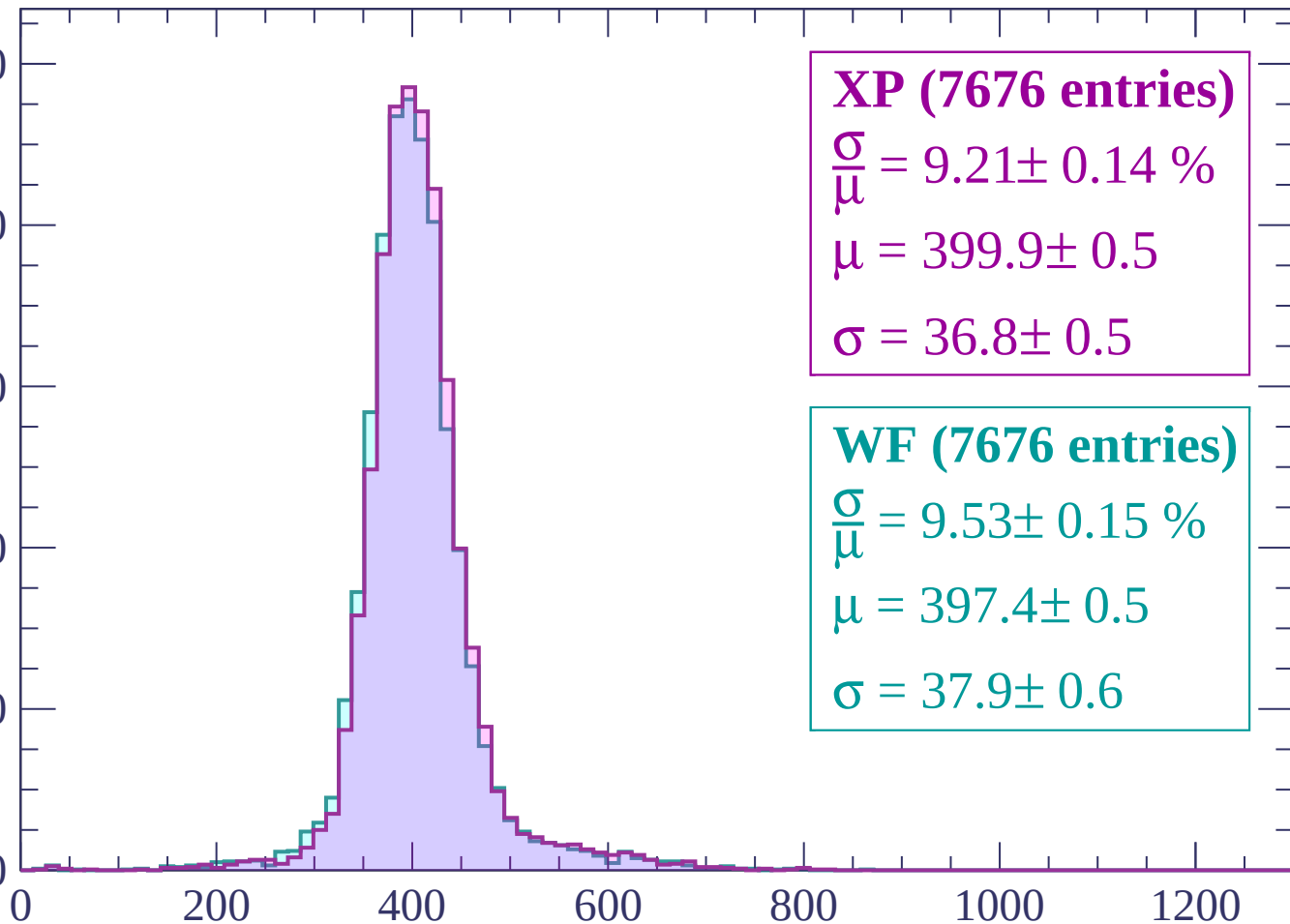
800

600

400

200

0



XP (7676 entries)

$\frac{\sigma}{\mu} = 9.21 \pm 0.14 \%$

$\mu = 399.9 \pm 0.5$

$\sigma = 36.8 \pm 0.5$

WF (7676 entries)

$\frac{\sigma}{\mu} = 9.53 \pm 0.15 \%$

$\mu = 397.4 \pm 0.5$

$\sigma = 37.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count

1000

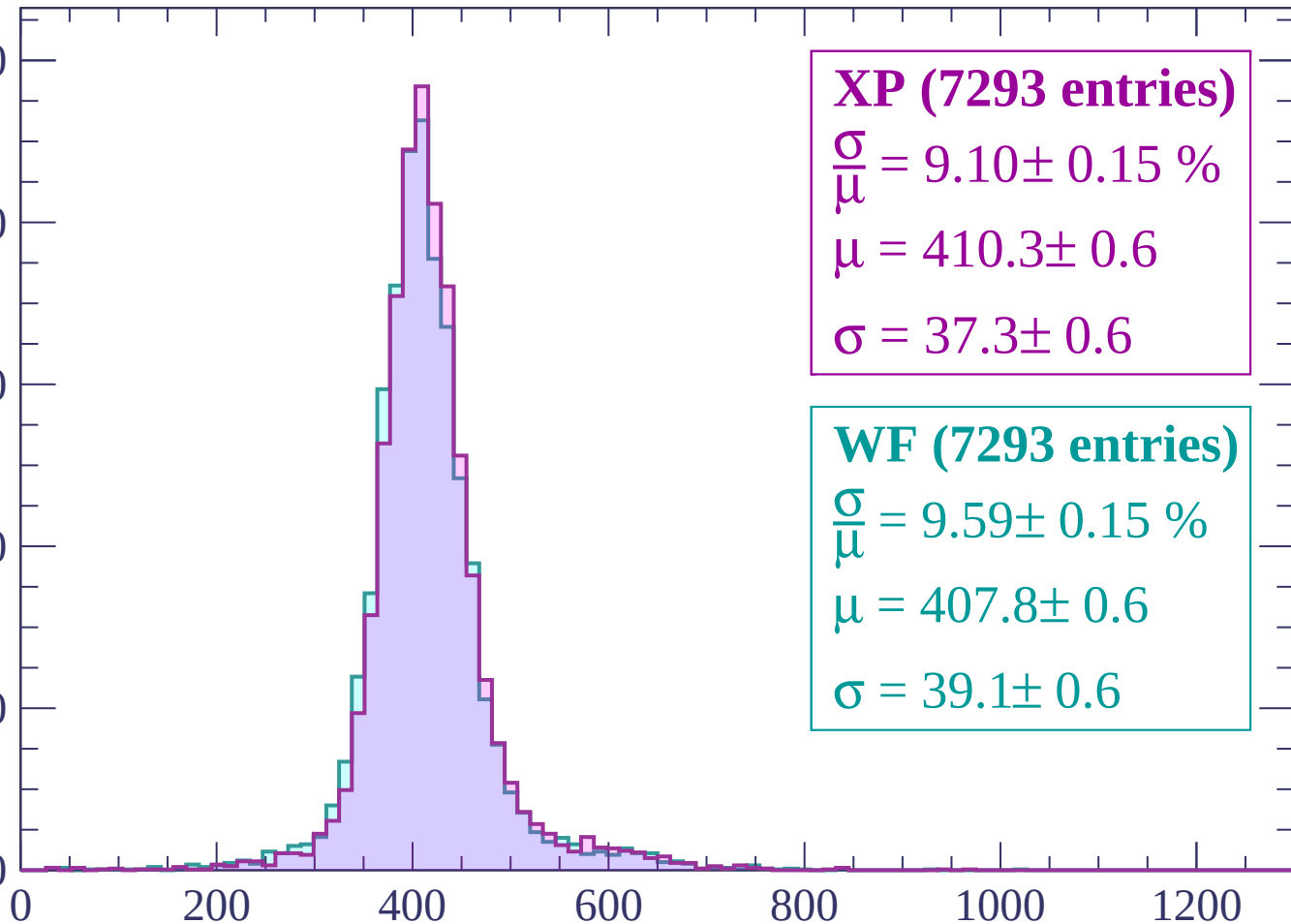
800

600

400

200

0



XP (7293 entries)

$\frac{\sigma}{\mu} = 9.10 \pm 0.15 \%$

$\mu = 410.3 \pm 0.6$

$\sigma = 37.3 \pm 0.6$

WF (7293 entries)

$\frac{\sigma}{\mu} = 9.59 \pm 0.15 \%$

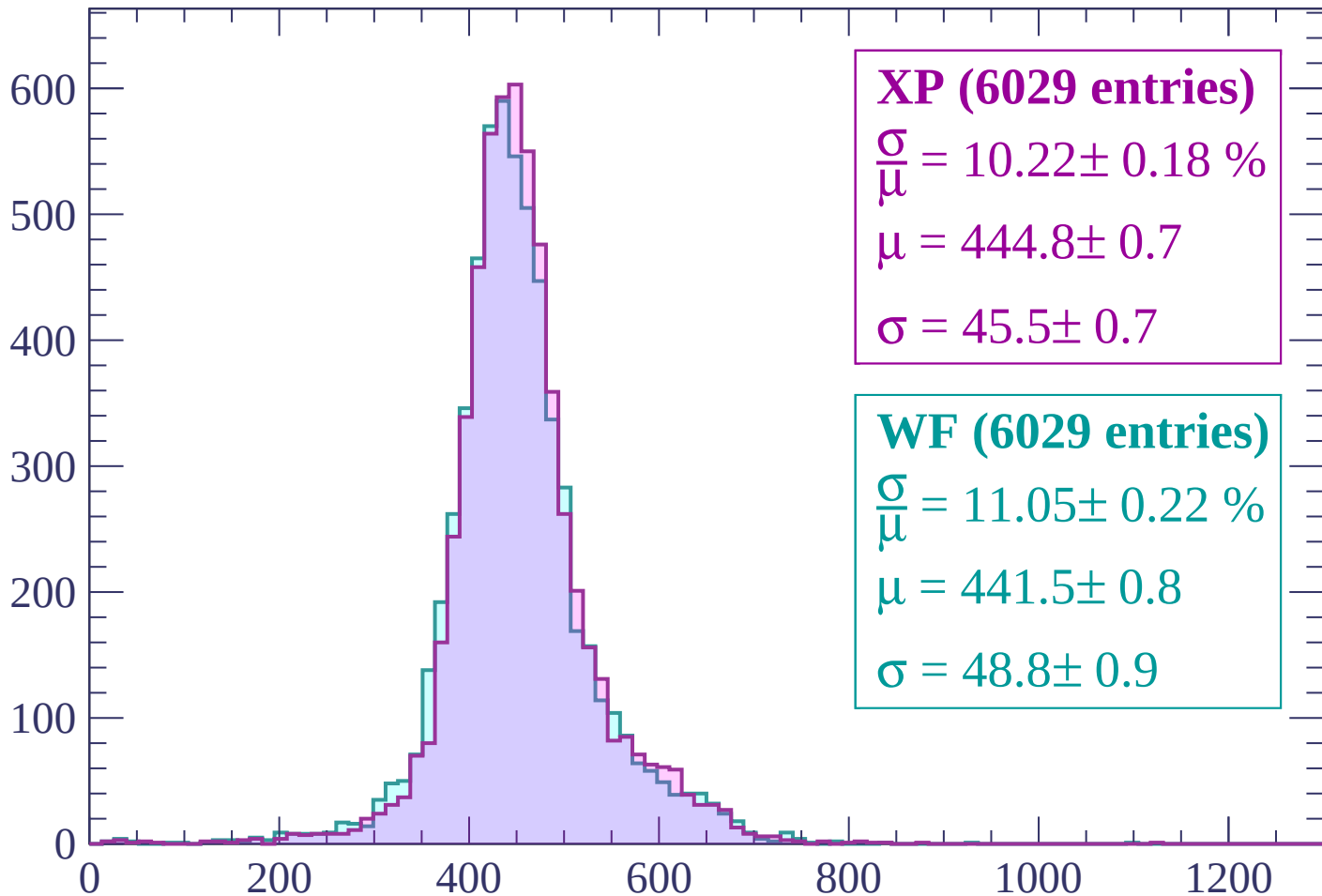
$\mu = 407.8 \pm 0.6$

$\sigma = 39.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (6029 entries)

$$\frac{\sigma}{\mu} = 10.22 \pm 0.18 \%$$

$$\mu = 444.8 \pm 0.7$$

$$\sigma = 45.5 \pm 0.7$$

WF (6029 entries)

$$\frac{\sigma}{\mu} = 11.05 \pm 0.22 \%$$

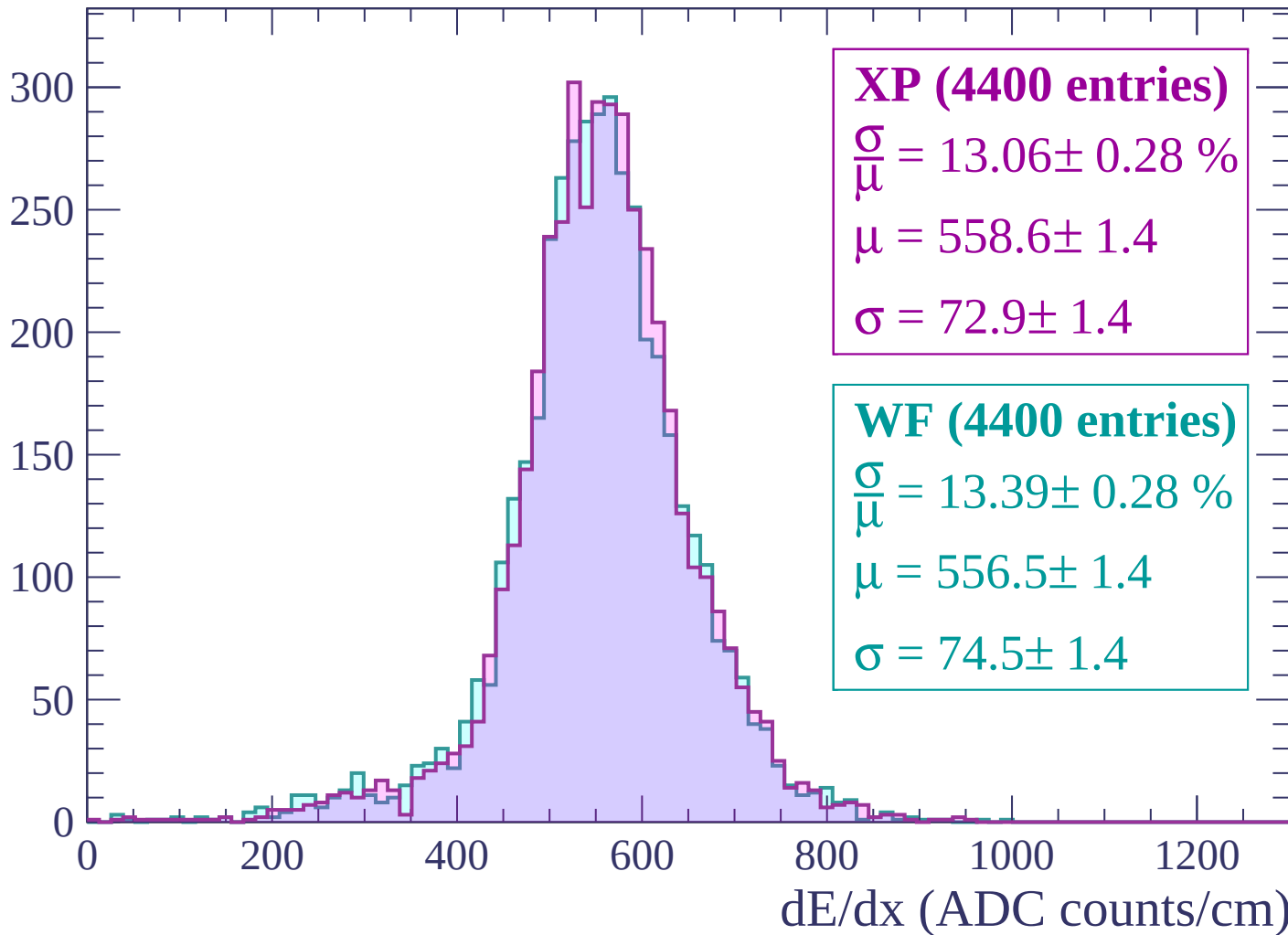
$$\mu = 441.5 \pm 0.8$$

$$\sigma = 48.8 \pm 0.9$$

dE/dx (ADC counts/cm)

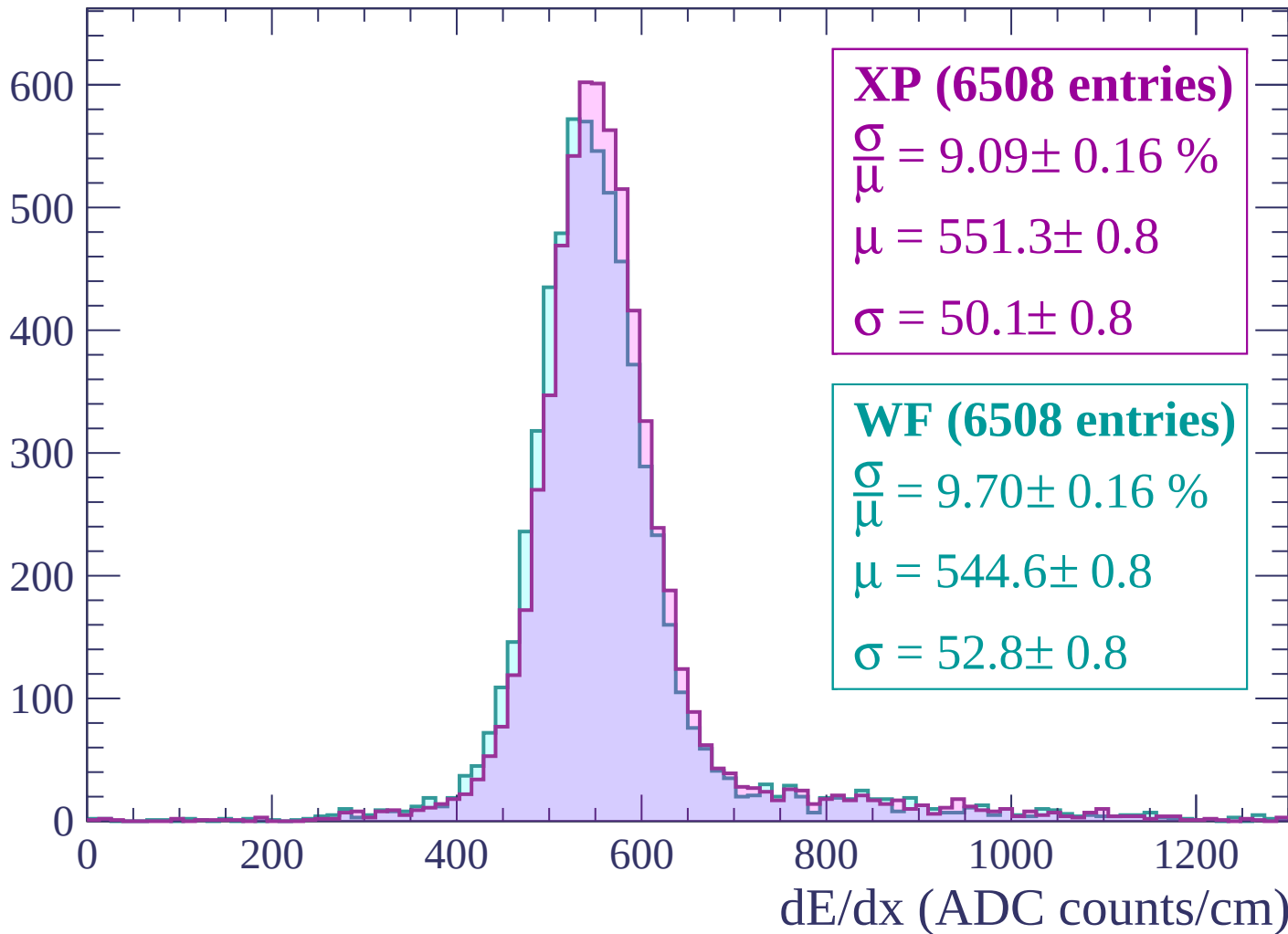
Energy loss | $-120 < p < -60$

Count



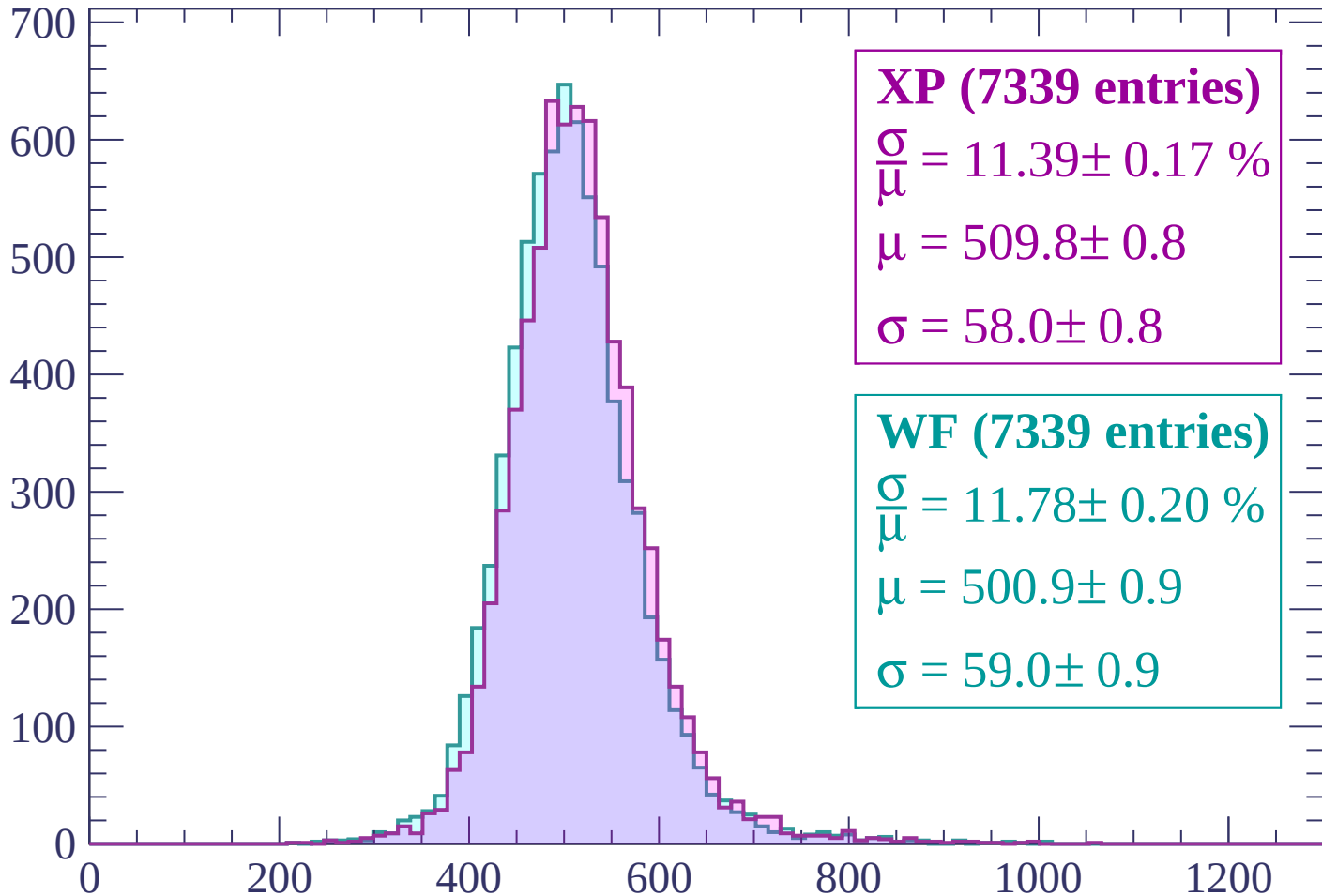
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP (7339 entries)

$\frac{\sigma}{\mu} = 11.39 \pm 0.17 \%$

$\mu = 509.8 \pm 0.8$

$\sigma = 58.0 \pm 0.8$

WF (7339 entries)

$\frac{\sigma}{\mu} = 11.78 \pm 0.20 \%$

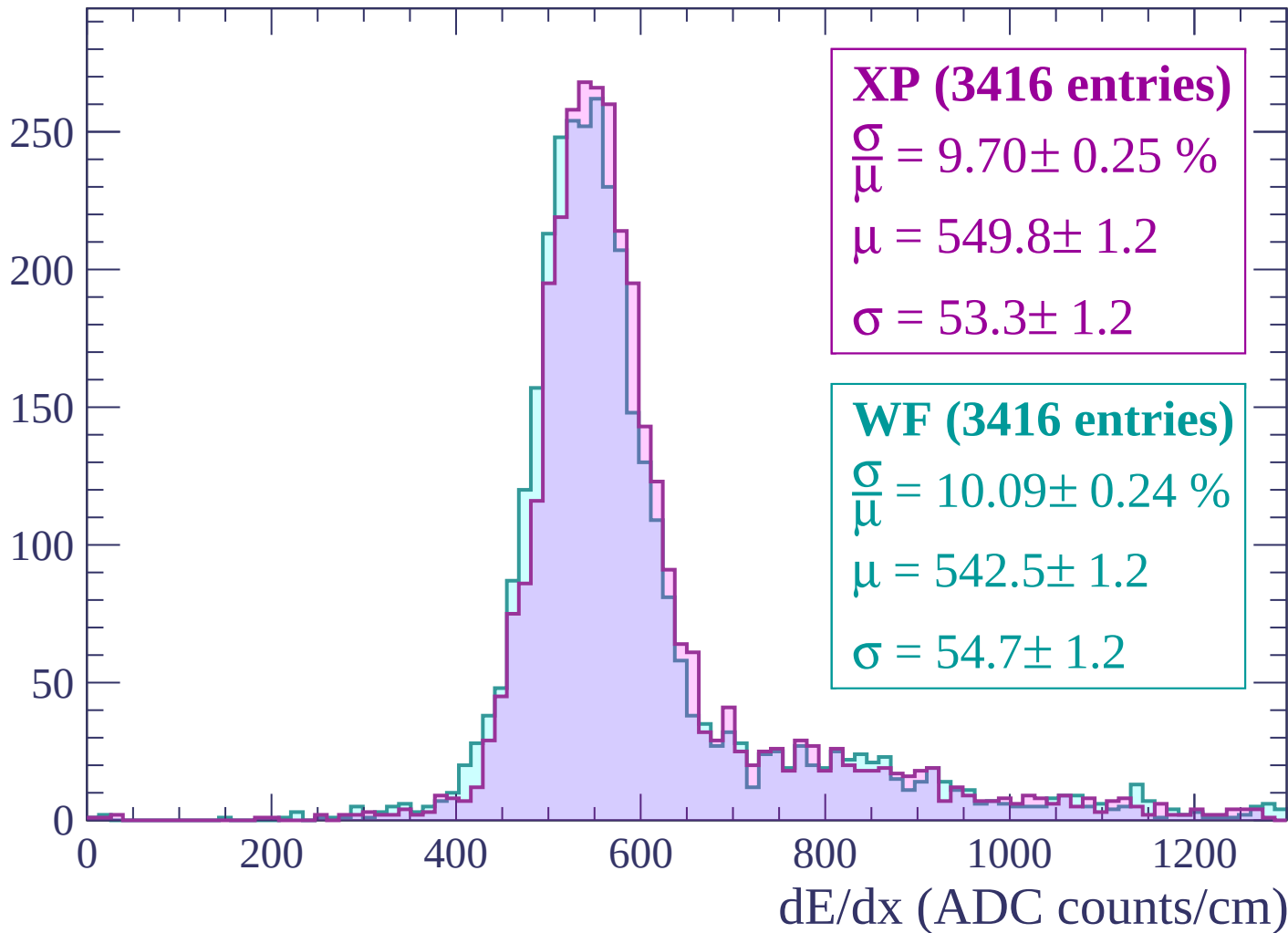
$\mu = 500.9 \pm 0.9$

$\sigma = 59.0 \pm 0.9$

dE/dx (ADC counts/cm)

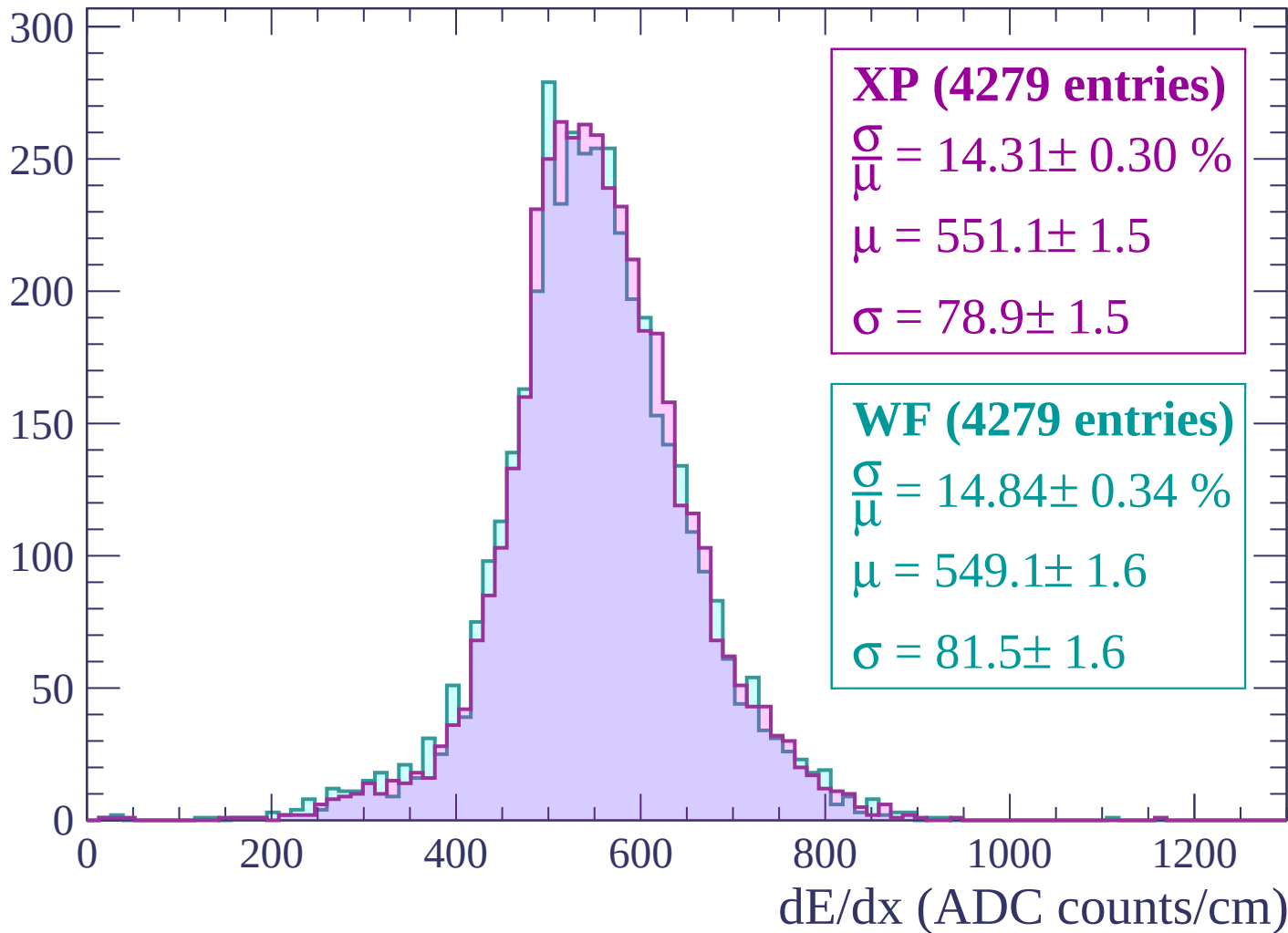
Energy loss | $60 < p < 120$

Count



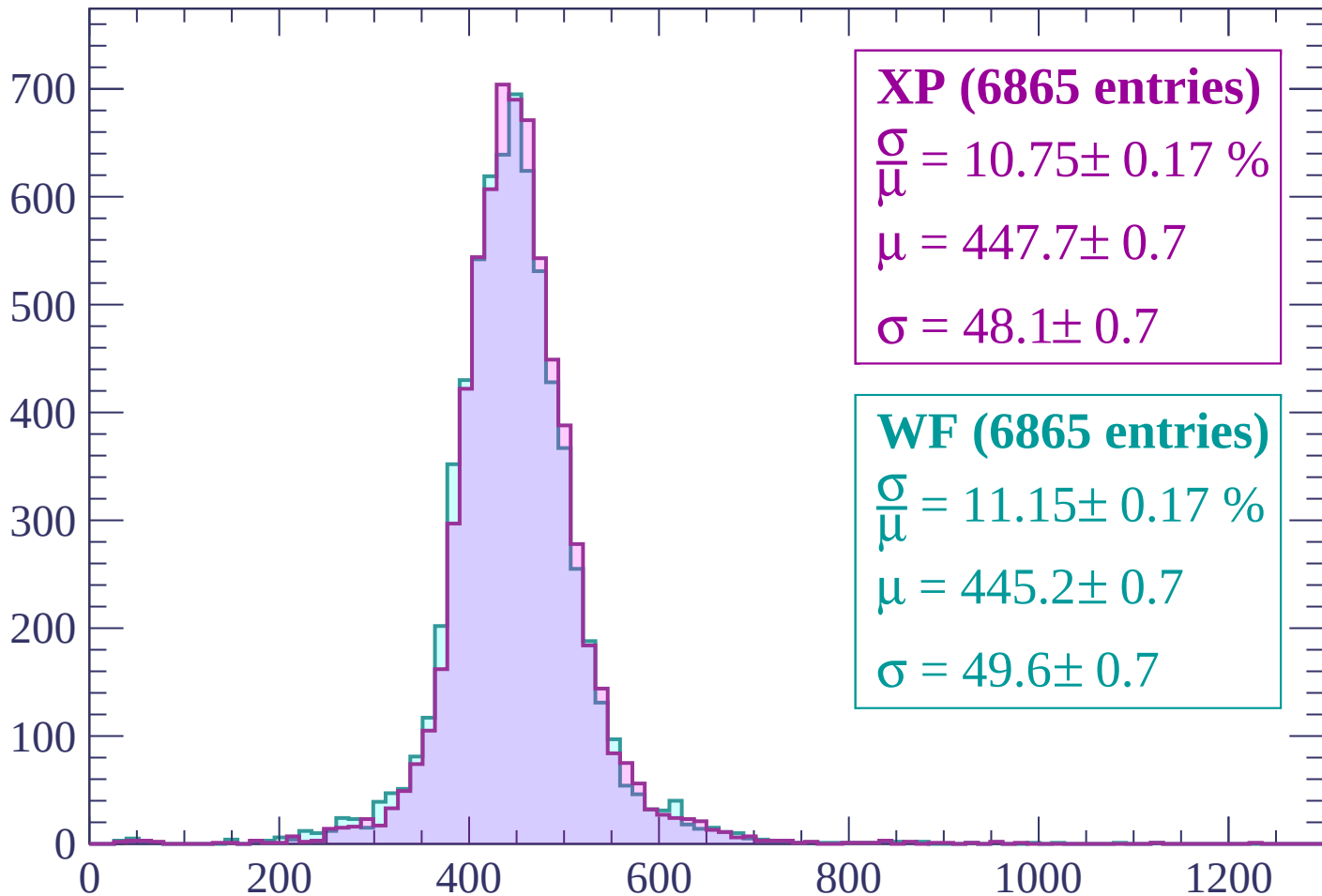
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (6865 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.17 \%$

$\mu = 447.7 \pm 0.7$

$\sigma = 48.1 \pm 0.7$

WF (6865 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.17 \%$

$\mu = 445.2 \pm 0.7$

$\sigma = 49.6 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

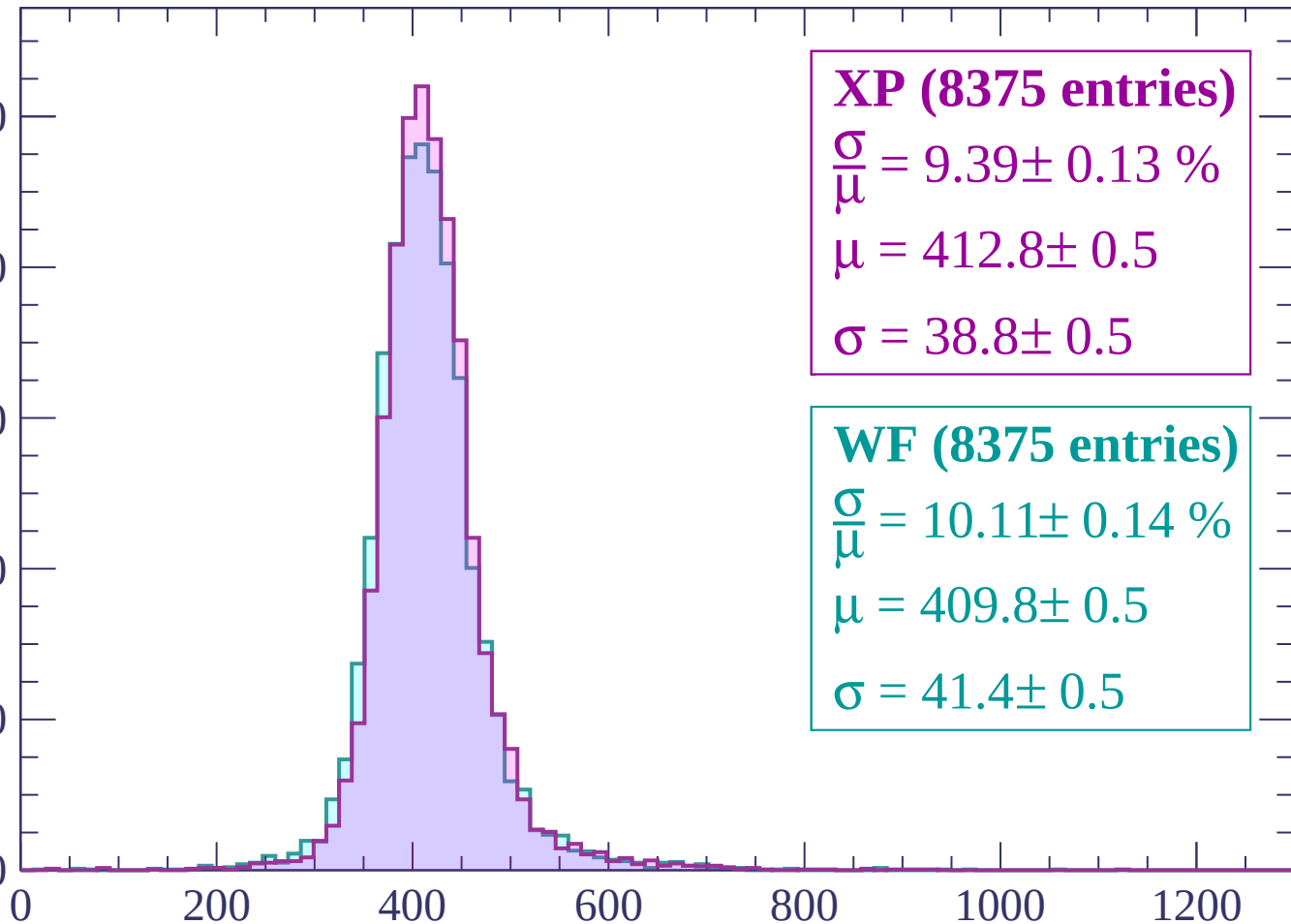
800

600

400

200

0



XP (8375 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.13 \%$

$\mu = 412.8 \pm 0.5$

$\sigma = 38.8 \pm 0.5$

WF (8375 entries)

$\frac{\sigma}{\mu} = 10.11 \pm 0.14 \%$

$\mu = 409.8 \pm 0.5$

$\sigma = 41.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9170 entries)

$$\frac{\sigma}{\mu} = 9.07 \pm 0.13 \%$$

$$\mu = 400.6 \pm 0.5$$

$$\sigma = 36.3 \pm 0.5$$

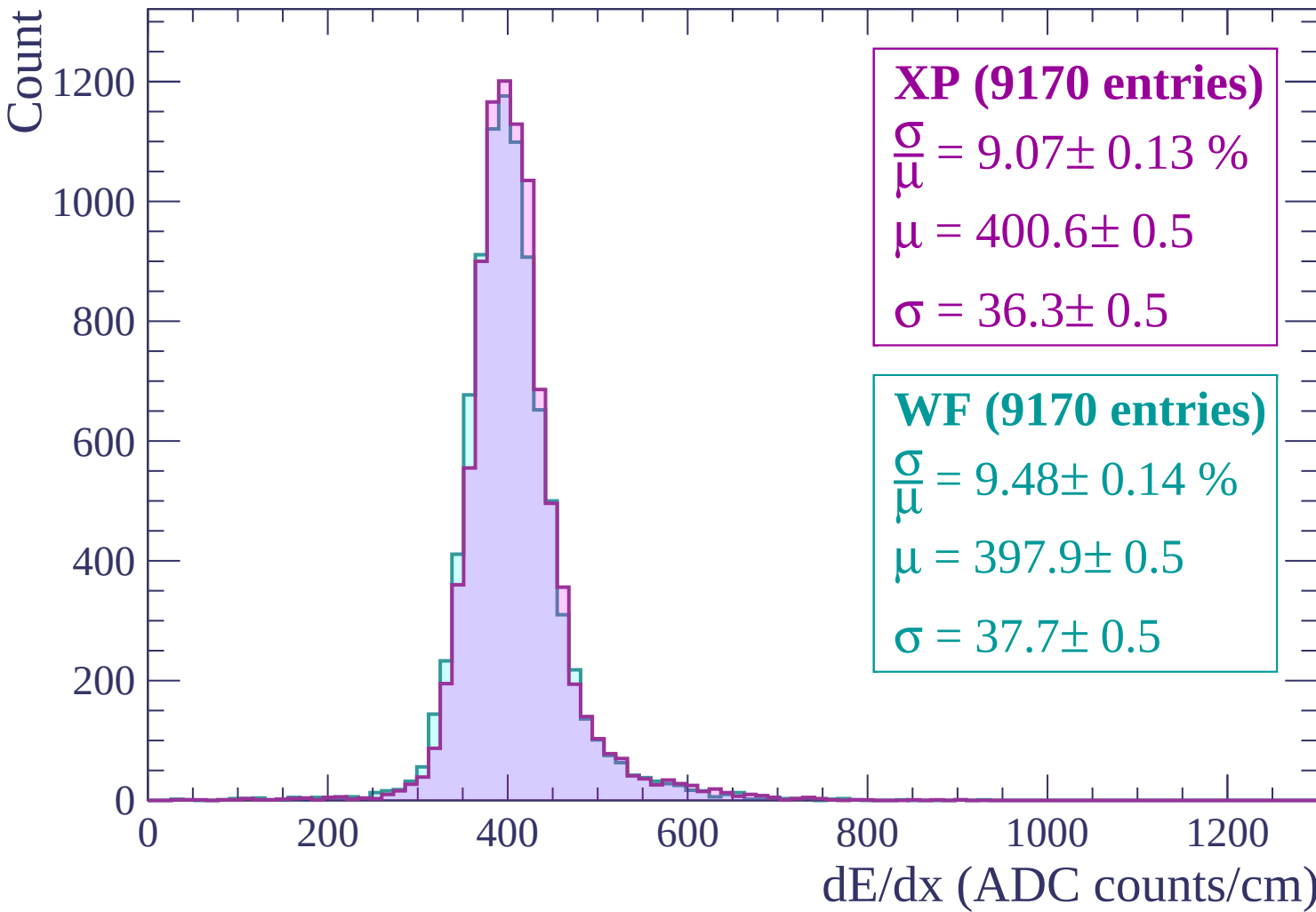
WF (9170 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.14 \%$$

$$\mu = 397.9 \pm 0.5$$

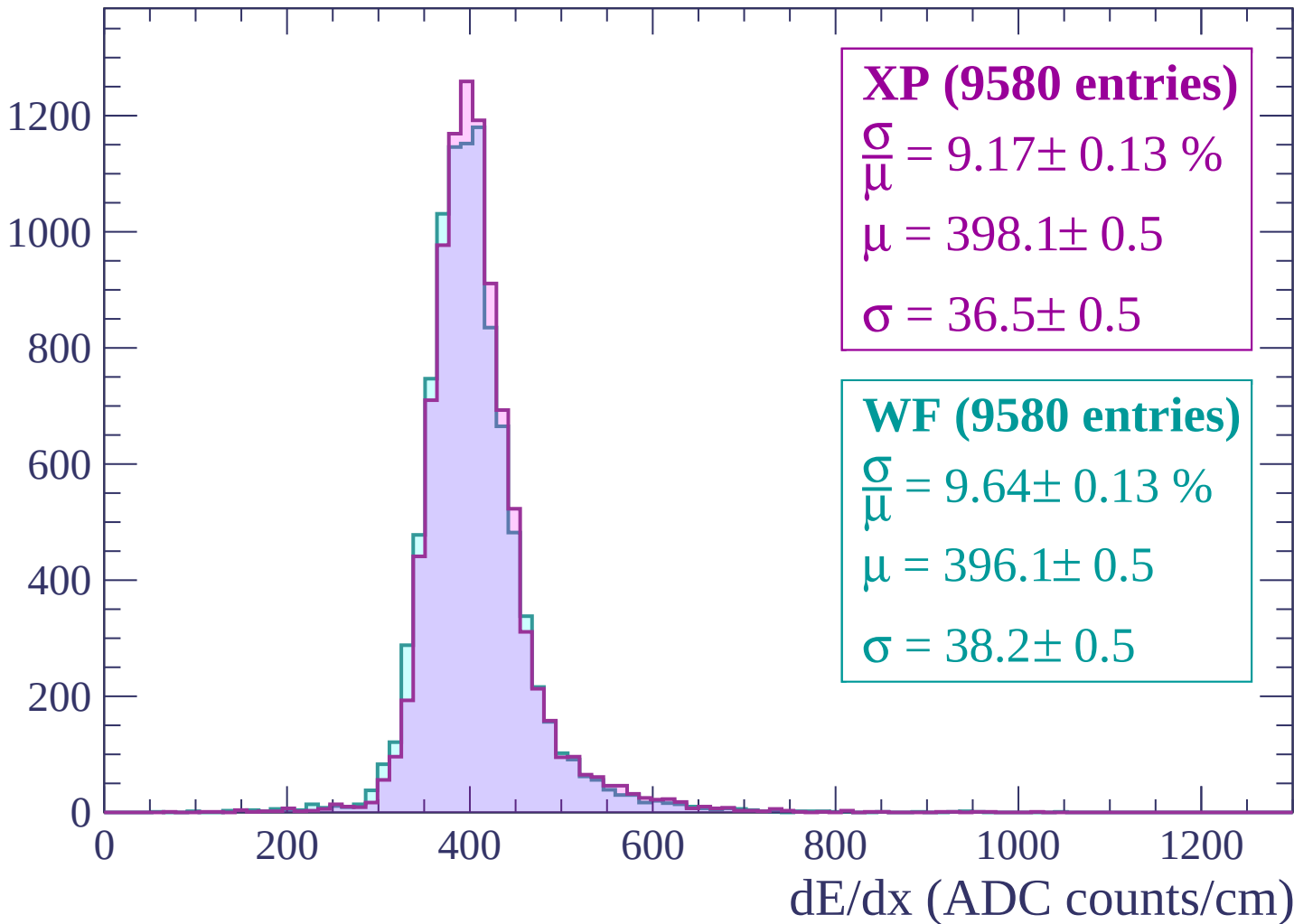
$$\sigma = 37.7 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$

Count



Energy loss | $420 < p < 480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (9744 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.13 \%$

$\mu = 399.9 \pm 0.5$

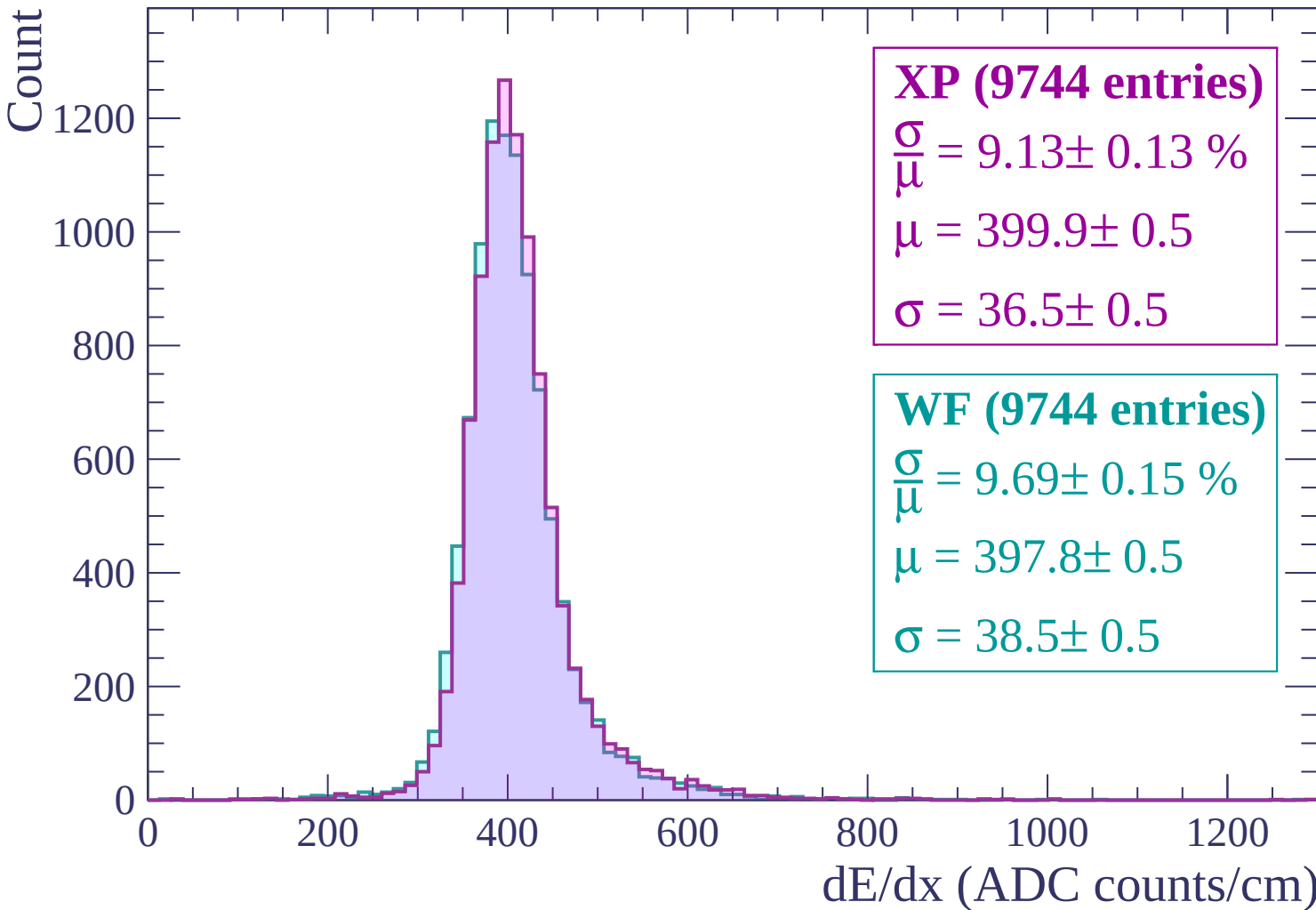
$\sigma = 36.5 \pm 0.5$

WF (9744 entries)

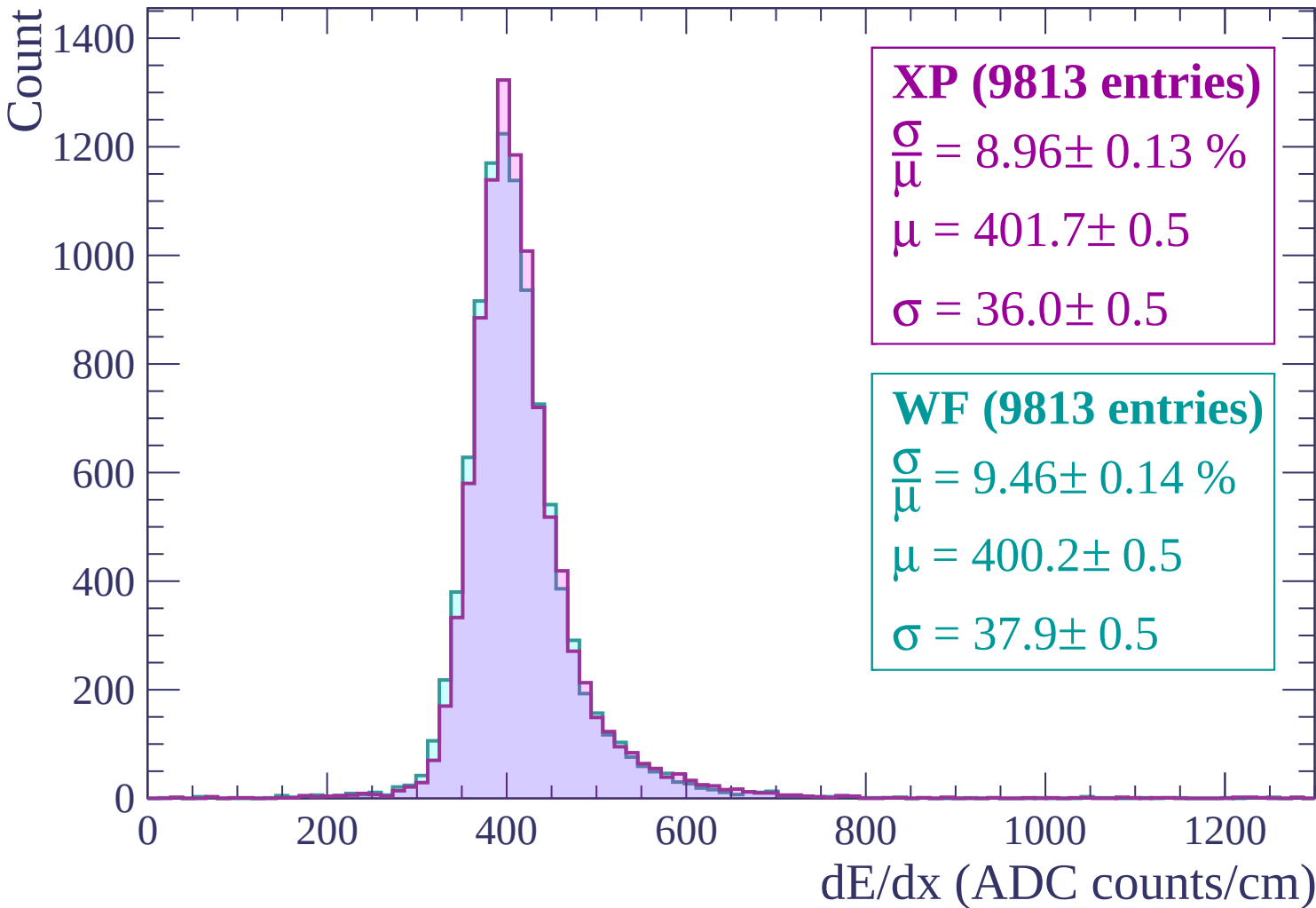
$\frac{\sigma}{\mu} = 9.69 \pm 0.15 \%$

$\mu = 397.8 \pm 0.5$

$\sigma = 38.5 \pm 0.5$



Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9776 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.12 \%$

$\mu = 406.8 \pm 0.5$

$\sigma = 36.9 \pm 0.4$

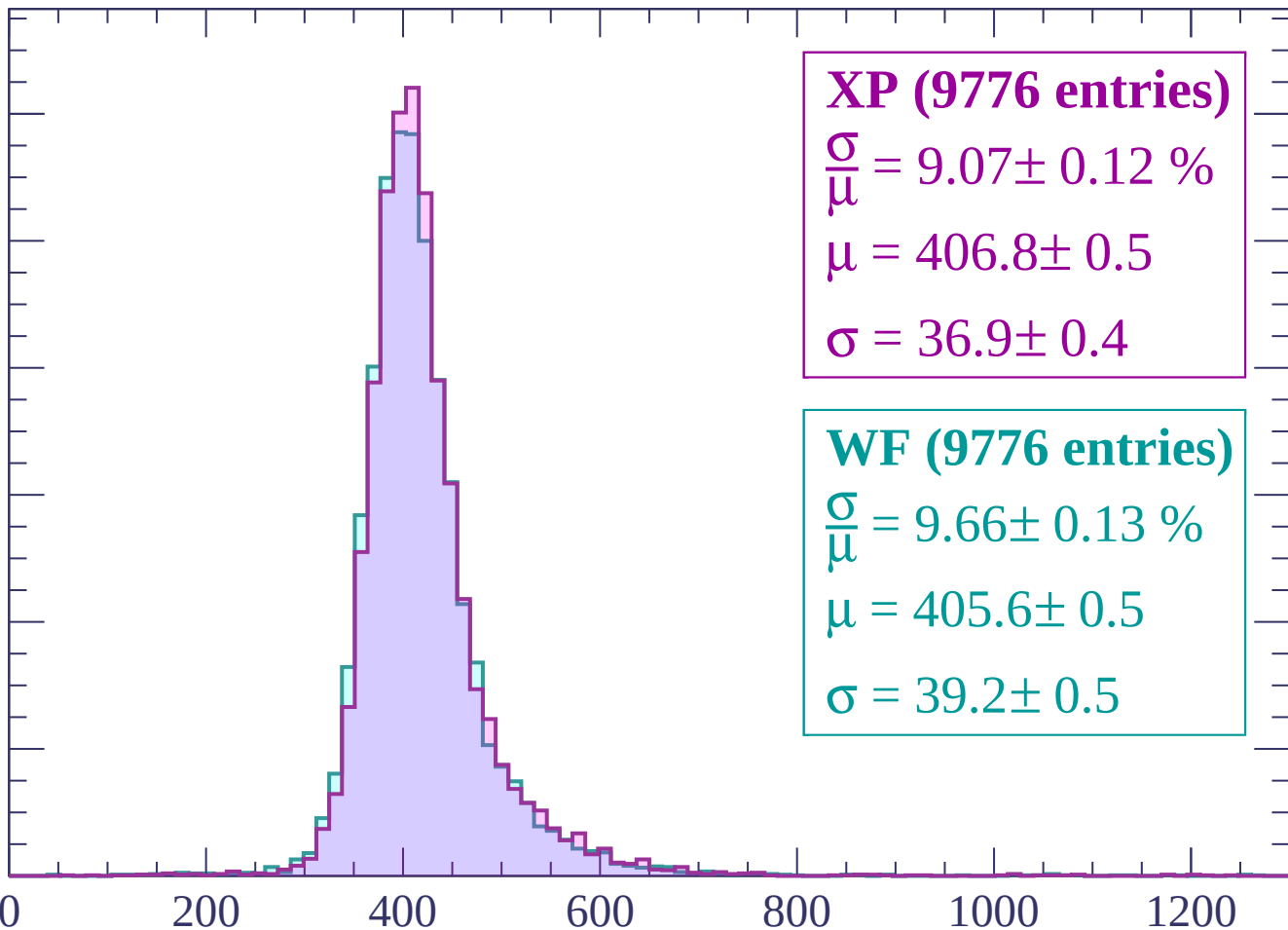
WF (9776 entries)

$\frac{\sigma}{\mu} = 9.66 \pm 0.13 \%$

$\mu = 405.6 \pm 0.5$

$\sigma = 39.2 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $600 < p < 660$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (9464 entries)

$\frac{\sigma}{\mu} = 9.67 \pm 0.12 \%$

$\mu = 411.6 \pm 0.5$

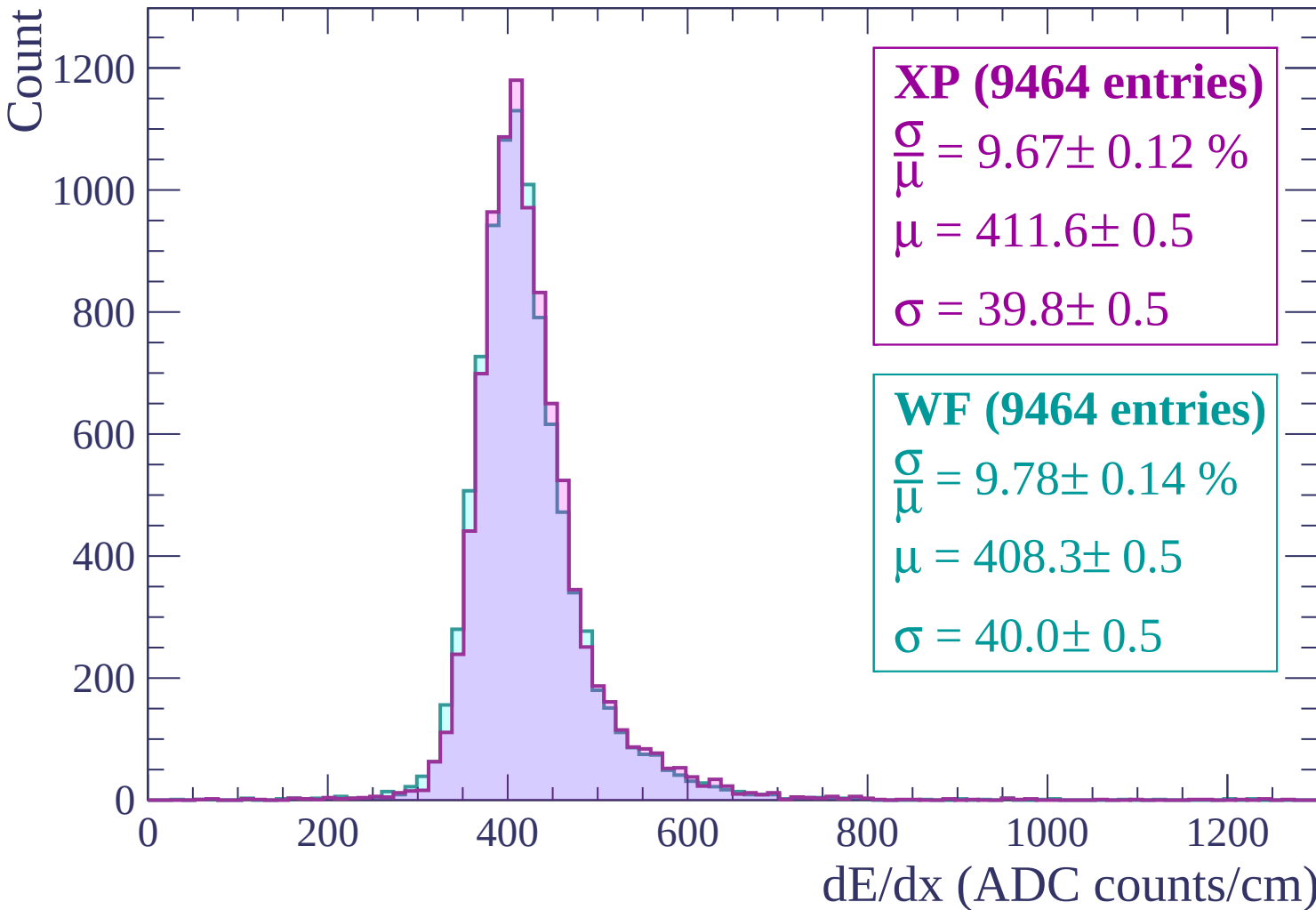
$\sigma = 39.8 \pm 0.5$

WF (9464 entries)

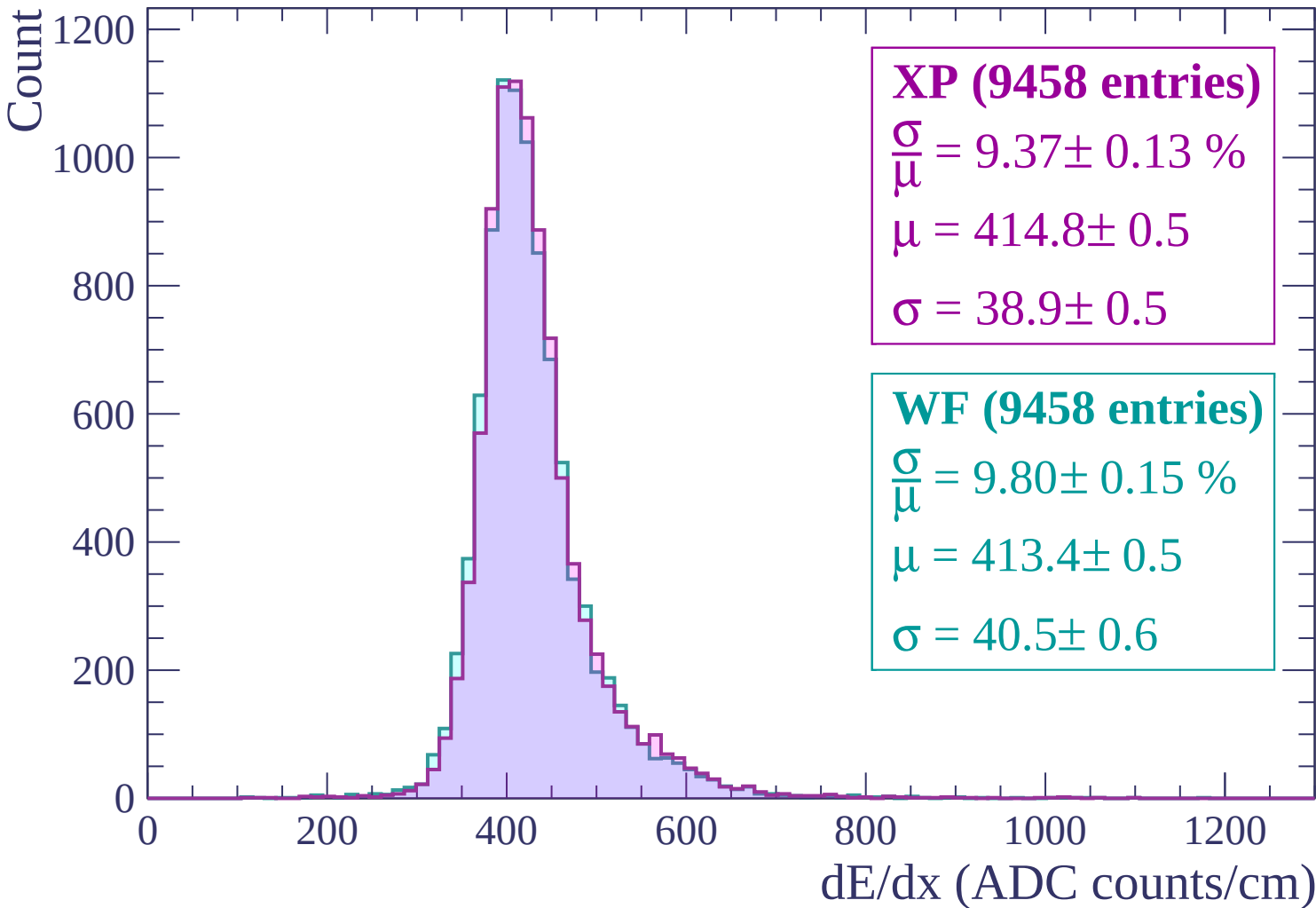
$\frac{\sigma}{\mu} = 9.78 \pm 0.14 \%$

$\mu = 408.3 \pm 0.5$

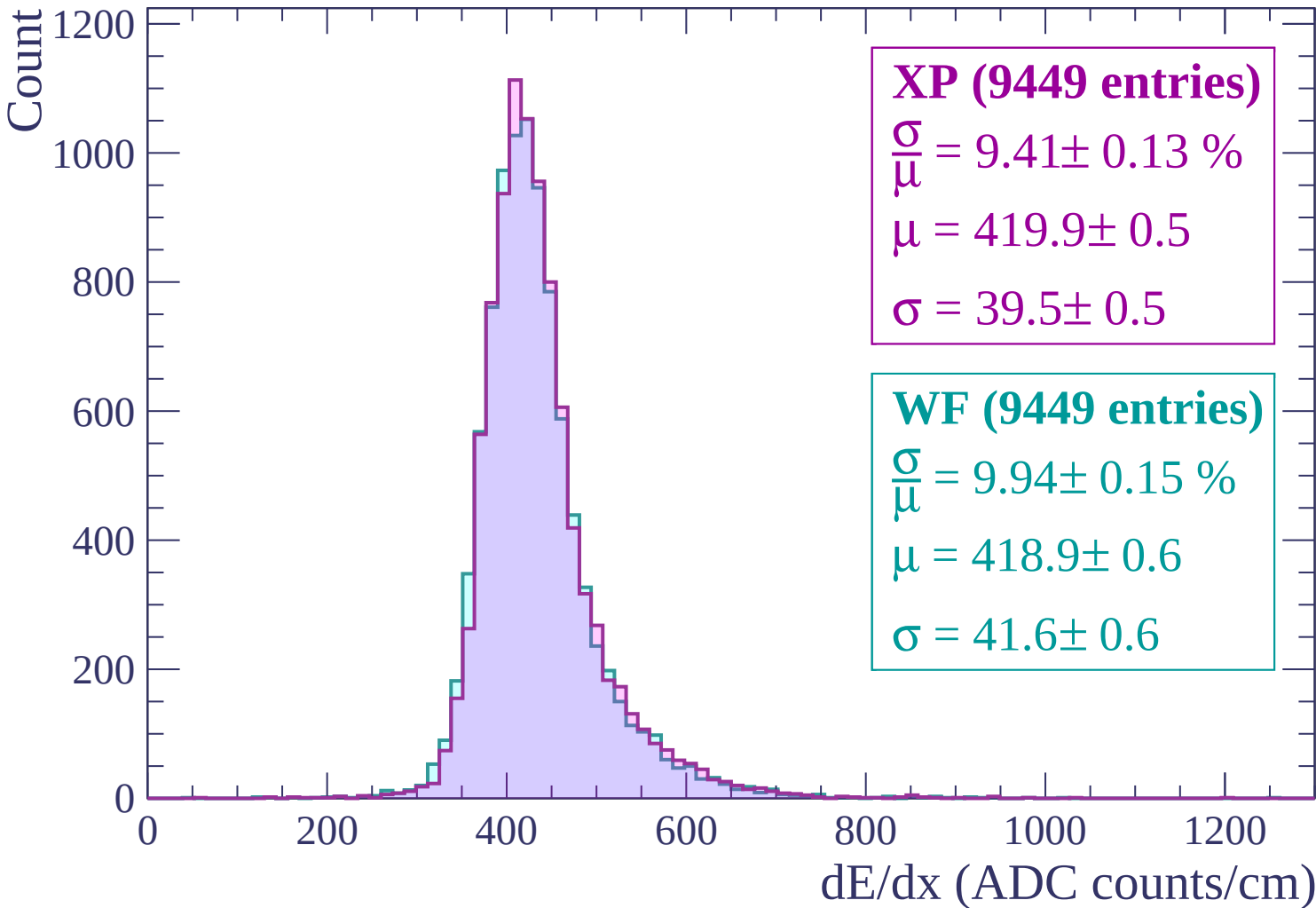
$\sigma = 40.0 \pm 0.5$



Energy loss | $660 < p < 720$

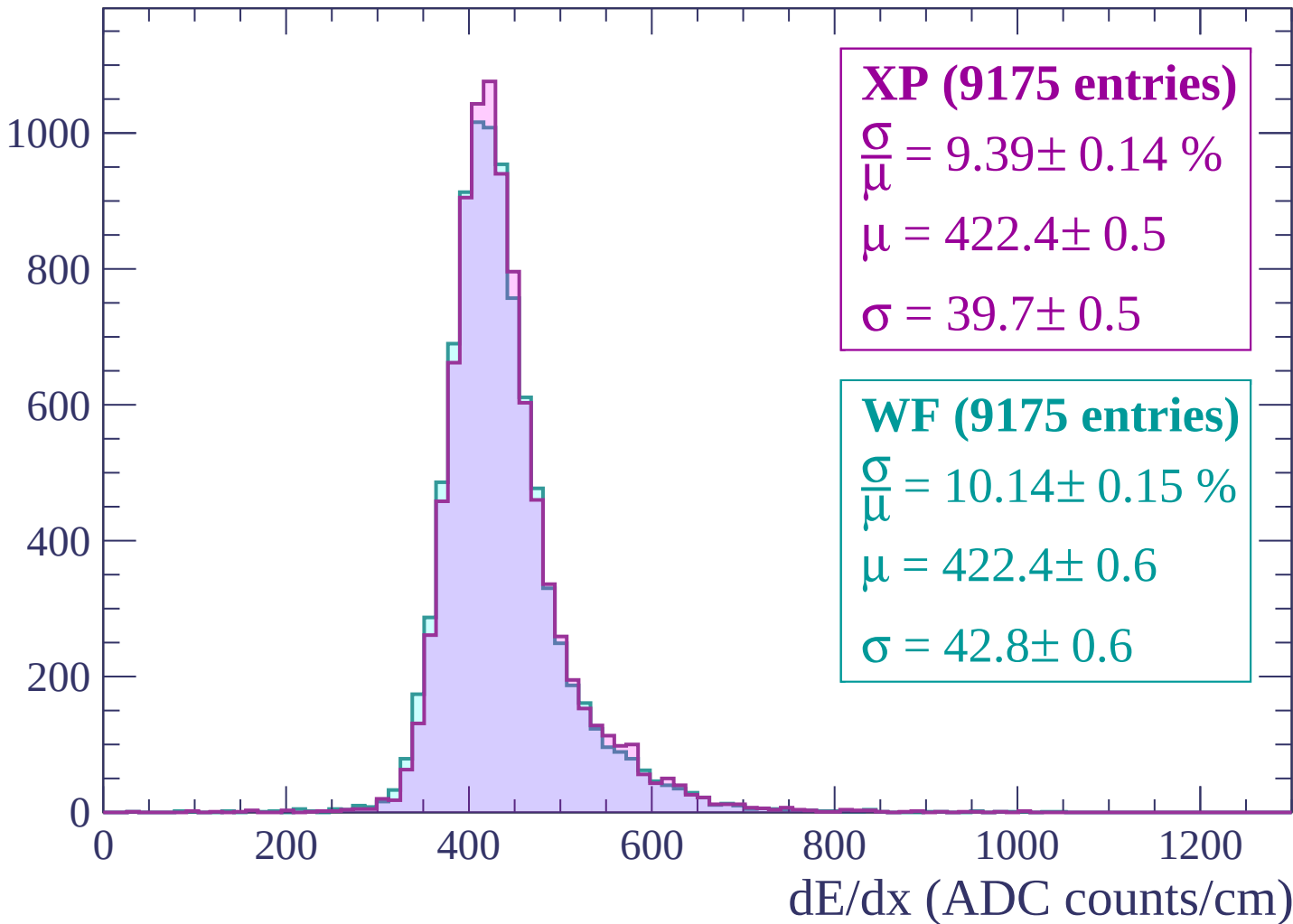


Energy loss | $720 < p < 780$



Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8916 entries)

$$\frac{\sigma}{\mu} = 9.80 \pm 0.16 \%$$

$$\mu = 427.2 \pm 0.6$$

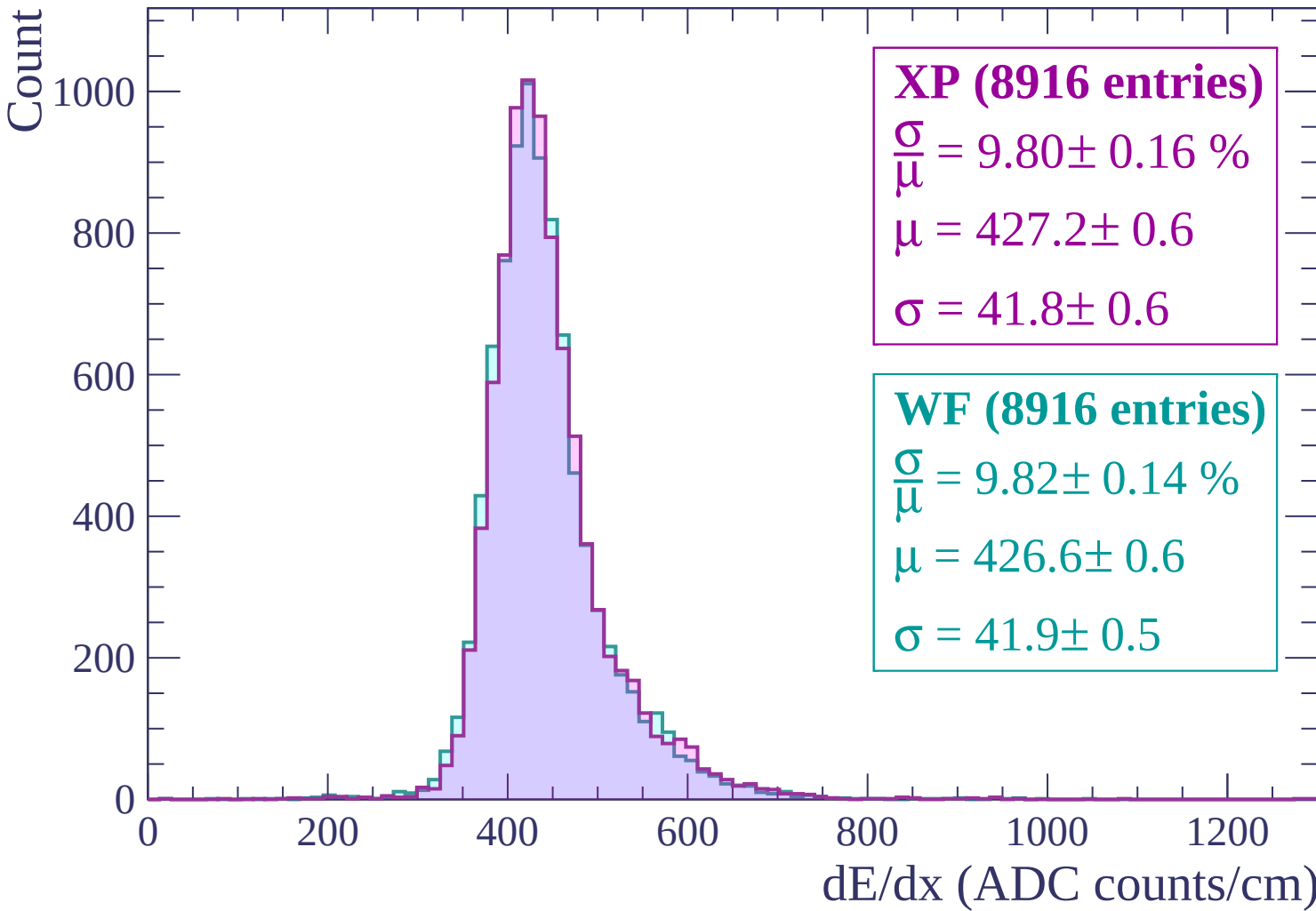
$$\sigma = 41.8 \pm 0.6$$

WF (8916 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.14 \%$$

$$\mu = 426.6 \pm 0.6$$

$$\sigma = 41.9 \pm 0.5$$



Energy loss | $900 < p < 960$

Count

1000

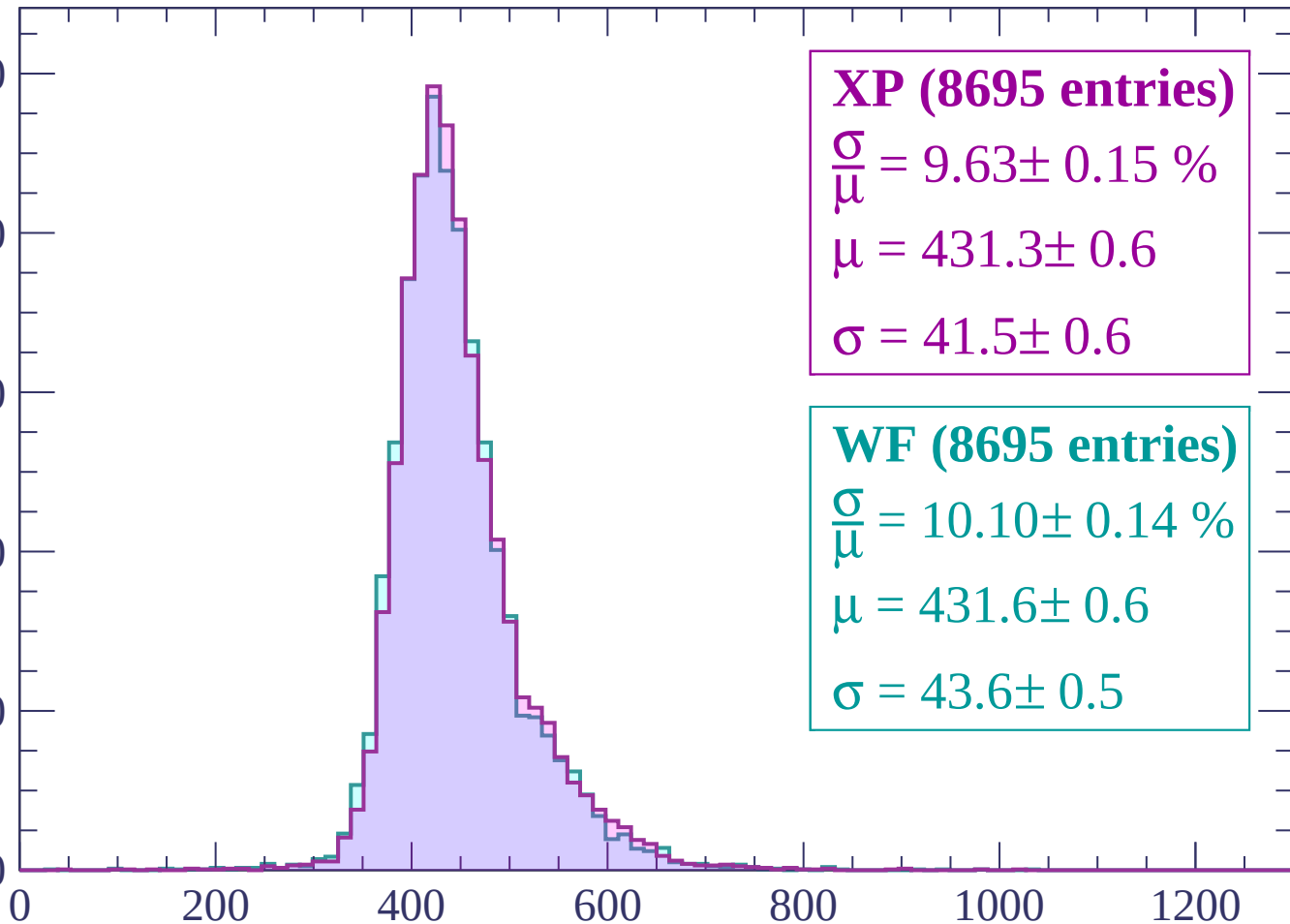
800

600

400

200

0



XP (8695 entries)

$\frac{\sigma}{\mu} = 9.63 \pm 0.15 \%$

$\mu = 431.3 \pm 0.6$

$\sigma = 41.5 \pm 0.6$

WF (8695 entries)

$\frac{\sigma}{\mu} = 10.10 \pm 0.14 \%$

$\mu = 431.6 \pm 0.6$

$\sigma = 43.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count

1000

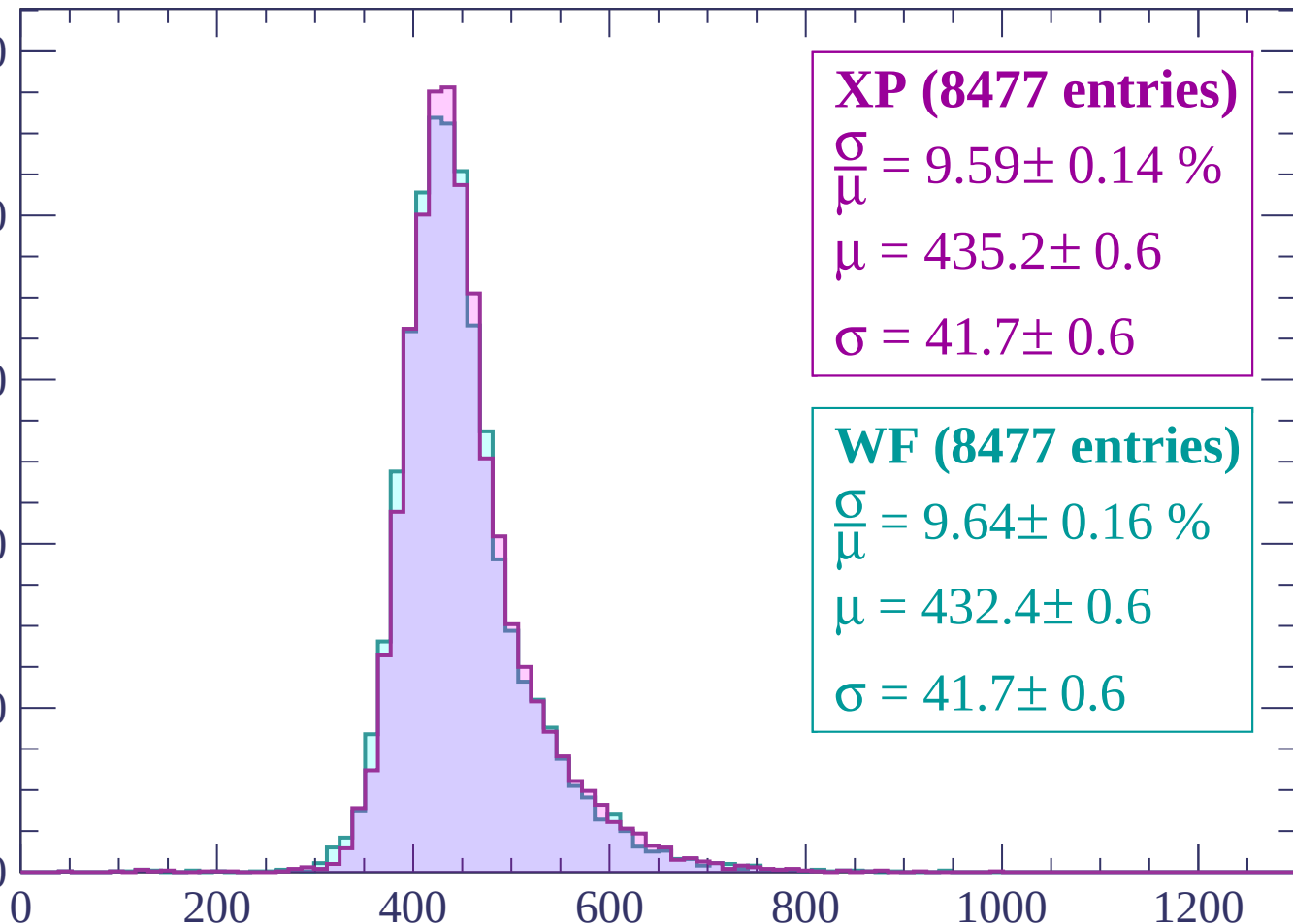
800

600

400

200

0



XP (8477 entries)

$\frac{\sigma}{\mu} = 9.59 \pm 0.14 \%$

$\mu = 435.2 \pm 0.6$

$\sigma = 41.7 \pm 0.6$

WF (8477 entries)

$\frac{\sigma}{\mu} = 9.64 \pm 0.16 \%$

$\mu = 432.4 \pm 0.6$

$\sigma = 41.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count

900
800
700
600
500
400
300
200
100
0

XP (8302 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.16 \%$$

$$\mu = 439.7 \pm 0.6$$

$$\sigma = 43.4 \pm 0.6$$

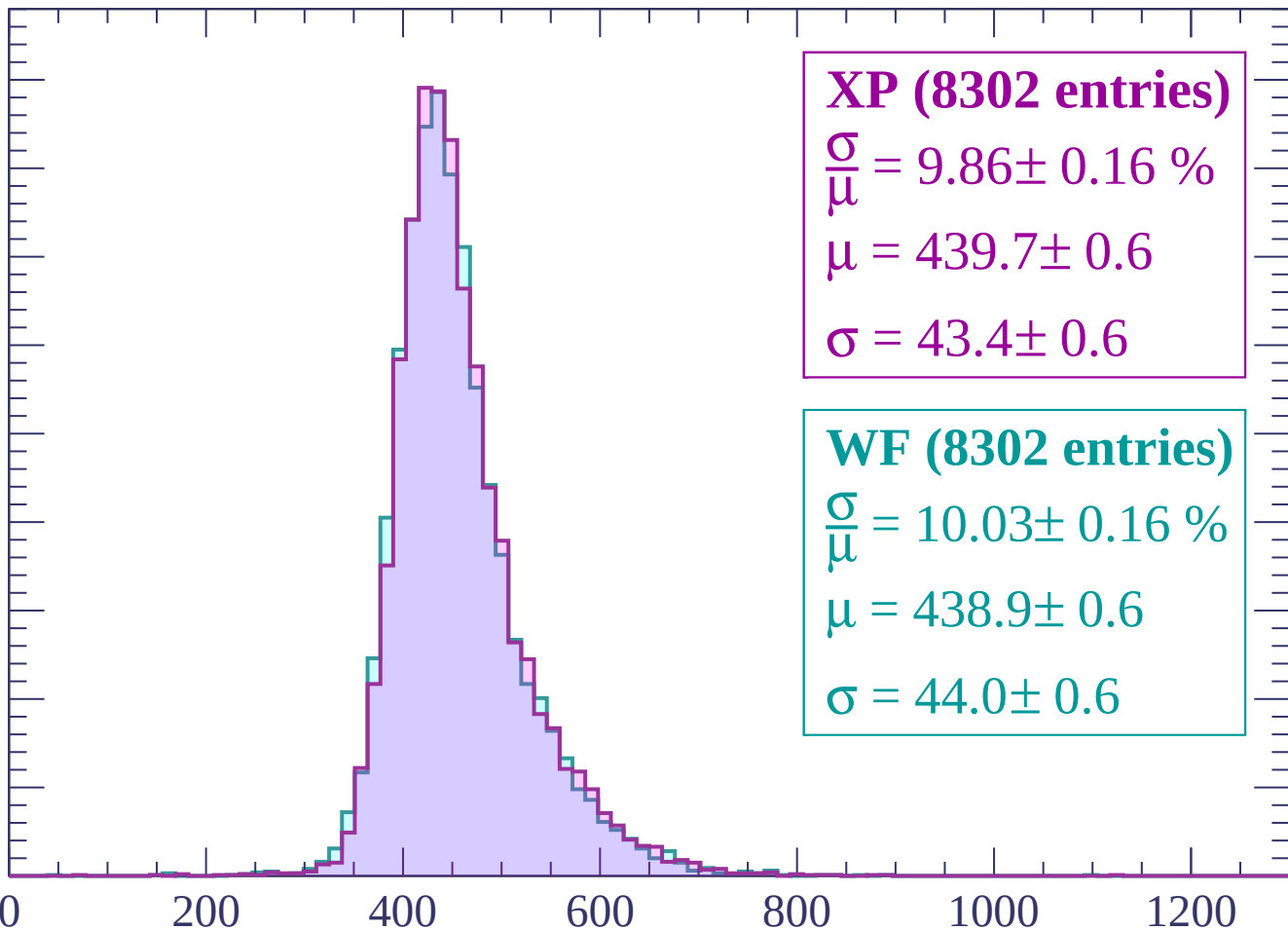
WF (8302 entries)

$$\frac{\sigma}{\mu} = 10.03 \pm 0.16 \%$$

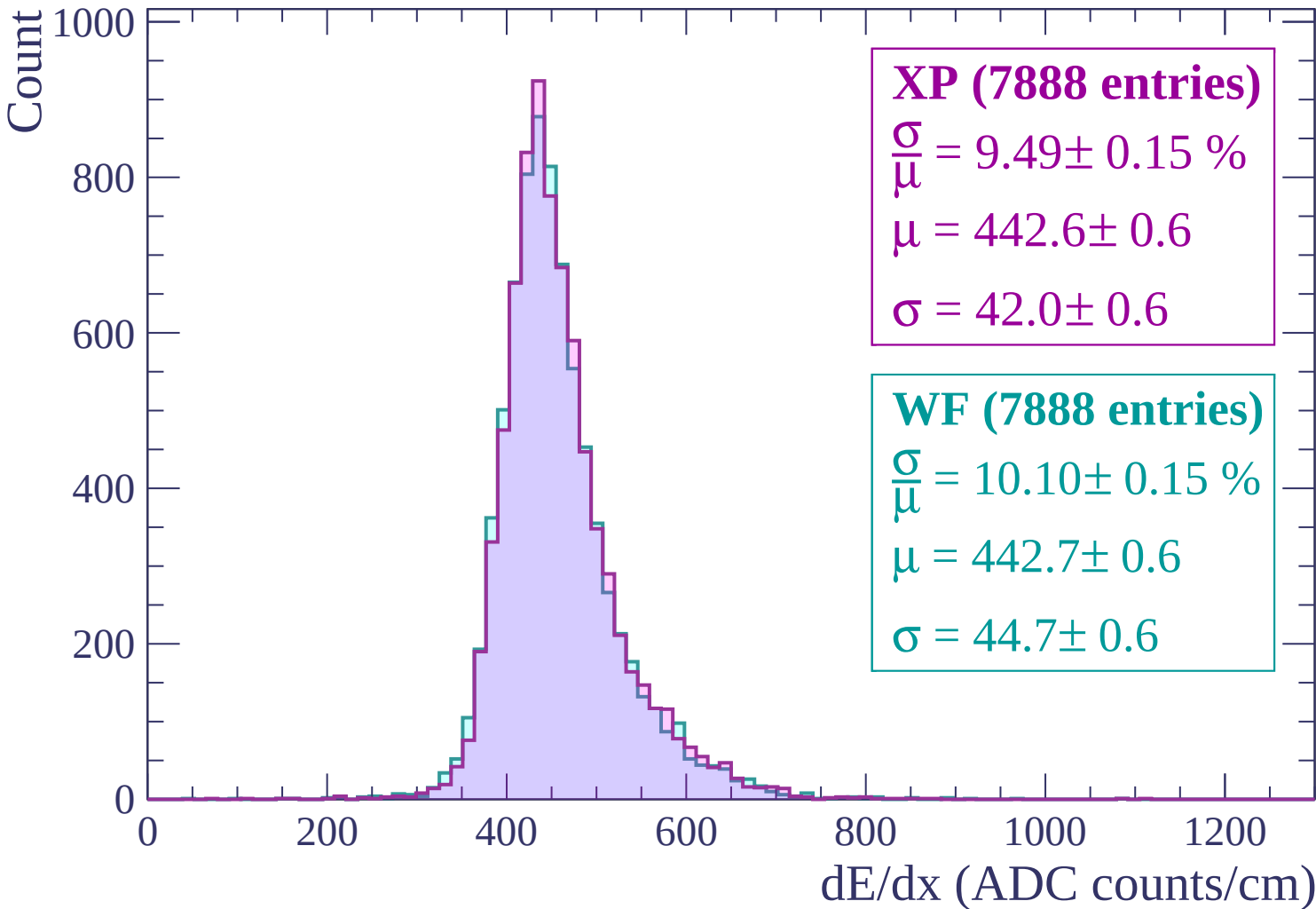
$$\mu = 438.9 \pm 0.6$$

$$\sigma = 44.0 \pm 0.6$$

dE/dx (ADC counts/cm)

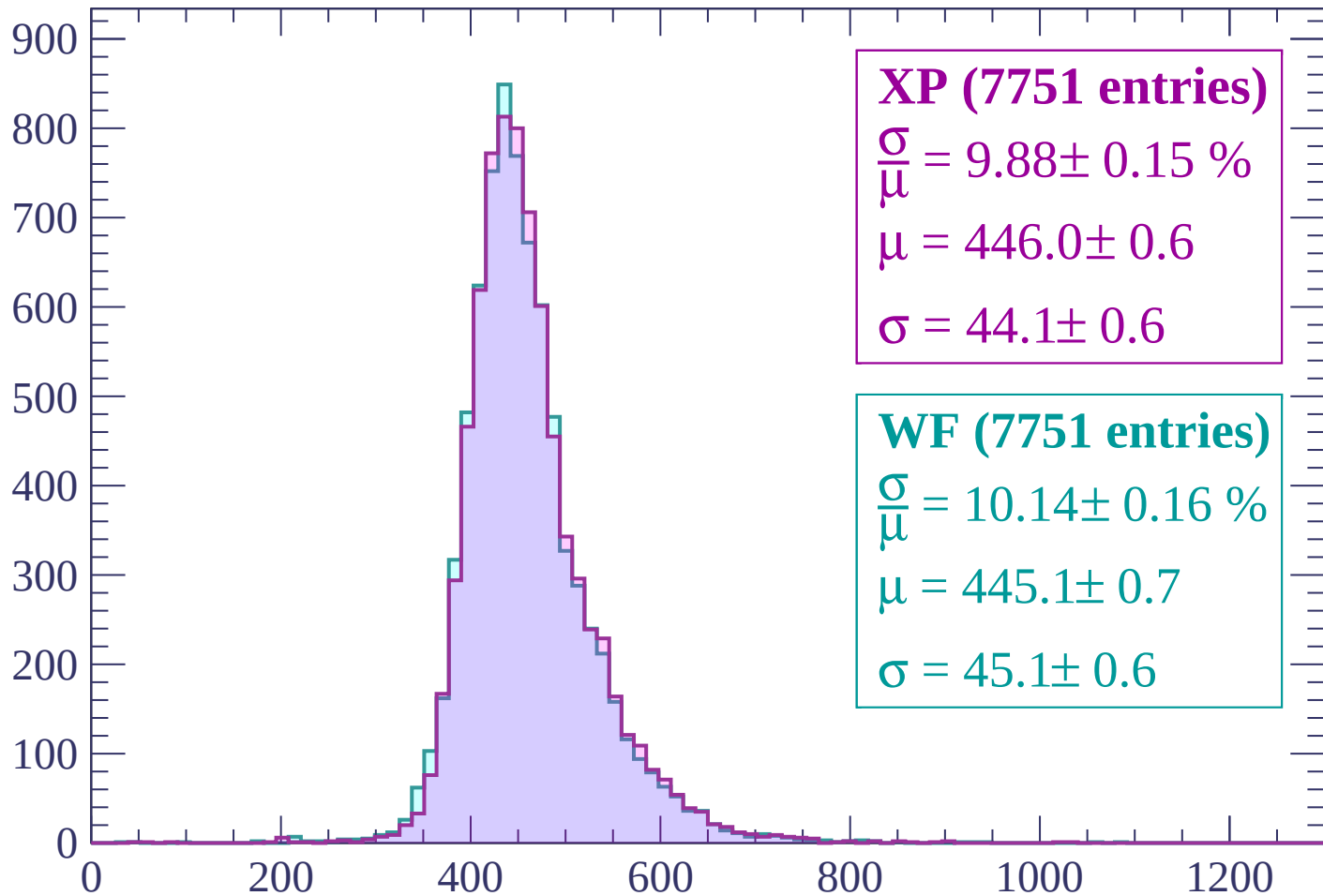


Energy loss | $1080 < p < 1140$



Energy loss | $1140 < p < 1200$

Count



XP (7751 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.15 \%$$

$$\mu = 446.0 \pm 0.6$$

$$\sigma = 44.1 \pm 0.6$$

WF (7751 entries)

$$\frac{\sigma}{\mu} = 10.14 \pm 0.16 \%$$

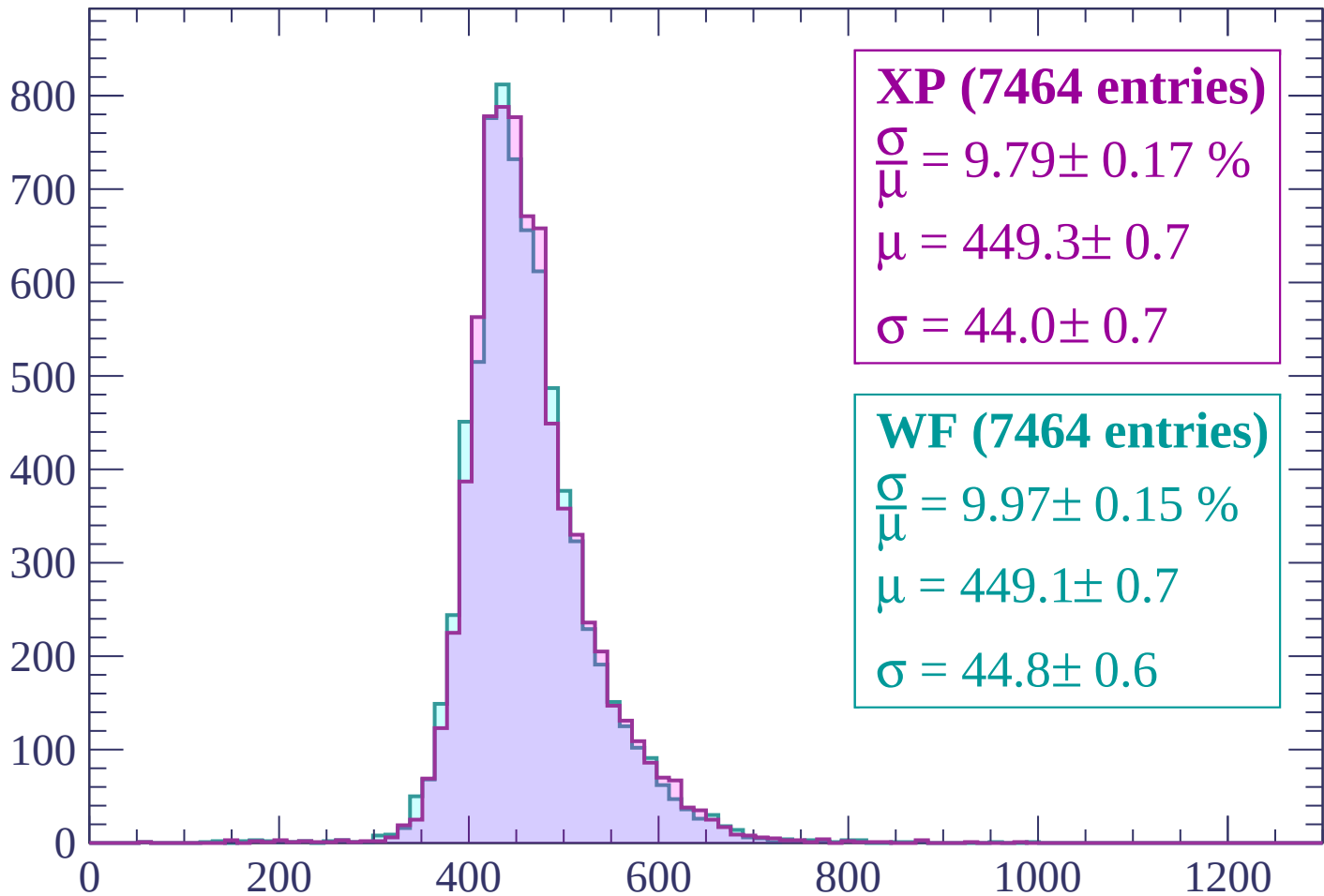
$$\mu = 445.1 \pm 0.7$$

$$\sigma = 45.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (7464 entries)

$\frac{\sigma}{\mu} = 9.79 \pm 0.17 \%$

$\mu = 449.3 \pm 0.7$

$\sigma = 44.0 \pm 0.7$

WF (7464 entries)

$\frac{\sigma}{\mu} = 9.97 \pm 0.15 \%$

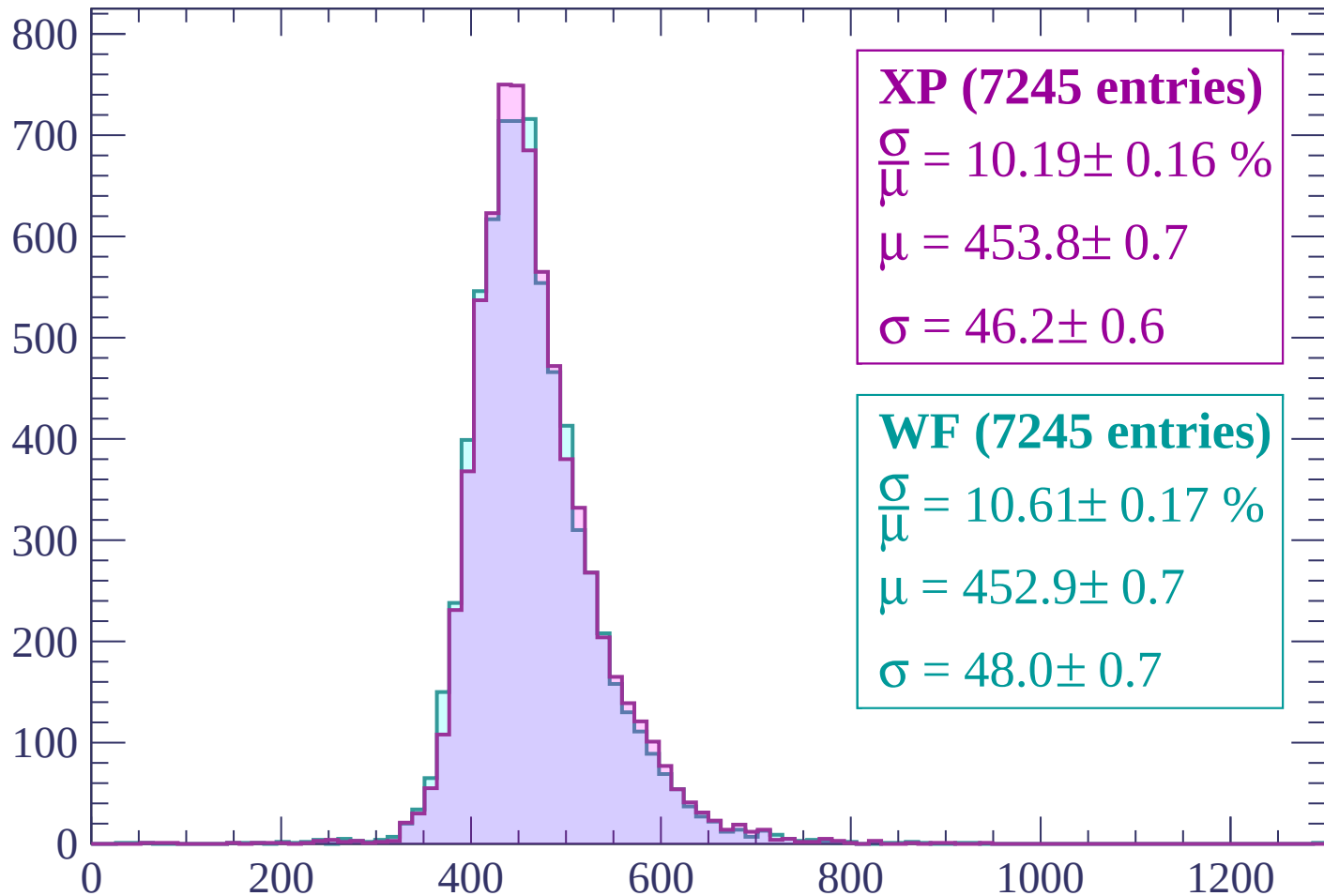
$\mu = 449.1 \pm 0.7$

$\sigma = 44.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (7245 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.16 \%$

$\mu = 453.8 \pm 0.7$

$\sigma = 46.2 \pm 0.6$

WF (7245 entries)

$\frac{\sigma}{\mu} = 10.61 \pm 0.17 \%$

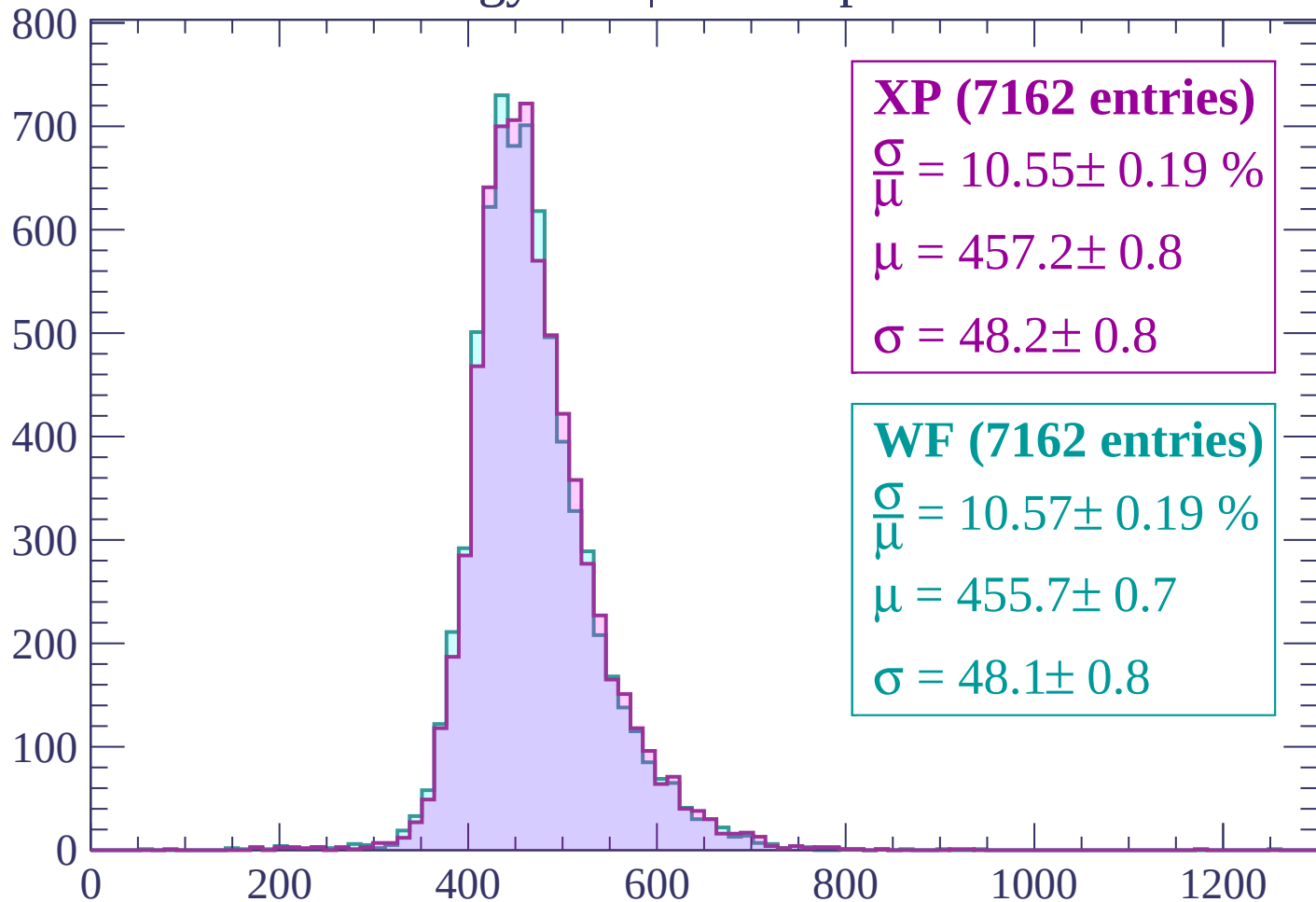
$\mu = 452.9 \pm 0.7$

$\sigma = 48.0 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (7162 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.19 \%$

$\mu = 457.2 \pm 0.8$

$\sigma = 48.2 \pm 0.8$

WF (7162 entries)

$\frac{\sigma}{\mu} = 10.57 \pm 0.19 \%$

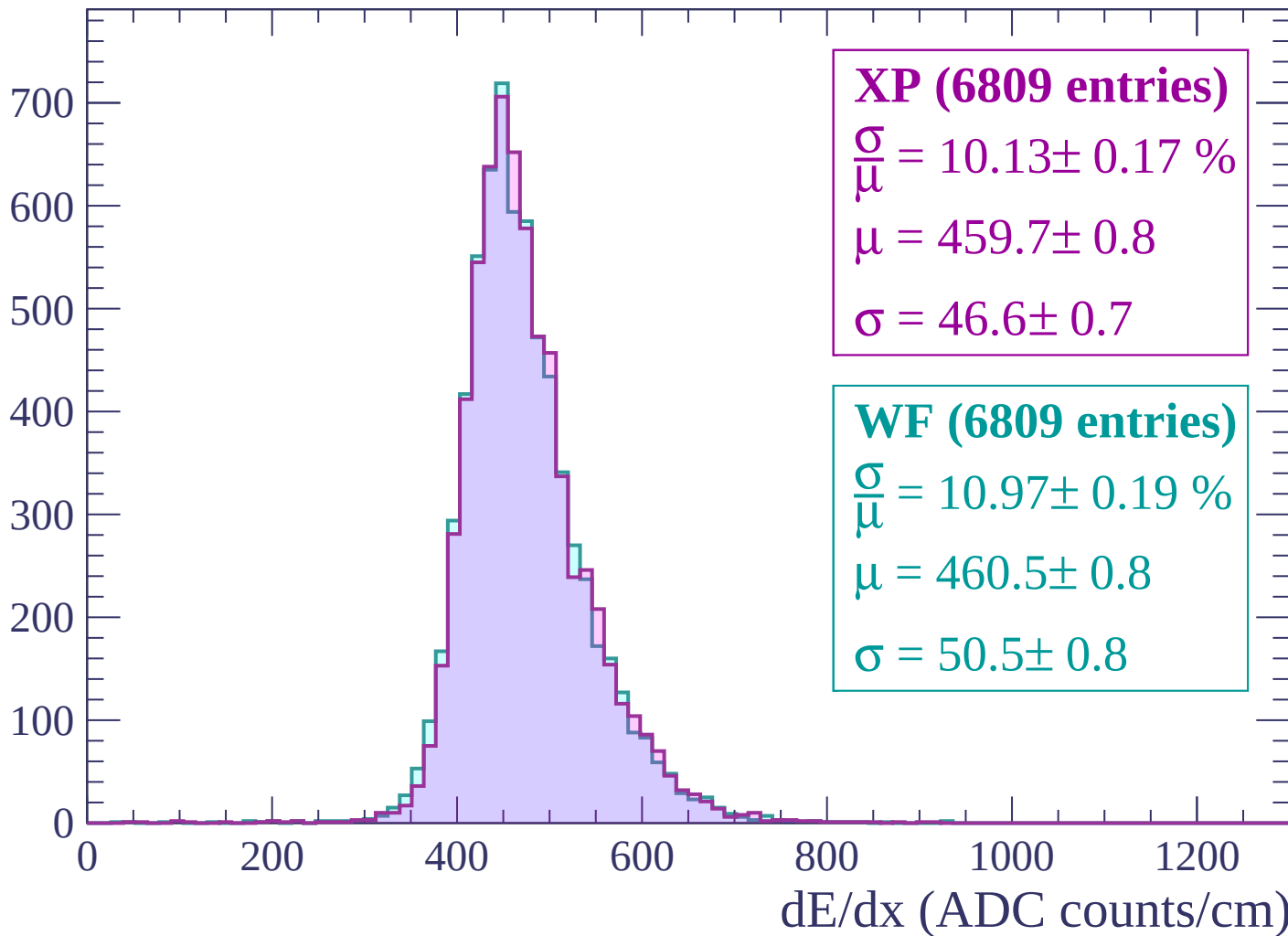
$\mu = 455.7 \pm 0.7$

$\sigma = 48.1 \pm 0.8$

dE/dx (ADC counts/cm)

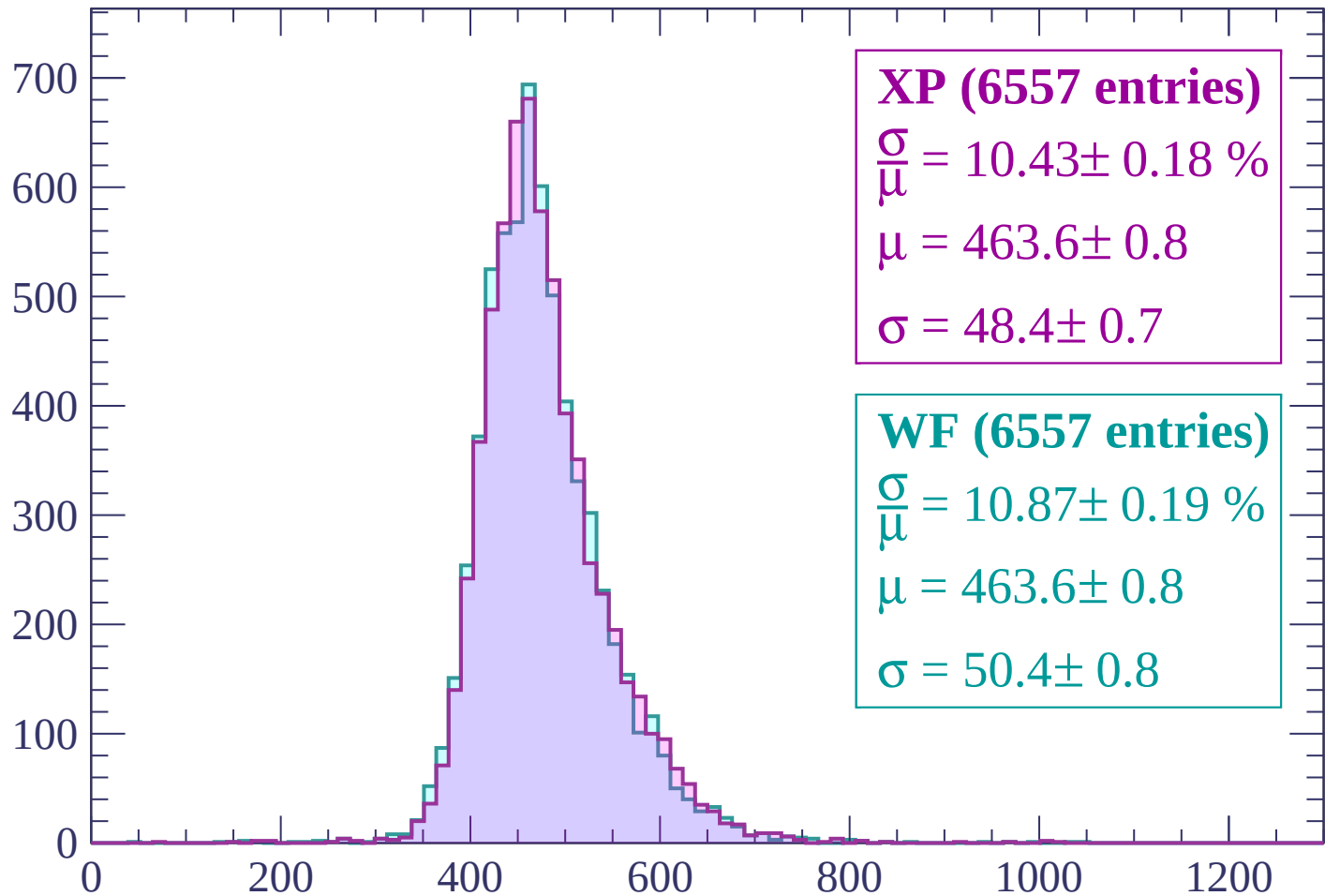
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP (6557 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.18 \%$

$\mu = 463.6 \pm 0.8$

$\sigma = 48.4 \pm 0.7$

WF (6557 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.19 \%$

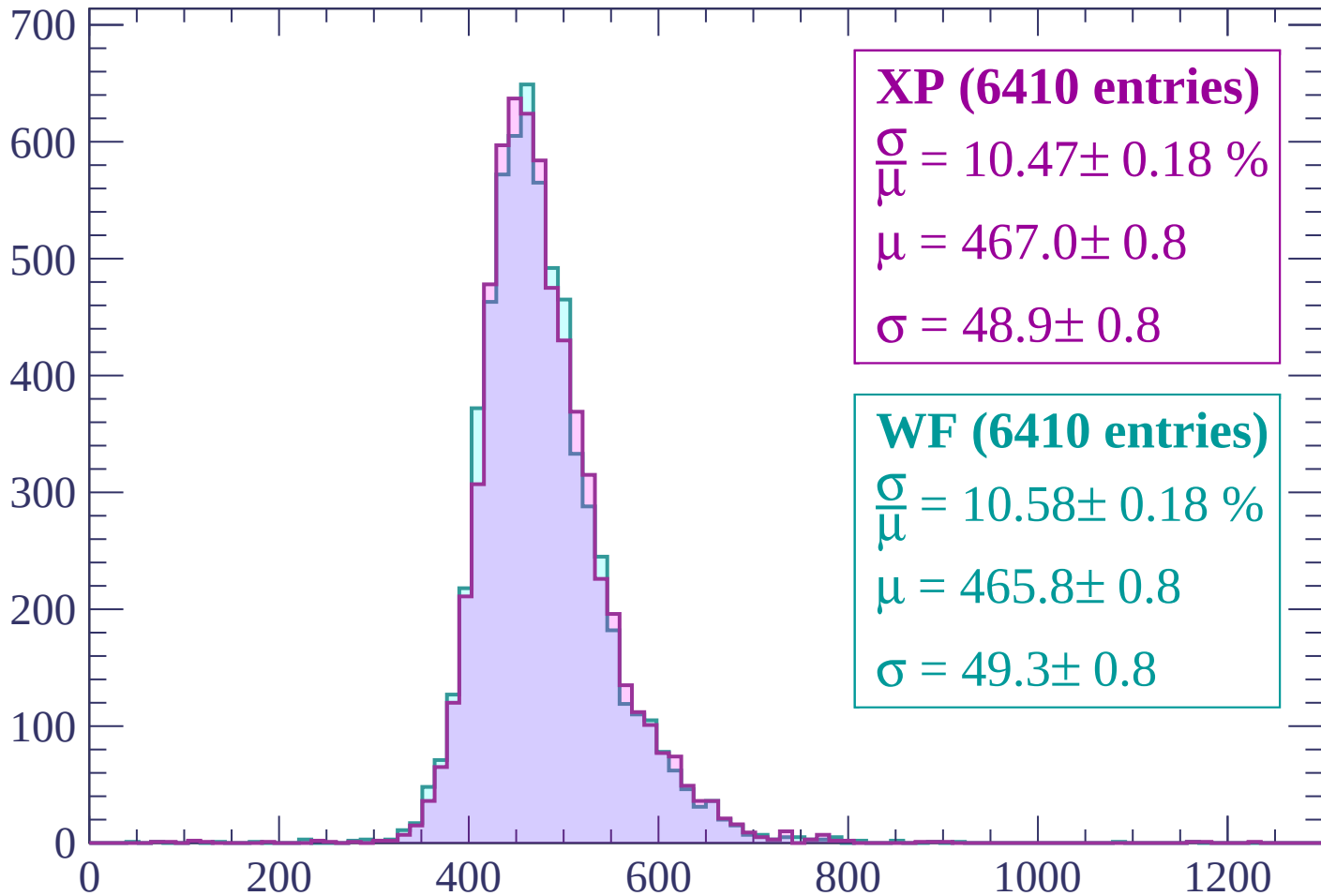
$\mu = 463.6 \pm 0.8$

$\sigma = 50.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (6410 entries)

$\frac{\sigma}{\mu} = 10.47 \pm 0.18 \%$

$\mu = 467.0 \pm 0.8$

$\sigma = 48.9 \pm 0.8$

WF (6410 entries)

$\frac{\sigma}{\mu} = 10.58 \pm 0.18 \%$

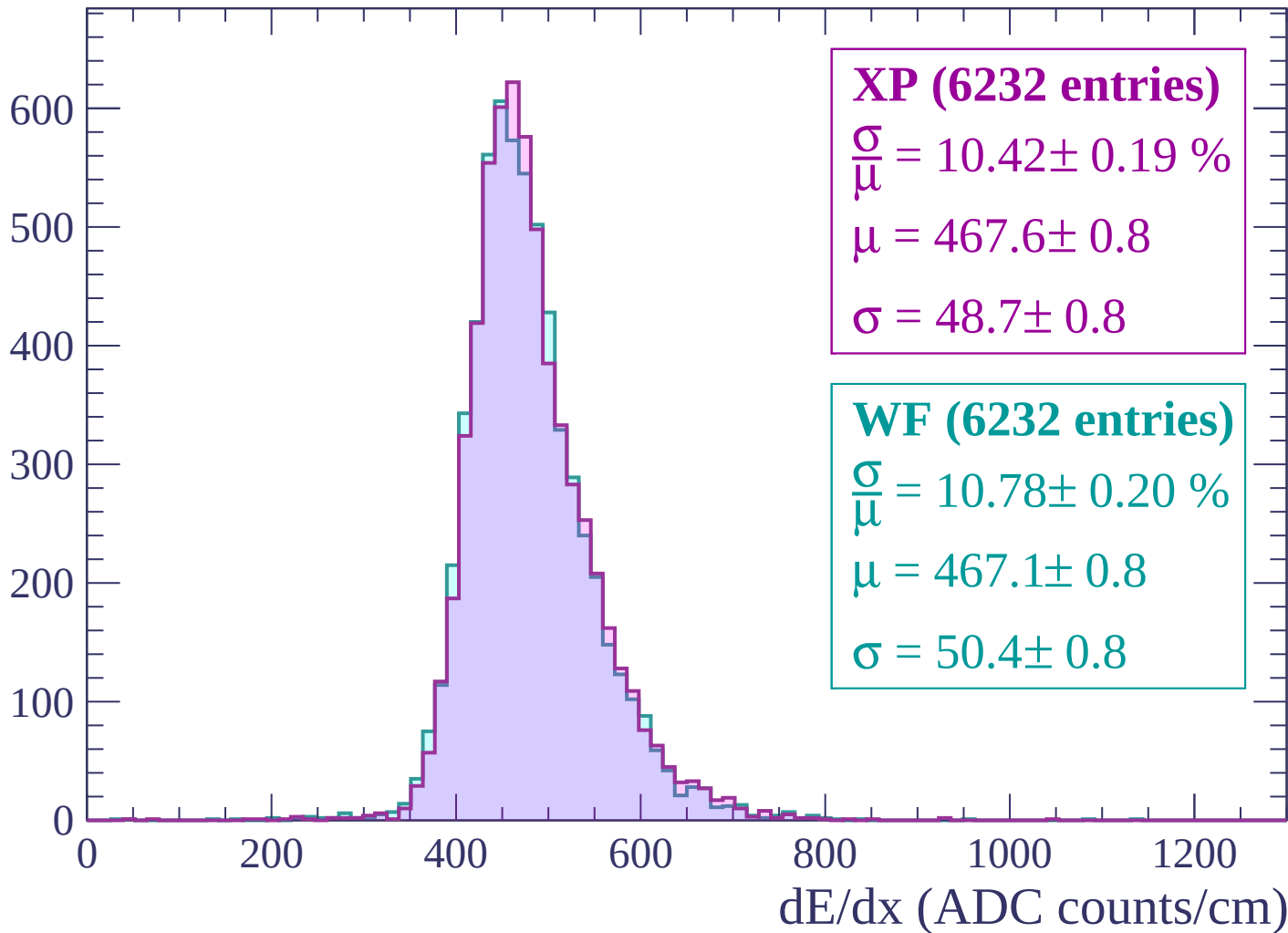
$\mu = 465.8 \pm 0.8$

$\sigma = 49.3 \pm 0.8$

dE/dx (ADC counts/cm)

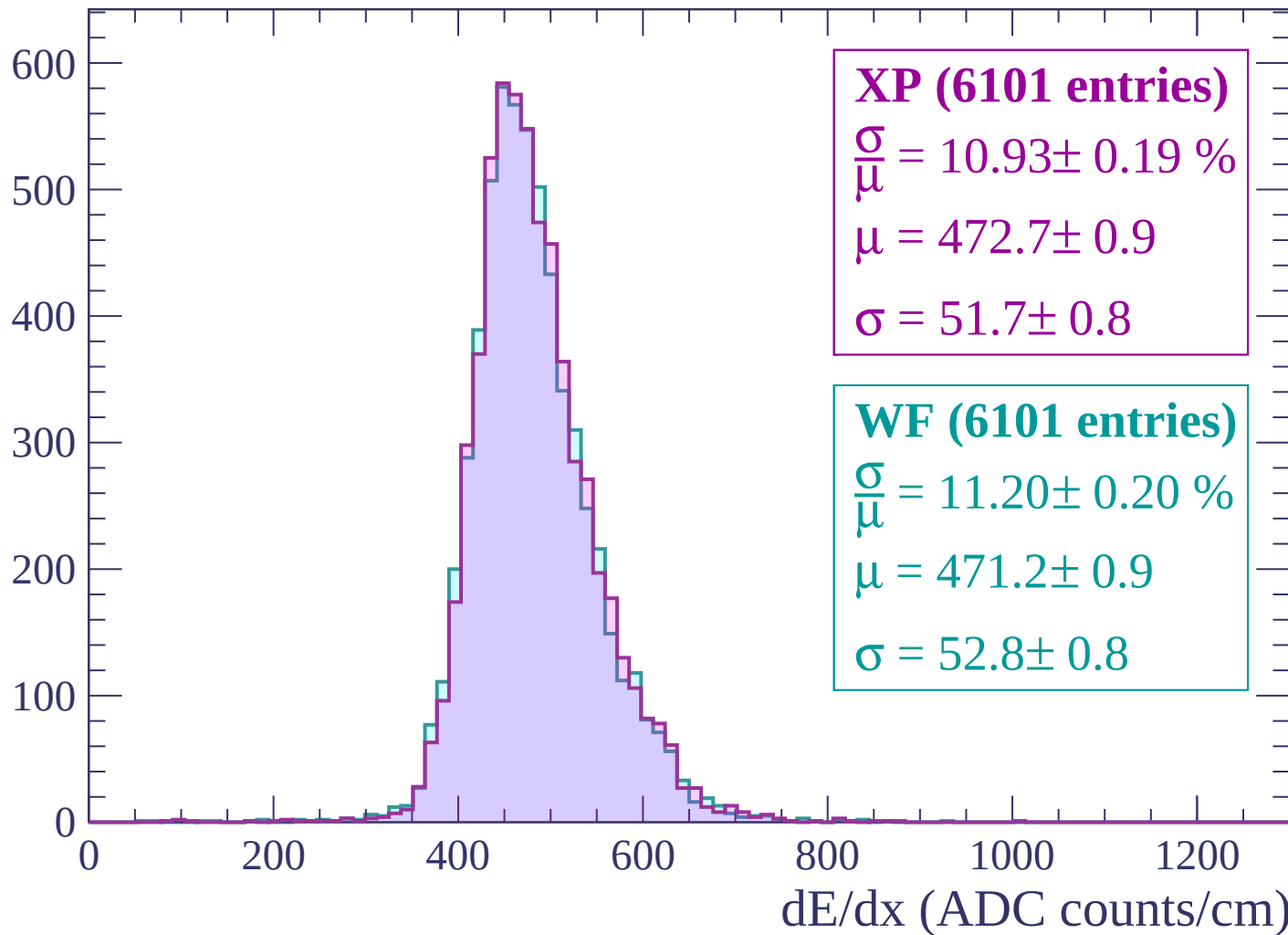
Energy loss | $1560 < p < 1620$

Count



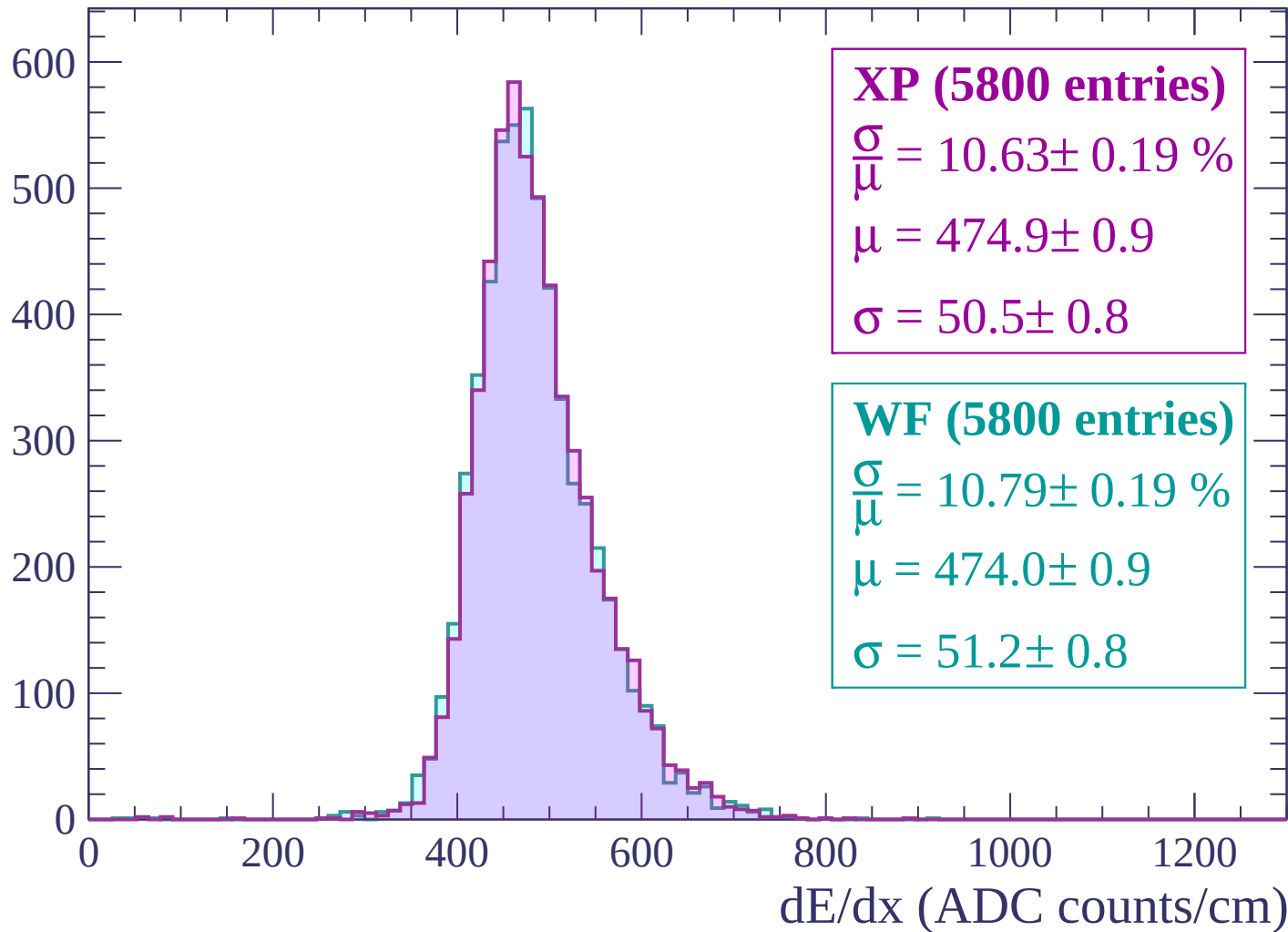
Energy loss | $1620 < p < 1680$

Count



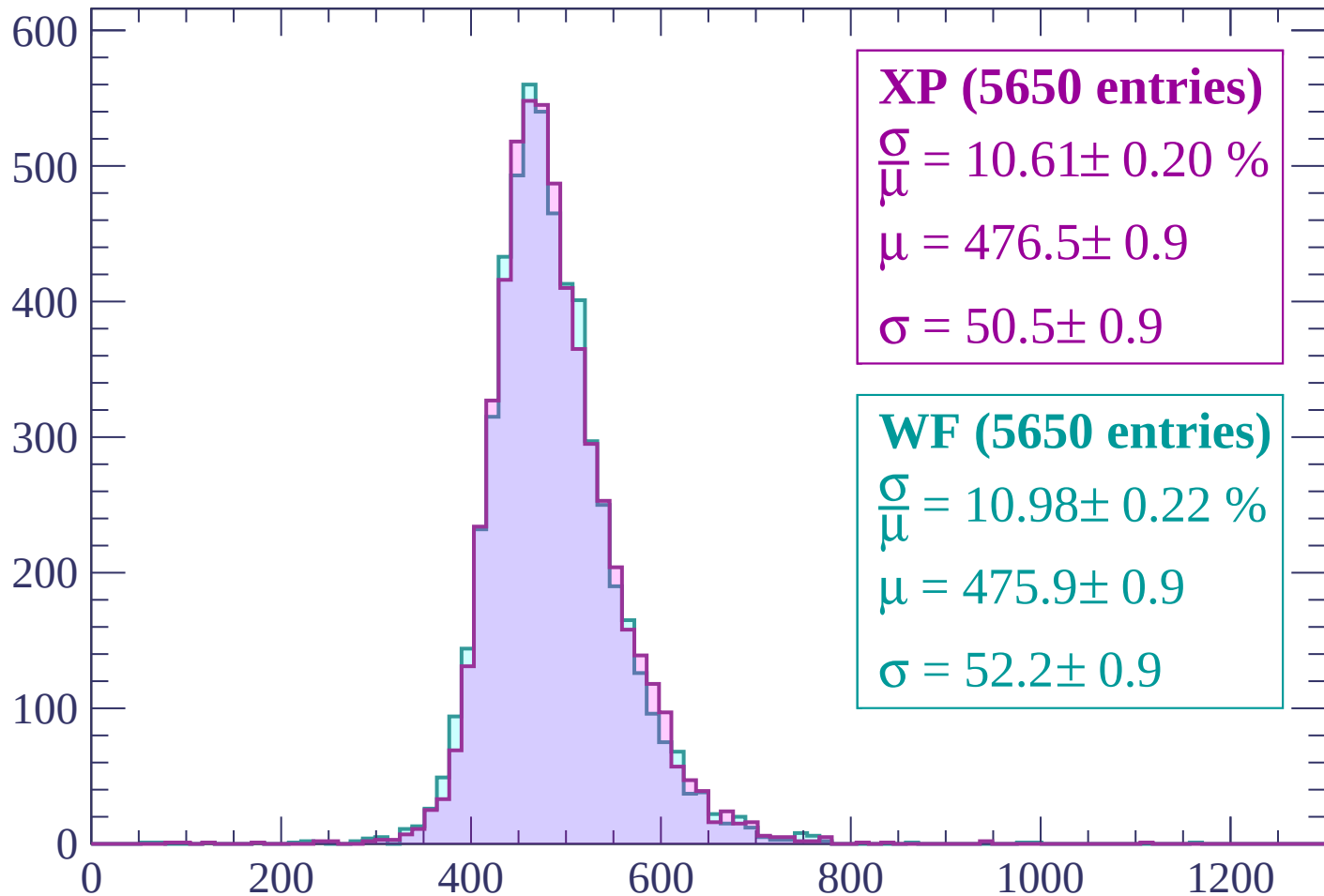
Energy loss | $1680 < p < 1740$

Count



Energy loss | $1740 < p < 1800$

Count



XP (5650 entries)

$$\frac{\sigma}{\mu} = 10.61 \pm 0.20 \%$$

$$\mu = 476.5 \pm 0.9$$

$$\sigma = 50.5 \pm 0.9$$

WF (5650 entries)

$$\frac{\sigma}{\mu} = 10.98 \pm 0.22 \%$$

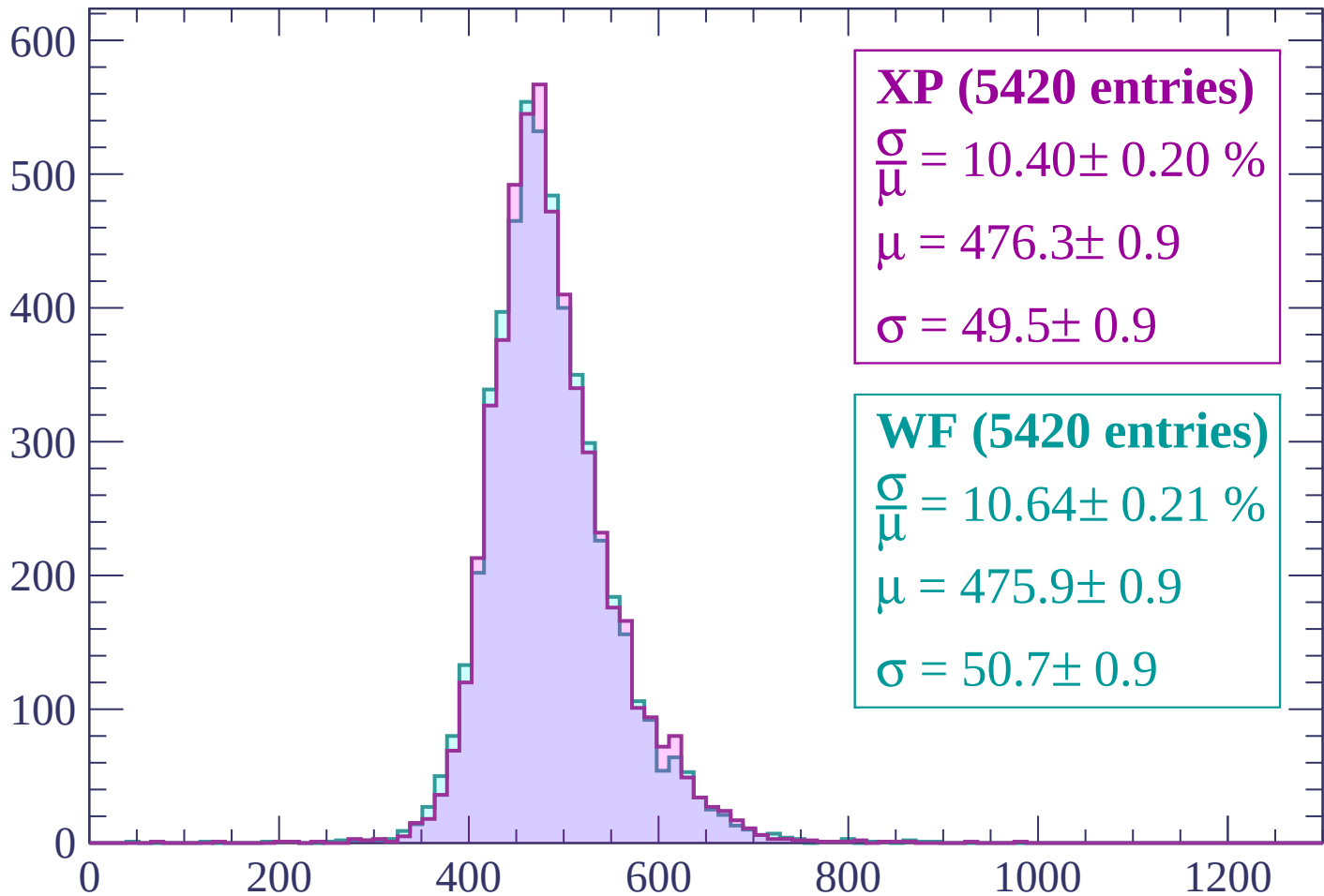
$$\mu = 475.9 \pm 0.9$$

$$\sigma = 52.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (5420 entries)

$\frac{\sigma}{\mu} = 10.40 \pm 0.20 \%$

$\mu = 476.3 \pm 0.9$

$\sigma = 49.5 \pm 0.9$

WF (5420 entries)

$\frac{\sigma}{\mu} = 10.64 \pm 0.21 \%$

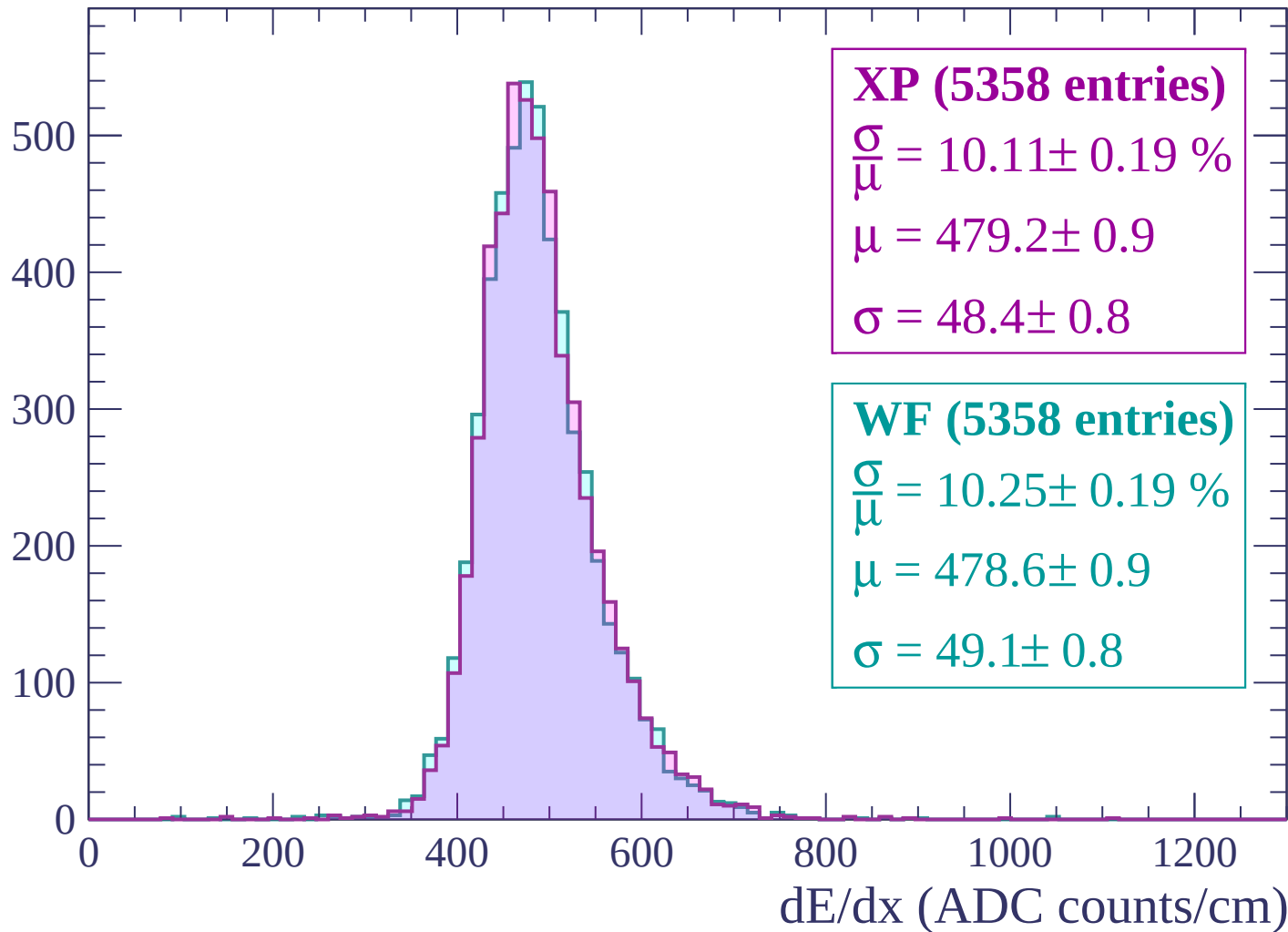
$\mu = 475.9 \pm 0.9$

$\sigma = 50.7 \pm 0.9$

dE/dx (ADC counts/cm)

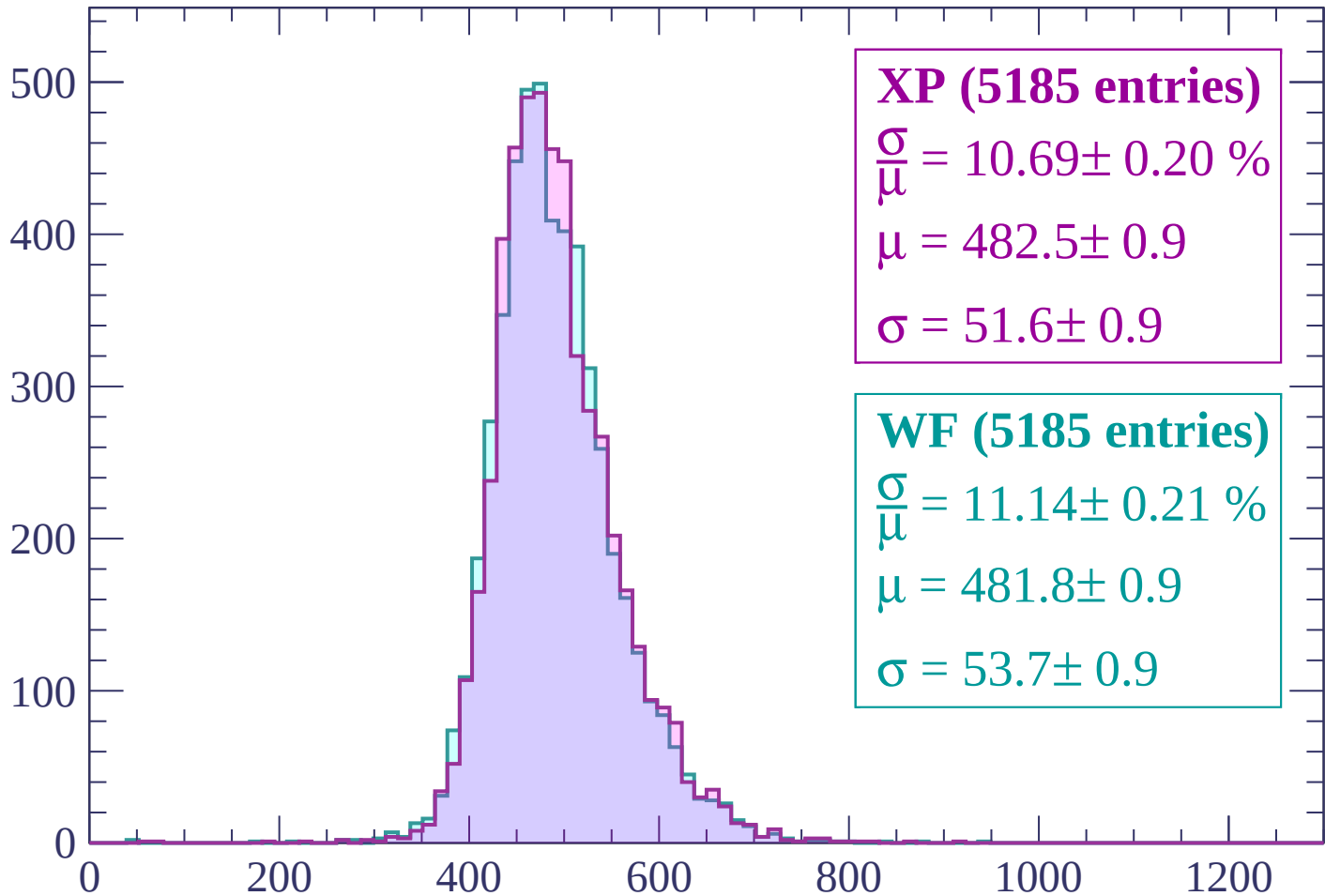
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (5185 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.20 \%$$

$$\mu = 482.5 \pm 0.9$$

$$\sigma = 51.6 \pm 0.9$$

WF (5185 entries)

$$\frac{\sigma}{\mu} = 11.14 \pm 0.21 \%$$

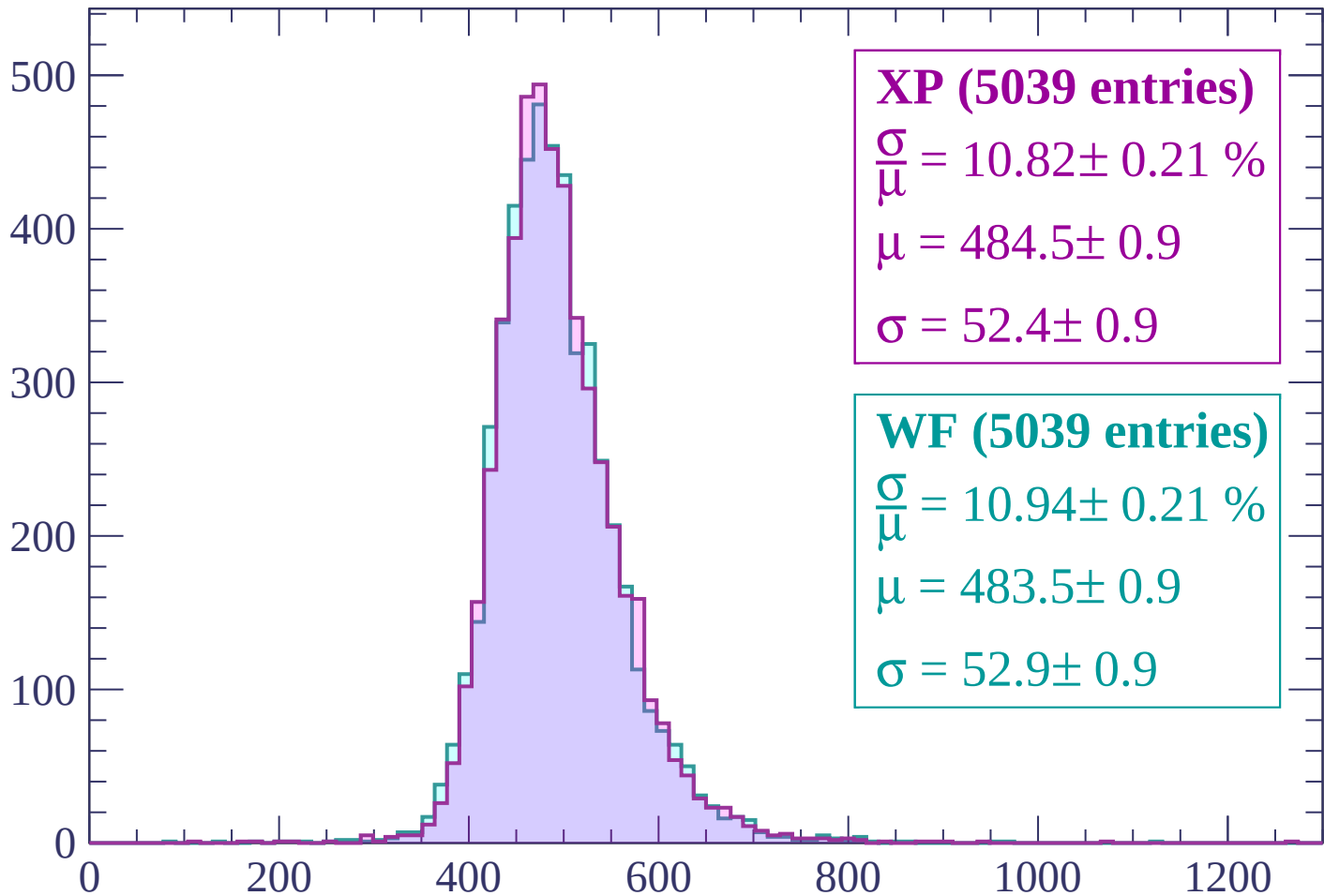
$$\mu = 481.8 \pm 0.9$$

$$\sigma = 53.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count



XP (5039 entries)

$$\frac{\sigma}{\mu} = 10.82 \pm 0.21 \%$$

$$\mu = 484.5 \pm 0.9$$

$$\sigma = 52.4 \pm 0.9$$

WF (5039 entries)

$$\frac{\sigma}{\mu} = 10.94 \pm 0.21 \%$$

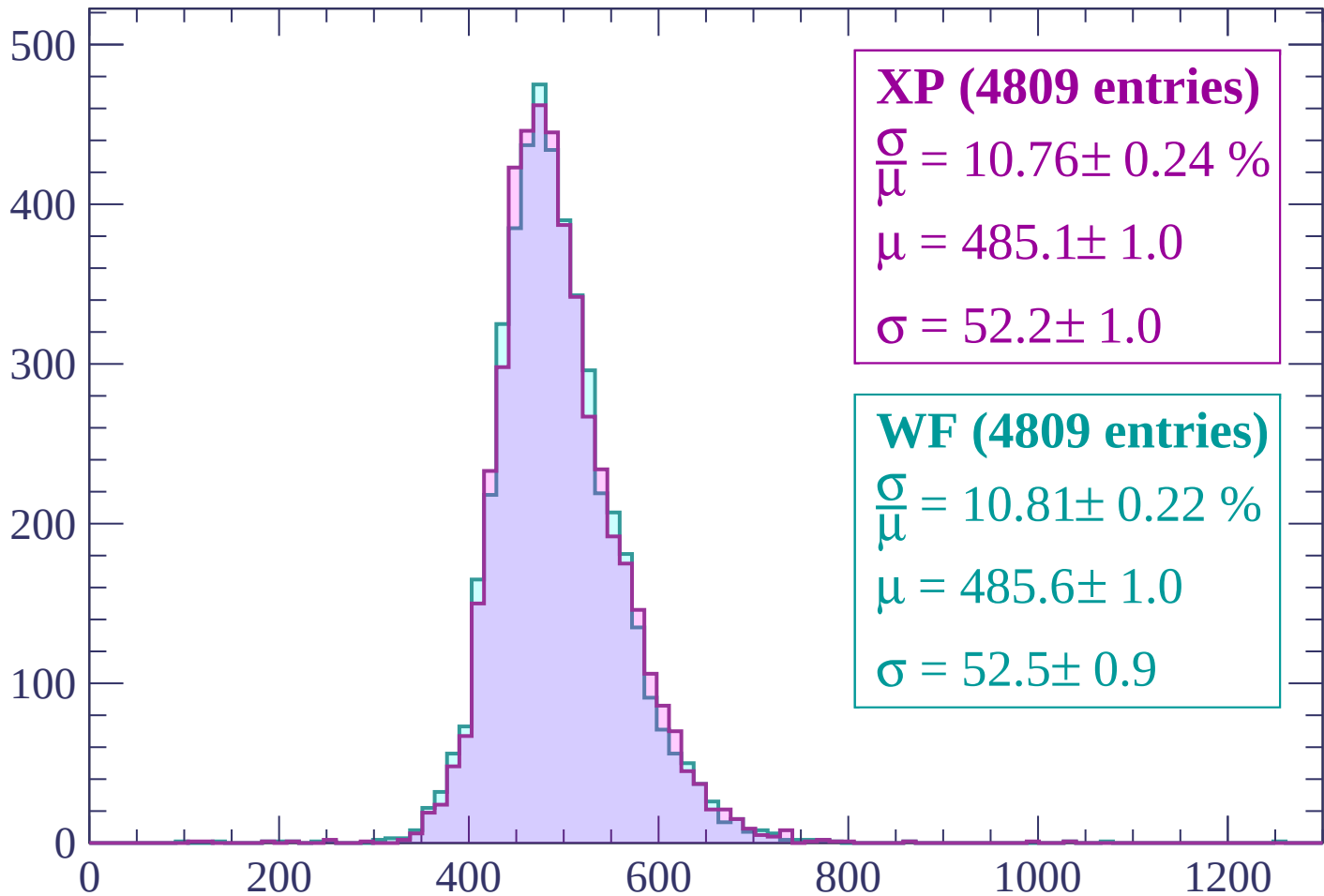
$$\mu = 483.5 \pm 0.9$$

$$\sigma = 52.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (4809 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.24 \%$$

$$\mu = 485.1 \pm 1.0$$

$$\sigma = 52.2 \pm 1.0$$

WF (4809 entries)

$$\frac{\sigma}{\mu} = 10.81 \pm 0.22 \%$$

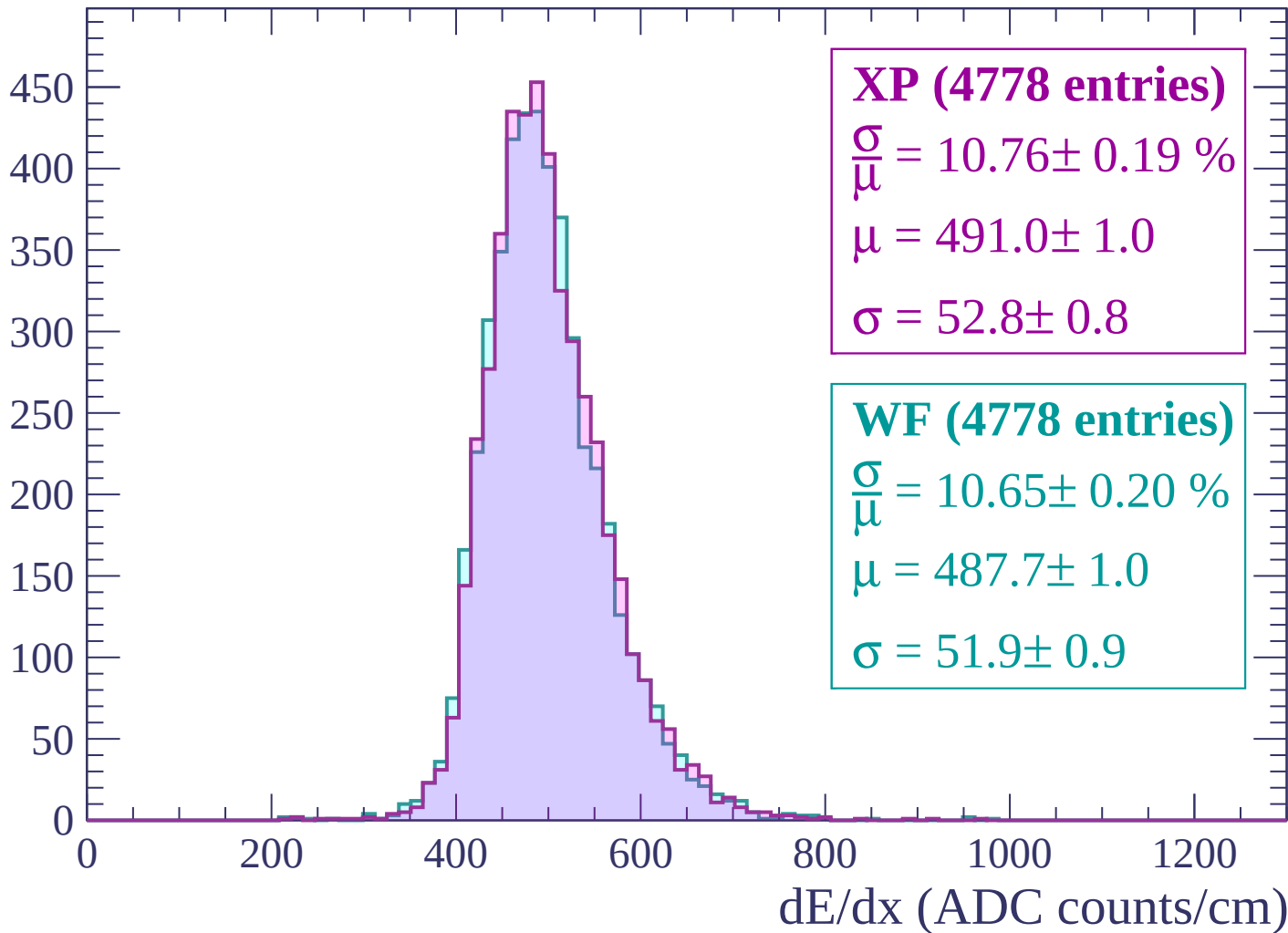
$$\mu = 485.6 \pm 1.0$$

$$\sigma = 52.5 \pm 0.9$$

dE/dx (ADC counts/cm)

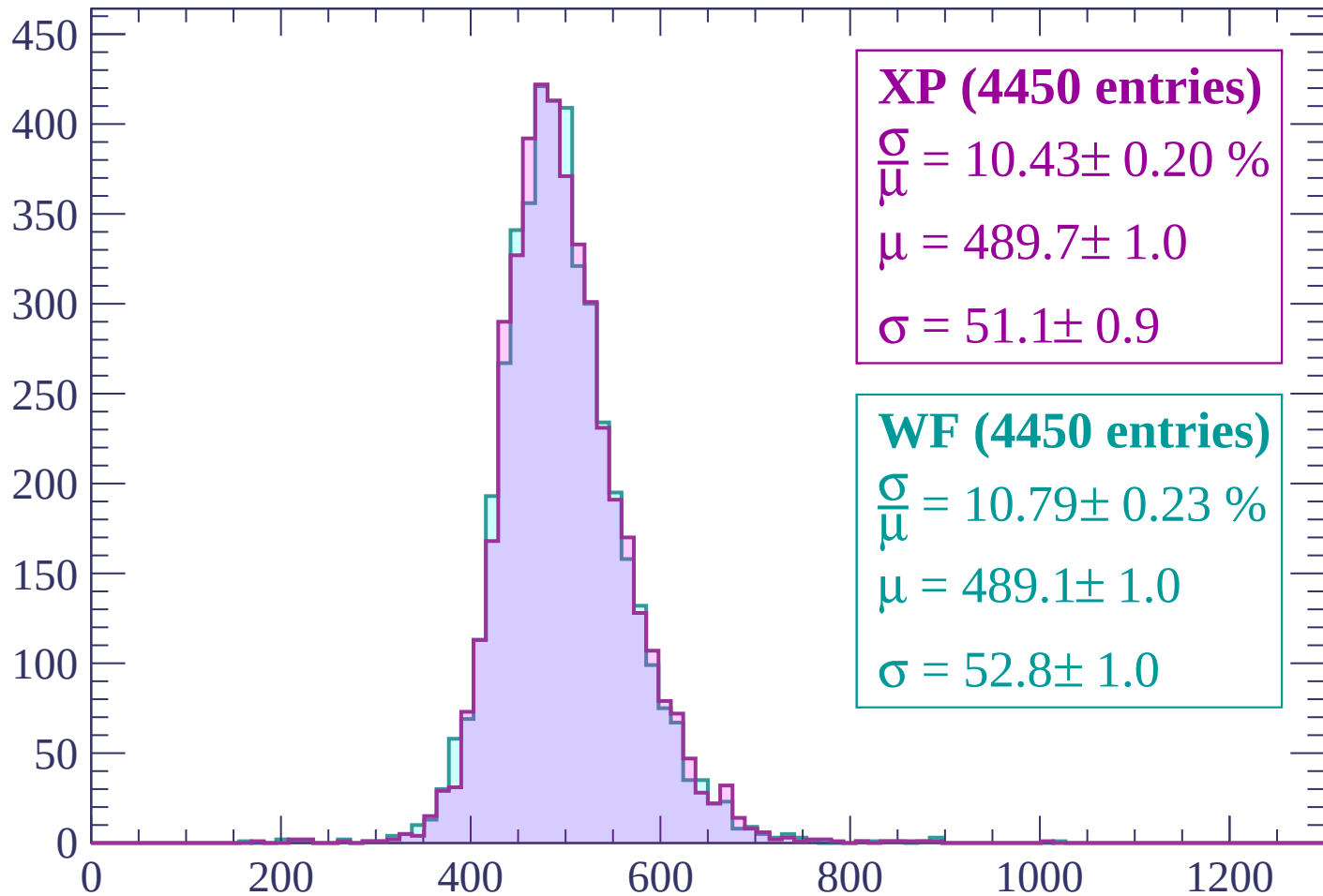
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (4450 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.20 \%$

$\mu = 489.7 \pm 1.0$

$\sigma = 51.1 \pm 0.9$

WF (4450 entries)

$\frac{\sigma}{\mu} = 10.79 \pm 0.23 \%$

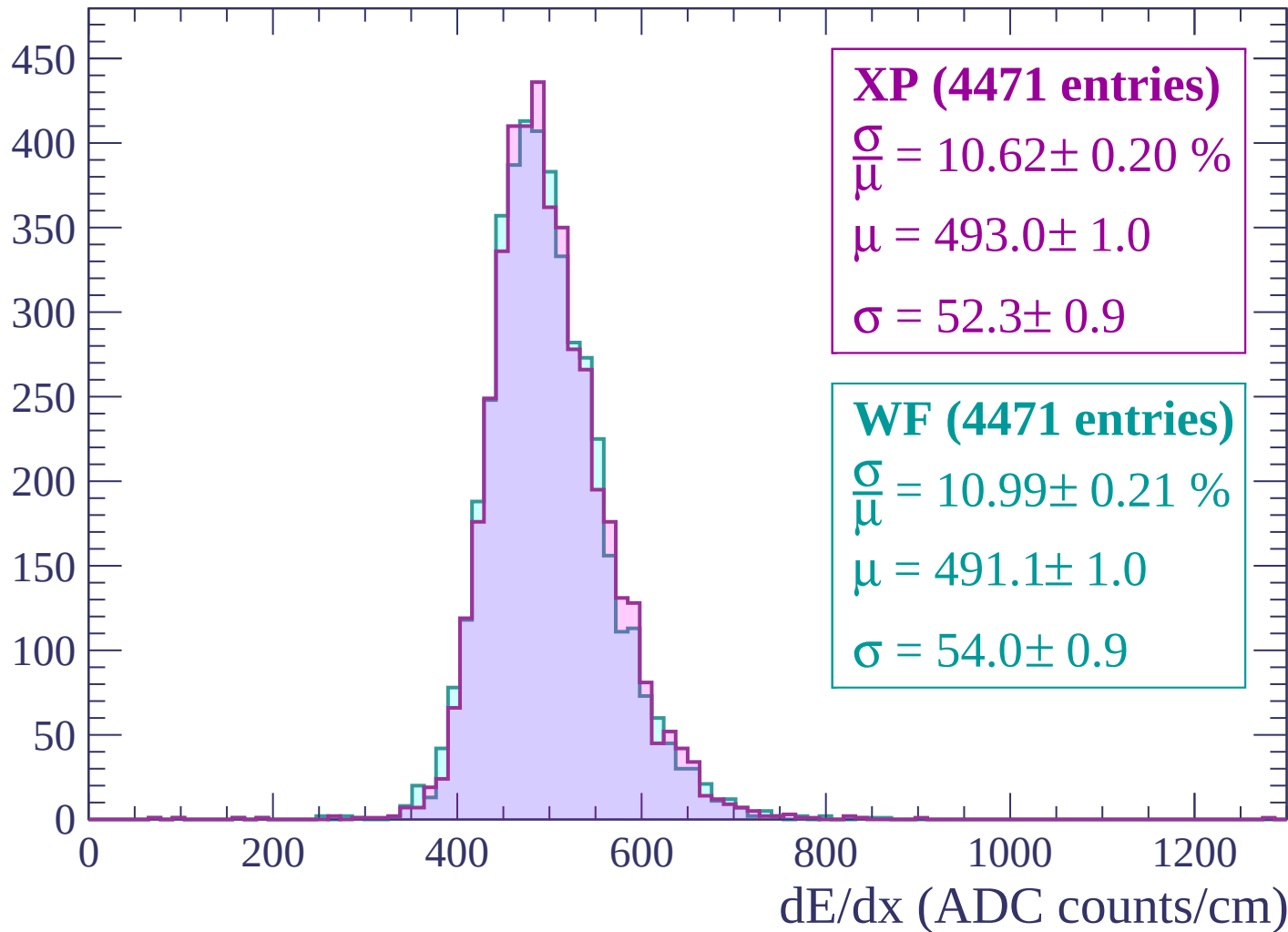
$\mu = 489.1 \pm 1.0$

$\sigma = 52.8 \pm 1.0$

dE/dx (ADC counts/cm)

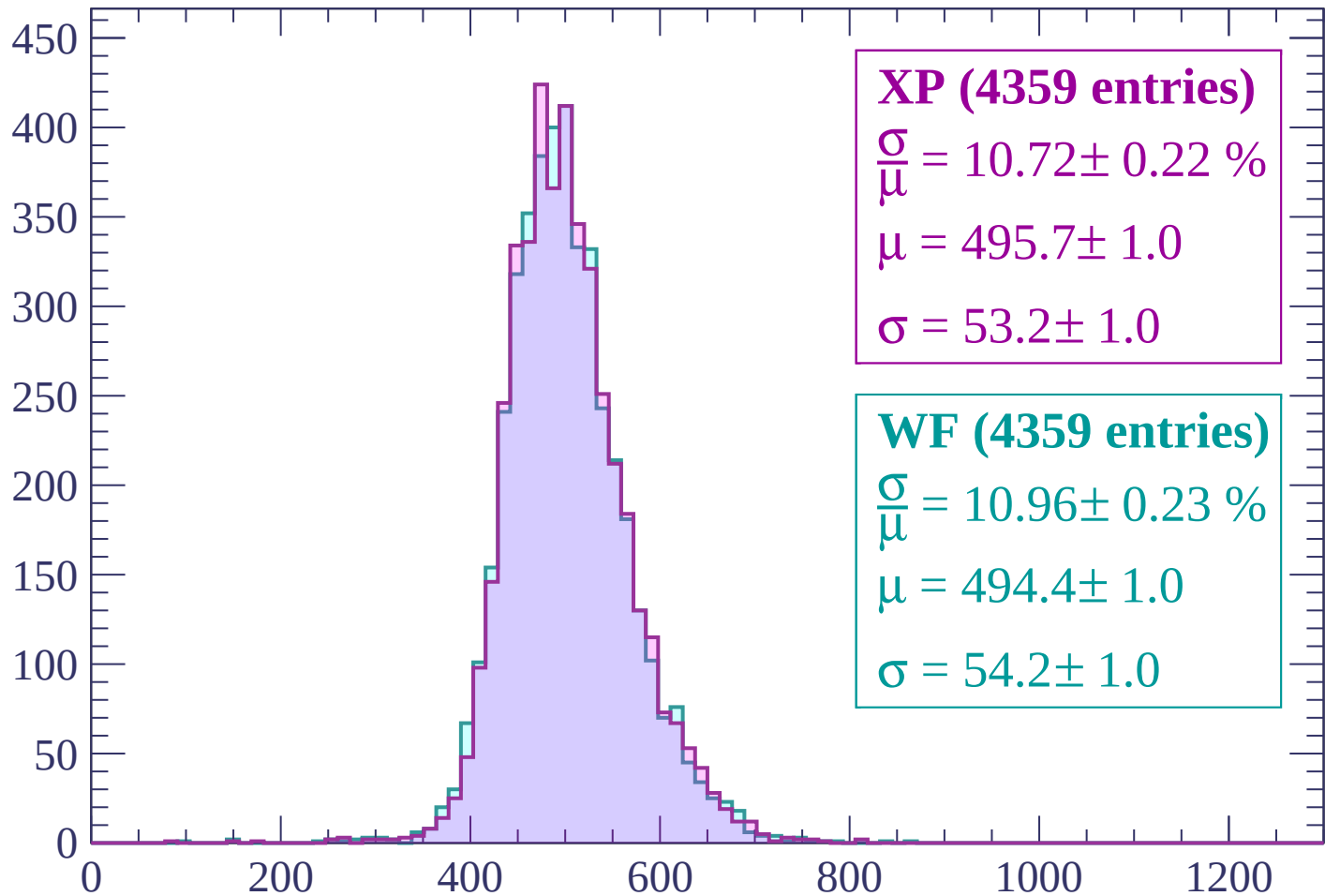
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

Count



XP (4359 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$

$\mu = 495.7 \pm 1.0$

$\sigma = 53.2 \pm 1.0$

WF (4359 entries)

$\frac{\sigma}{\mu} = 10.96 \pm 0.23 \%$

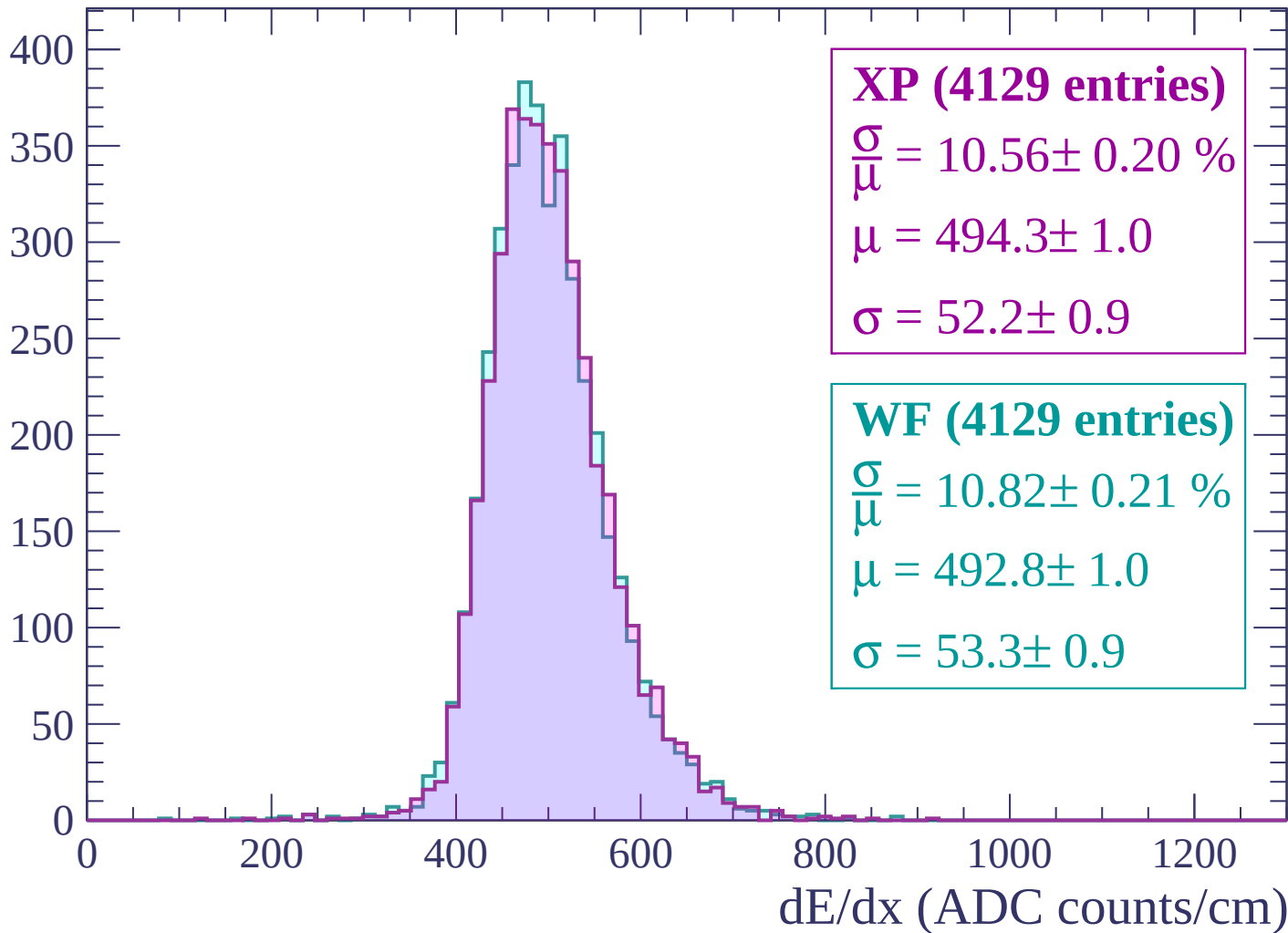
$\mu = 494.4 \pm 1.0$

$\sigma = 54.2 \pm 1.0$

dE/dx (ADC counts/cm)

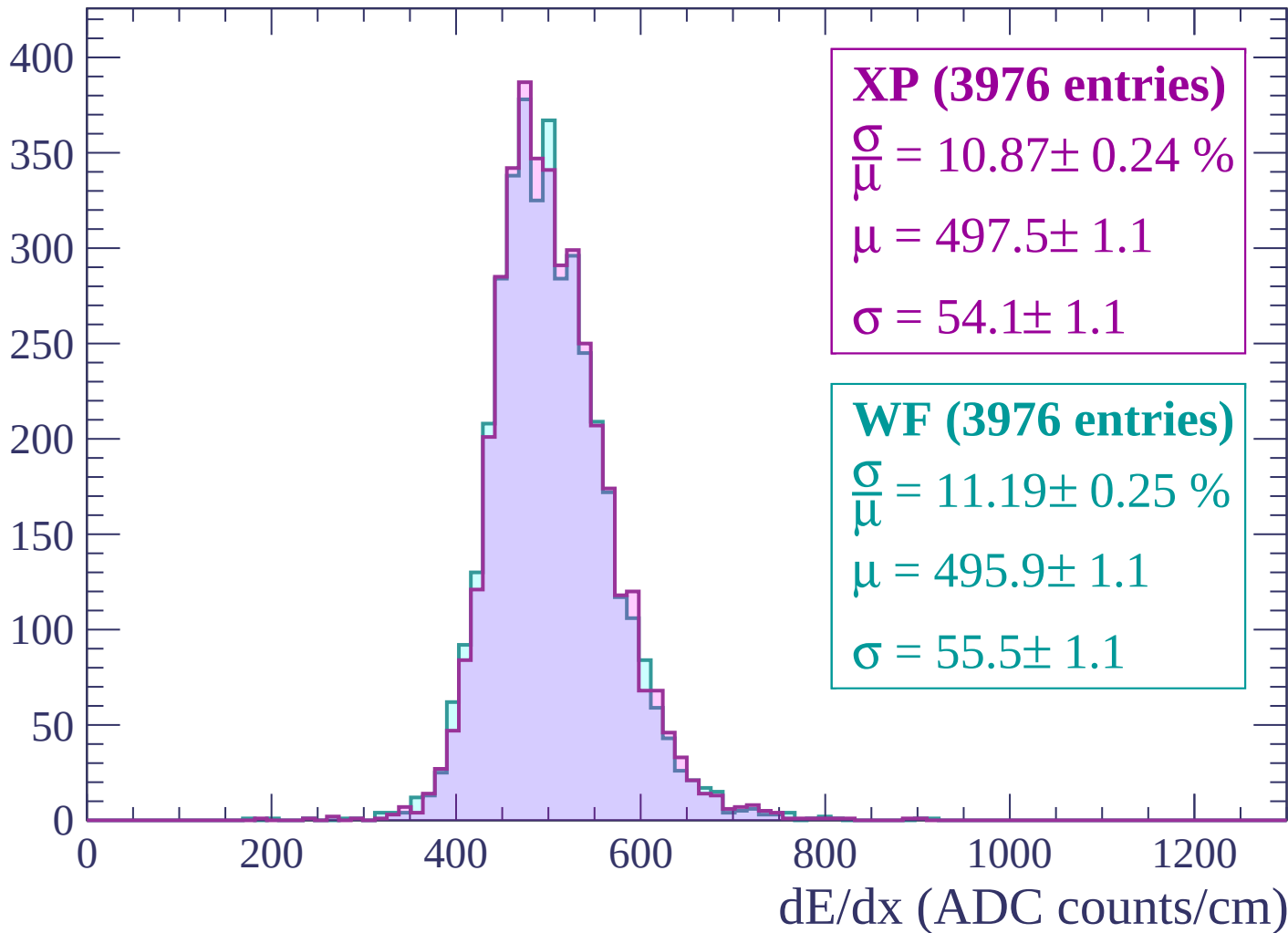
Energy loss | $2340 < p < 2400$

Count



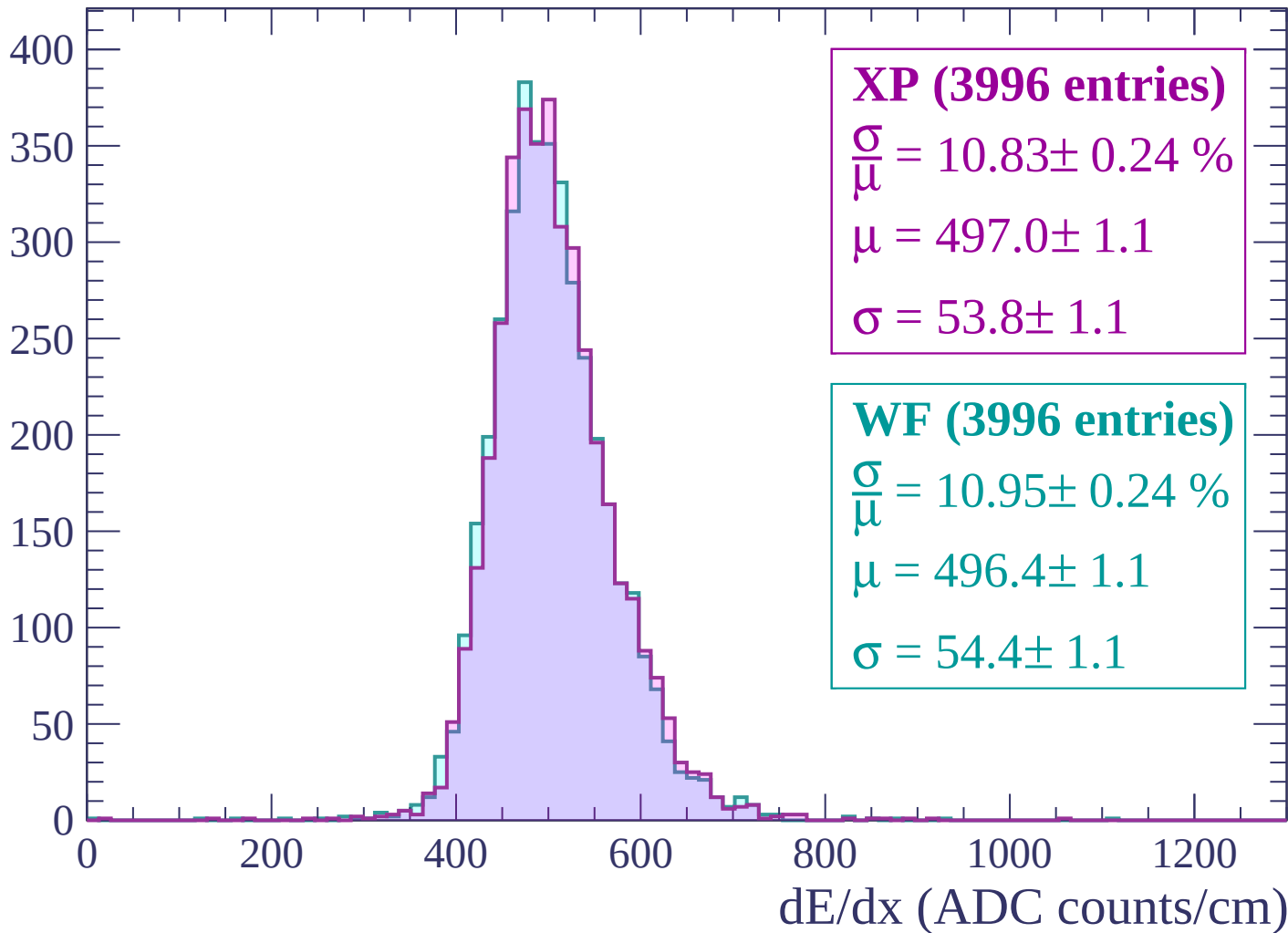
Energy loss | $2400 < p < 2460$

Count



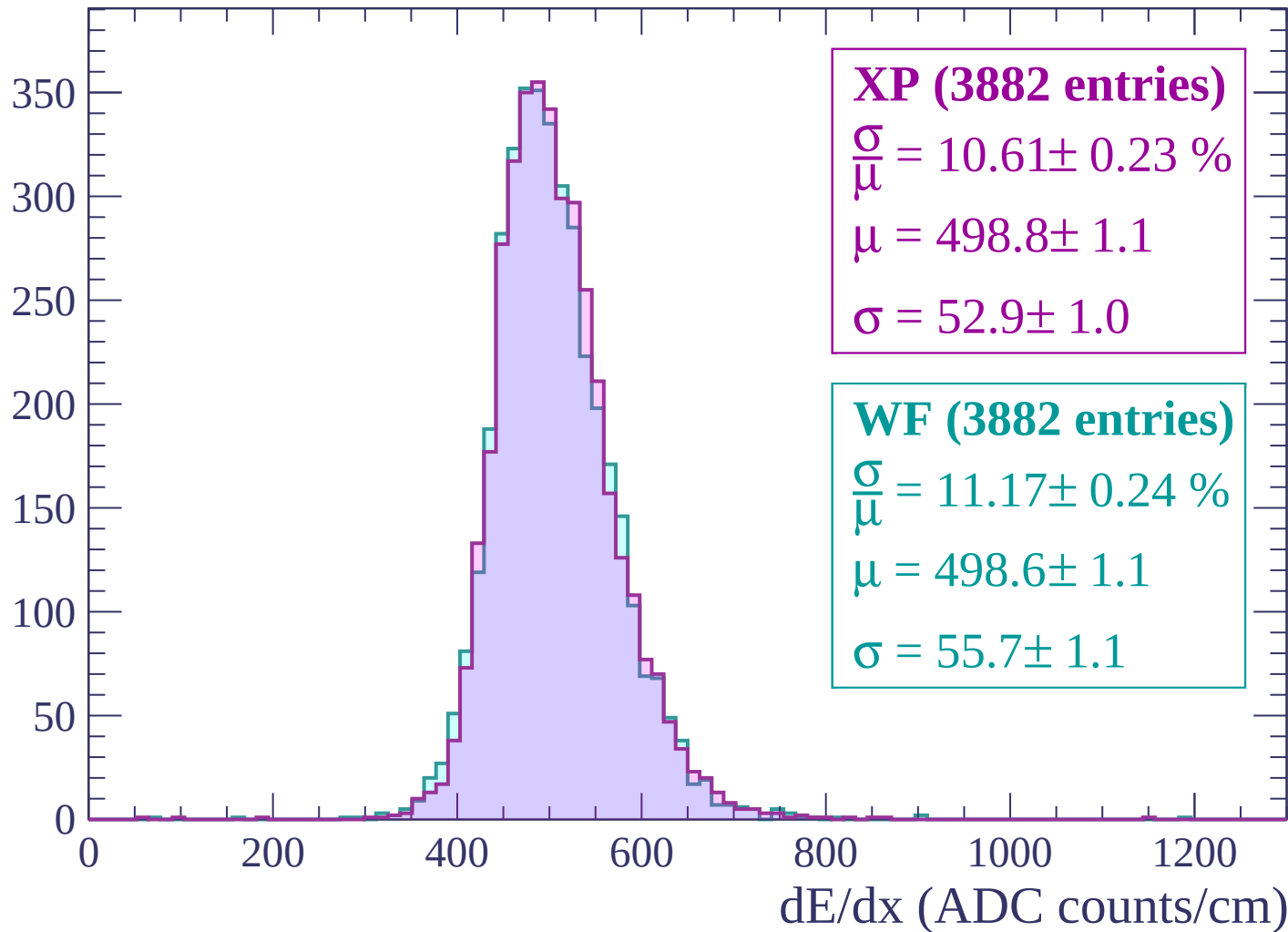
Energy loss | $2460 < p < 2520$

Count



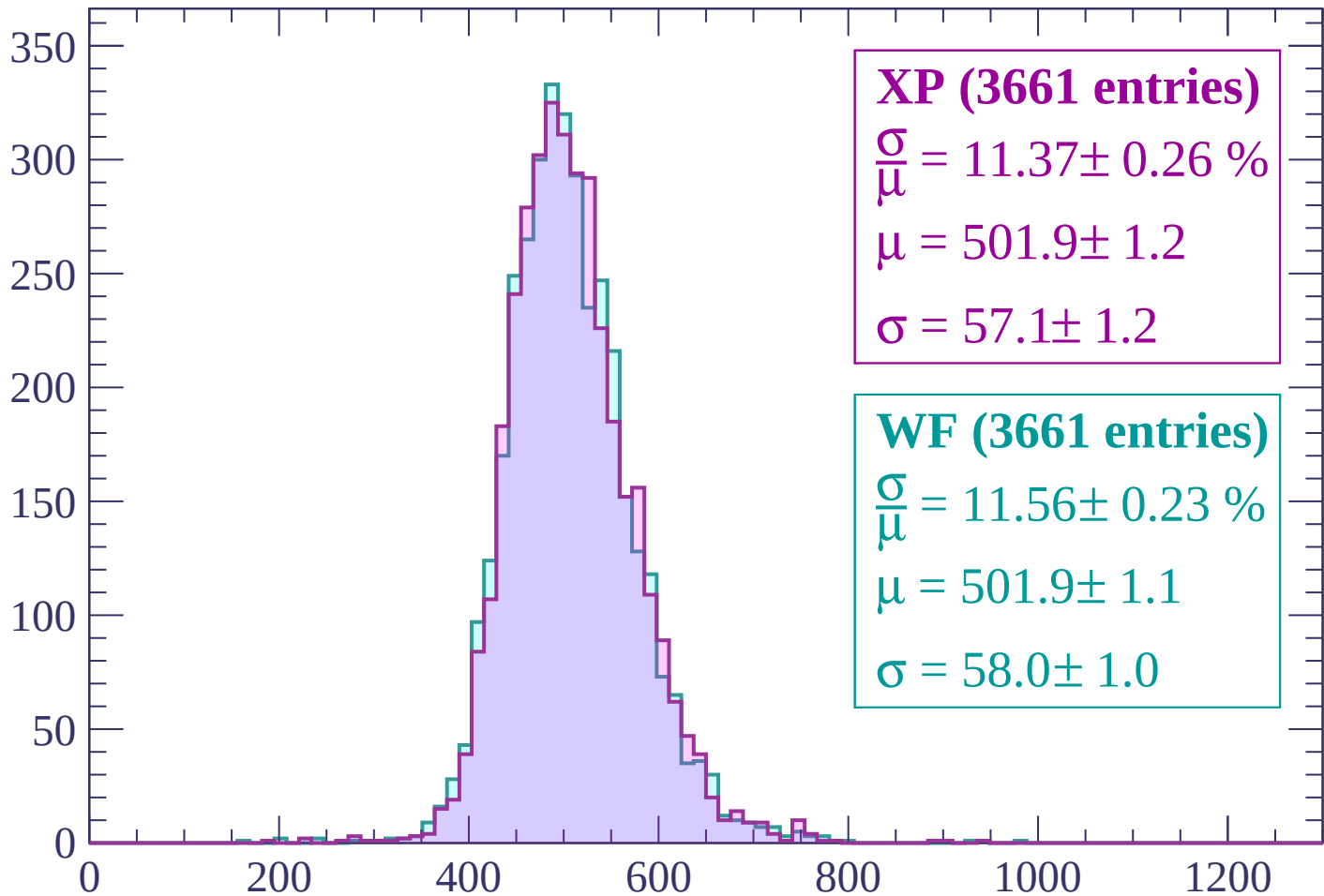
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP (3661 entries)

$\frac{\sigma}{\mu} = 11.37 \pm 0.26 \%$

$\mu = 501.9 \pm 1.2$

$\sigma = 57.1 \pm 1.2$

WF (3661 entries)

$\frac{\sigma}{\mu} = 11.56 \pm 0.23 \%$

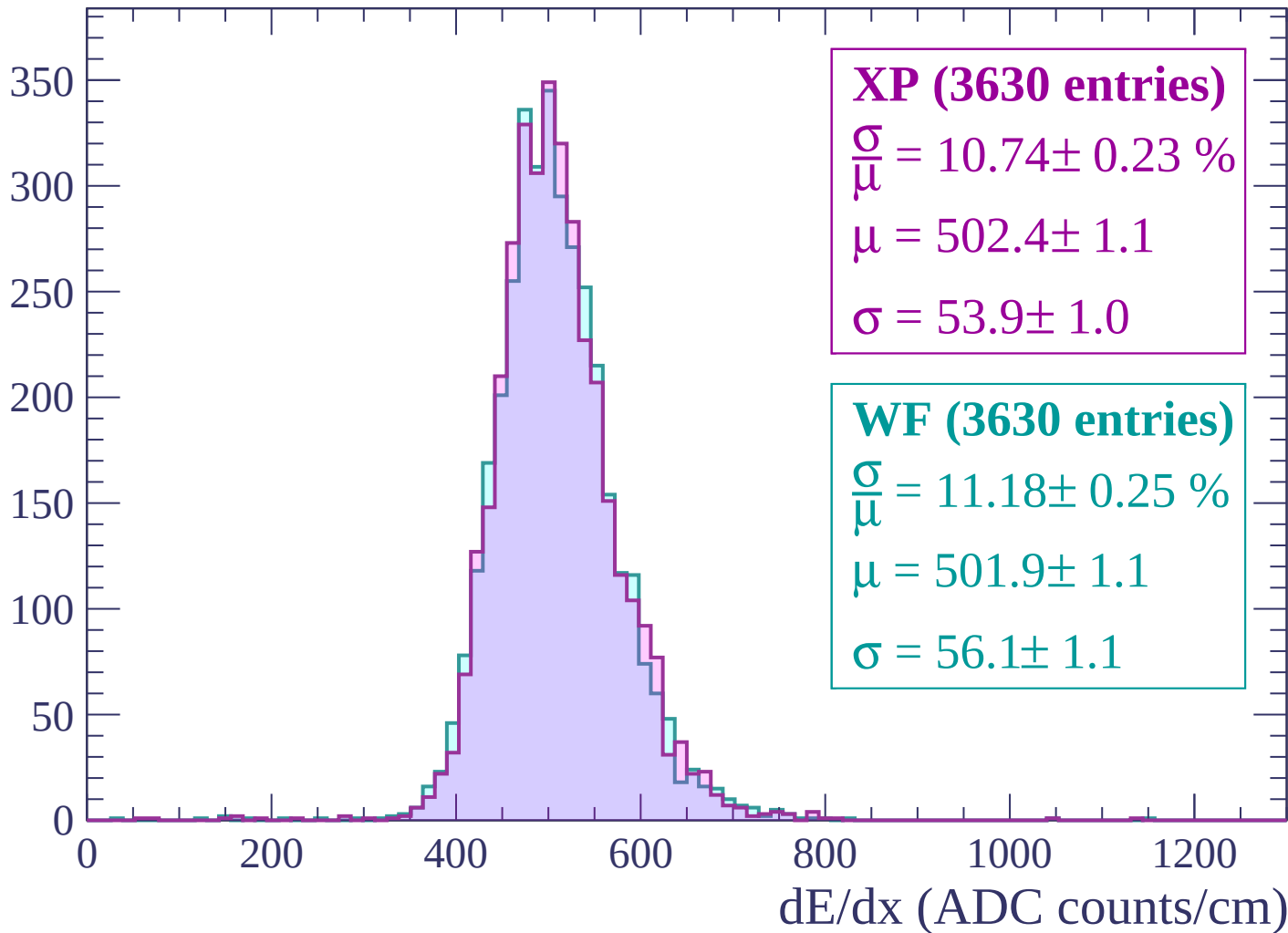
$\mu = 501.9 \pm 1.1$

$\sigma = 58.0 \pm 1.0$

dE/dx (ADC counts/cm)

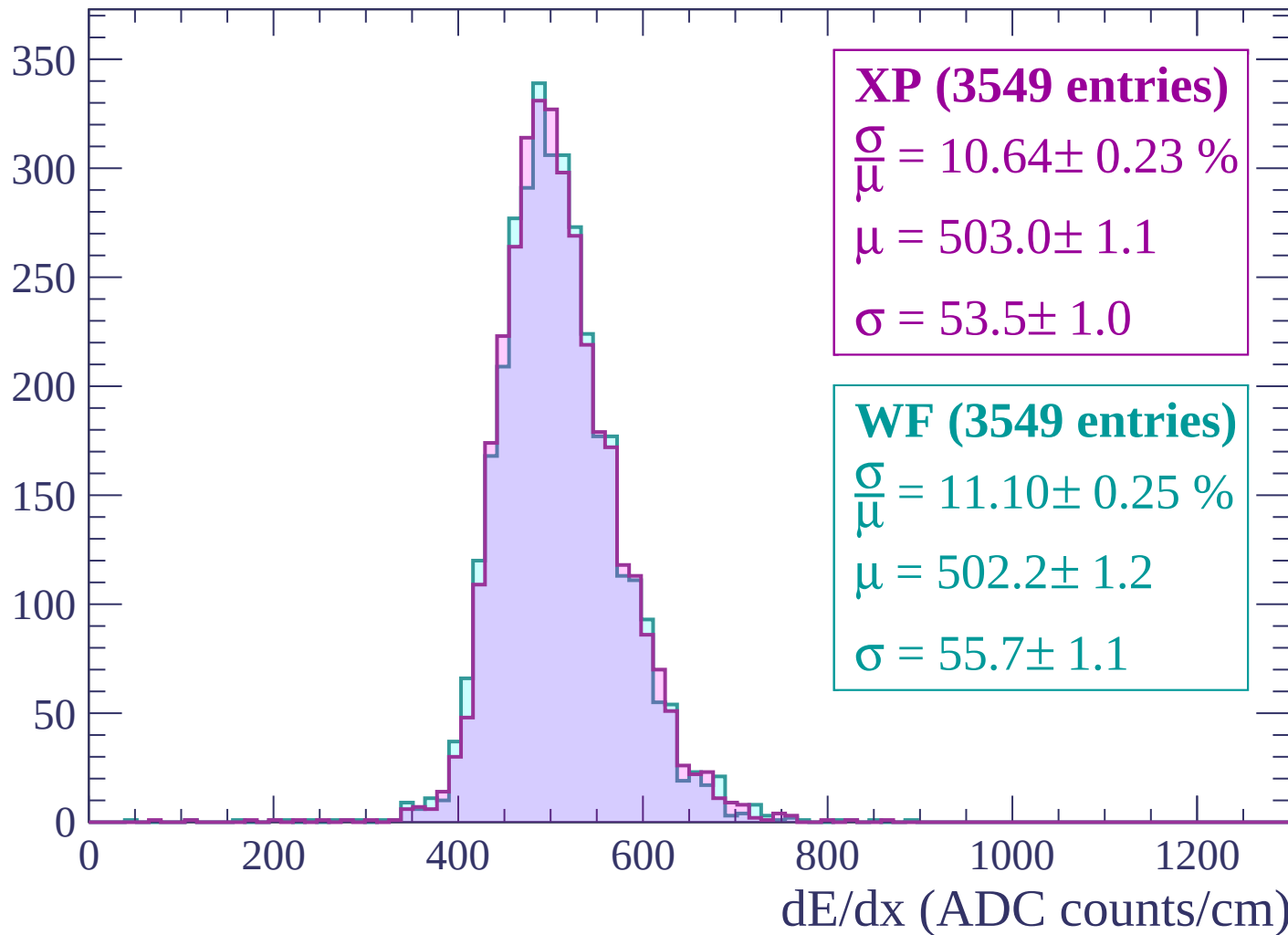
Energy loss | $2640 < p < 2700$

Count



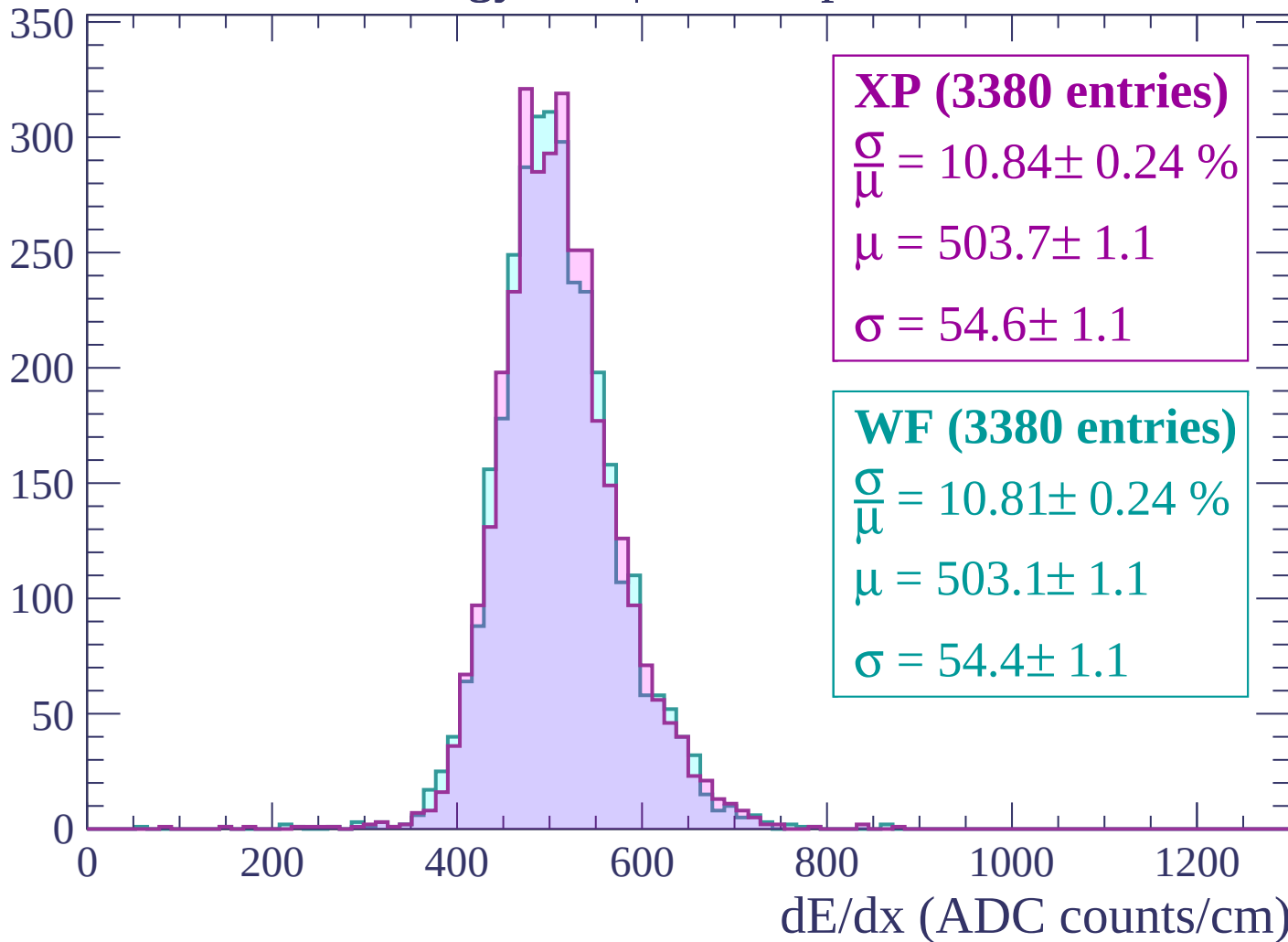
Energy loss | $2700 < p < 2760$

Count



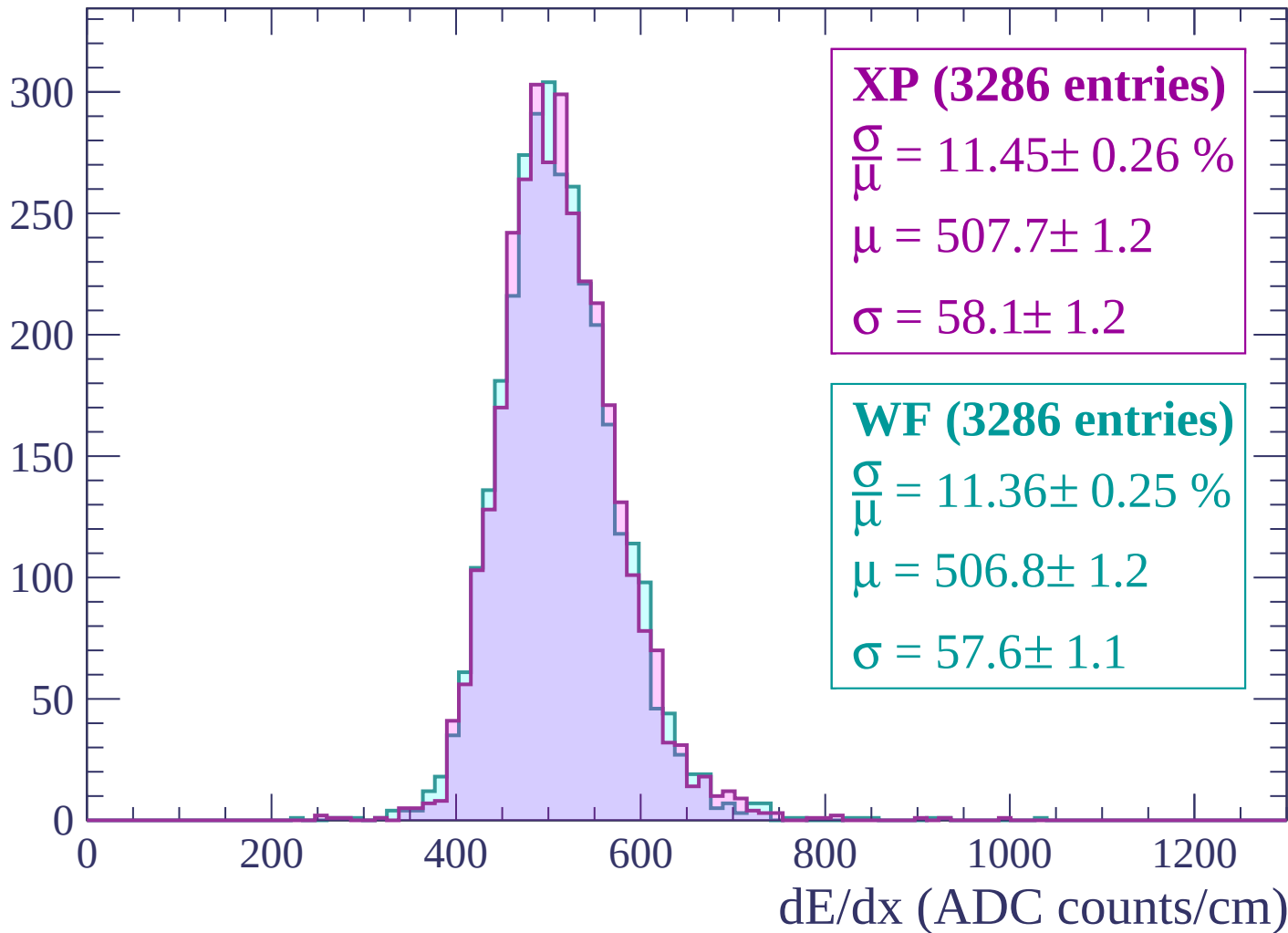
Energy loss | $2760 < p < 2820$

Count



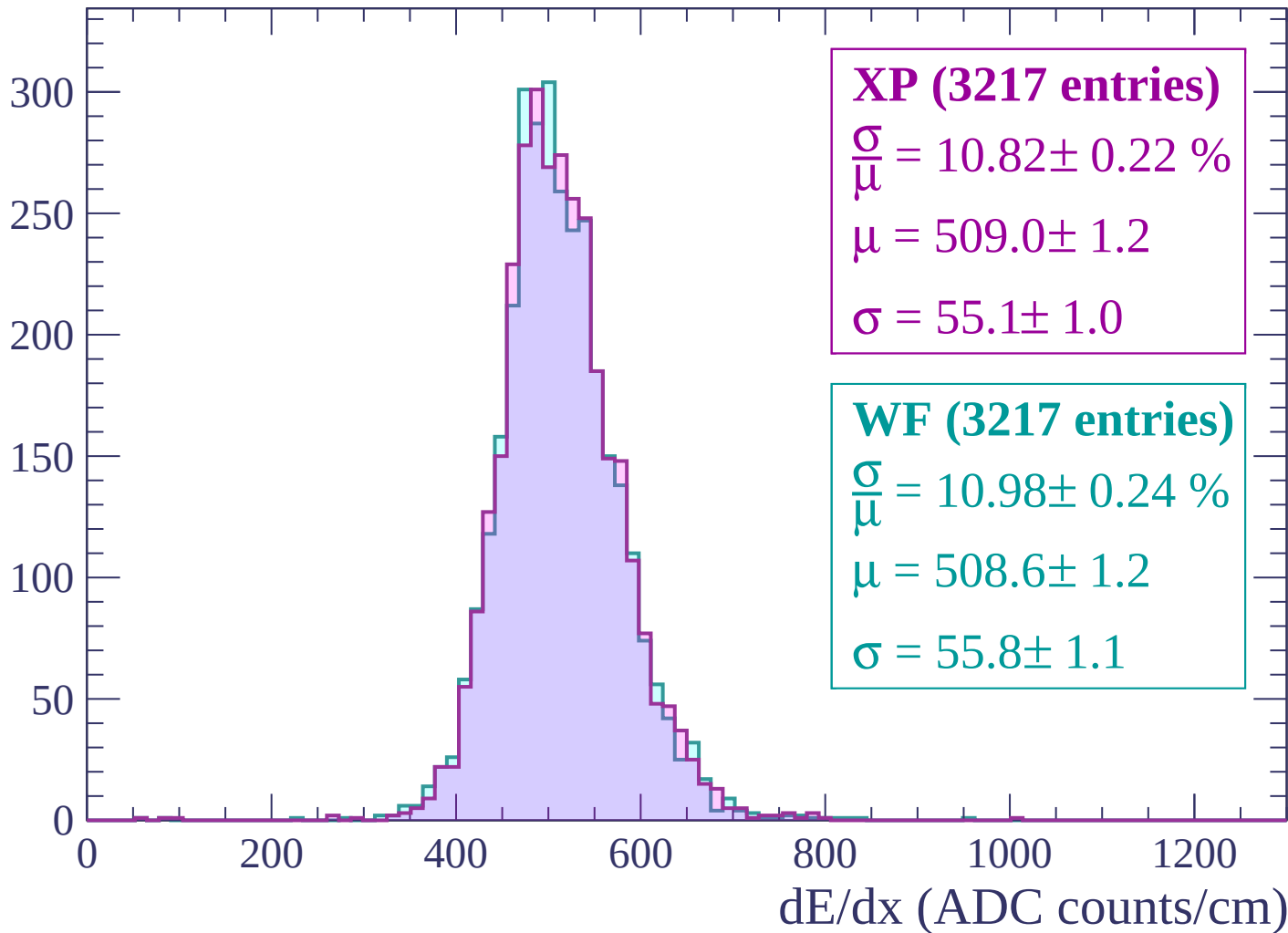
Energy loss | $2820 < p < 2880$

Count



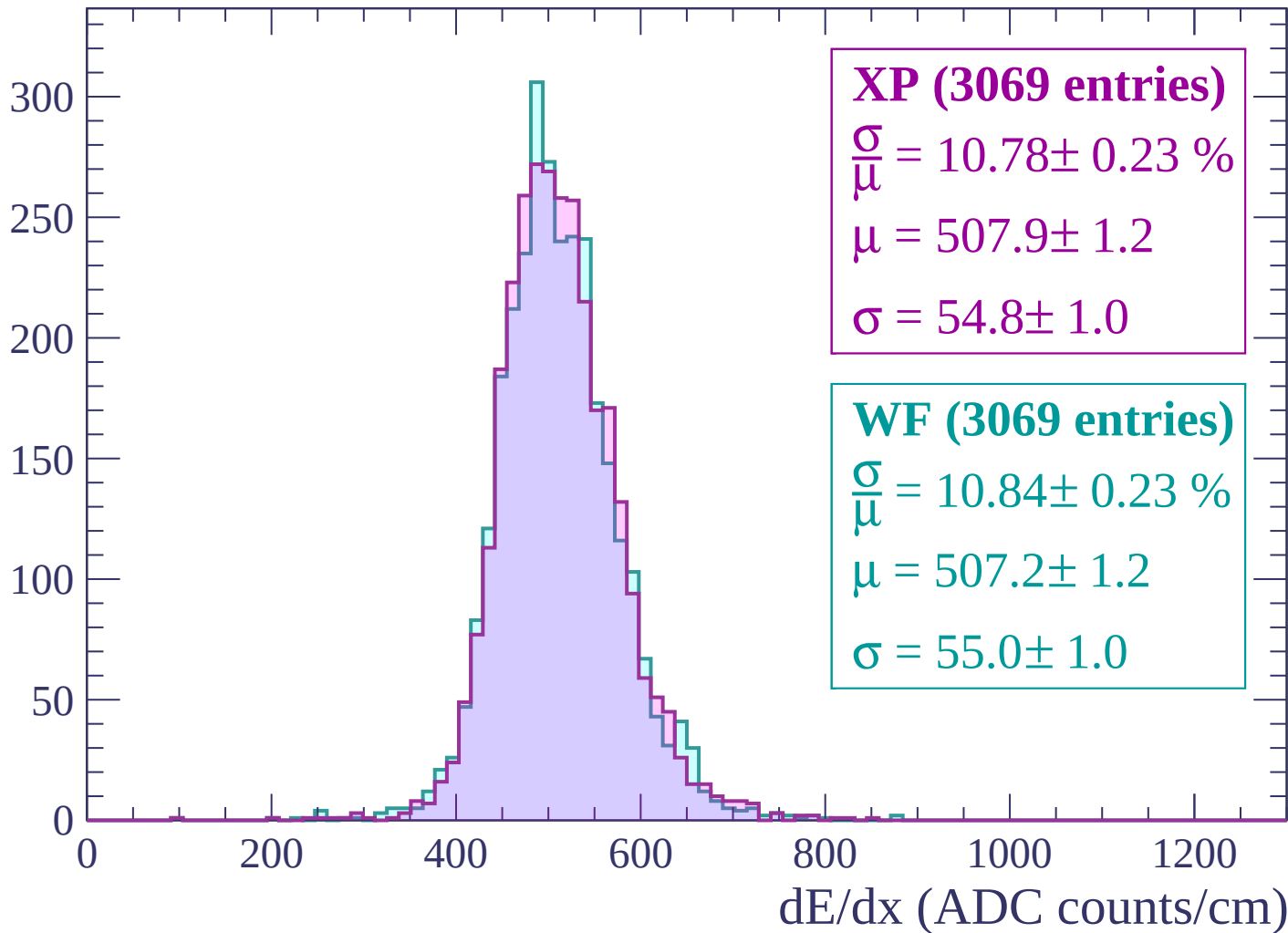
Energy loss | $2880 < p < 2940$

Count



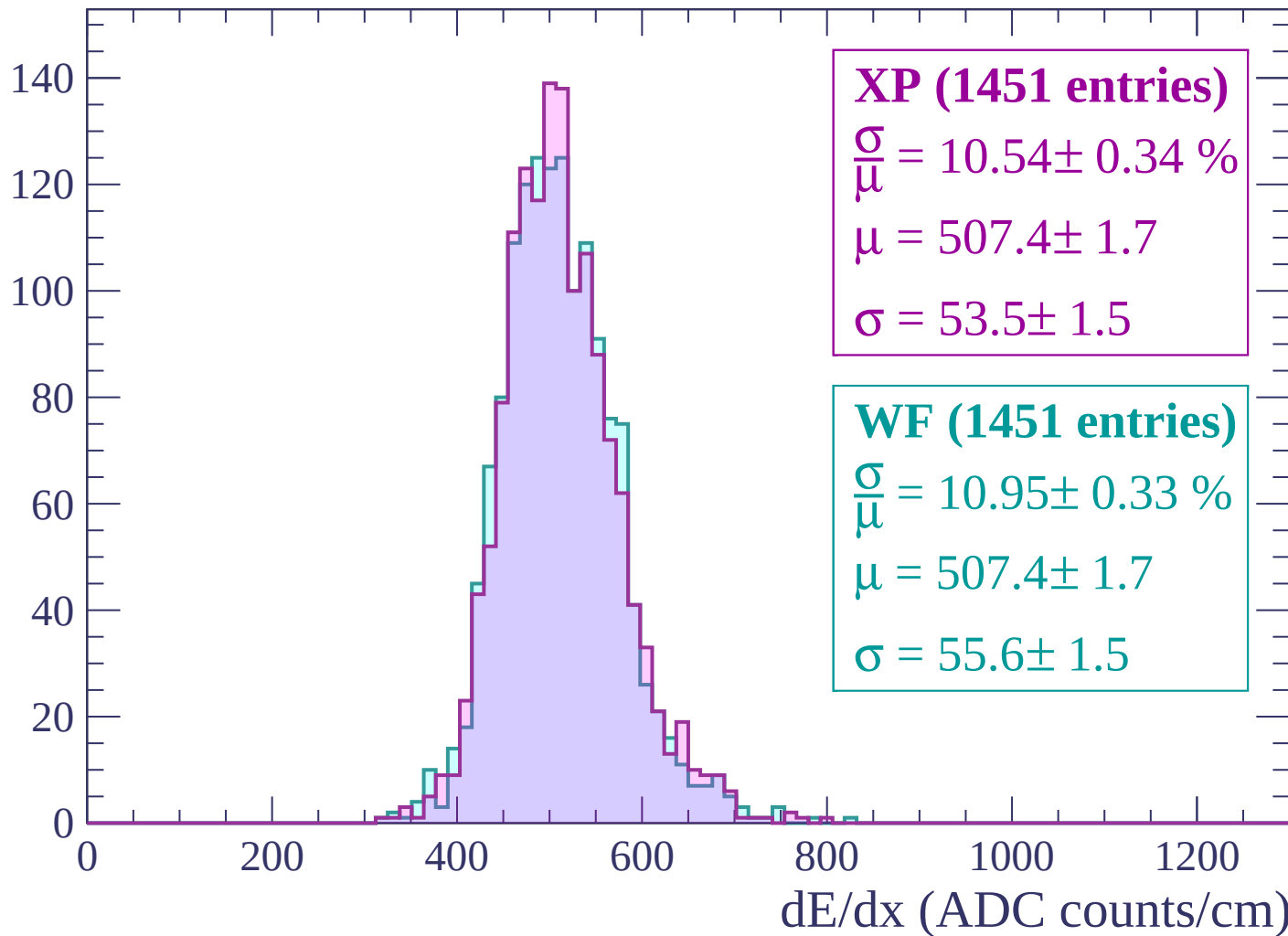
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$

$\mu = 6.2 \pm 2.4$

$\sigma = 0.3 \pm 1.3$

WF (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

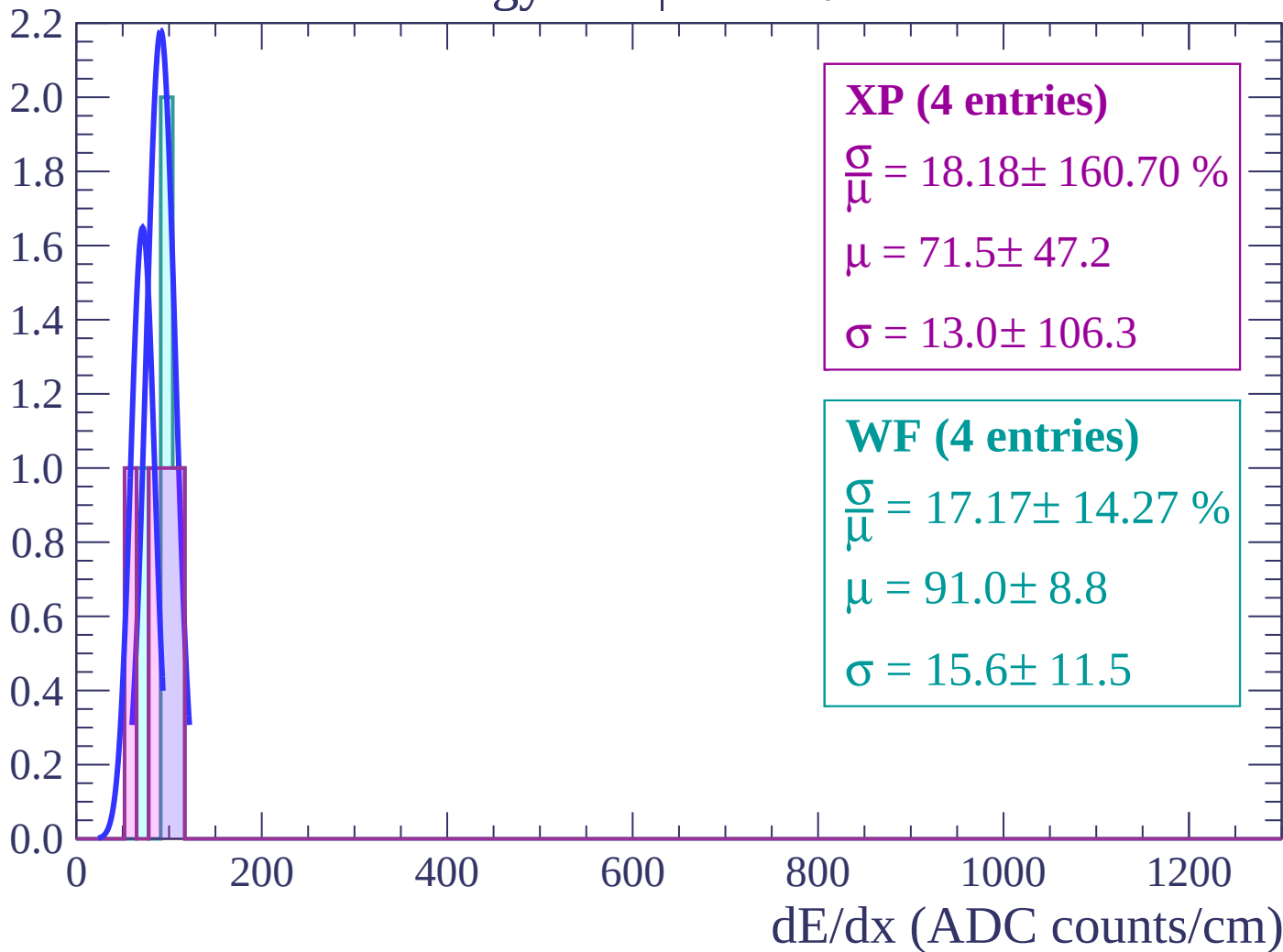
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

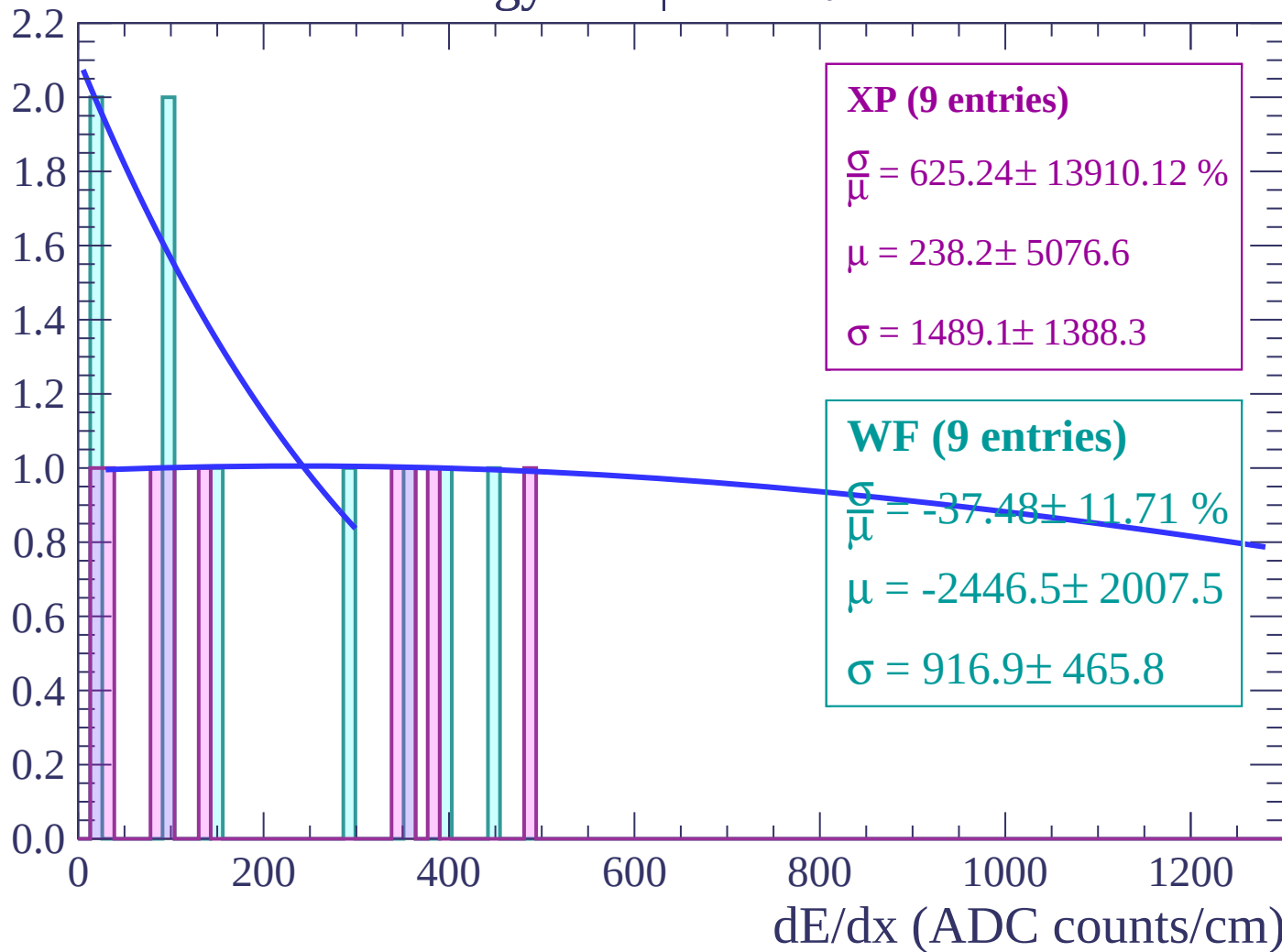
Energy loss | $-78 < \theta < -76$

Count



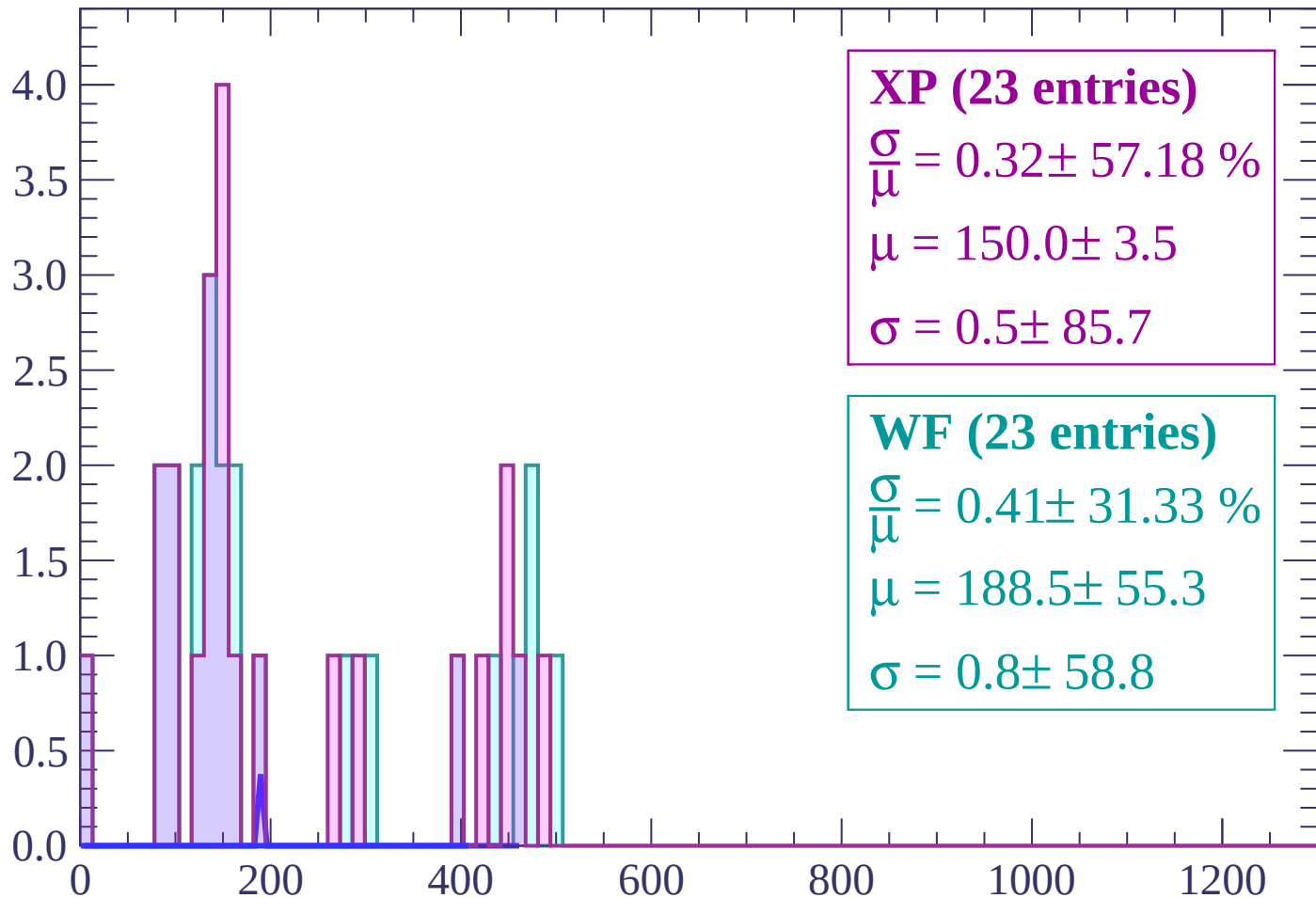
Energy loss | $-76 < \theta < -74$

Count



Energy loss | $-74 < \theta < -72$

Count



XP (23 entries)

$$\frac{\sigma}{\mu} = 0.32 \pm 57.18 \%$$

$$\mu = 150.0 \pm 3.5$$

$$\sigma = 0.5 \pm 85.7$$

WF (23 entries)

$$\frac{\sigma}{\mu} = 0.41 \pm 31.33 \%$$

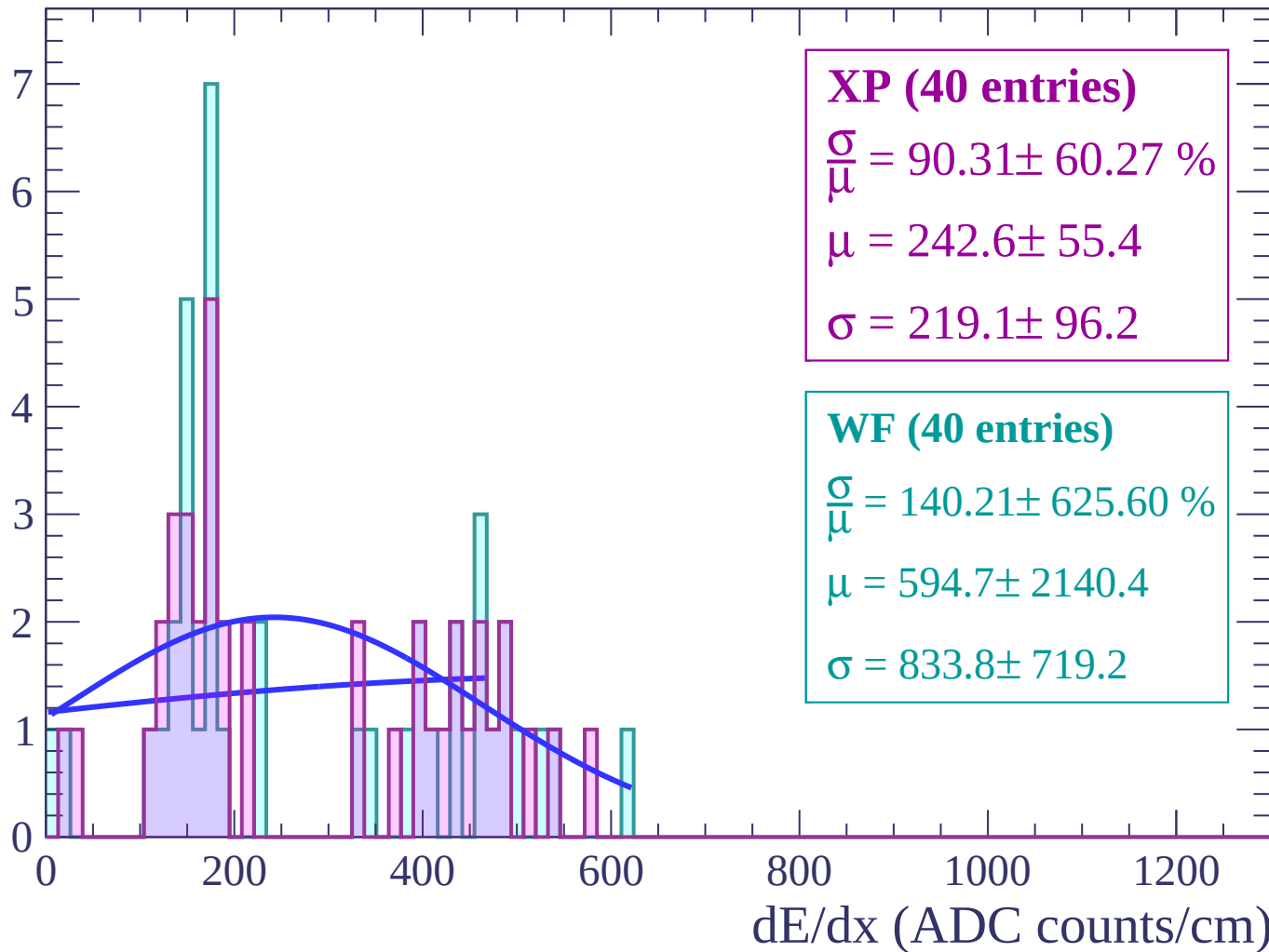
$$\mu = 188.5 \pm 55.3$$

$$\sigma = 0.8 \pm 58.8$$

dE/dx (ADC counts/cm)

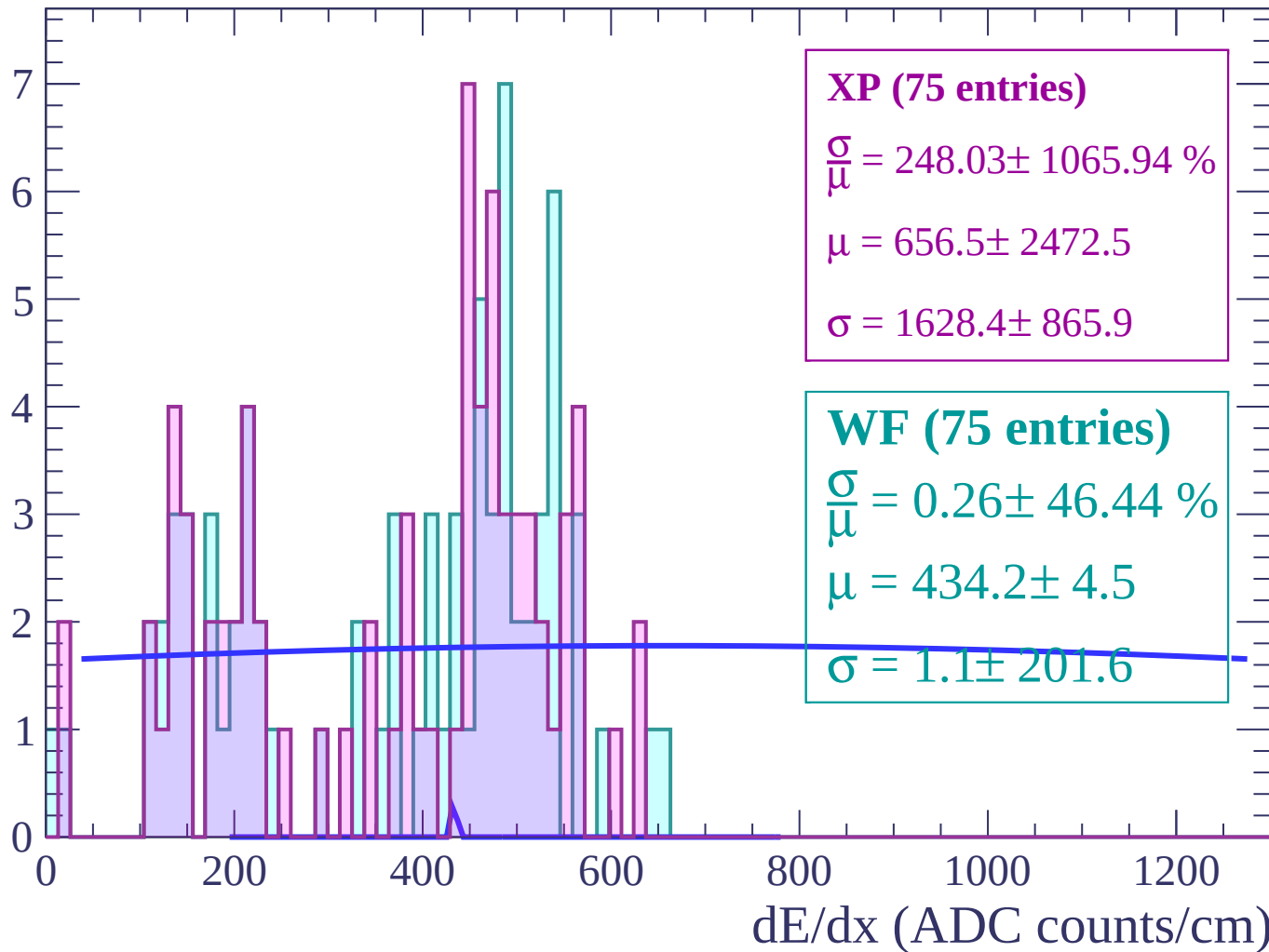
Energy loss | $-72 < \theta < -70$

Count



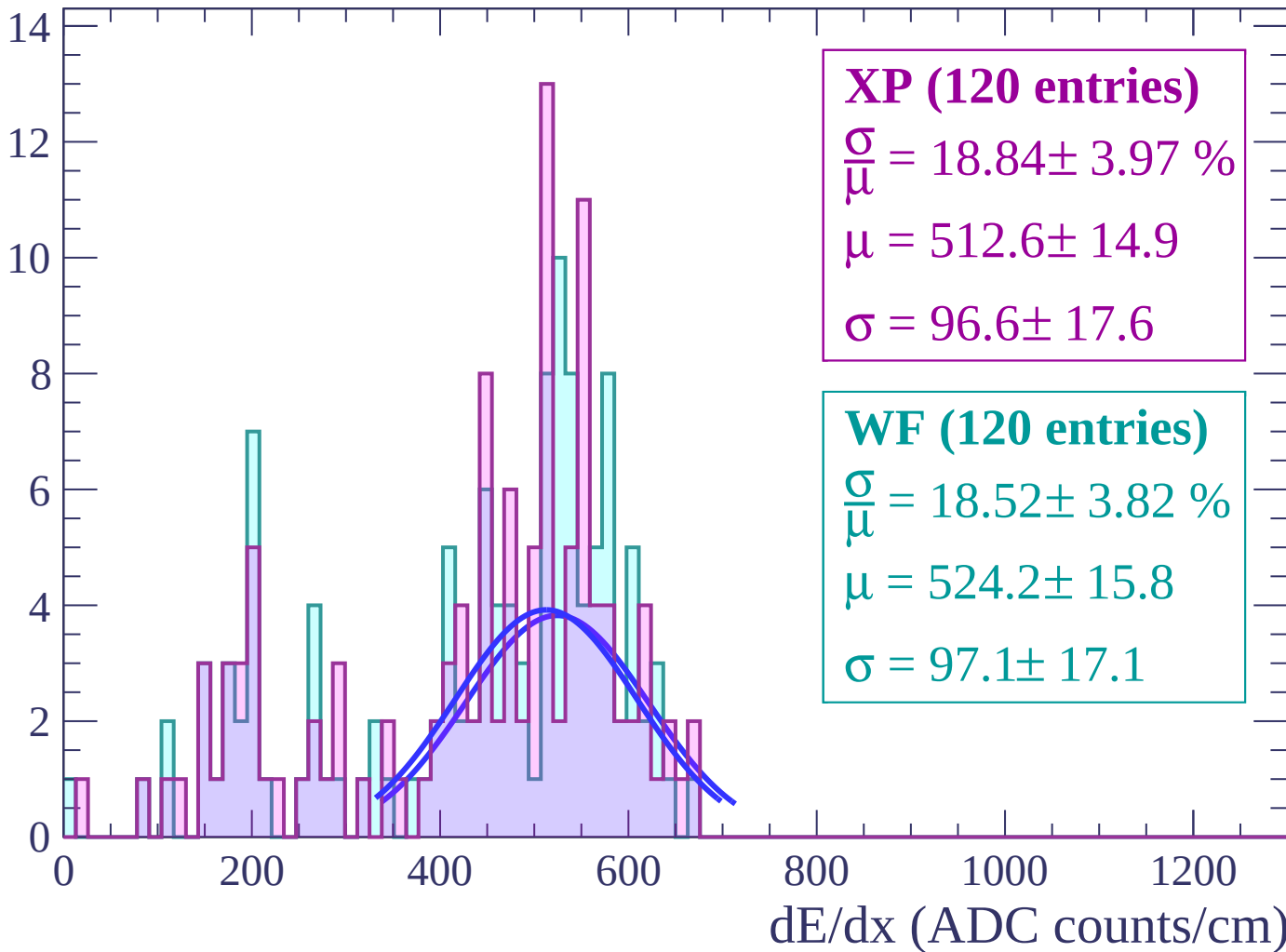
Energy loss | $-70 < \theta < -68$

Count



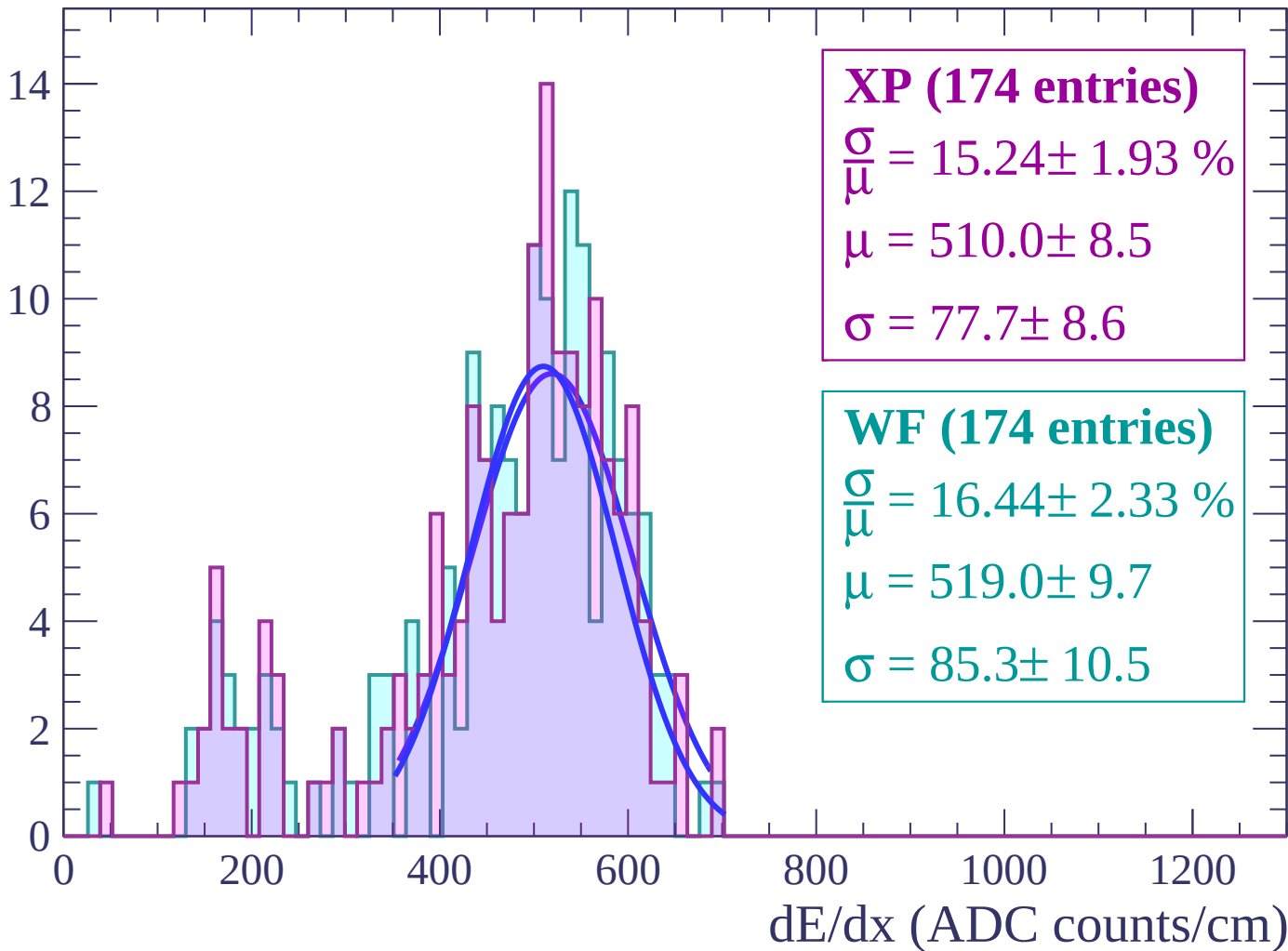
Energy loss | $-68 < \theta < -66$

Count



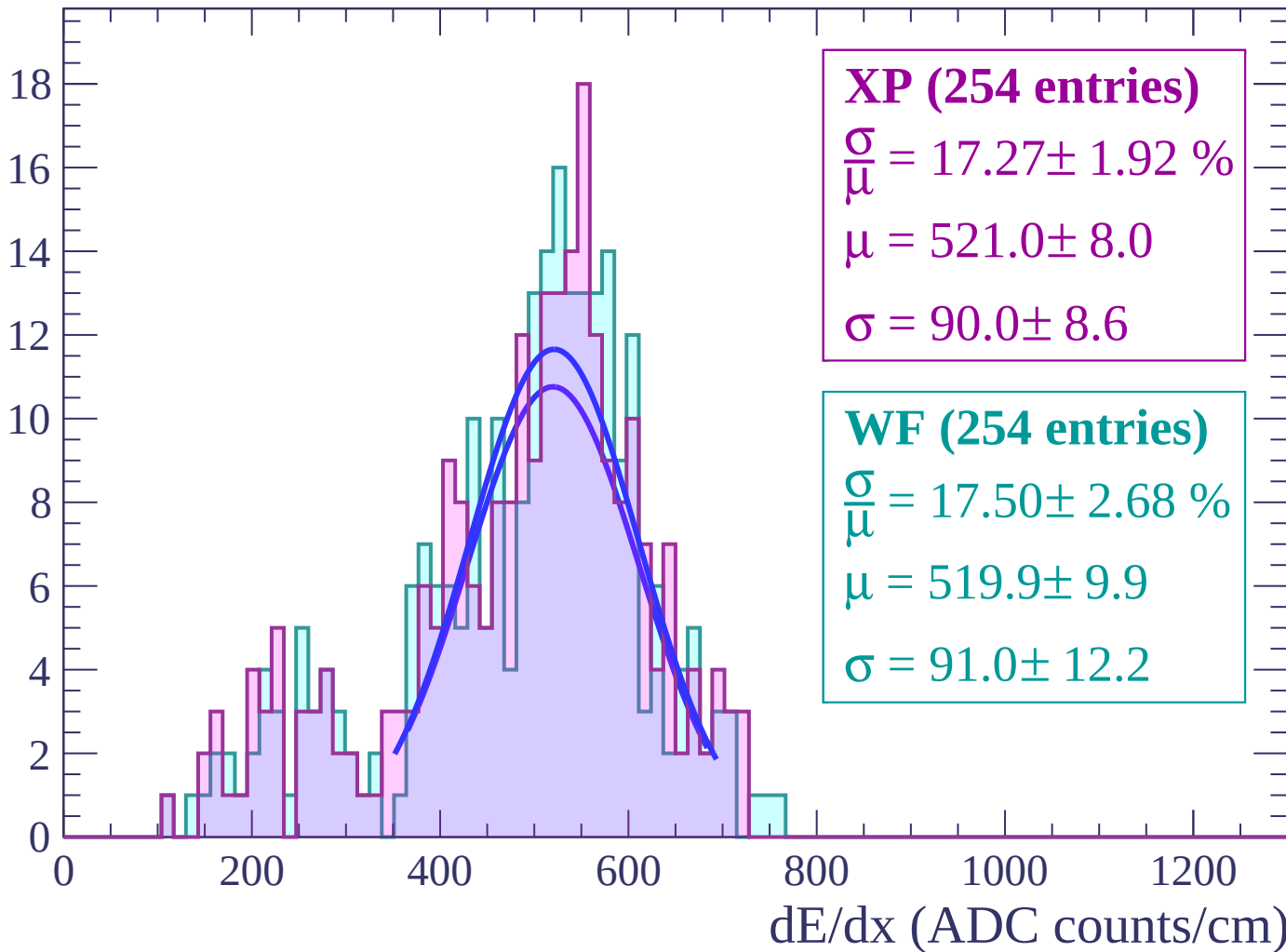
Energy loss | $-66 < \theta < -64$

Count



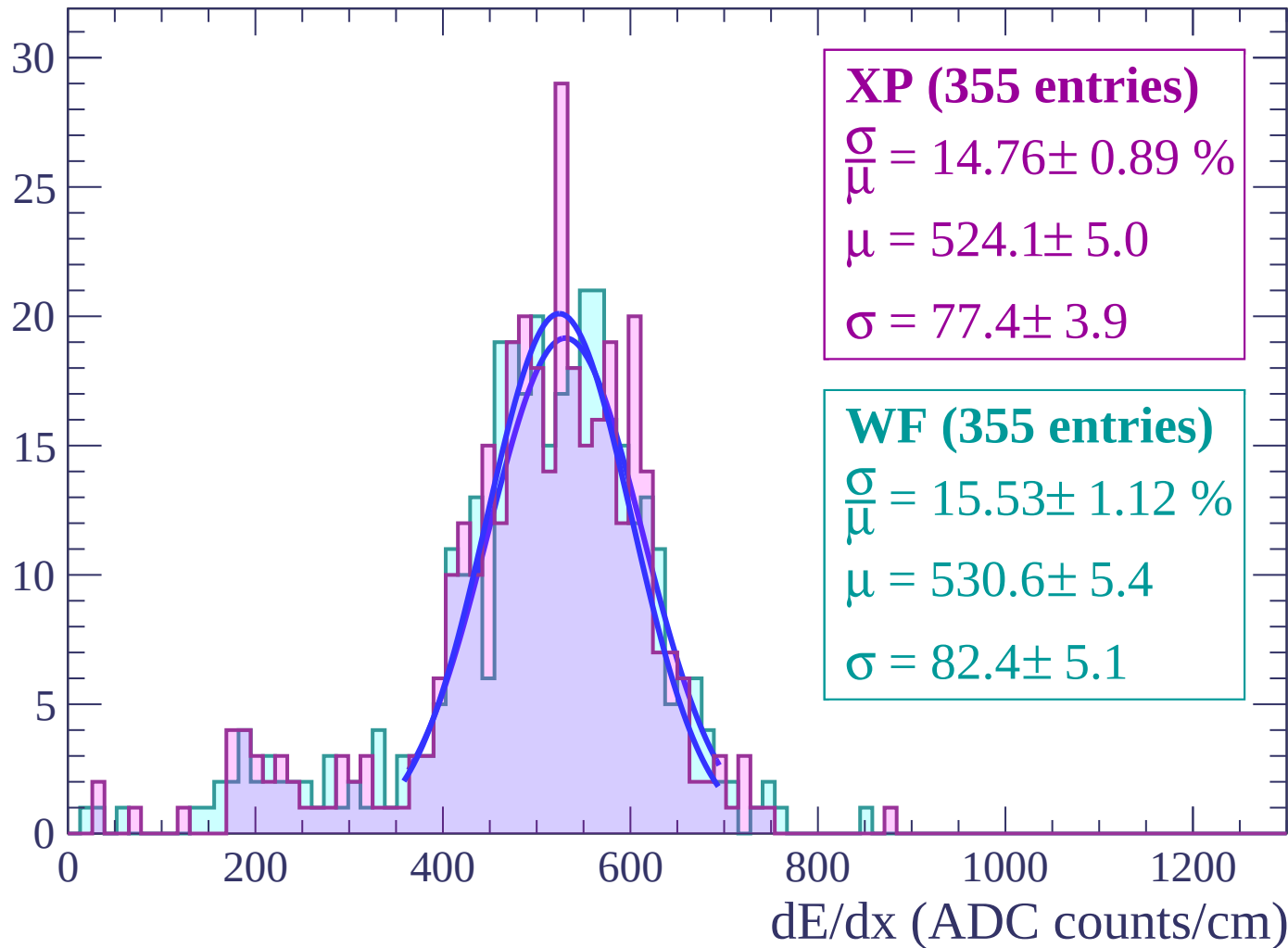
Energy loss | $-64 < \theta < -62$

Count



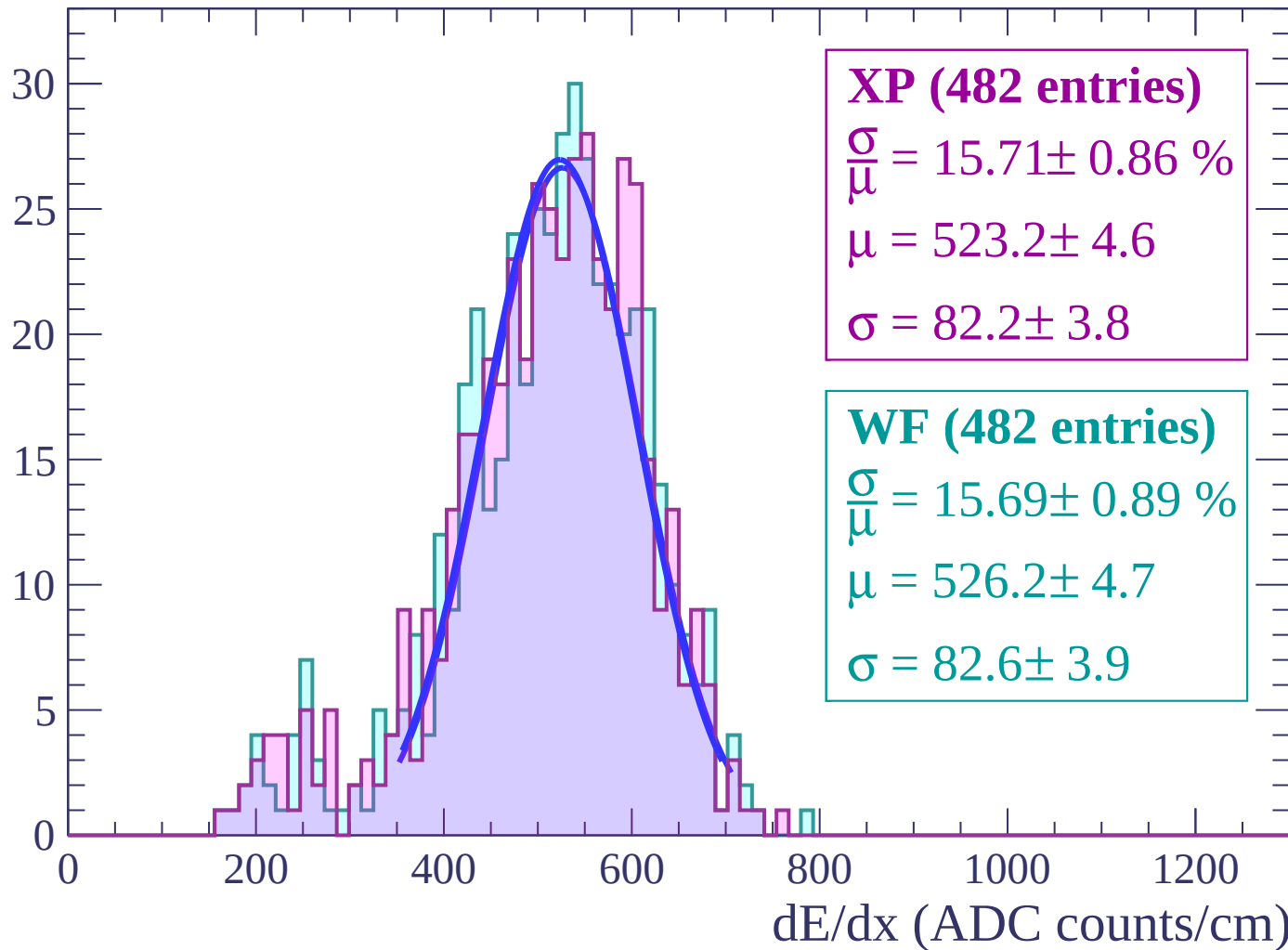
Energy loss | $-62 < \theta < -60$

Count



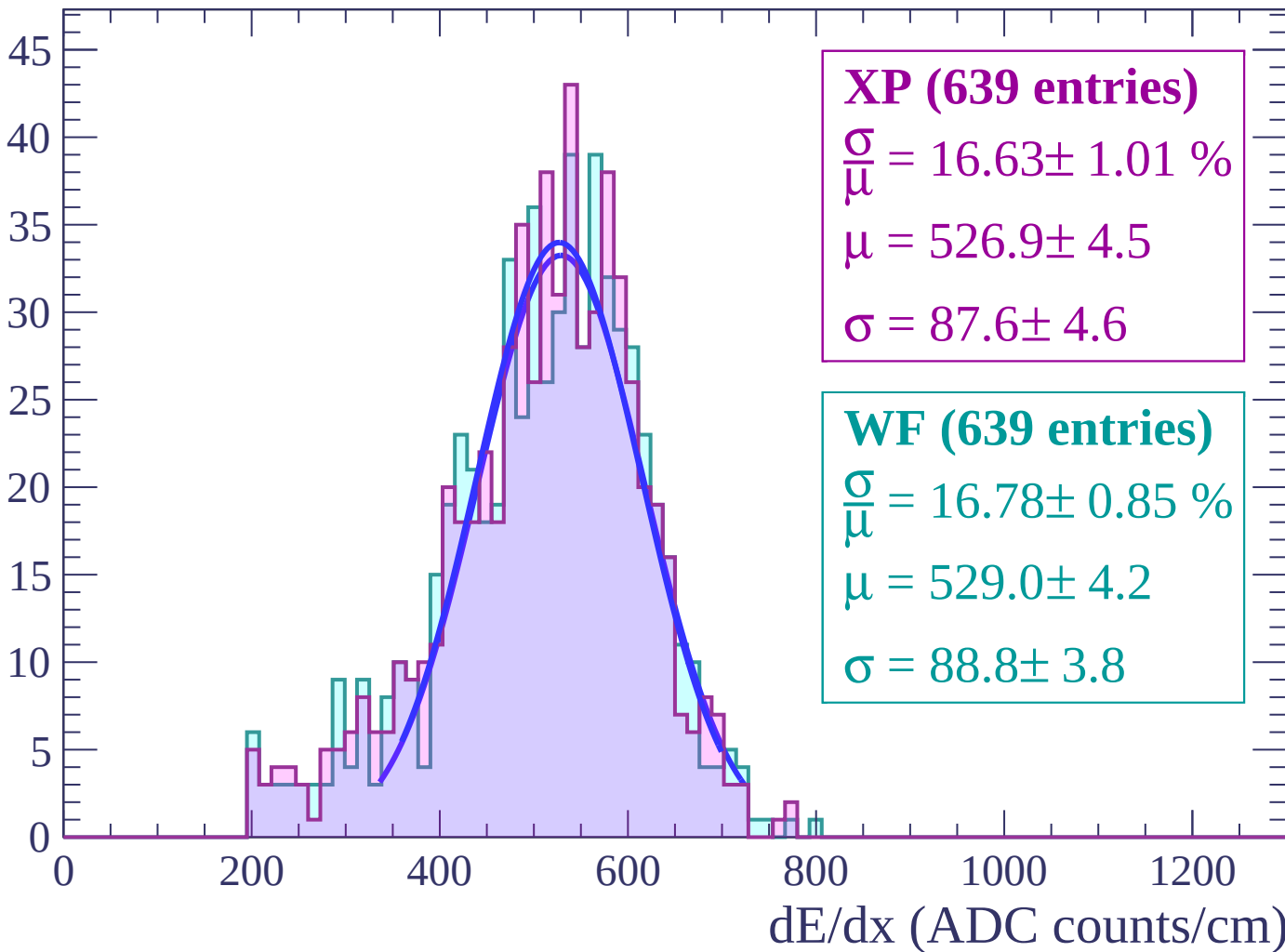
Energy loss | $-60 < \theta < -58$

Count



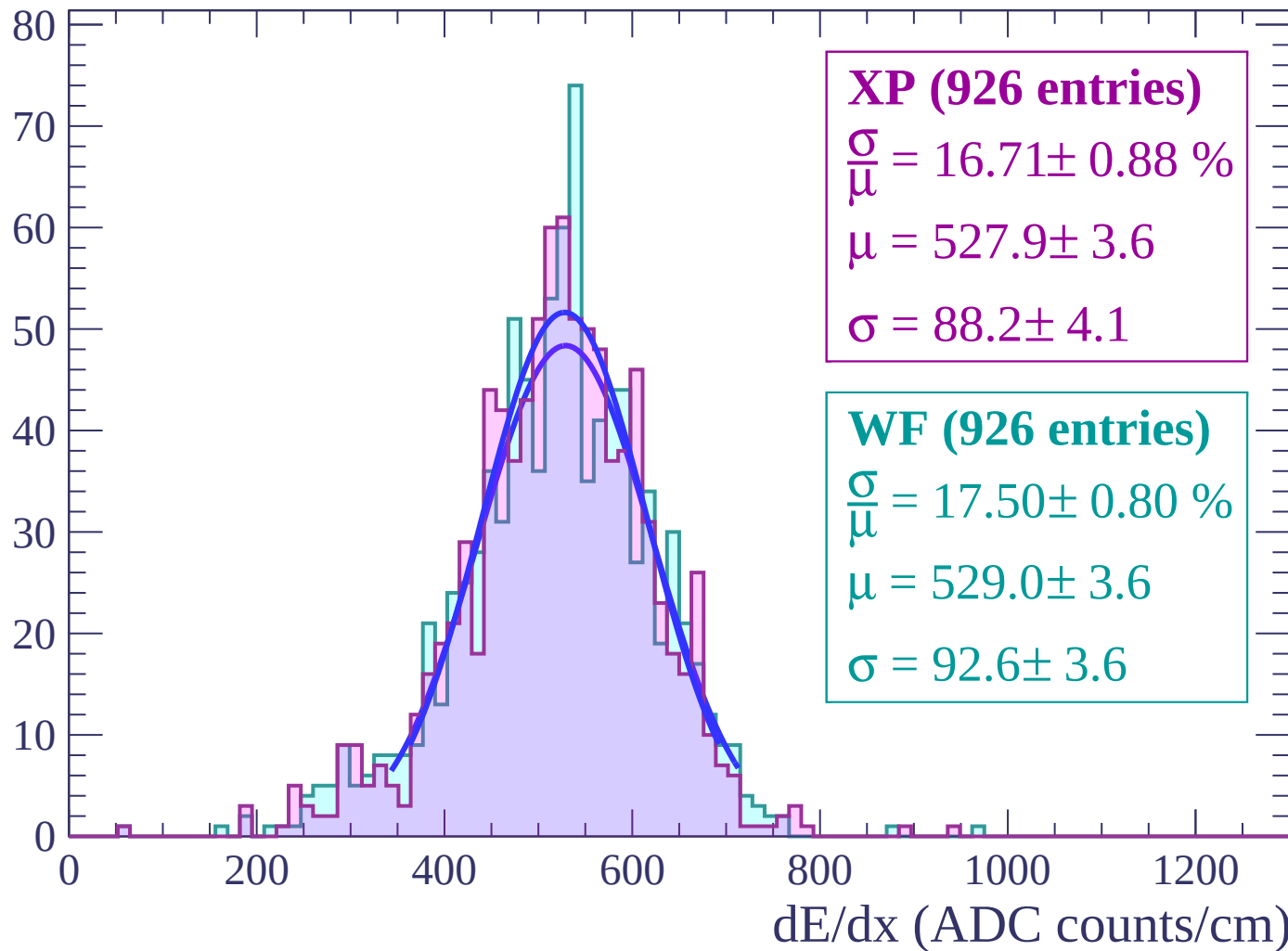
Energy loss | $-58 < \theta < -56$

Count



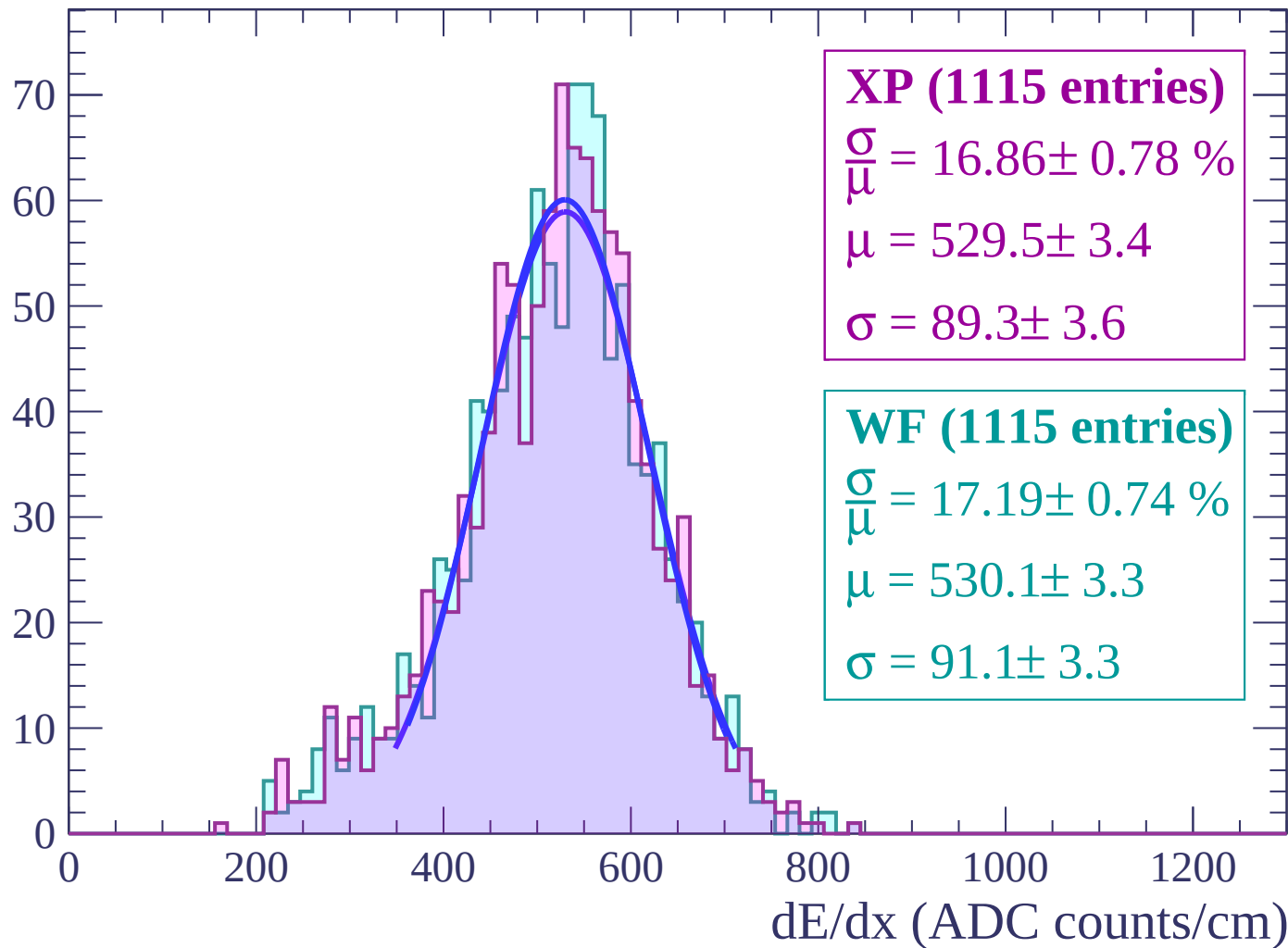
Energy loss | $-56 < \theta < -54$

Count



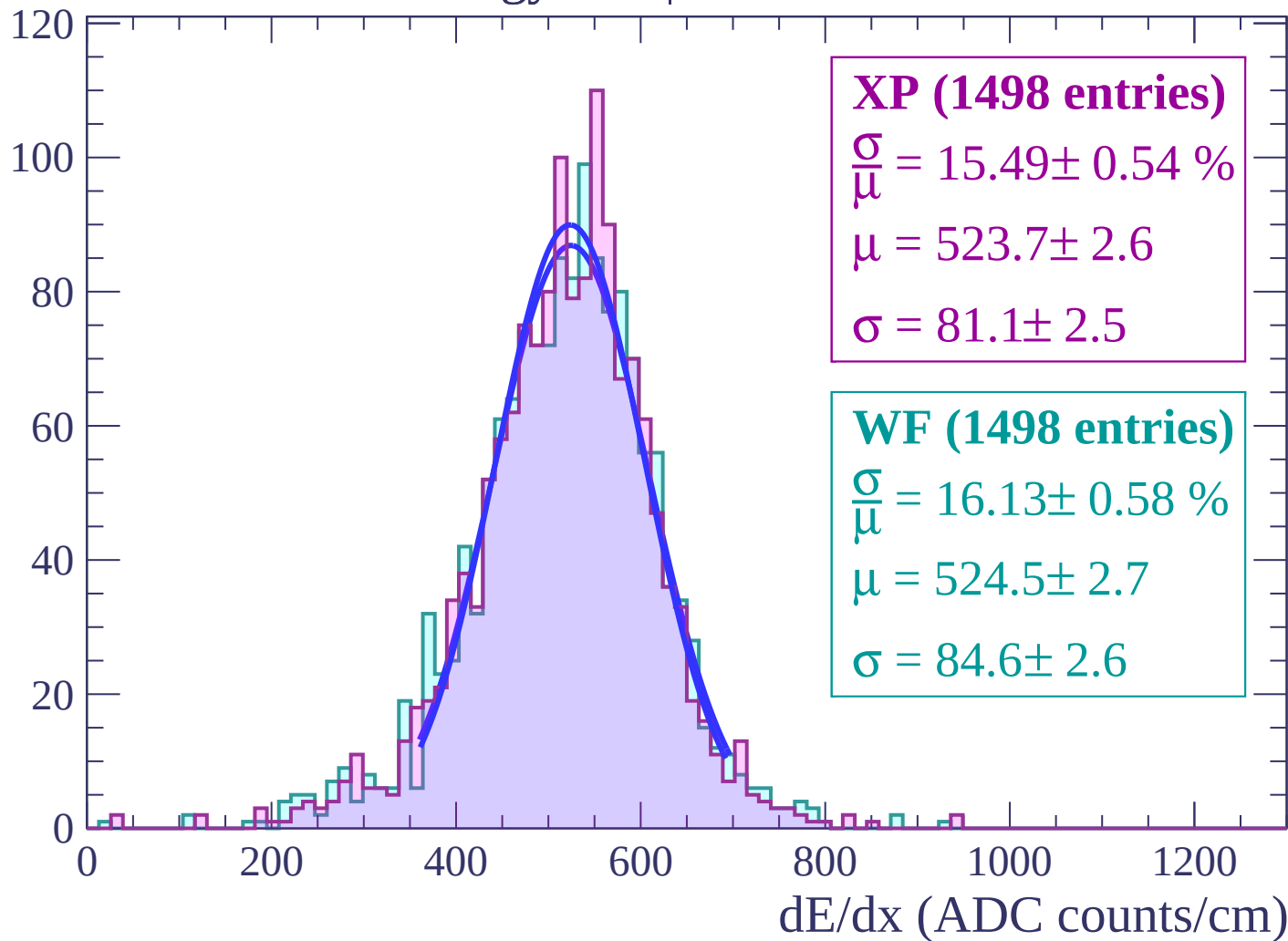
Energy loss | $-54 < \theta < -52$

Count



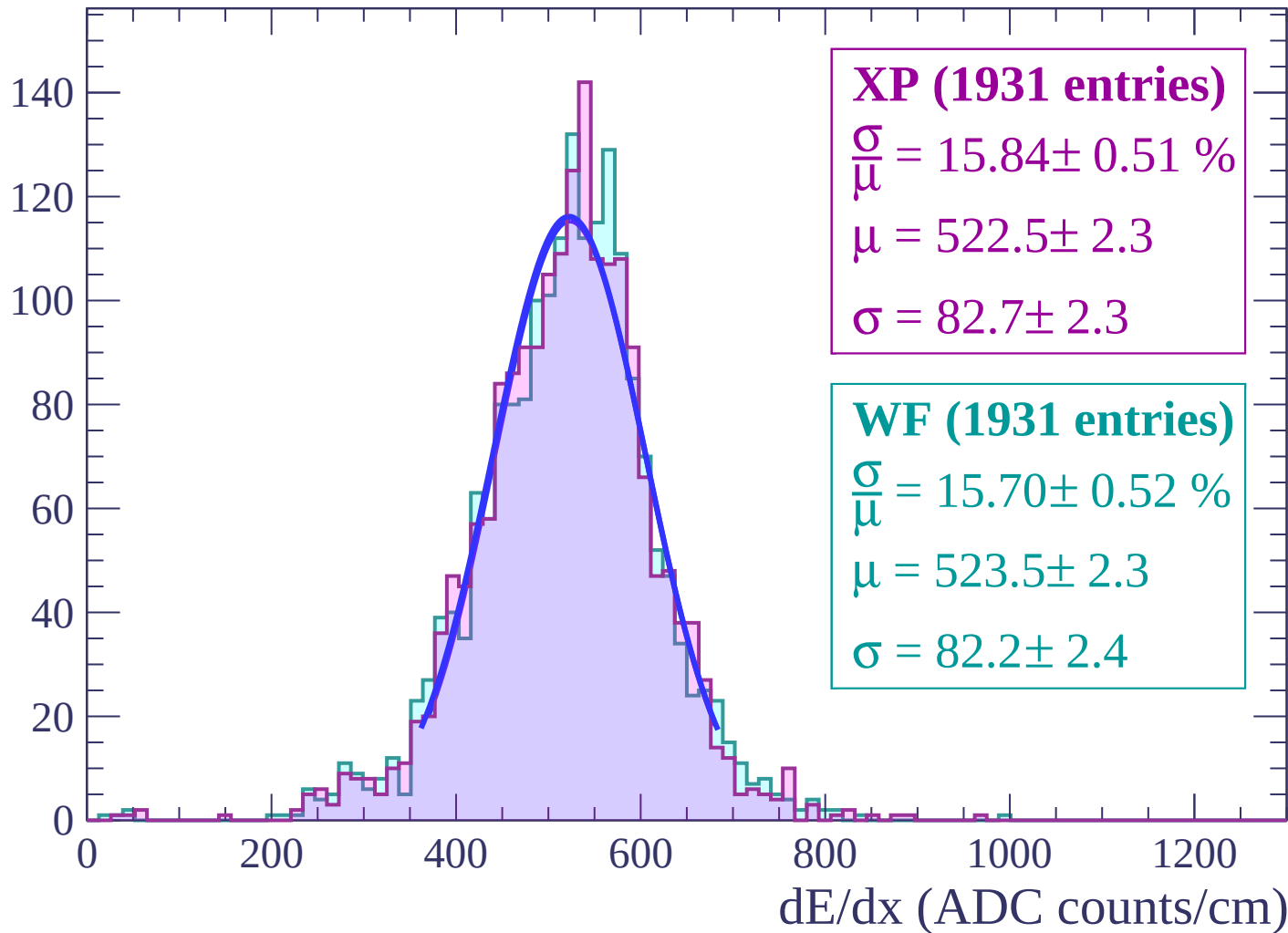
Energy loss | $-52 < \theta < -50$

Count



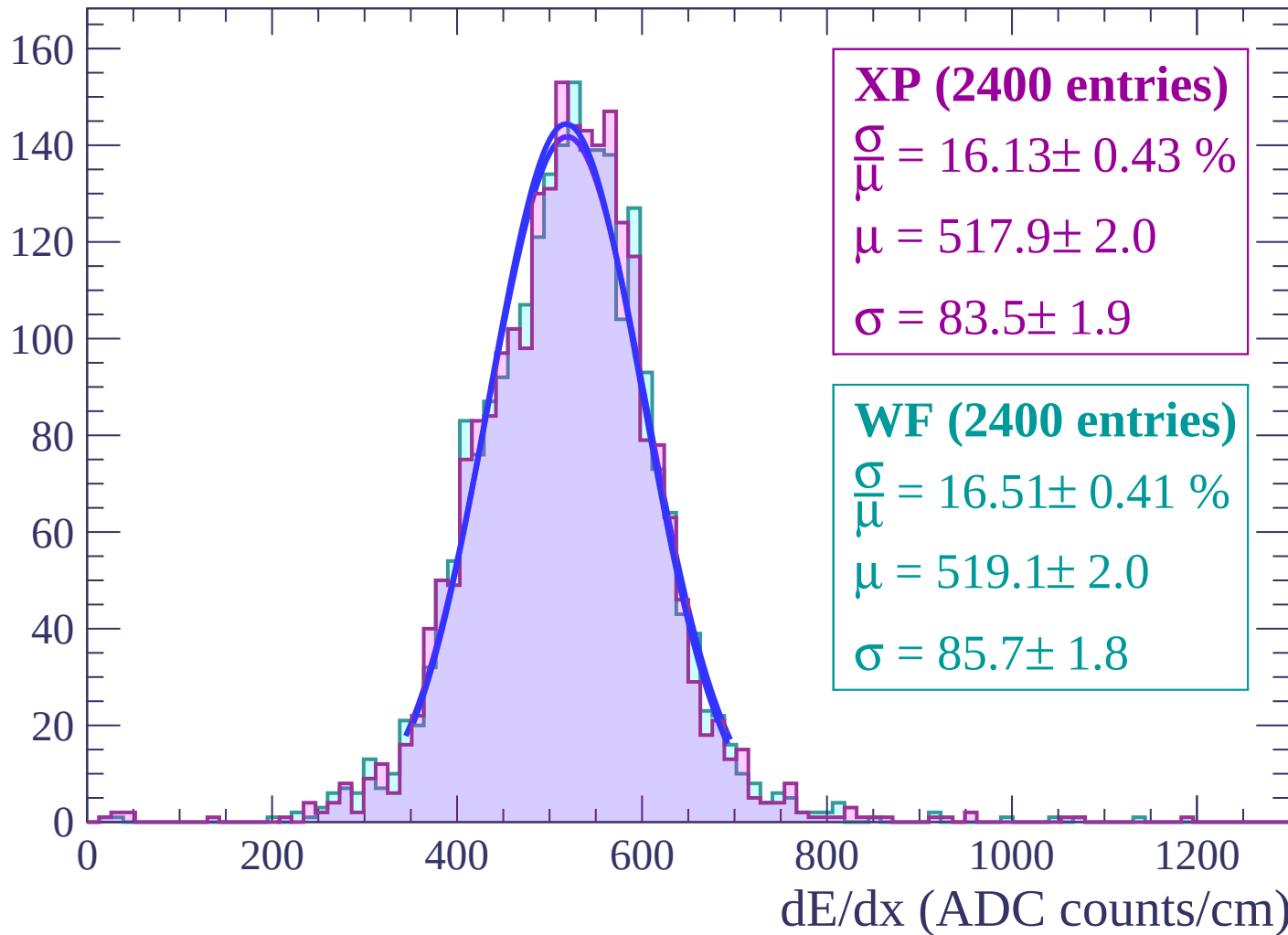
Energy loss | $-50 < \theta < -48$

Count

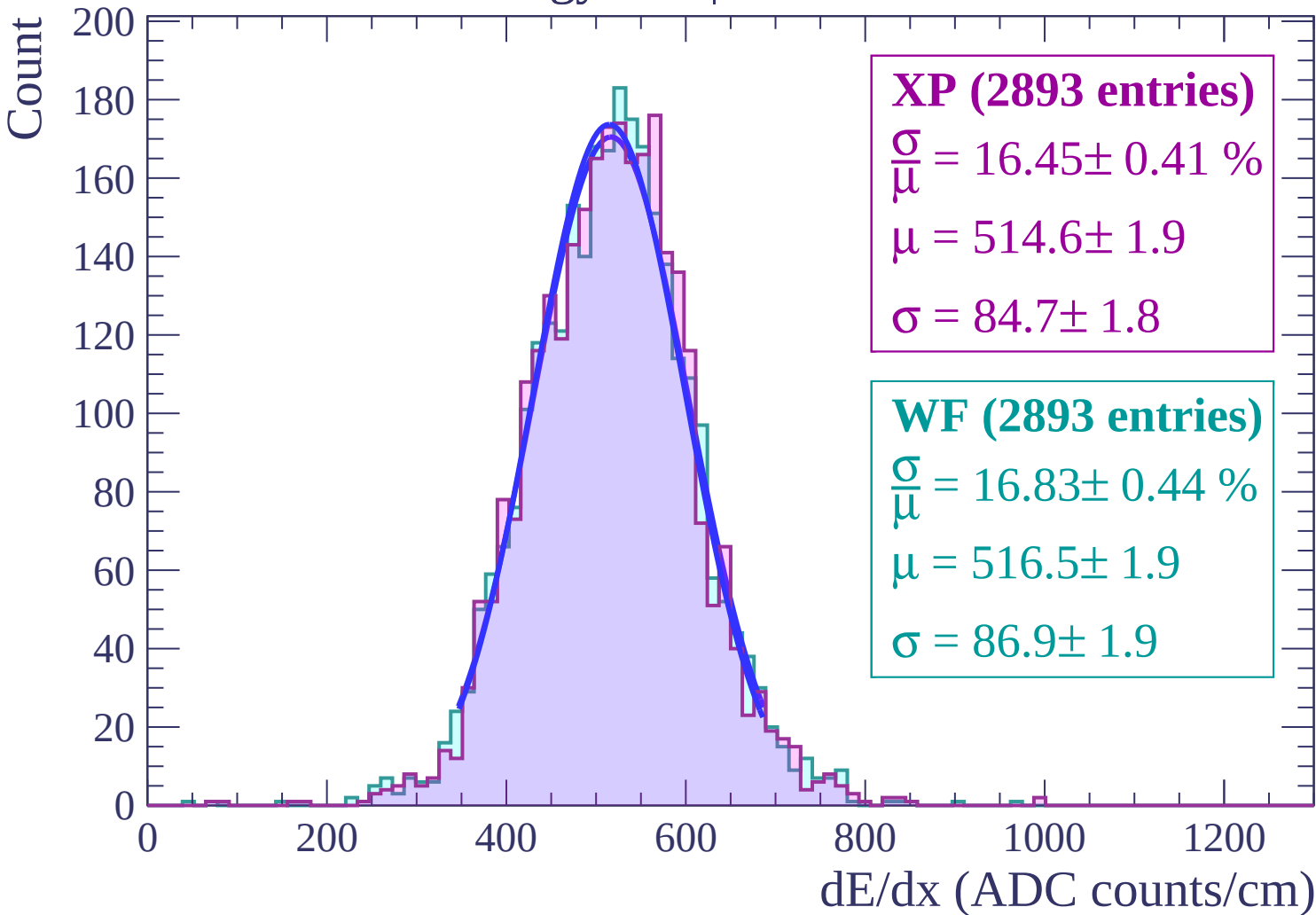


Energy loss | $-48 < \theta < -46$

Count

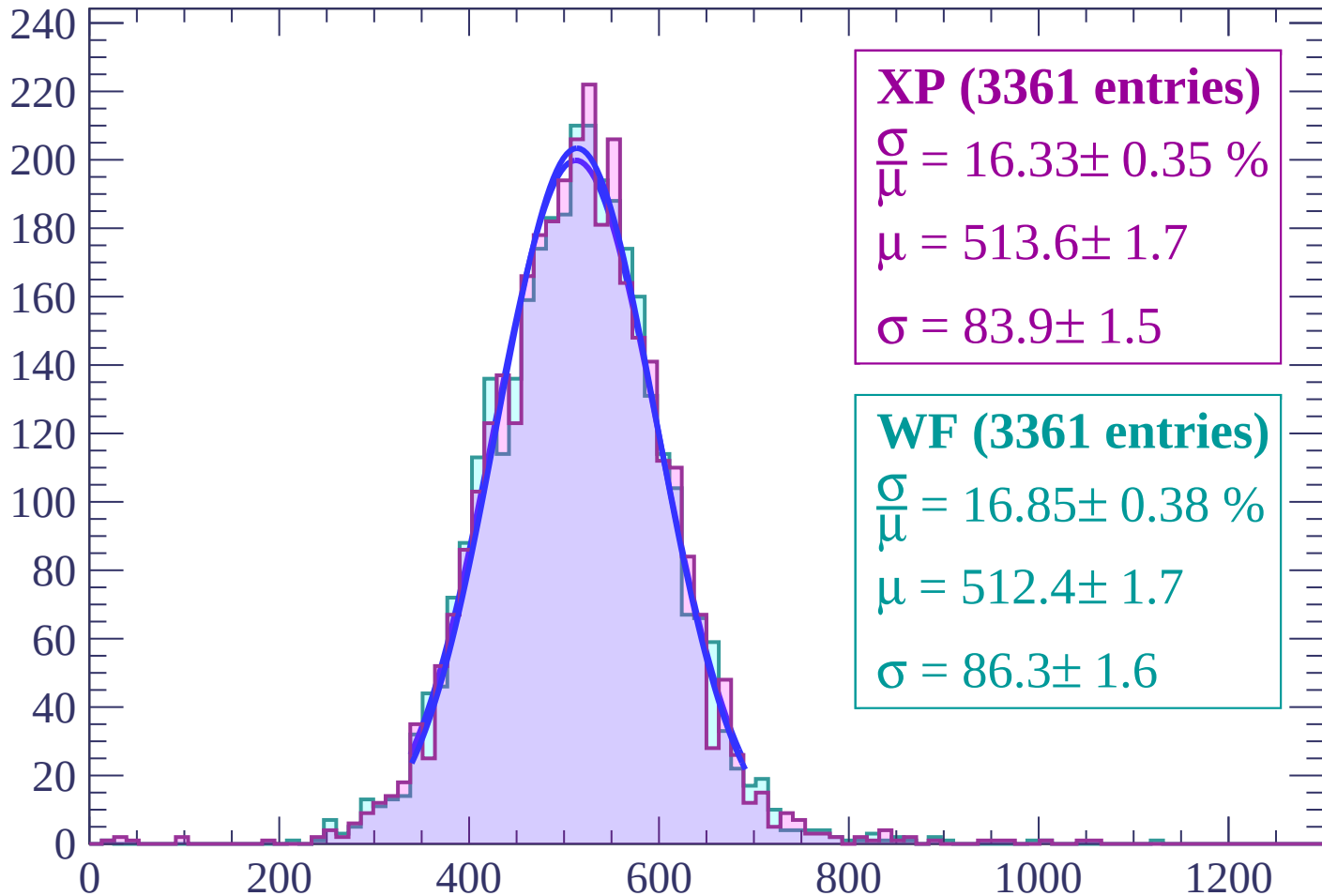


Energy loss | $-46 < \theta < -44$



Energy loss | $-44 < \theta < -42$

Count



XP (3361 entries)

$\frac{\sigma}{\mu} = 16.33 \pm 0.35 \%$

$\mu = 513.6 \pm 1.7$

$\sigma = 83.9 \pm 1.5$

WF (3361 entries)

$\frac{\sigma}{\mu} = 16.85 \pm 0.38 \%$

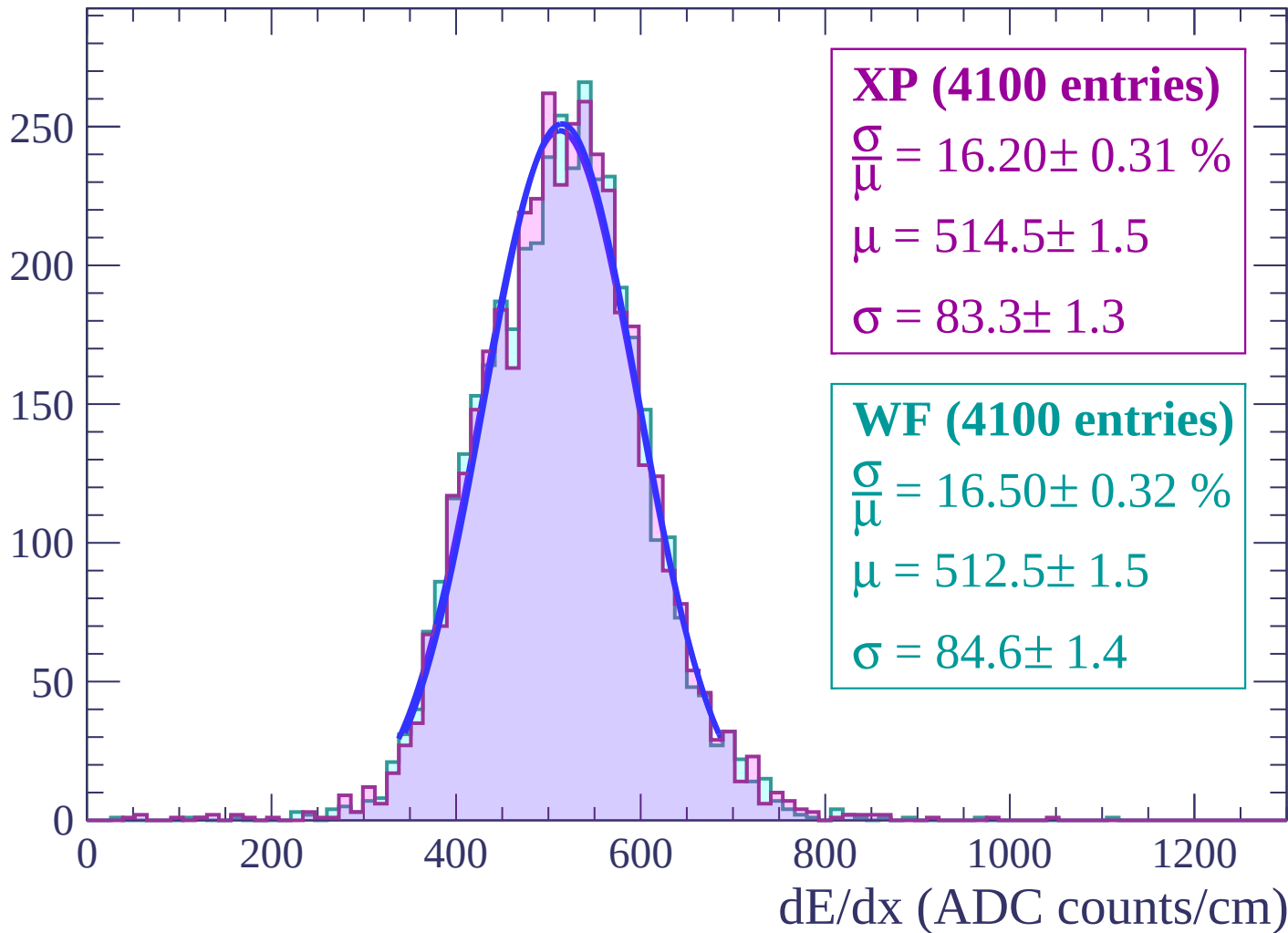
$\mu = 512.4 \pm 1.7$

$\sigma = 86.3 \pm 1.6$

dE/dx (ADC counts/cm)

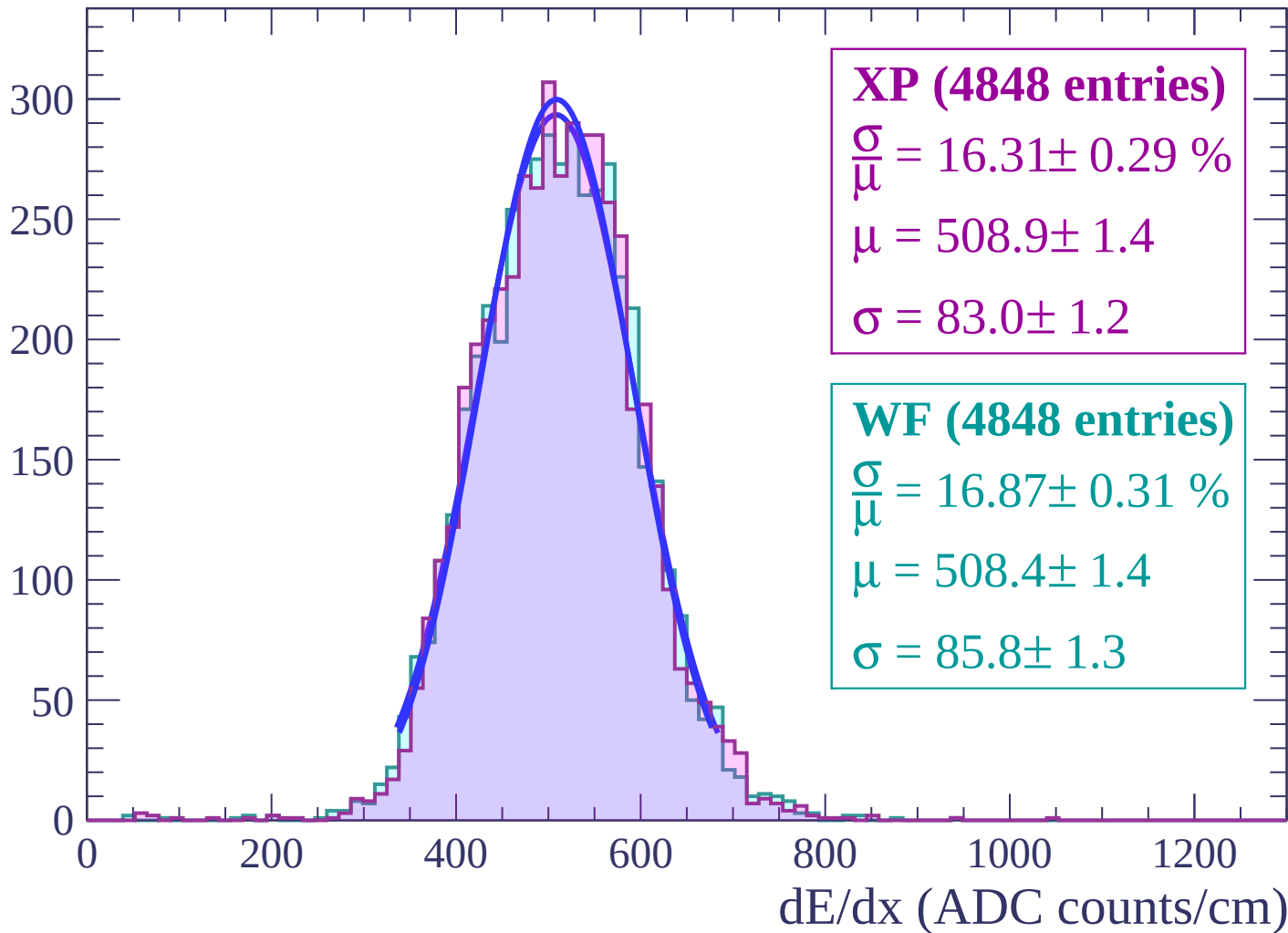
Energy loss | $-42 < \theta < -40$

Count



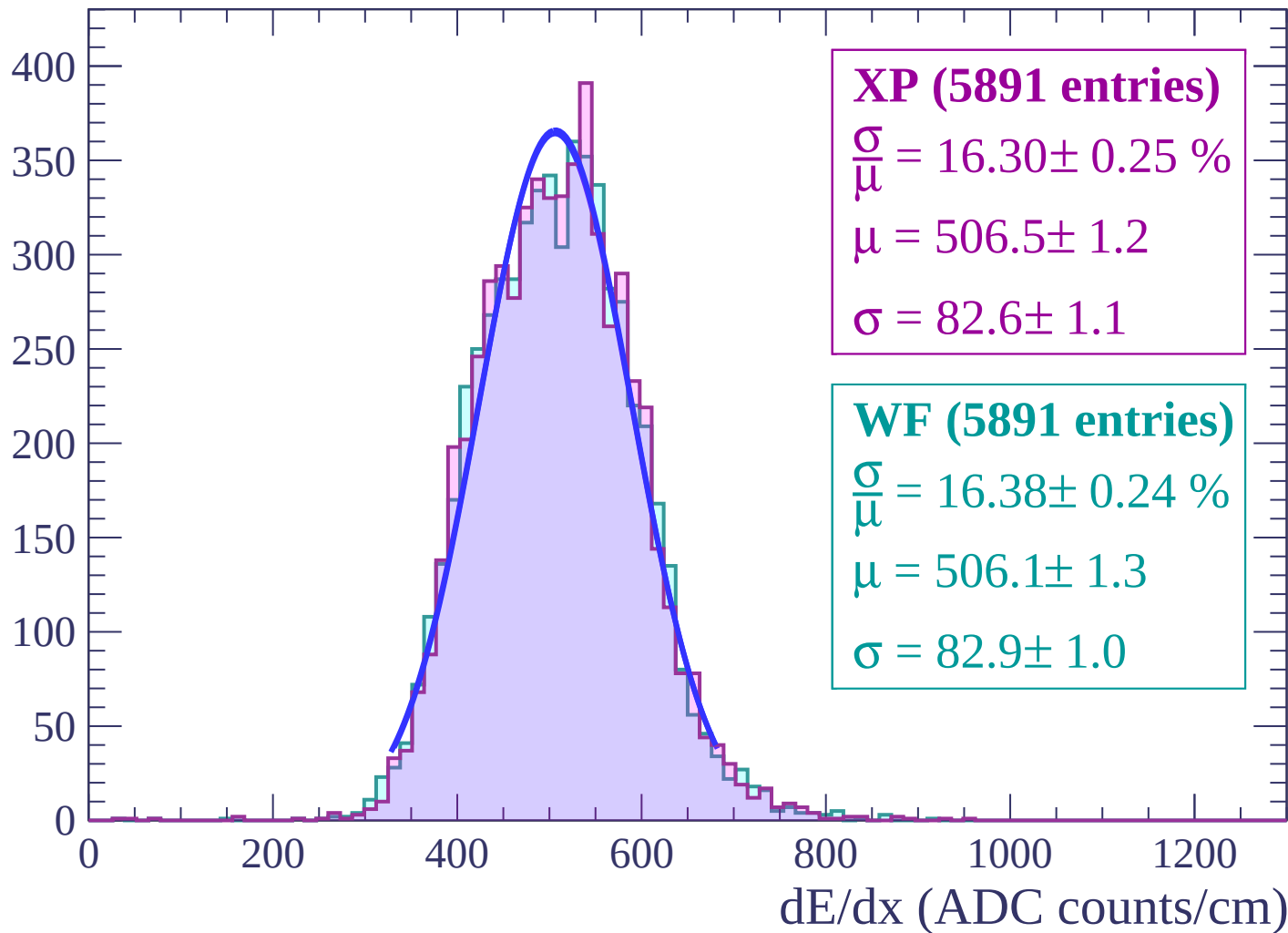
Energy loss | $-40 < \theta < -38$

Count



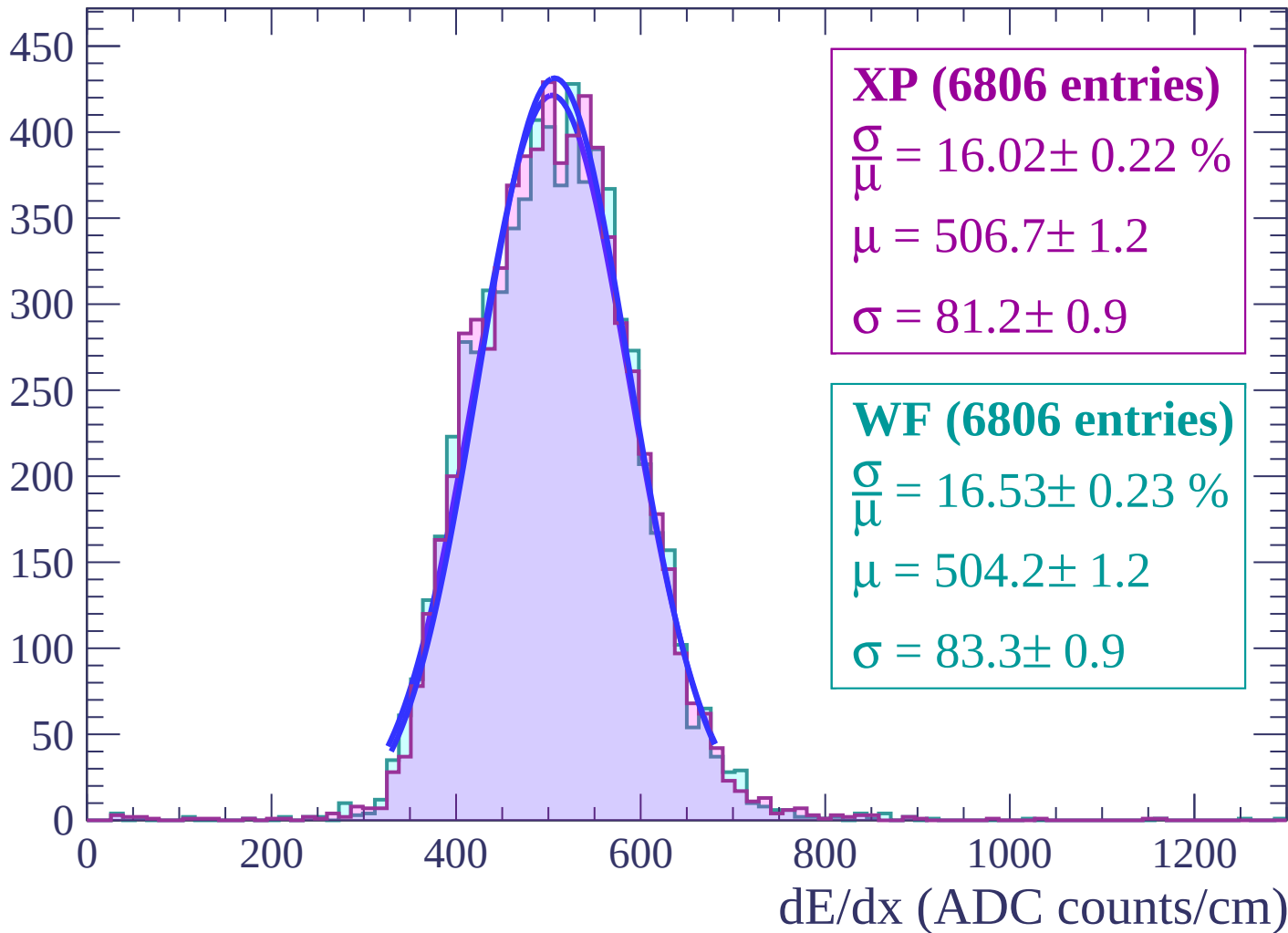
Energy loss | $-38 < \theta < -36$

Count



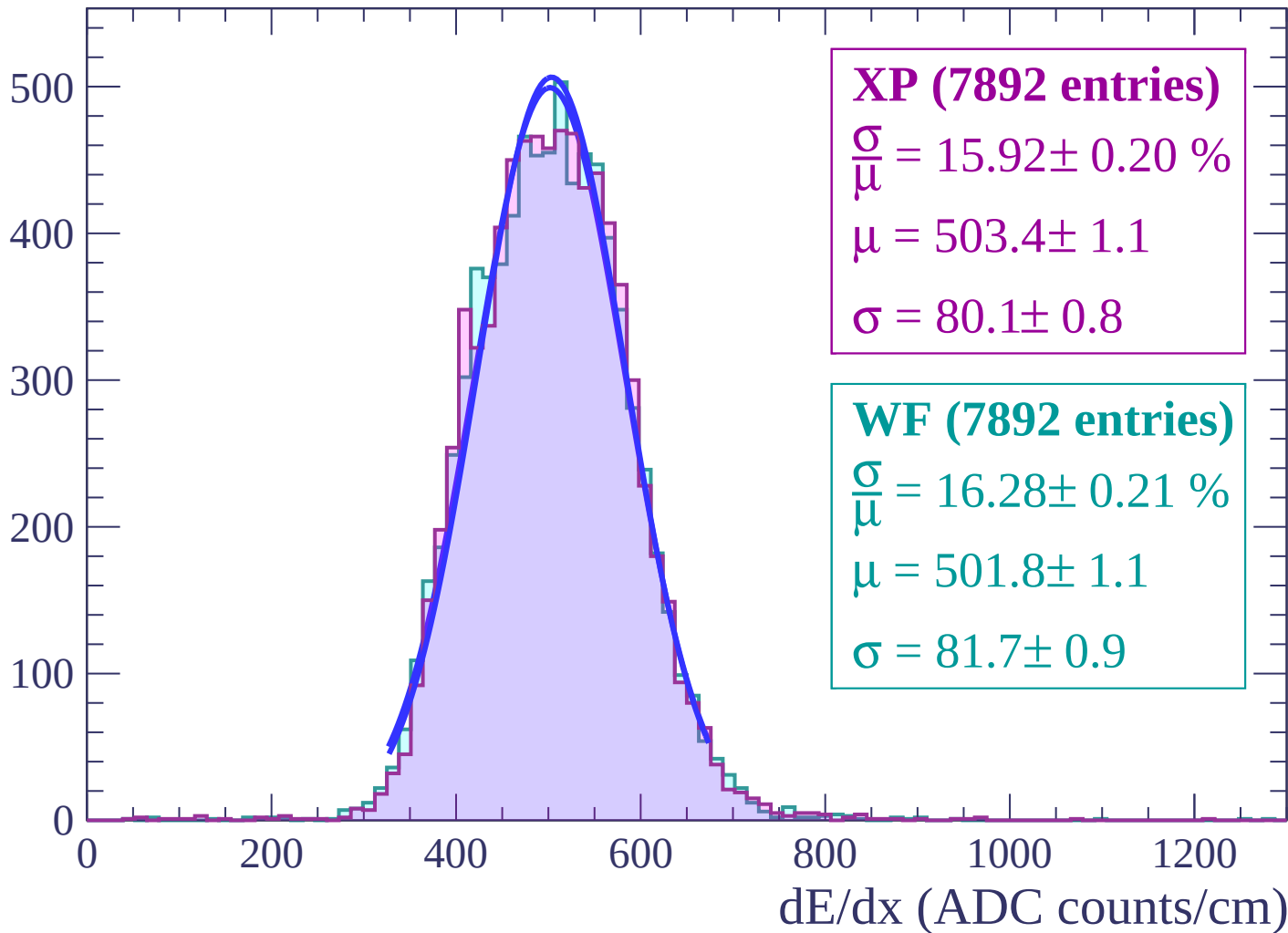
Energy loss | $-36 < \theta < -34$

Count



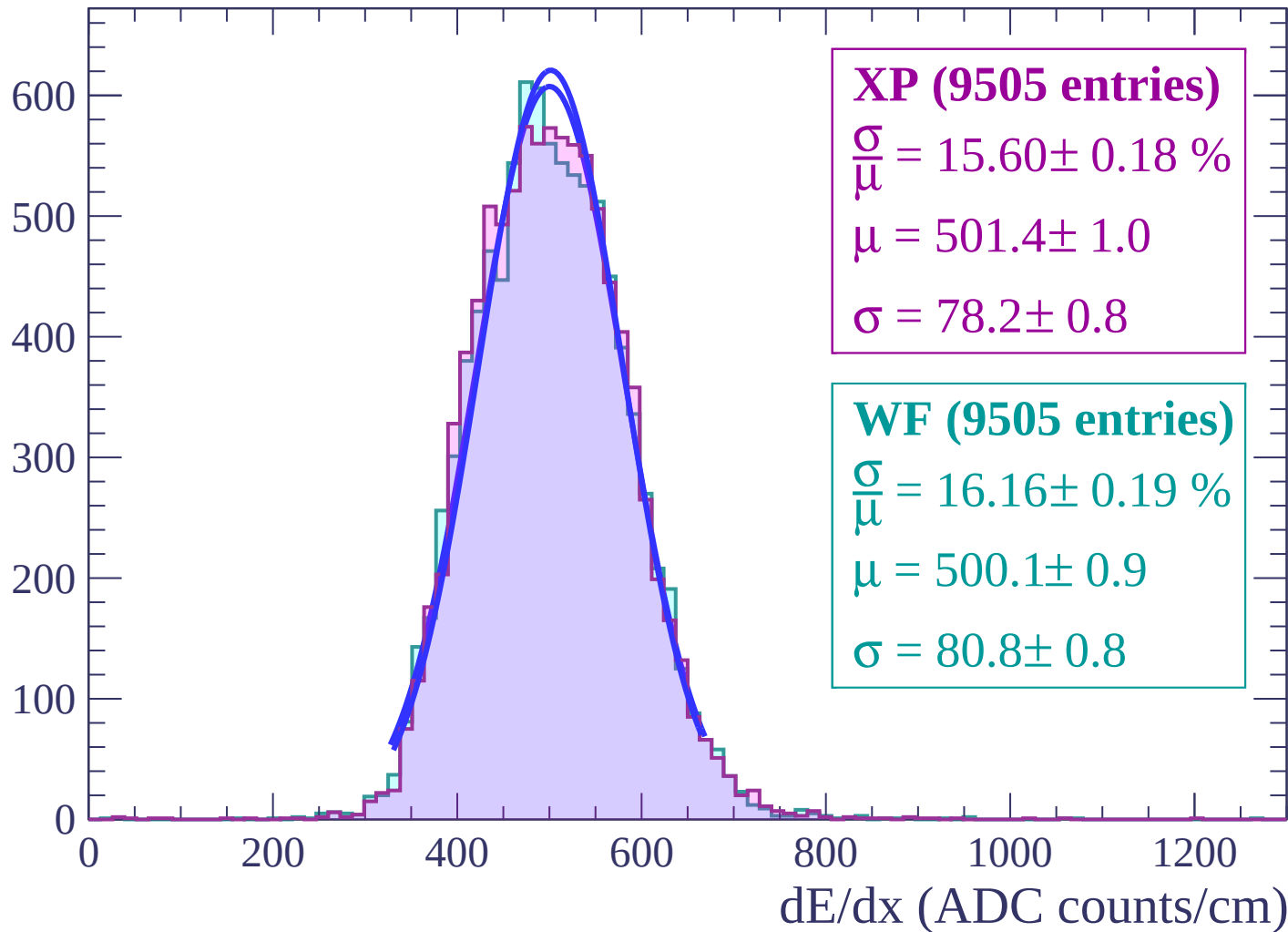
Energy loss | $-34 < \theta < -32$

Count



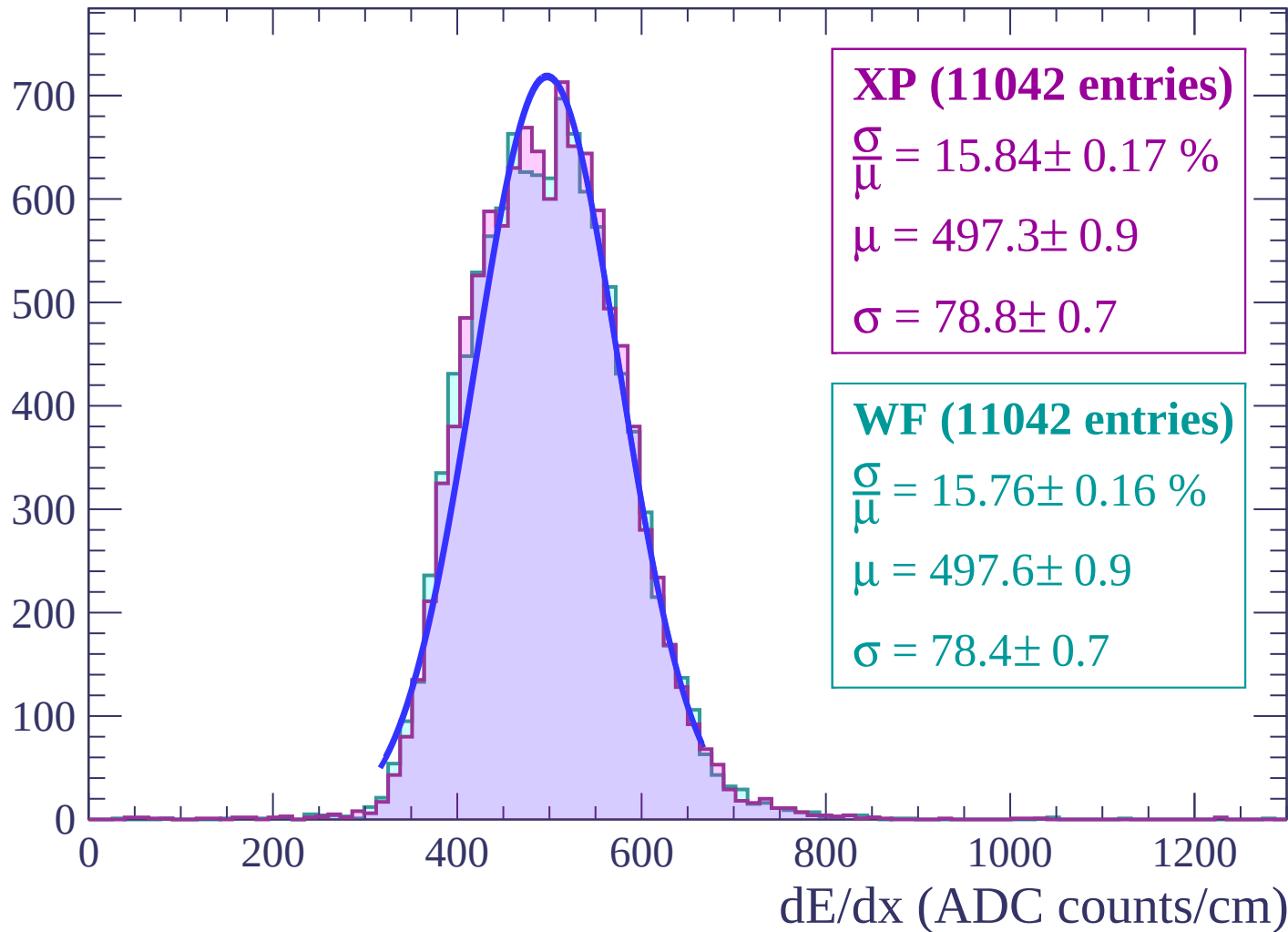
Energy loss | $-32 < \theta < -30$

Count



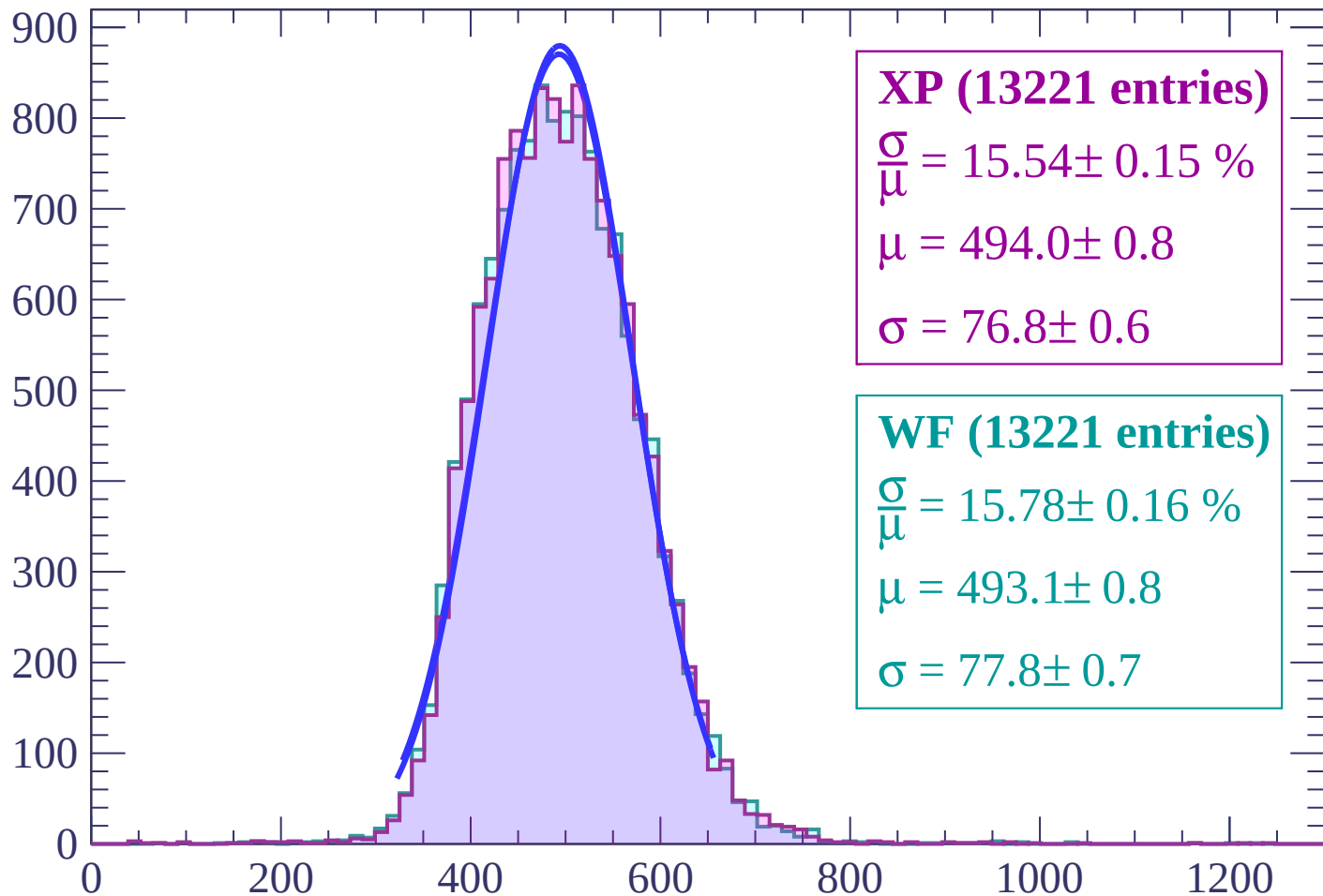
Energy loss | $-30 < \theta < -28$

Count



Energy loss | $-28 < \theta < -26$

Count



XP (13221 entries)

$$\frac{\sigma}{\mu} = 15.54 \pm 0.15 \%$$

$$\mu = 494.0 \pm 0.8$$

$$\sigma = 76.8 \pm 0.6$$

WF (13221 entries)

$$\frac{\sigma}{\mu} = 15.78 \pm 0.16 \%$$

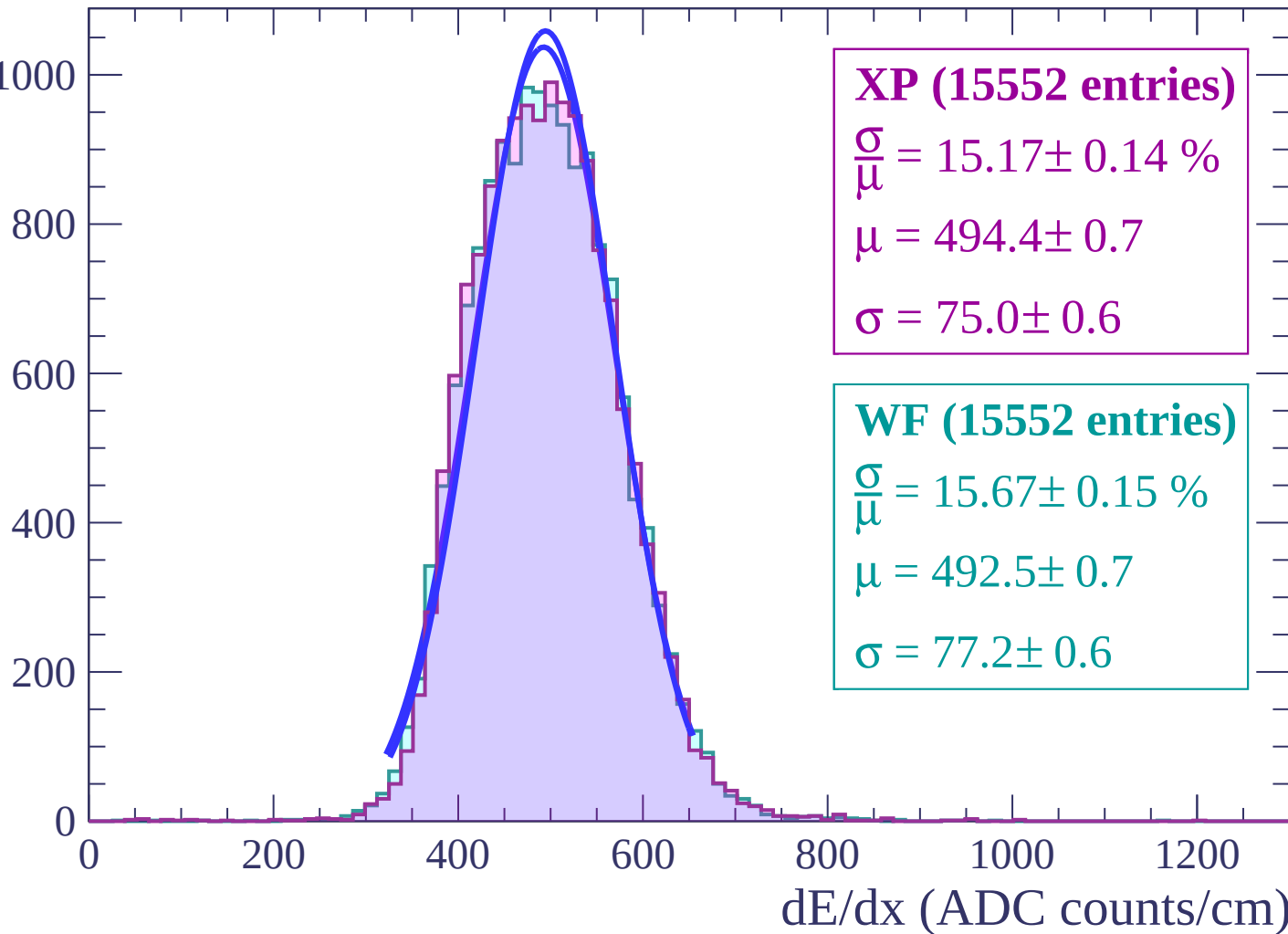
$$\mu = 493.1 \pm 0.8$$

$$\sigma = 77.8 \pm 0.7$$

dE/dx (ADC counts/cm)

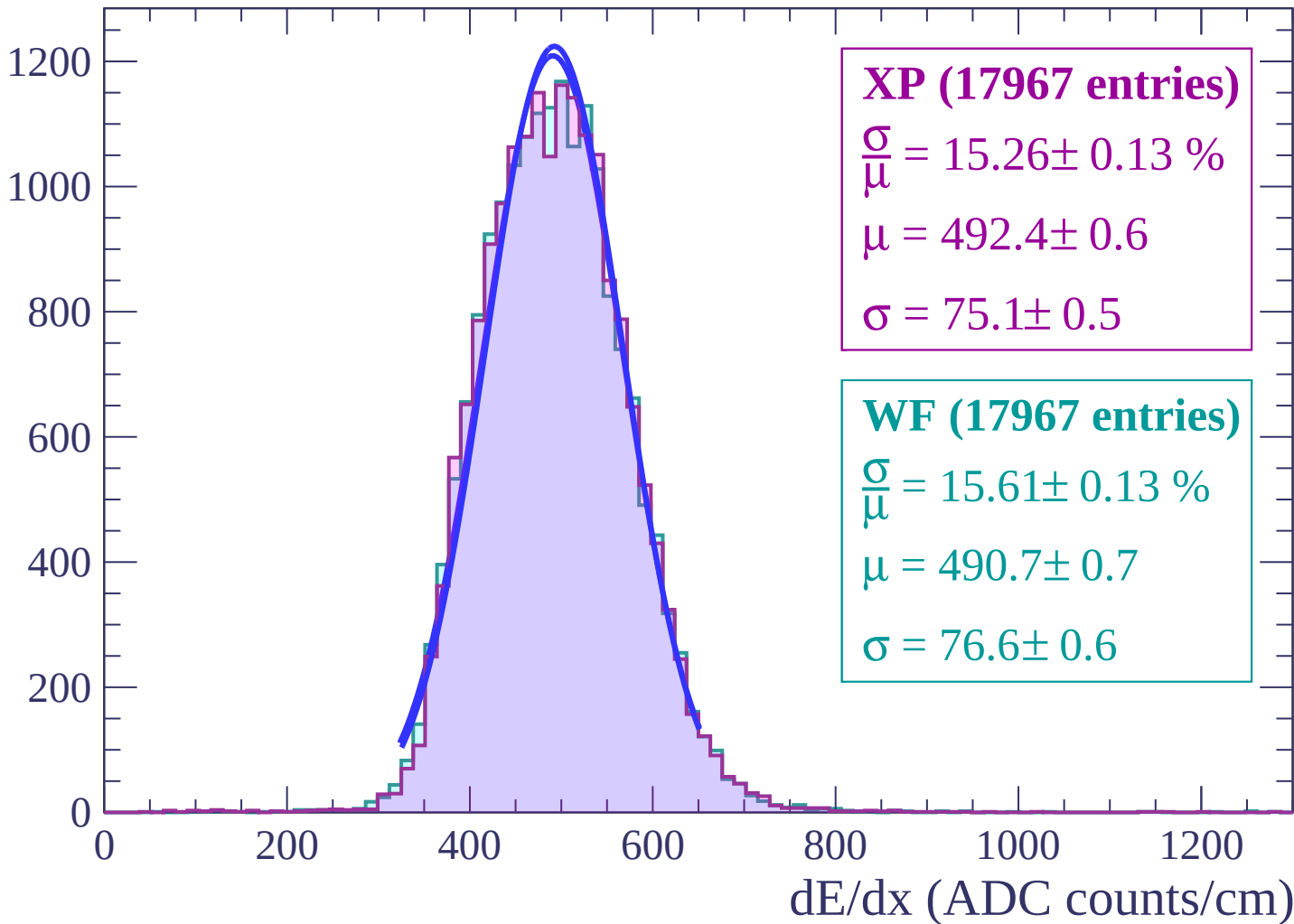
Energy loss | $-26 < \theta < -24$

Count

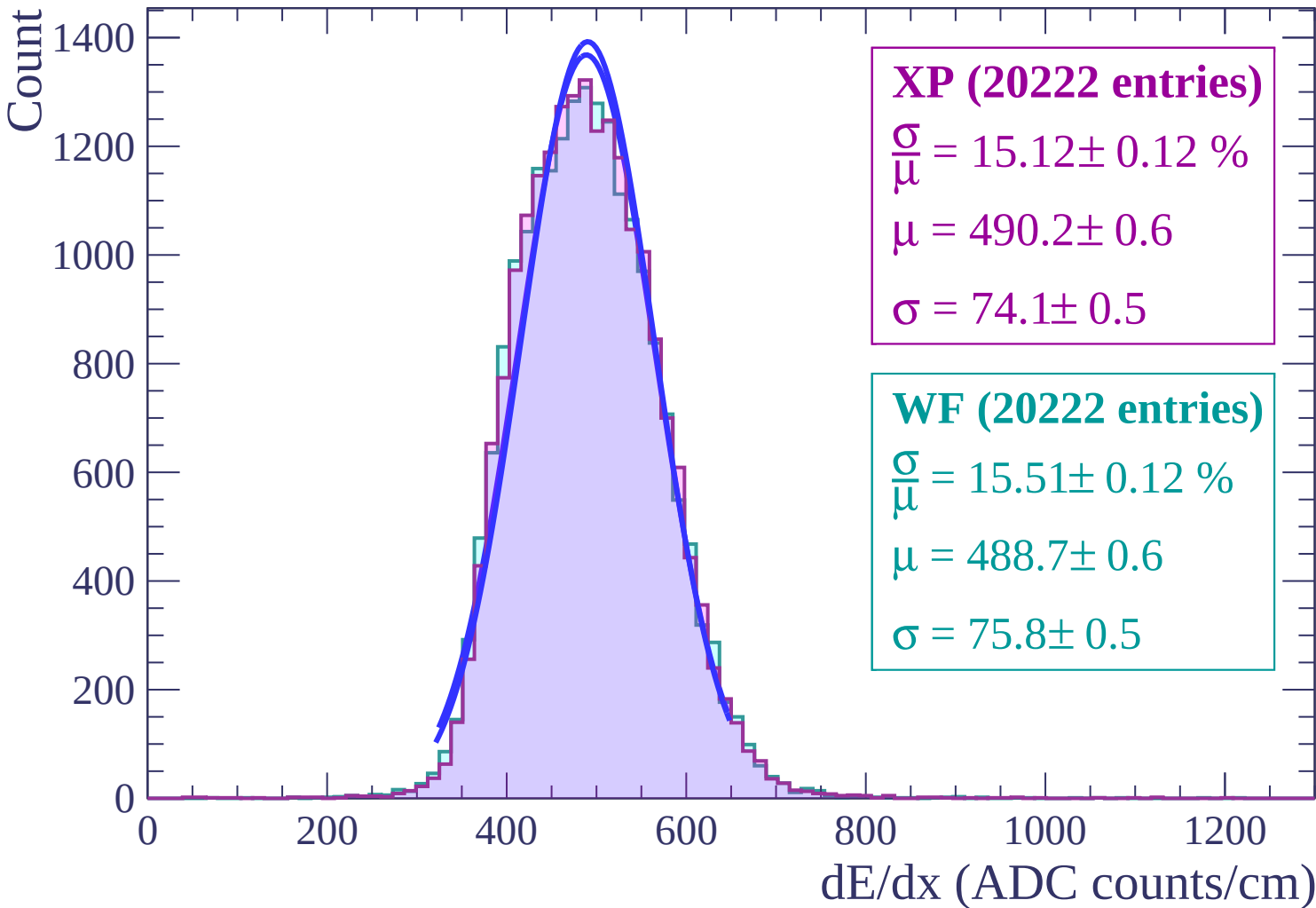


Energy loss | $-24 < \theta < -22$

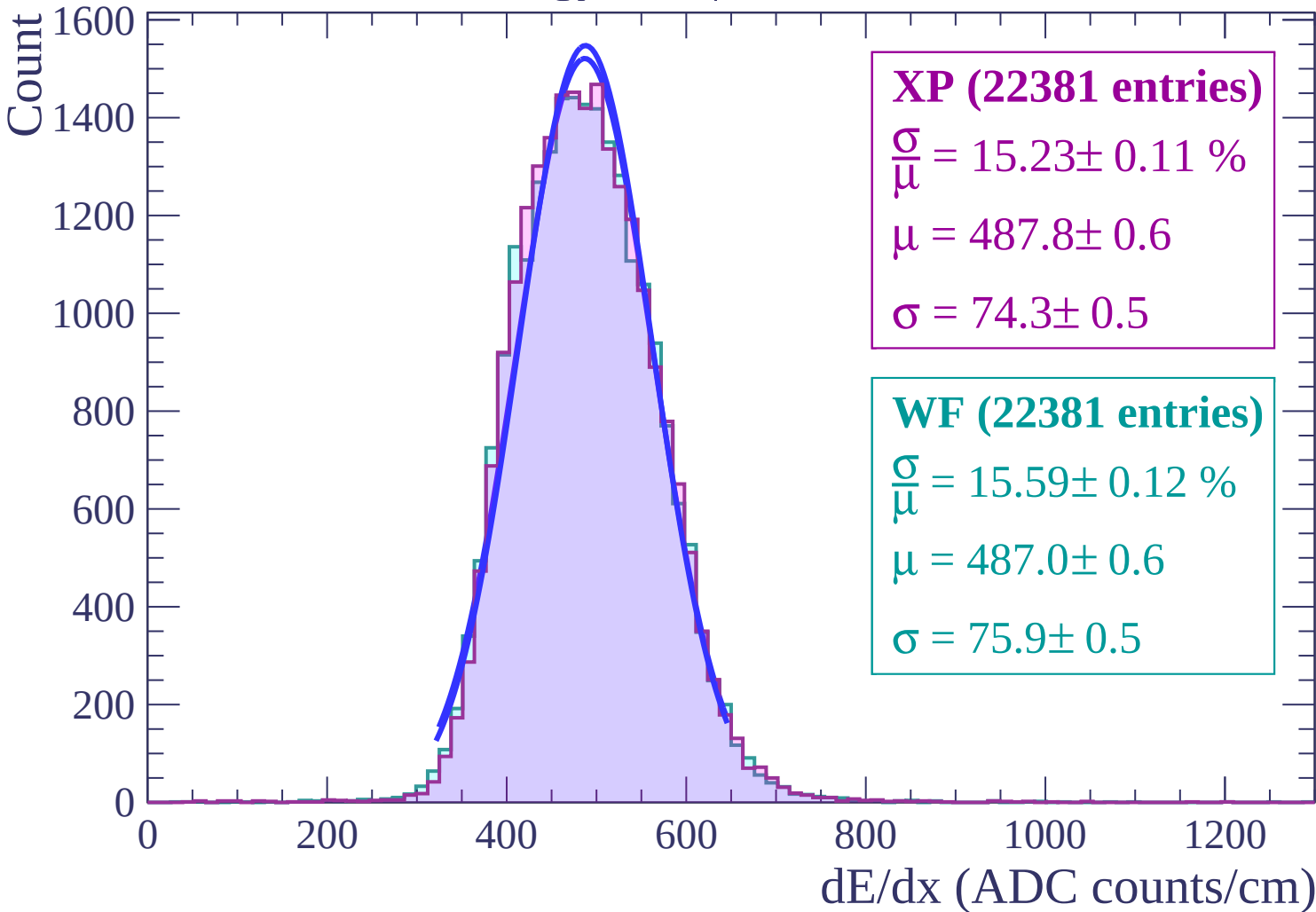
Count



Energy loss | $-22 < \theta < -20$



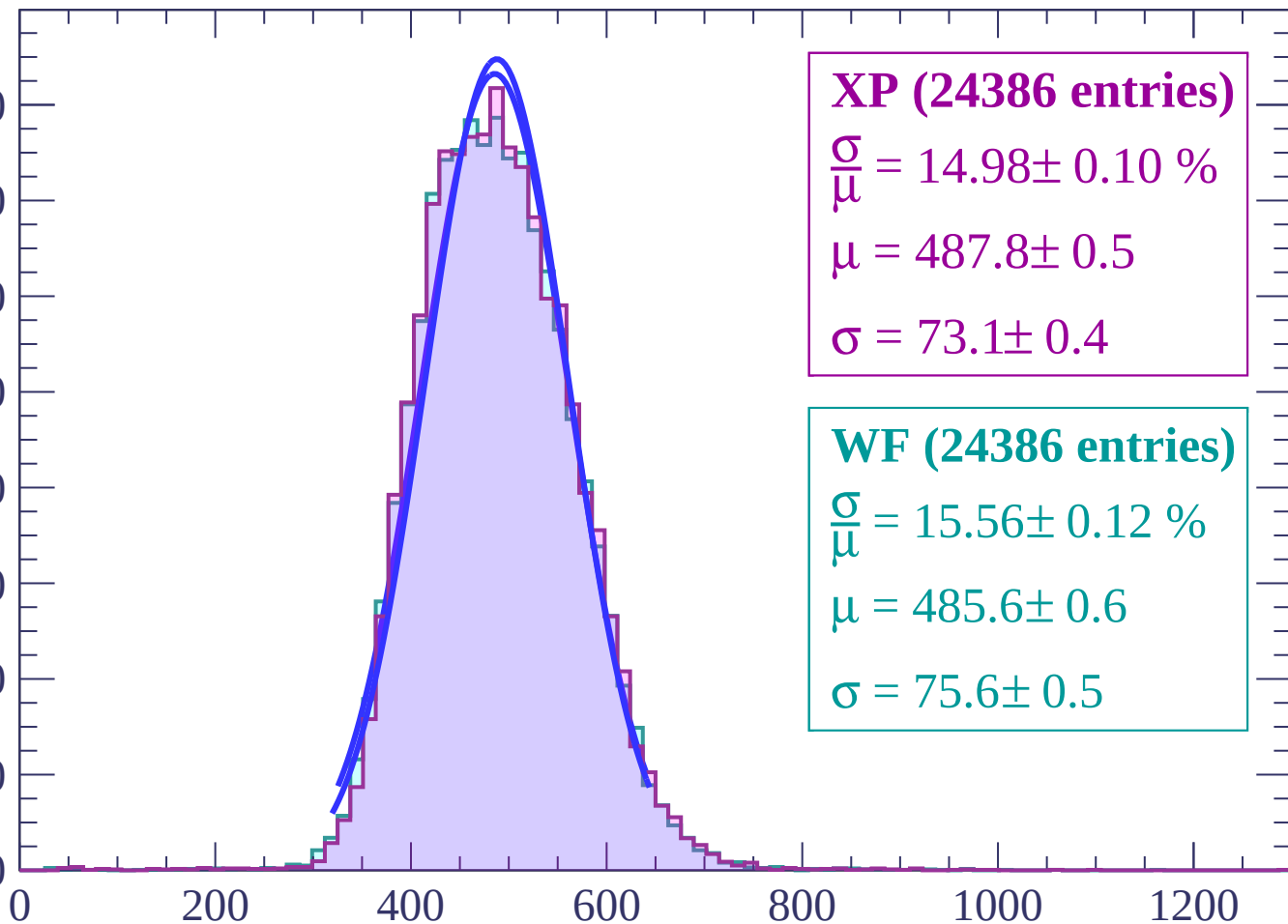
Energy loss | $-20 < \theta < -18$



Energy loss | $-18 < \theta < -16$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (24386 entries)

$\frac{\sigma}{\mu} = 14.98 \pm 0.10 \%$

$\mu = 487.8 \pm 0.5$

$\sigma = 73.1 \pm 0.4$

WF (24386 entries)

$\frac{\sigma}{\mu} = 15.56 \pm 0.12 \%$

$\mu = 485.6 \pm 0.6$

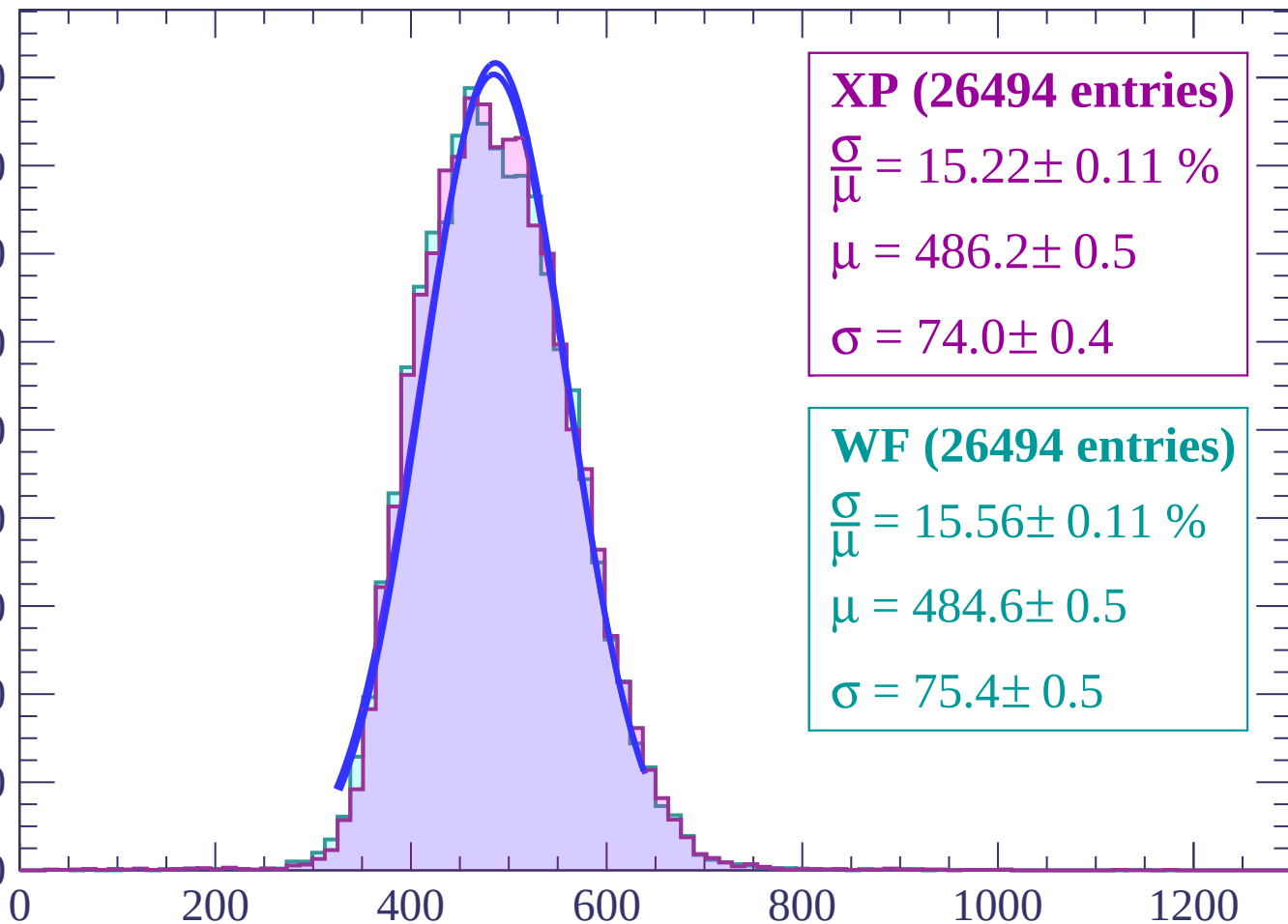
$\sigma = 75.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-16 < \theta < -14$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26494 entries)

$$\frac{\sigma}{\mu} = 15.22 \pm 0.11 \%$$

$$\mu = 486.2 \pm 0.5$$

$$\sigma = 74.0 \pm 0.4$$

WF (26494 entries)

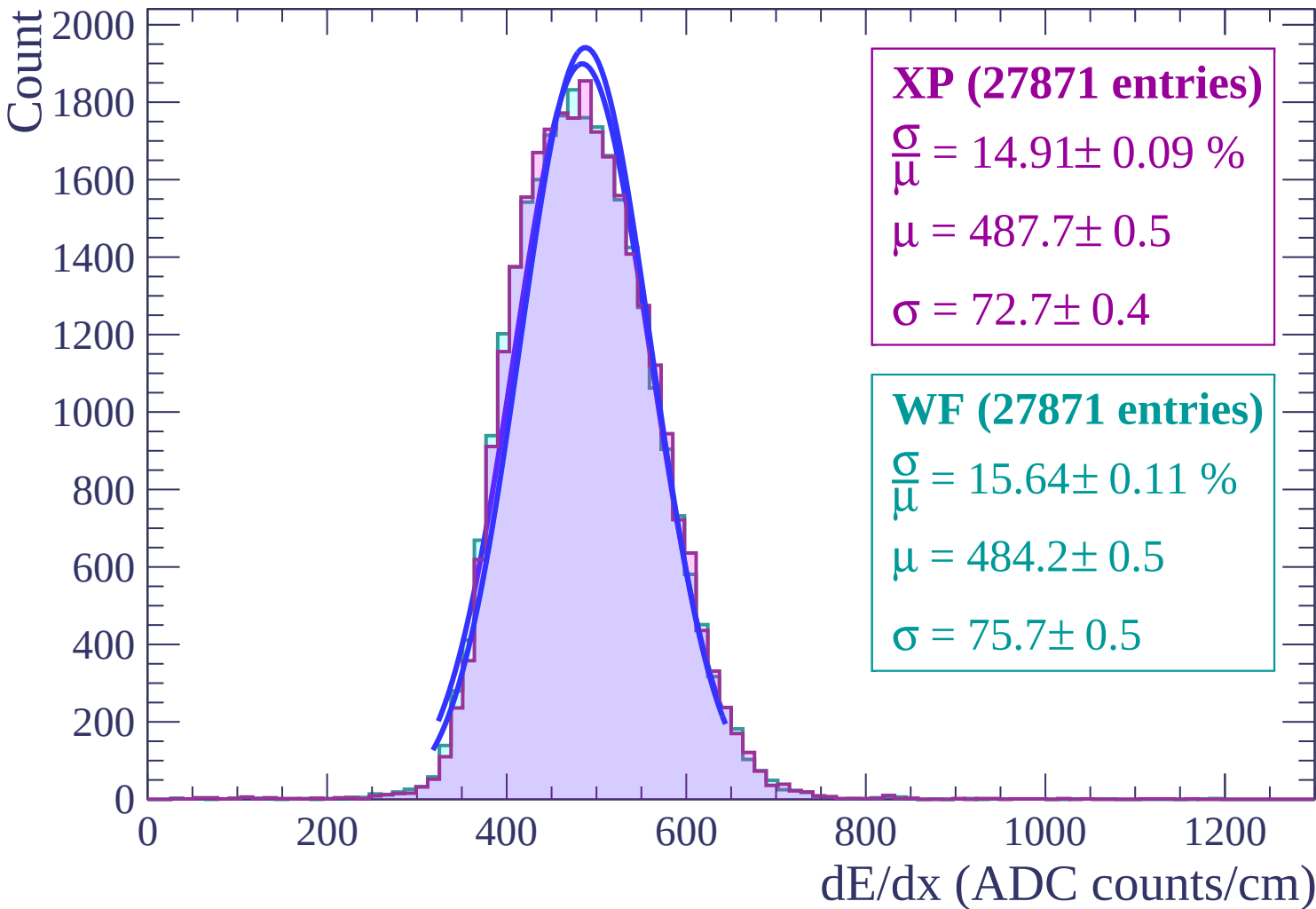
$$\frac{\sigma}{\mu} = 15.56 \pm 0.11 \%$$

$$\mu = 484.6 \pm 0.5$$

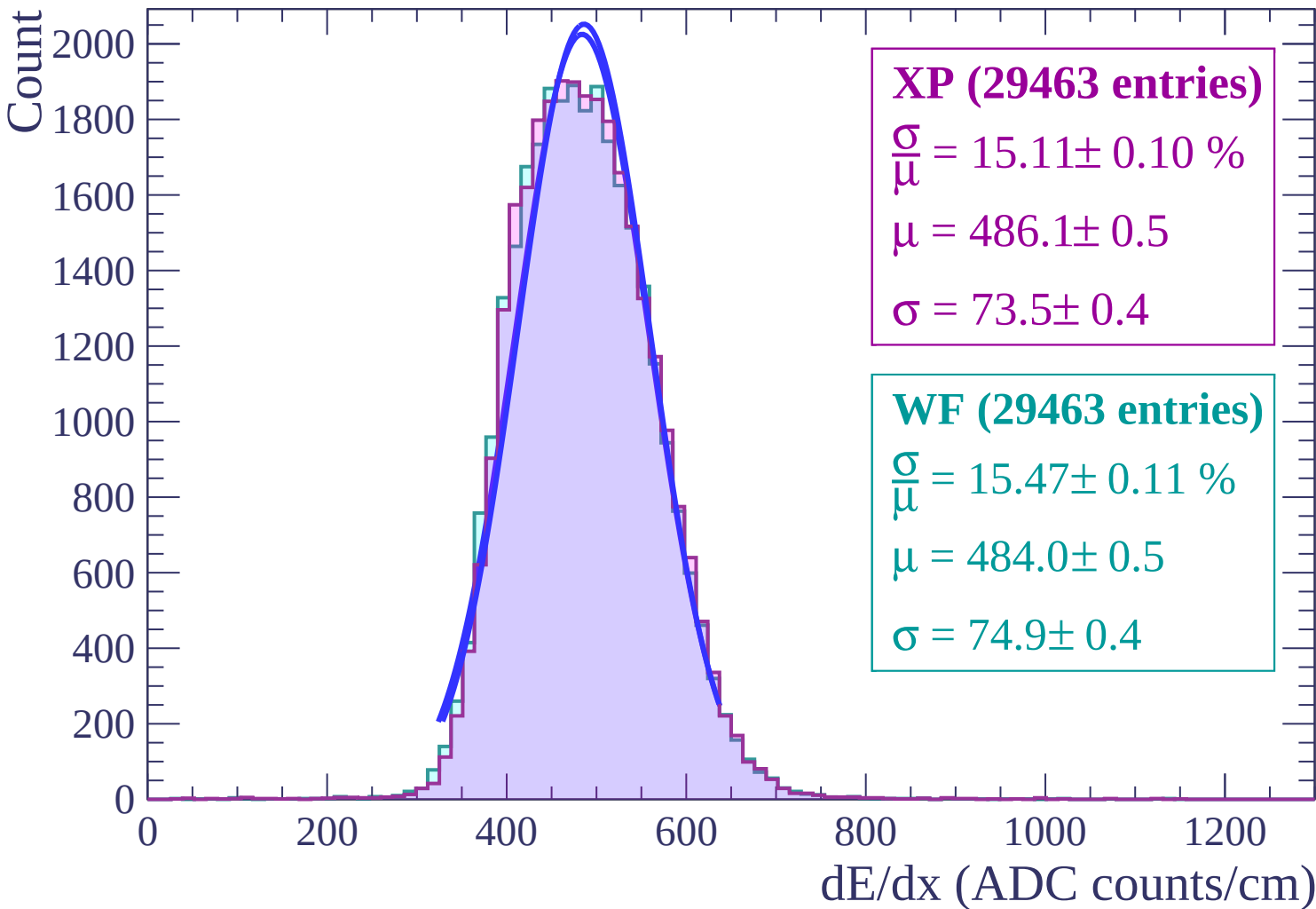
$$\sigma = 75.4 \pm 0.5$$

dE/dx (ADC counts/cm)

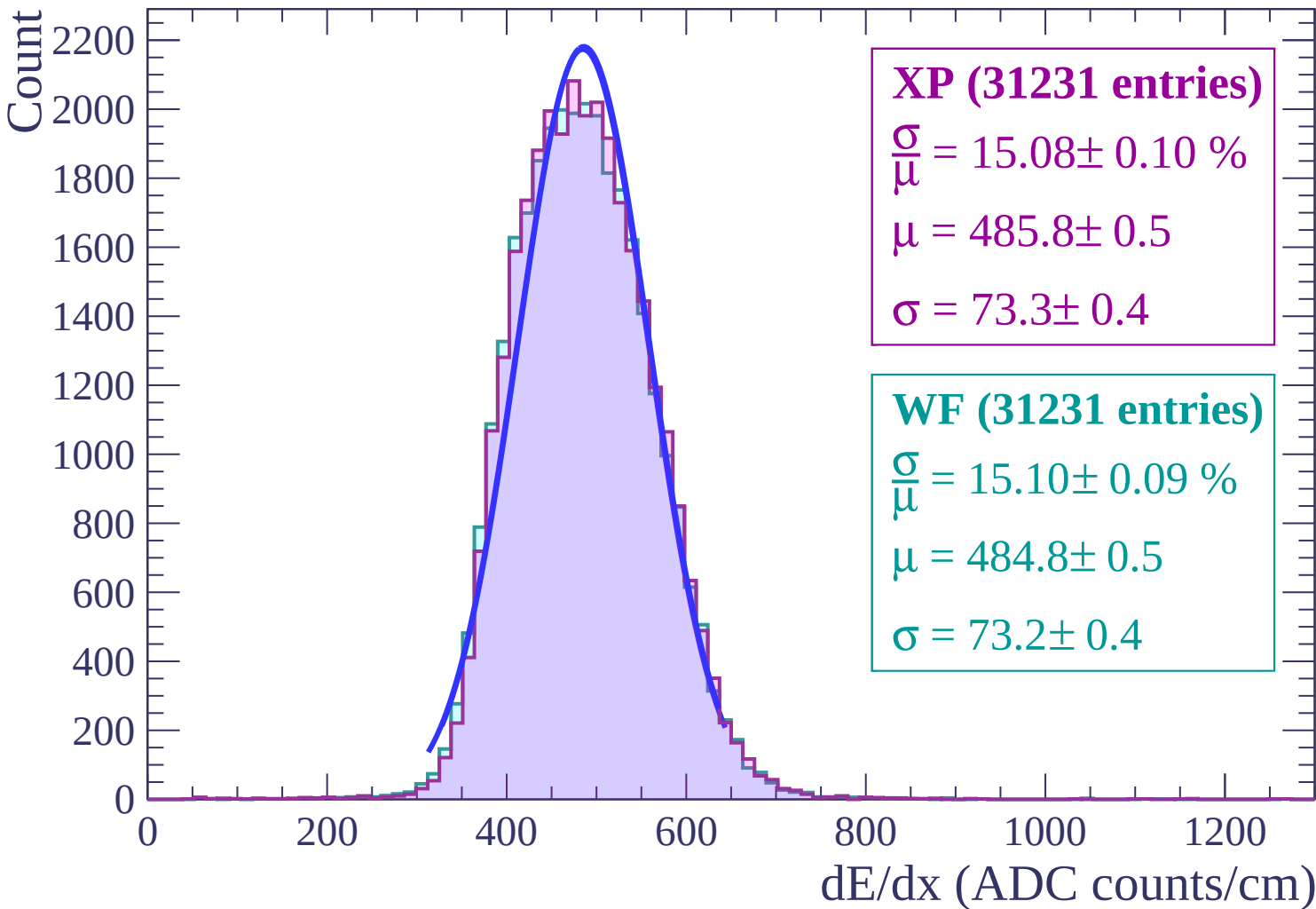
Energy loss | $-14 < \theta < -12$



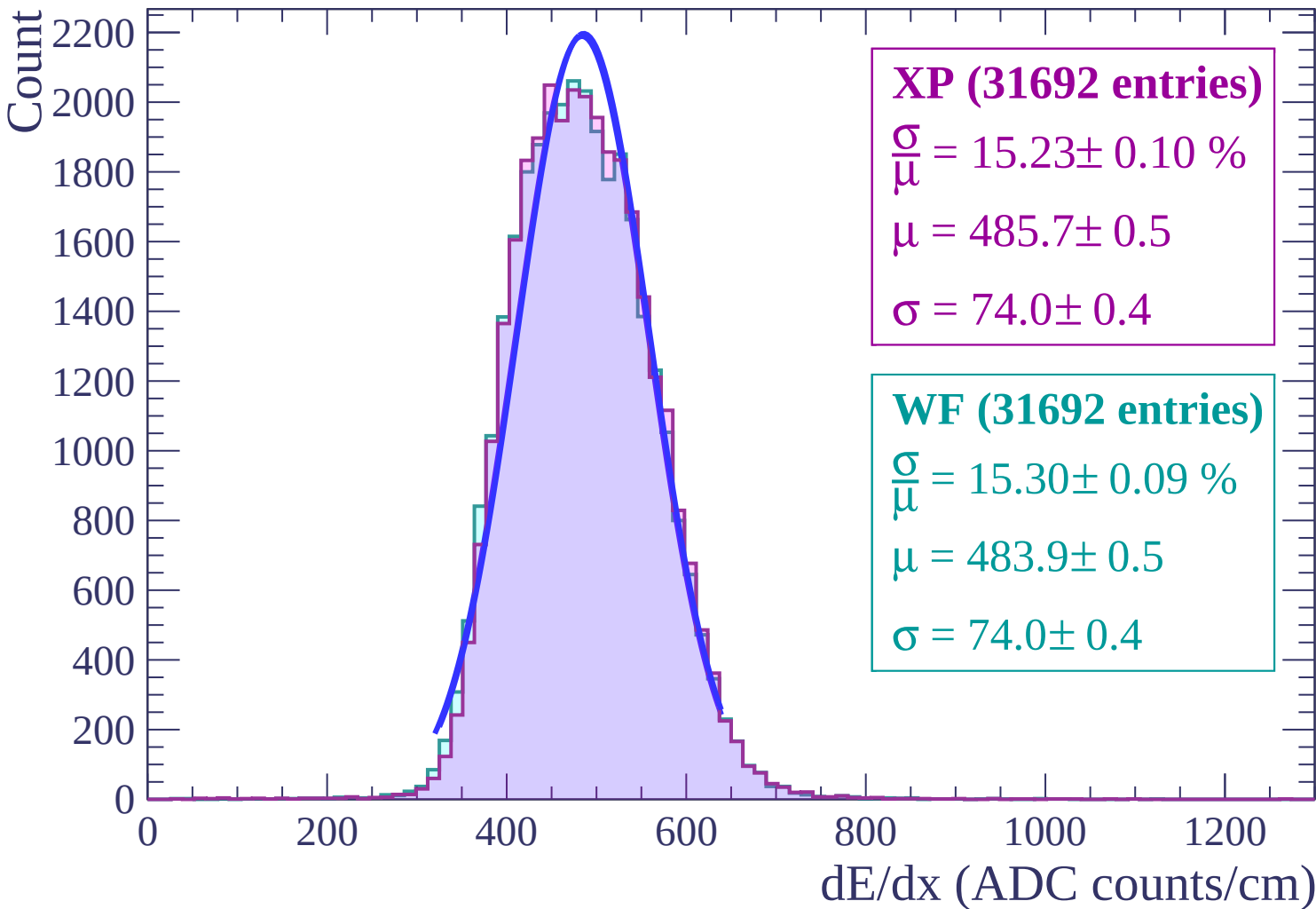
Energy loss | $-12 < \theta < -10$



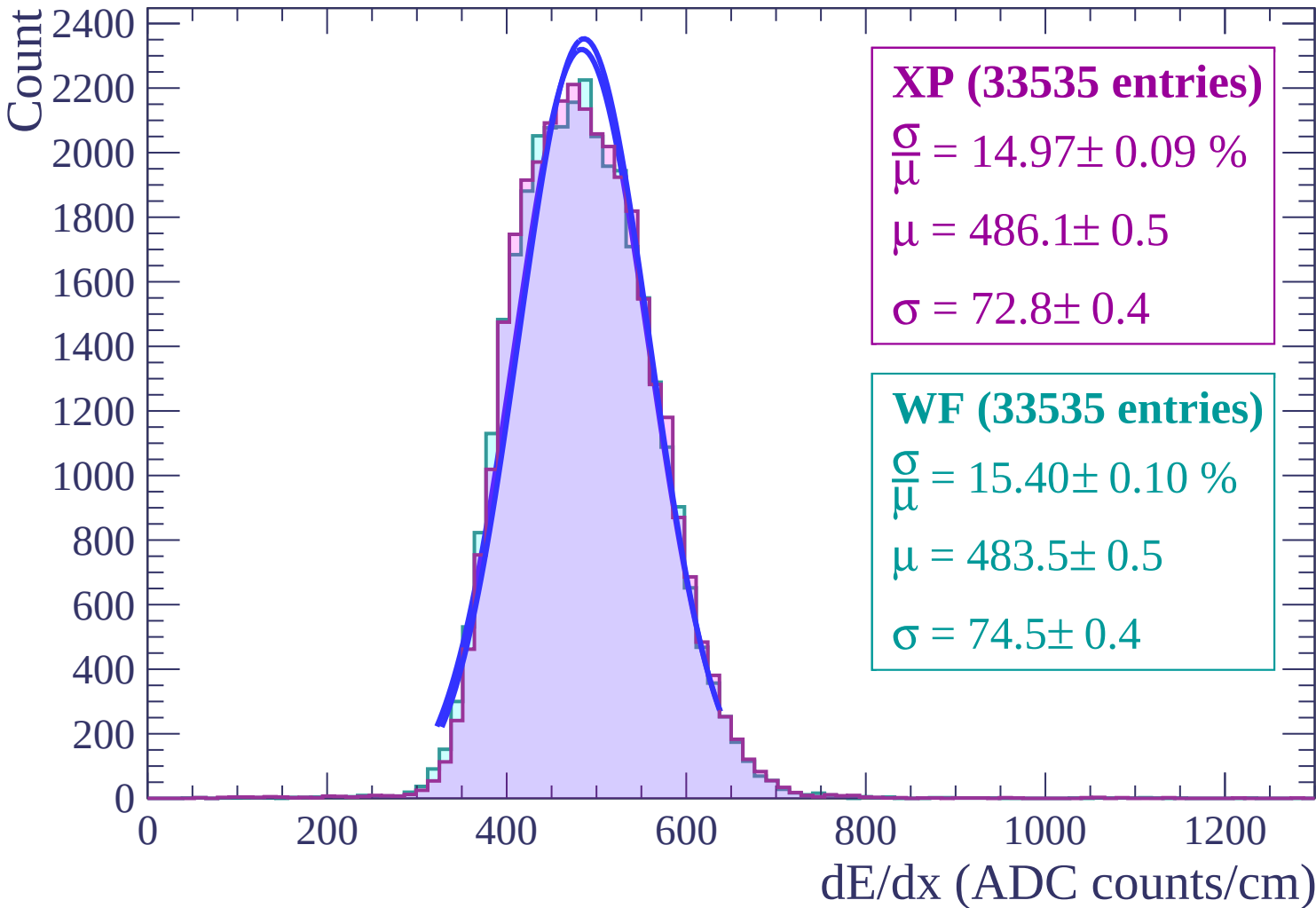
Energy loss | $-10 < \theta < -8$



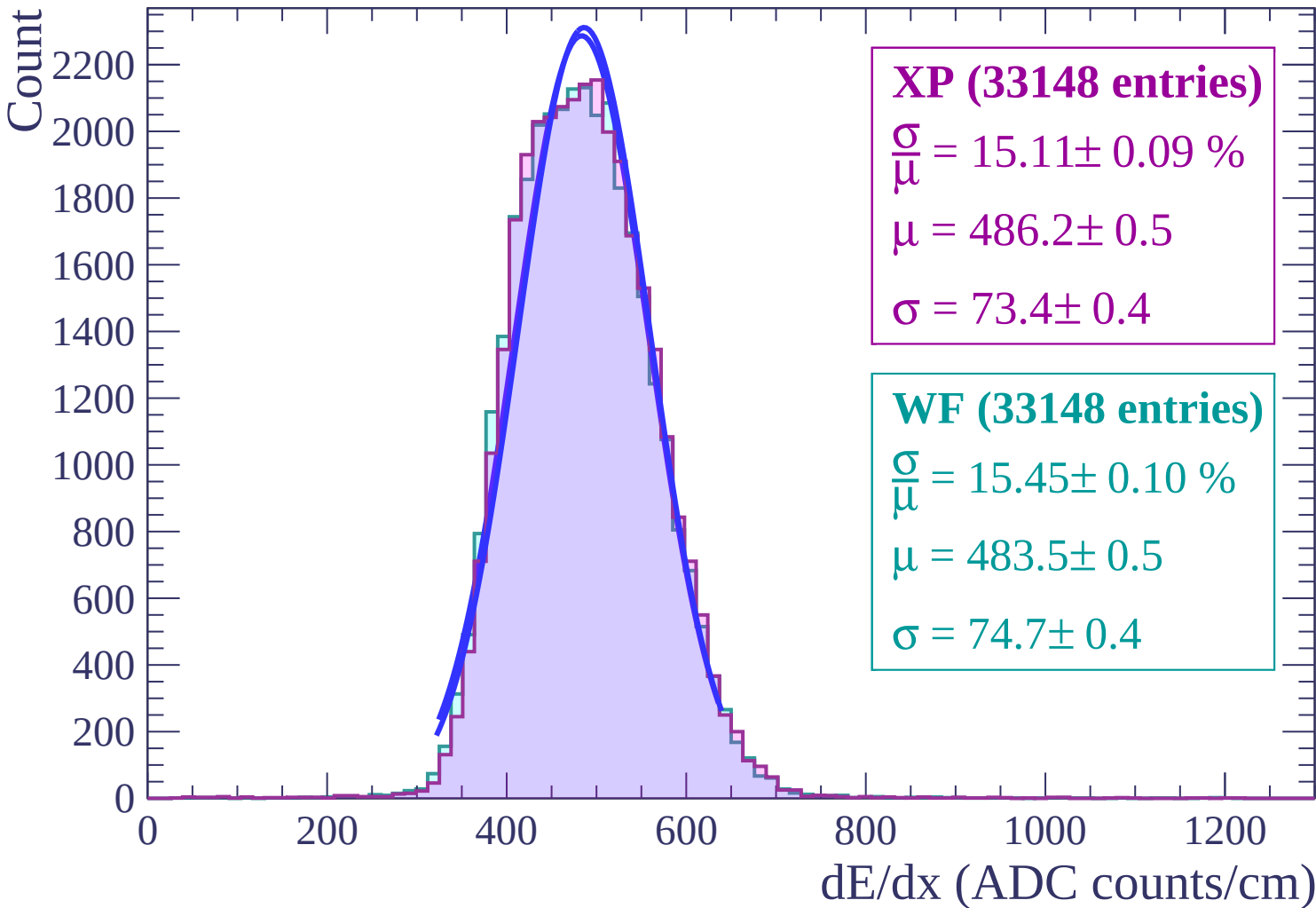
Energy loss | $-8 < \theta < -6$



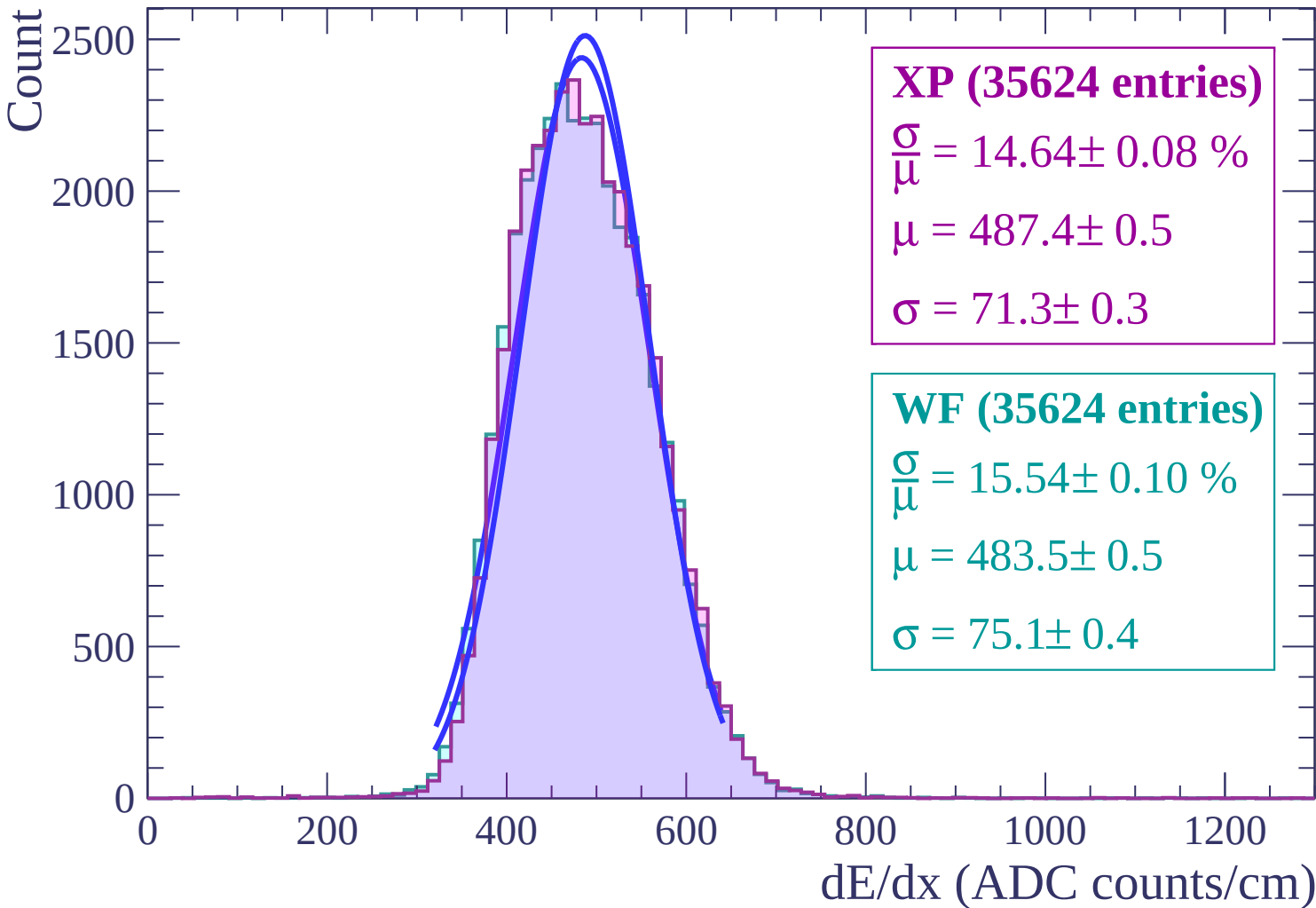
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$



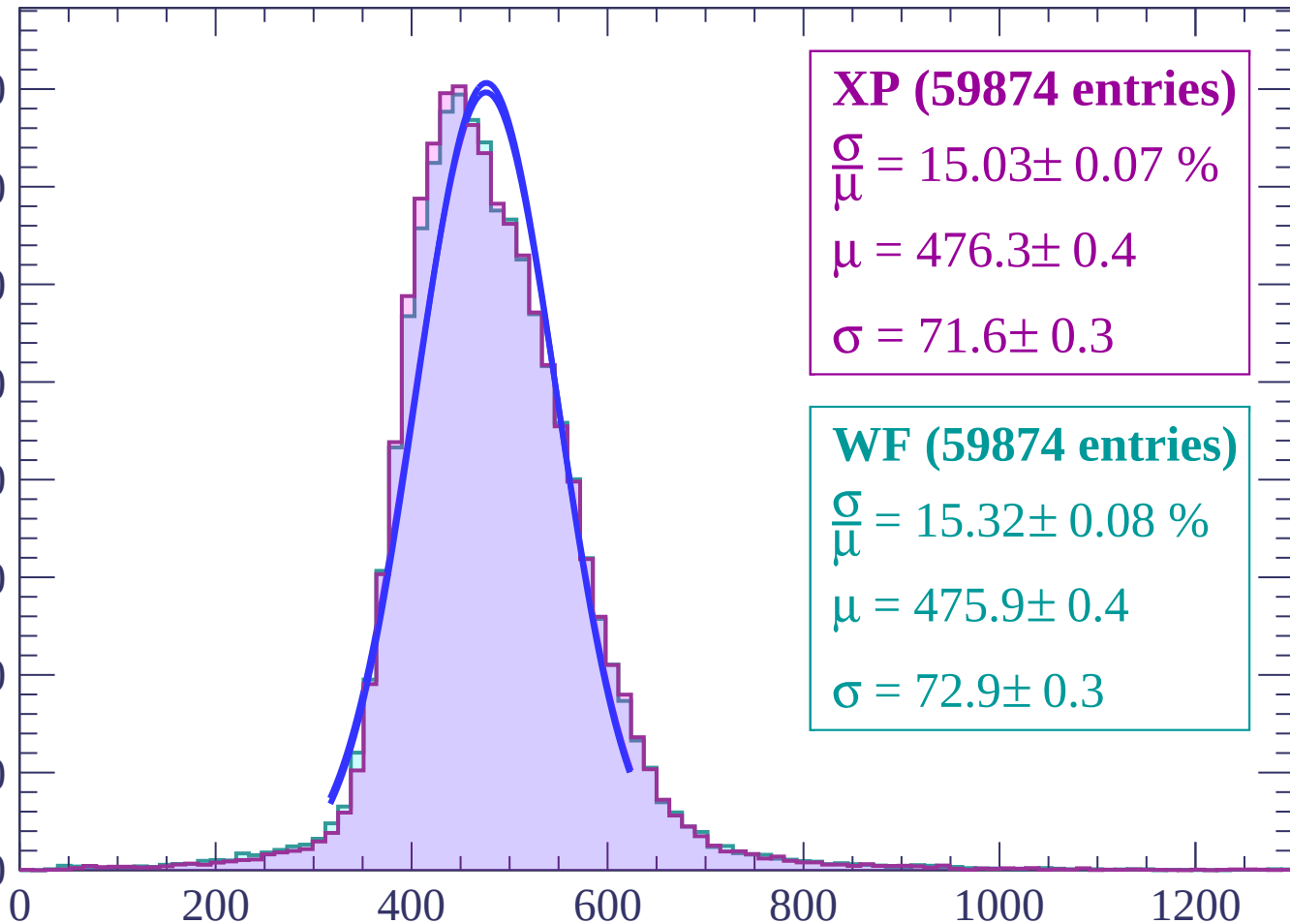
Energy loss | $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

4000
3500
3000
2500
2000
1500
1000
500
0



XP (59874 entries)

$\frac{\sigma}{\mu} = 15.03 \pm 0.07 \%$

$\mu = 476.3 \pm 0.4$

$\sigma = 71.6 \pm 0.3$

WF (59874 entries)

$\frac{\sigma}{\mu} = 15.32 \pm 0.08 \%$

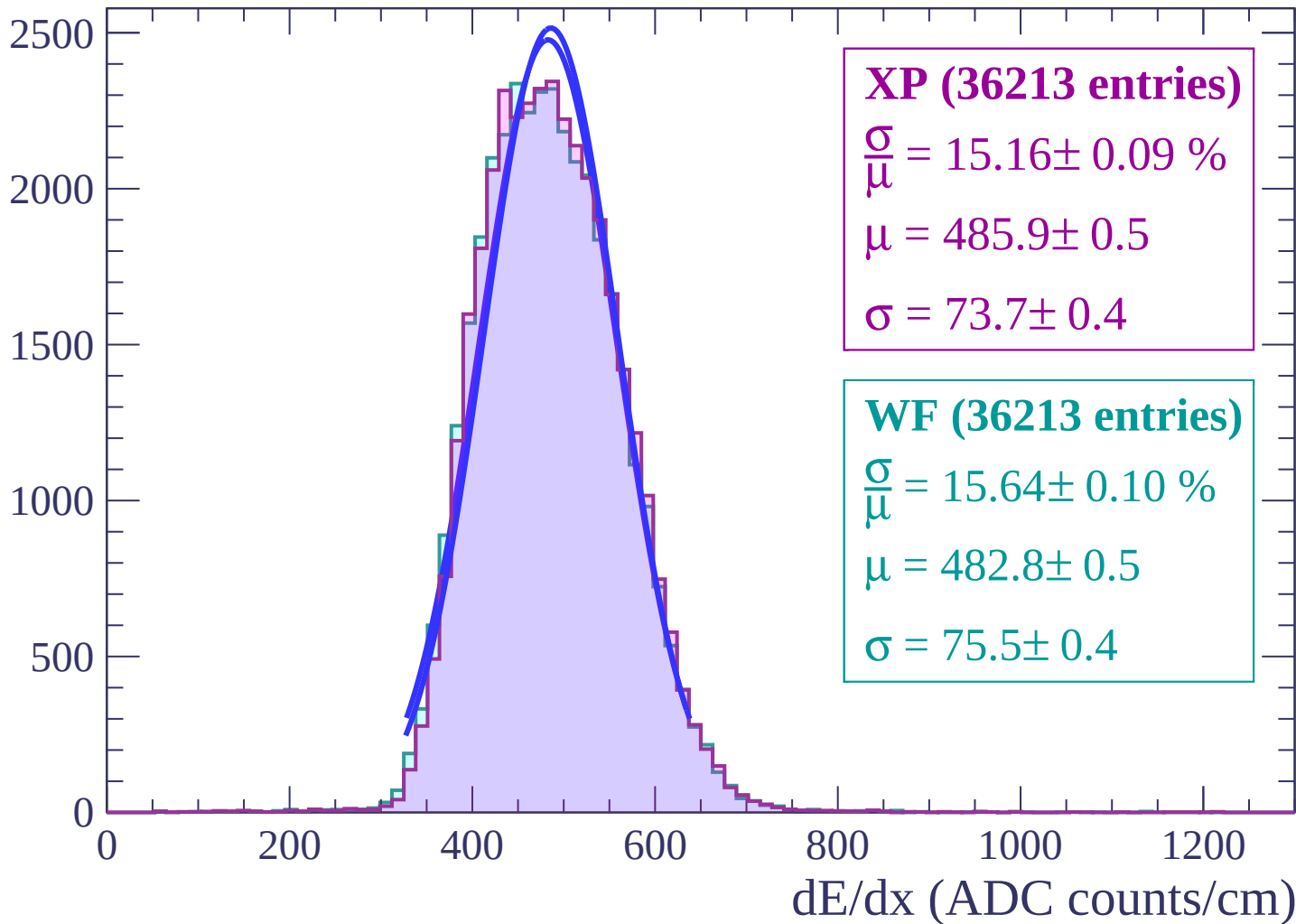
$\mu = 475.9 \pm 0.4$

$\sigma = 72.9 \pm 0.3$

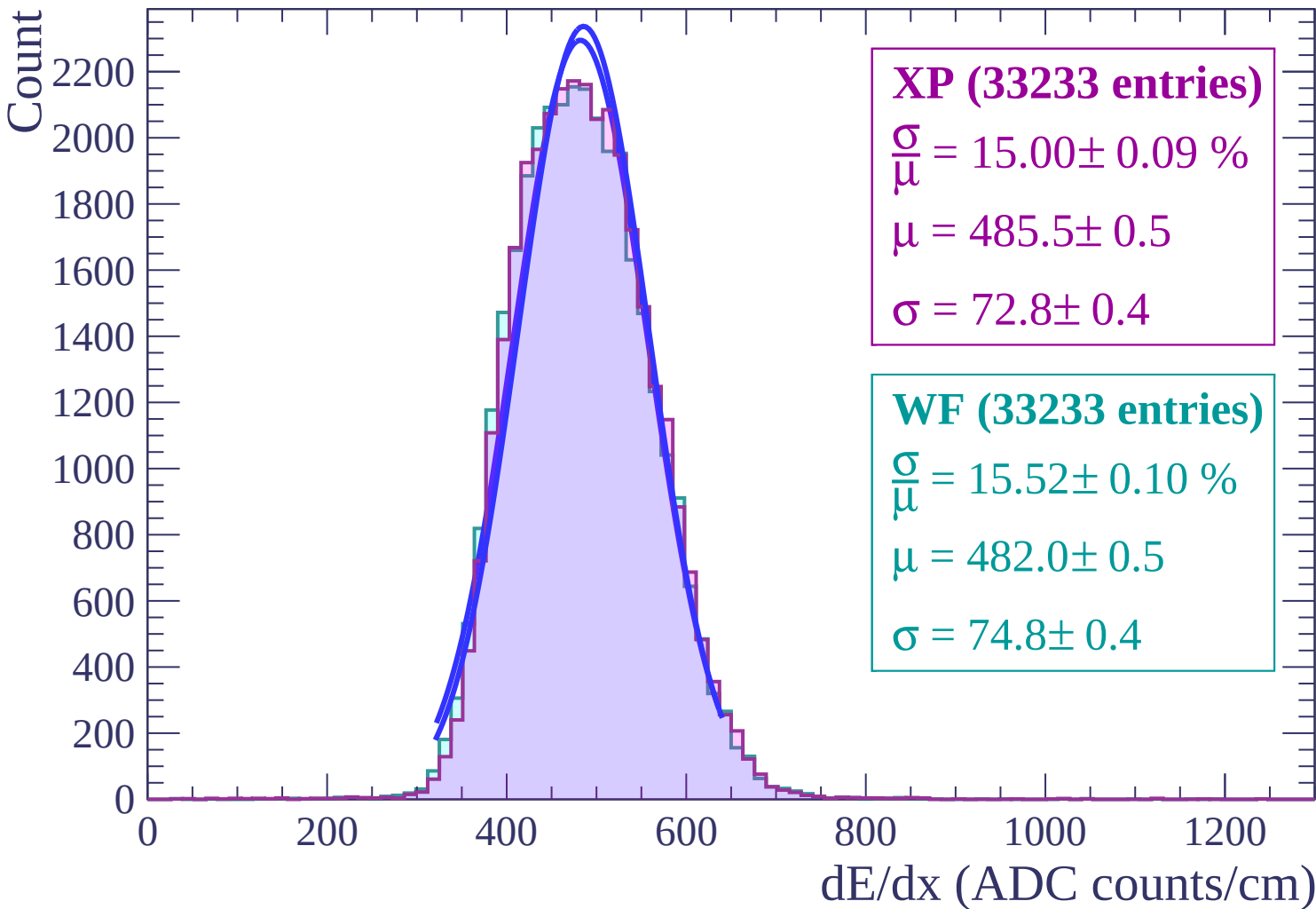
dE/dx (ADC counts/cm)

Energy loss | $2 < \theta < 4$

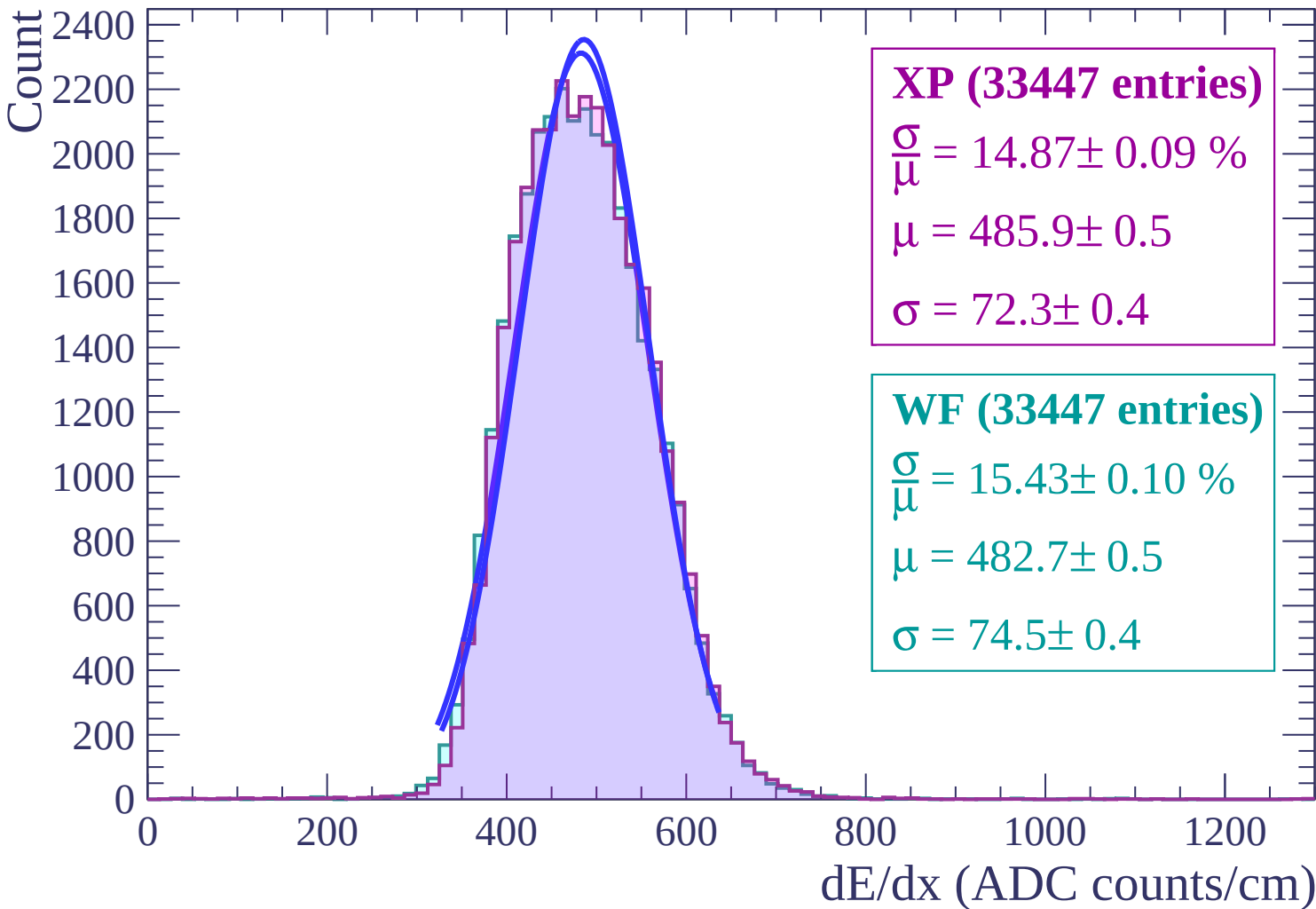
Count



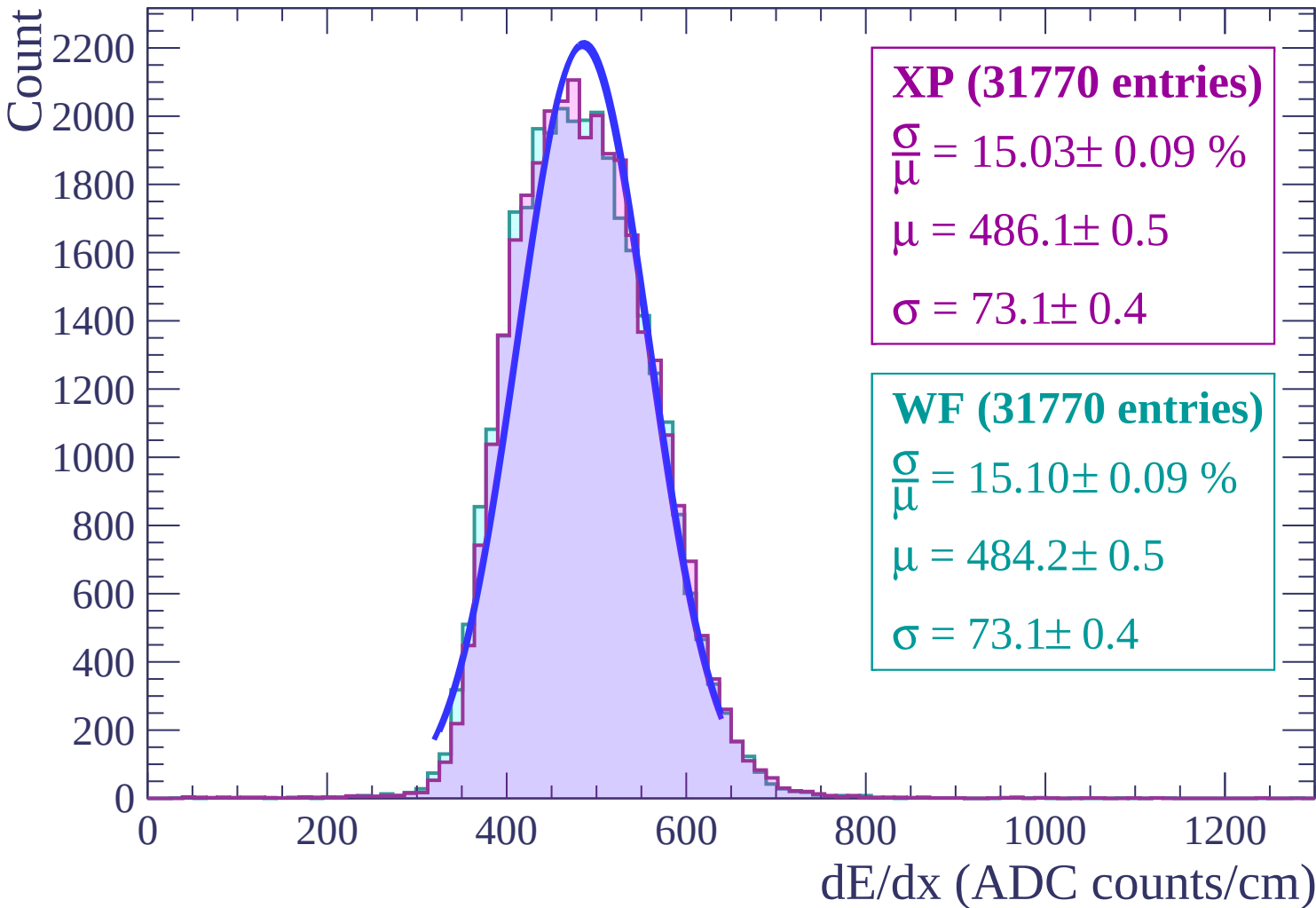
Energy loss | $4 < \theta < 6$



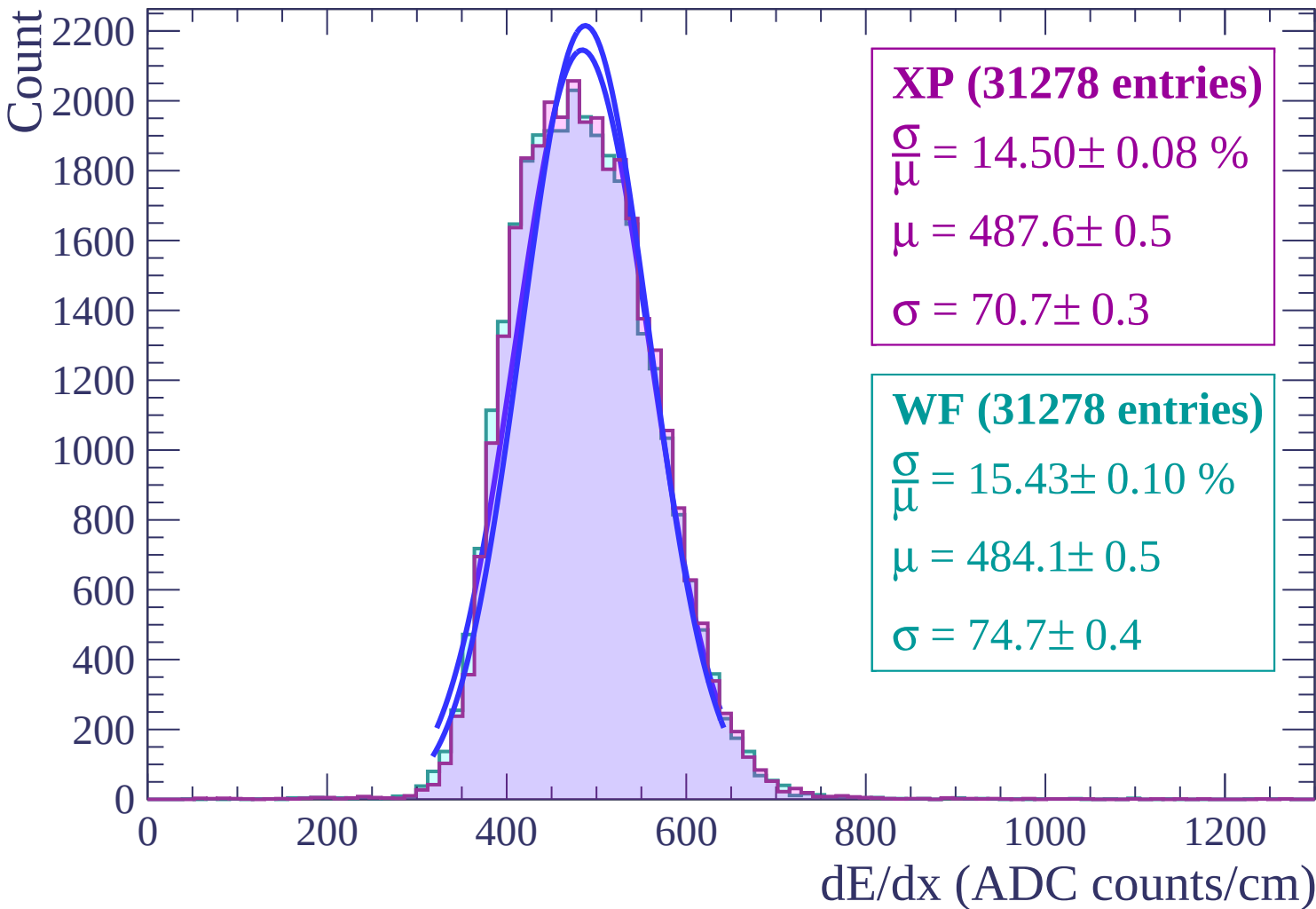
Energy loss | $6 < \theta < 8$



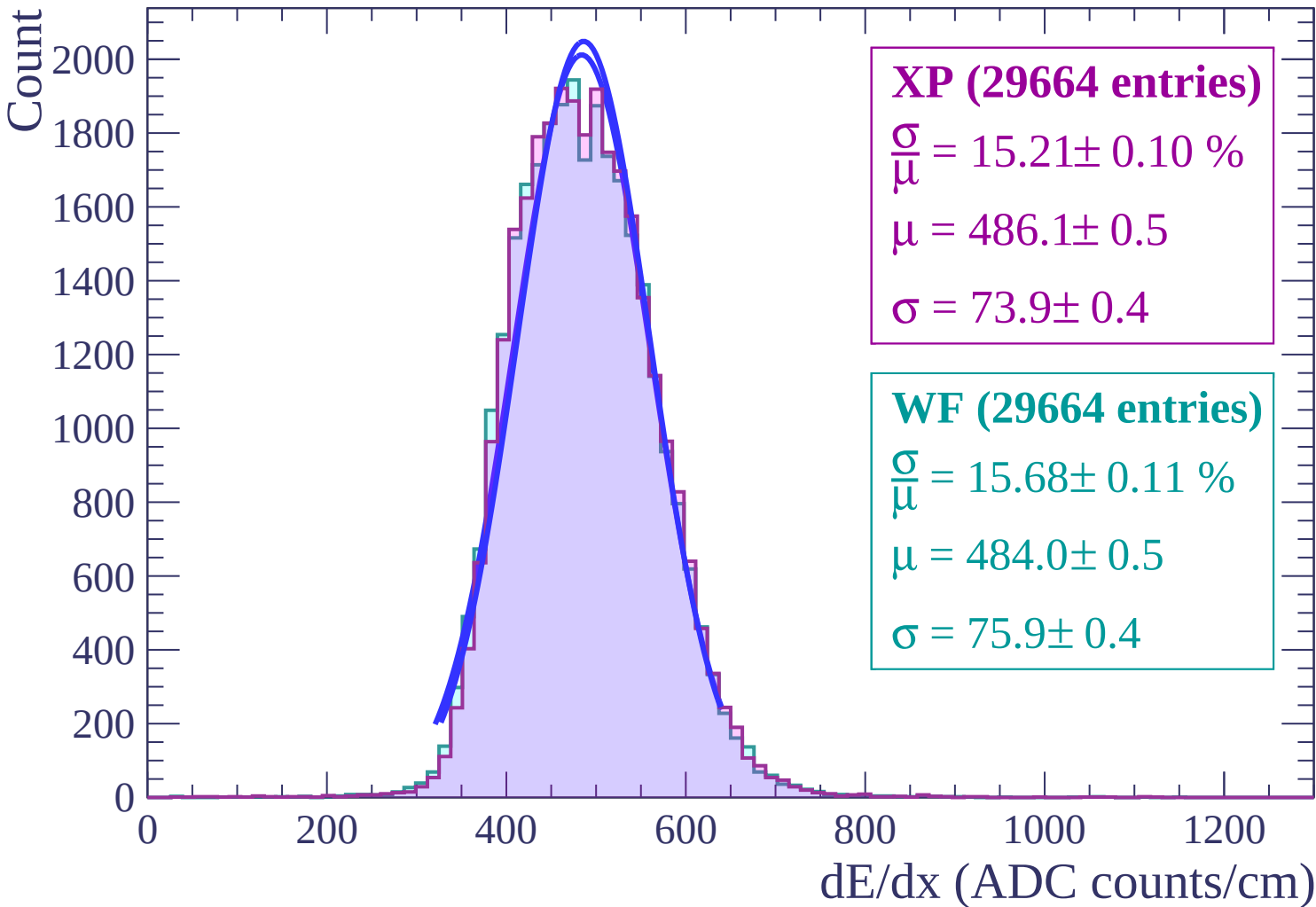
Energy loss | $8 < \theta < 10$



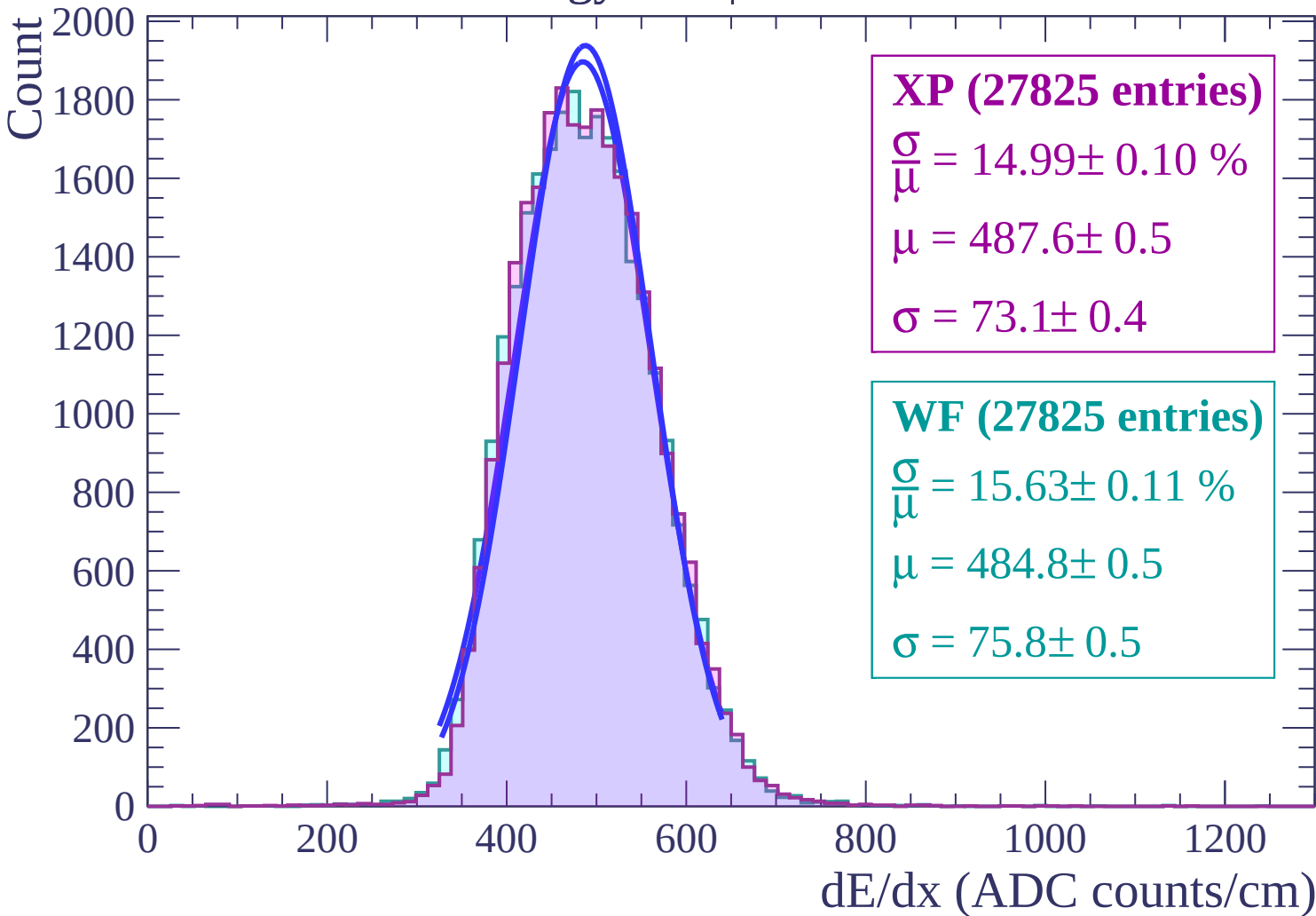
Energy loss | $10 < \theta < 12$



Energy loss | $12 < \theta < 14$



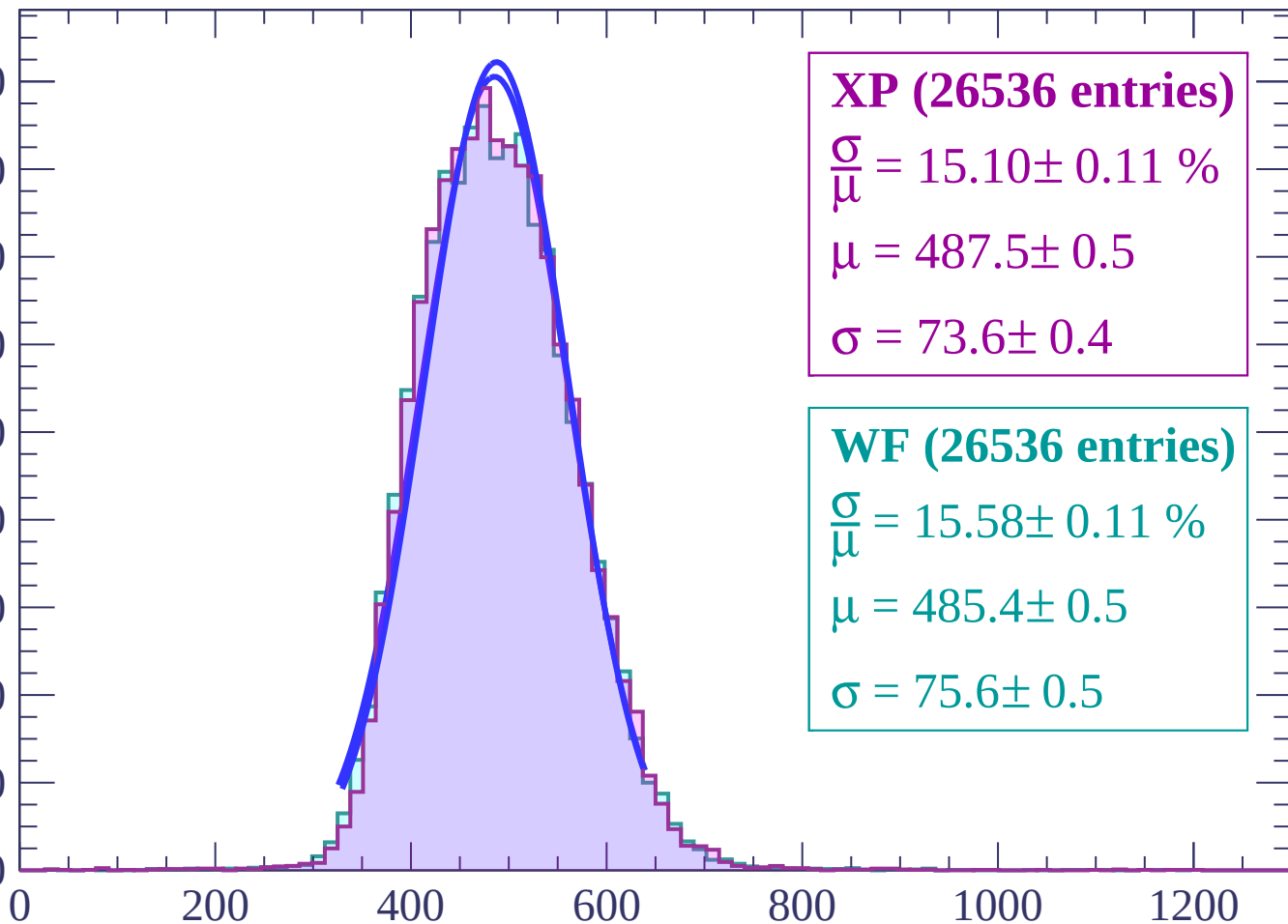
Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26536 entries)

$$\frac{\sigma}{\mu} = 15.10 \pm 0.11 \%$$

$$\mu = 487.5 \pm 0.5$$

$$\sigma = 73.6 \pm 0.4$$

WF (26536 entries)

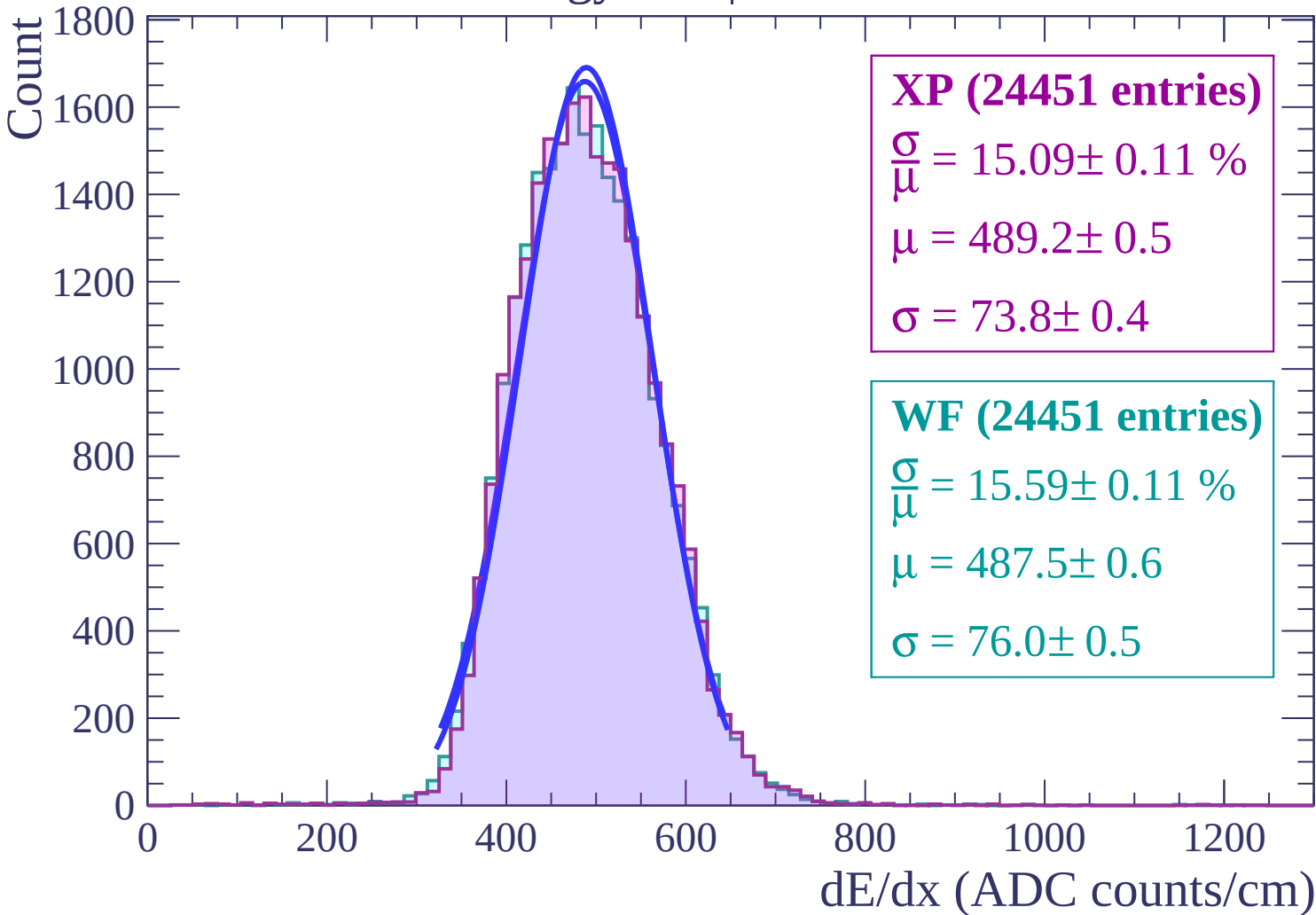
$$\frac{\sigma}{\mu} = 15.58 \pm 0.11 \%$$

$$\mu = 485.4 \pm 0.5$$

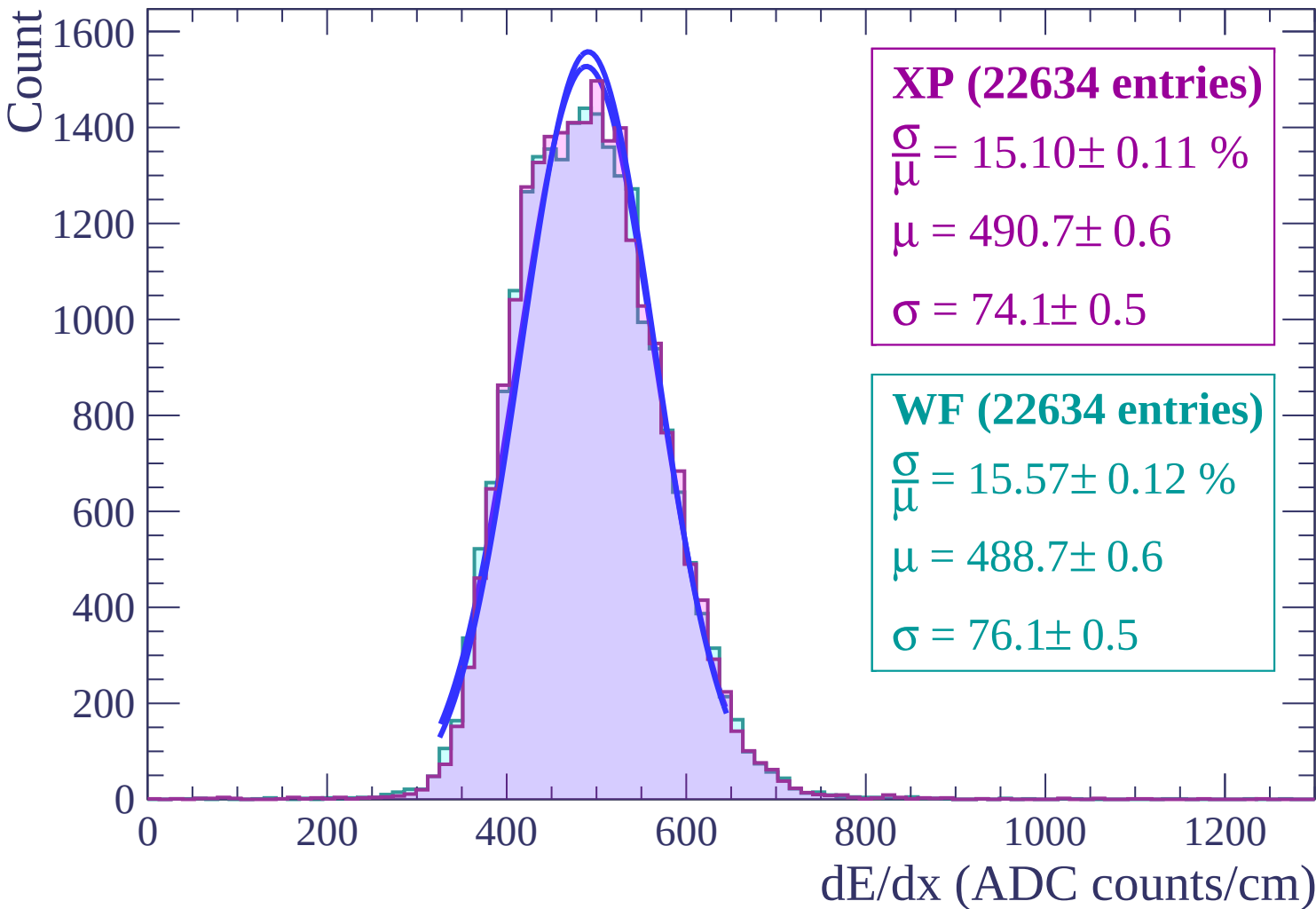
$$\sigma = 75.6 \pm 0.5$$

dE/dx (ADC counts/cm)

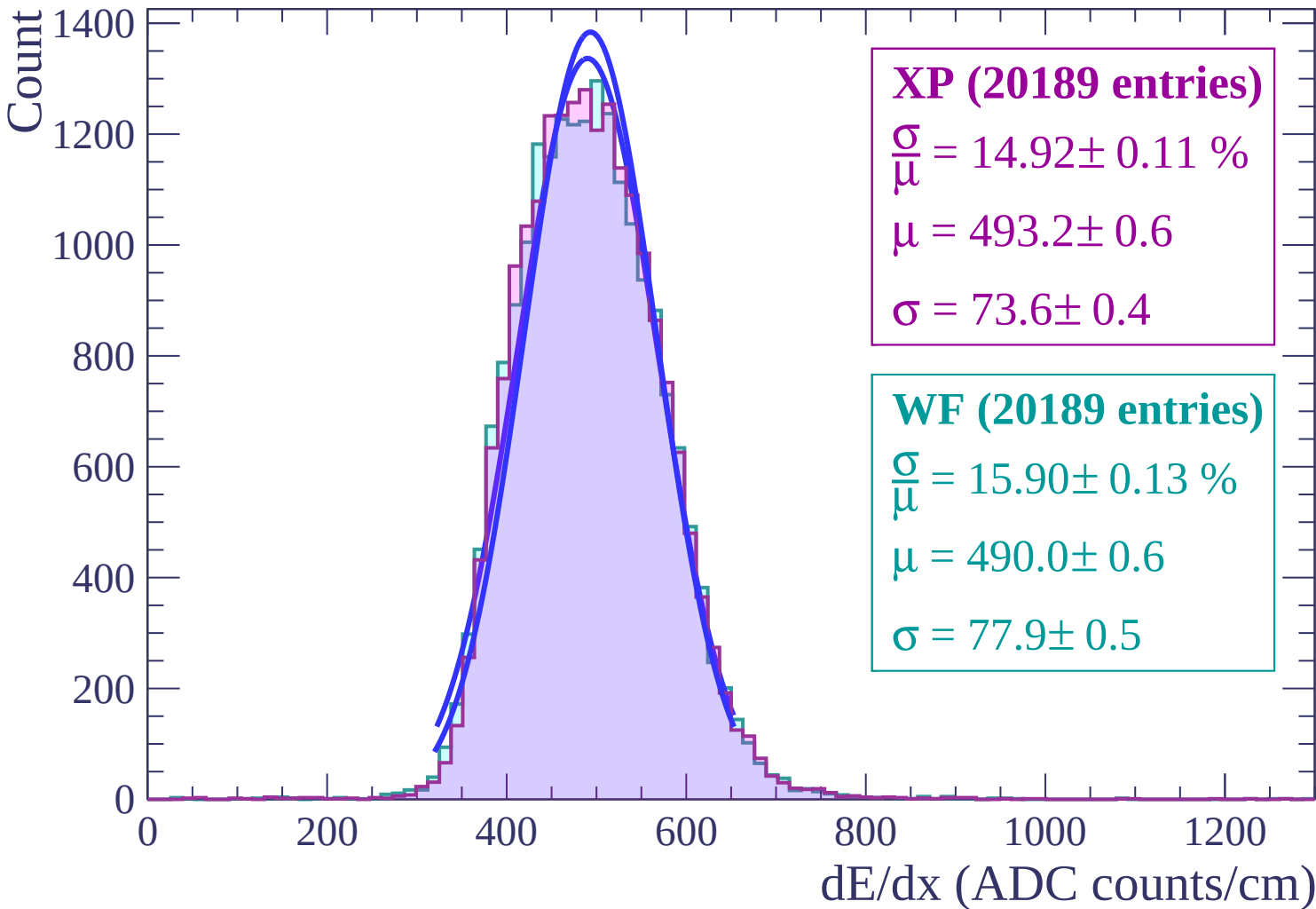
Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$



Energy loss | $24 < \theta < 26$

Count

1200

1000

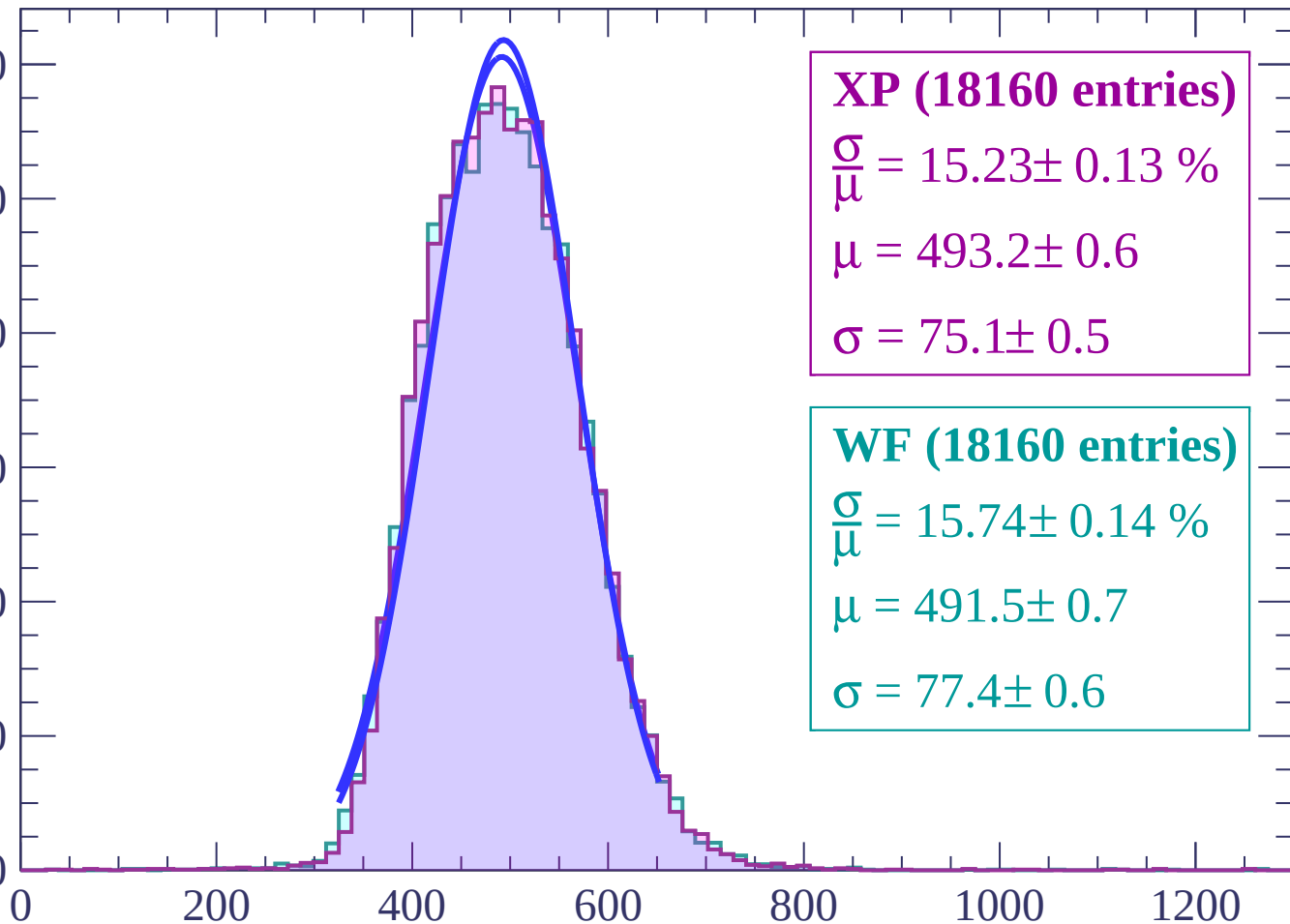
800

600

400

200

0



XP (18160 entries)

$\frac{\sigma}{\mu} = 15.23 \pm 0.13 \%$

$\mu = 493.2 \pm 0.6$

$\sigma = 75.1 \pm 0.5$

WF (18160 entries)

$\frac{\sigma}{\mu} = 15.74 \pm 0.14 \%$

$\mu = 491.5 \pm 0.7$

$\sigma = 77.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15775 entries)

$\frac{\sigma}{\mu} = 15.19 \pm 0.12 \%$

$\mu = 496.1 \pm 0.7$

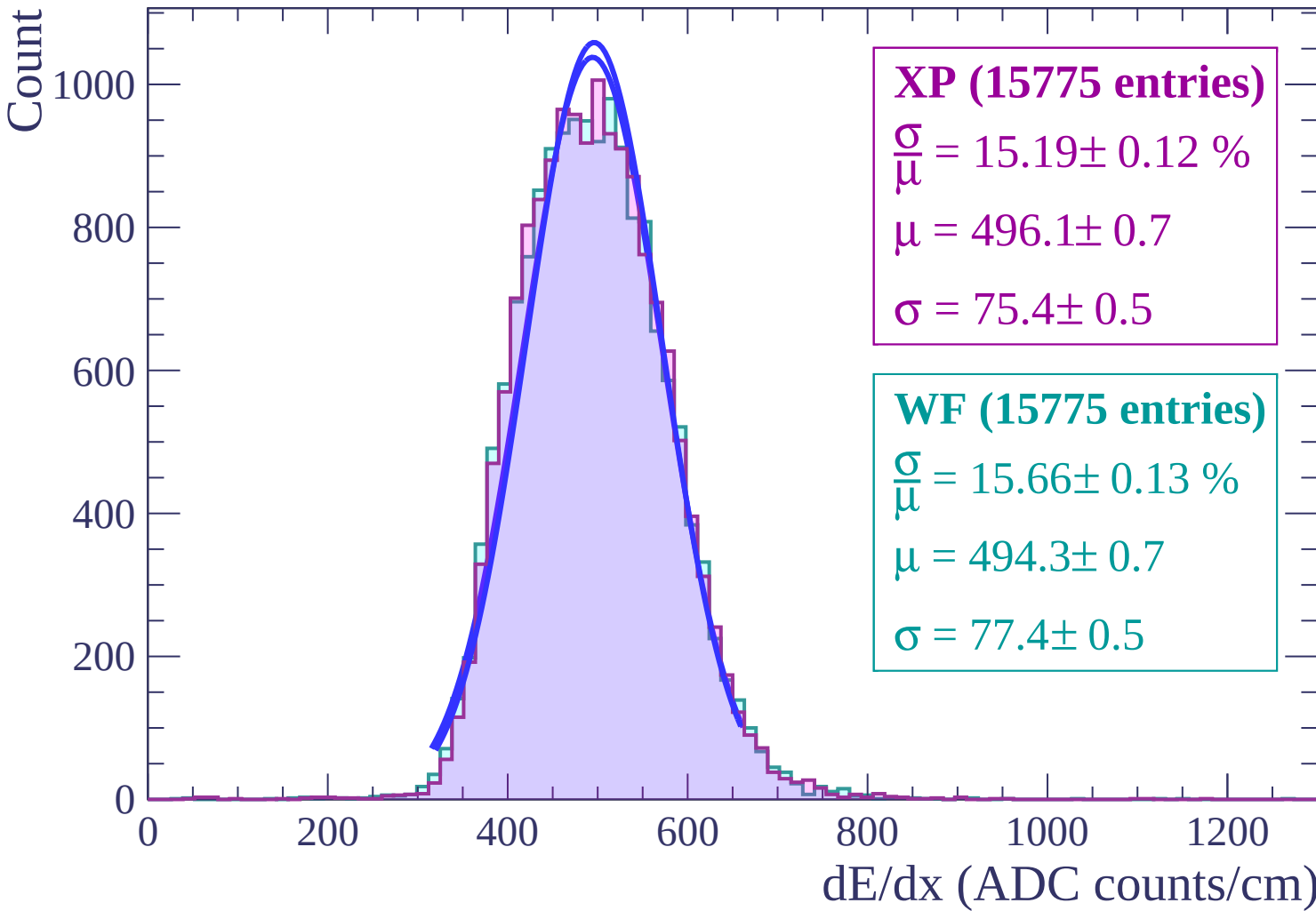
$\sigma = 75.4 \pm 0.5$

WF (15775 entries)

$\frac{\sigma}{\mu} = 15.66 \pm 0.13 \%$

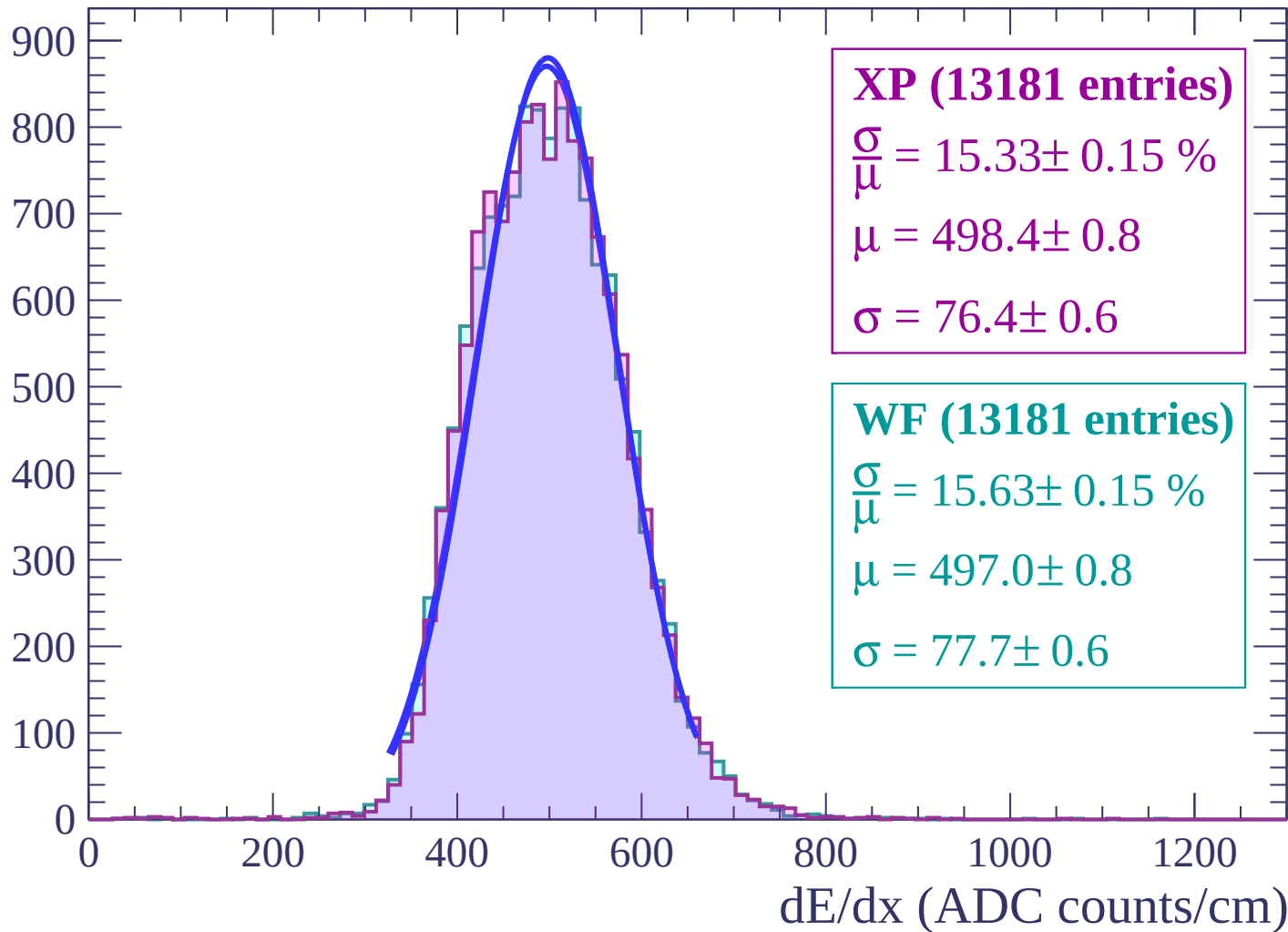
$\mu = 494.3 \pm 0.7$

$\sigma = 77.4 \pm 0.5$



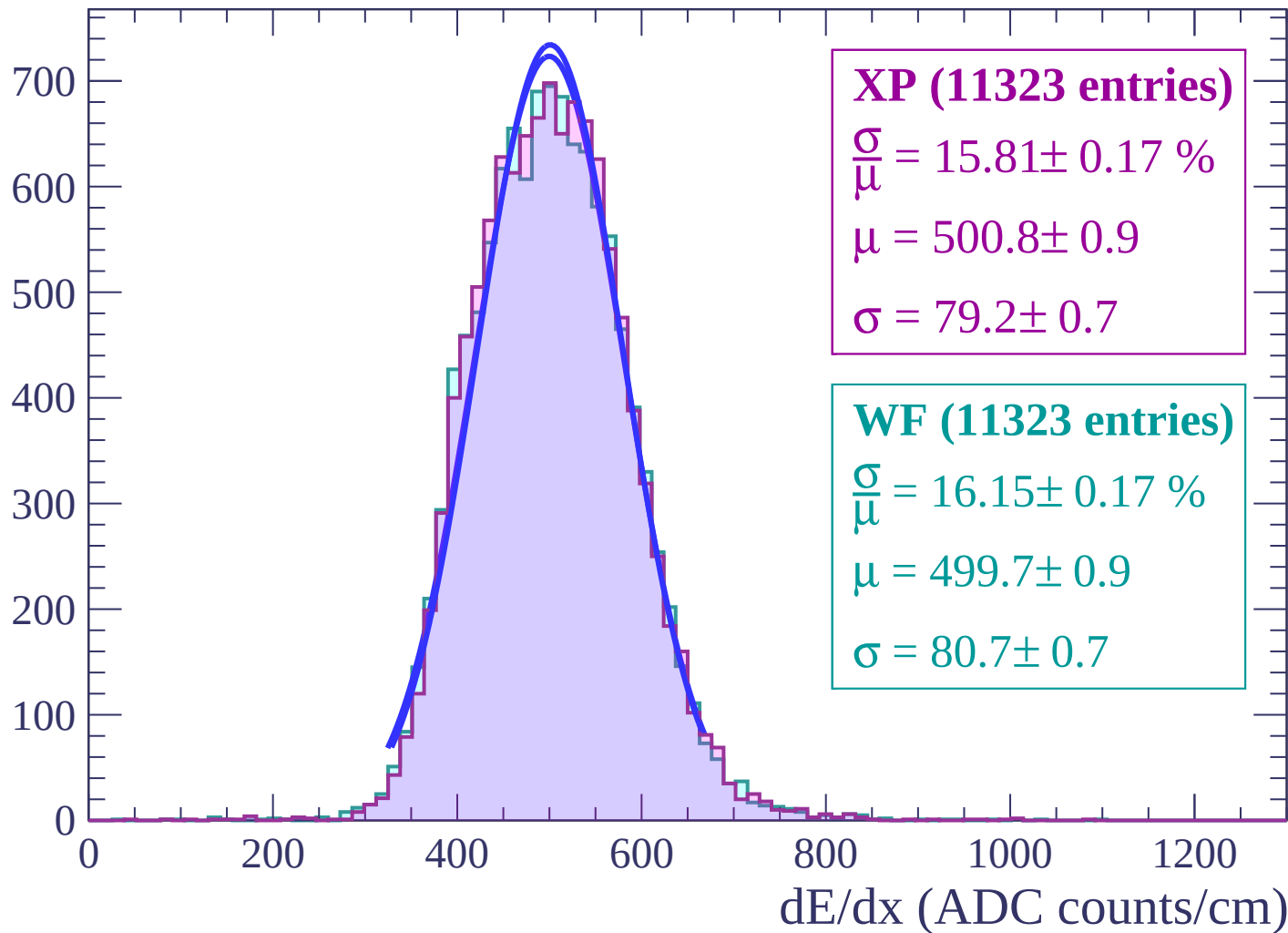
Energy loss | $28 < \theta < 30$

Count



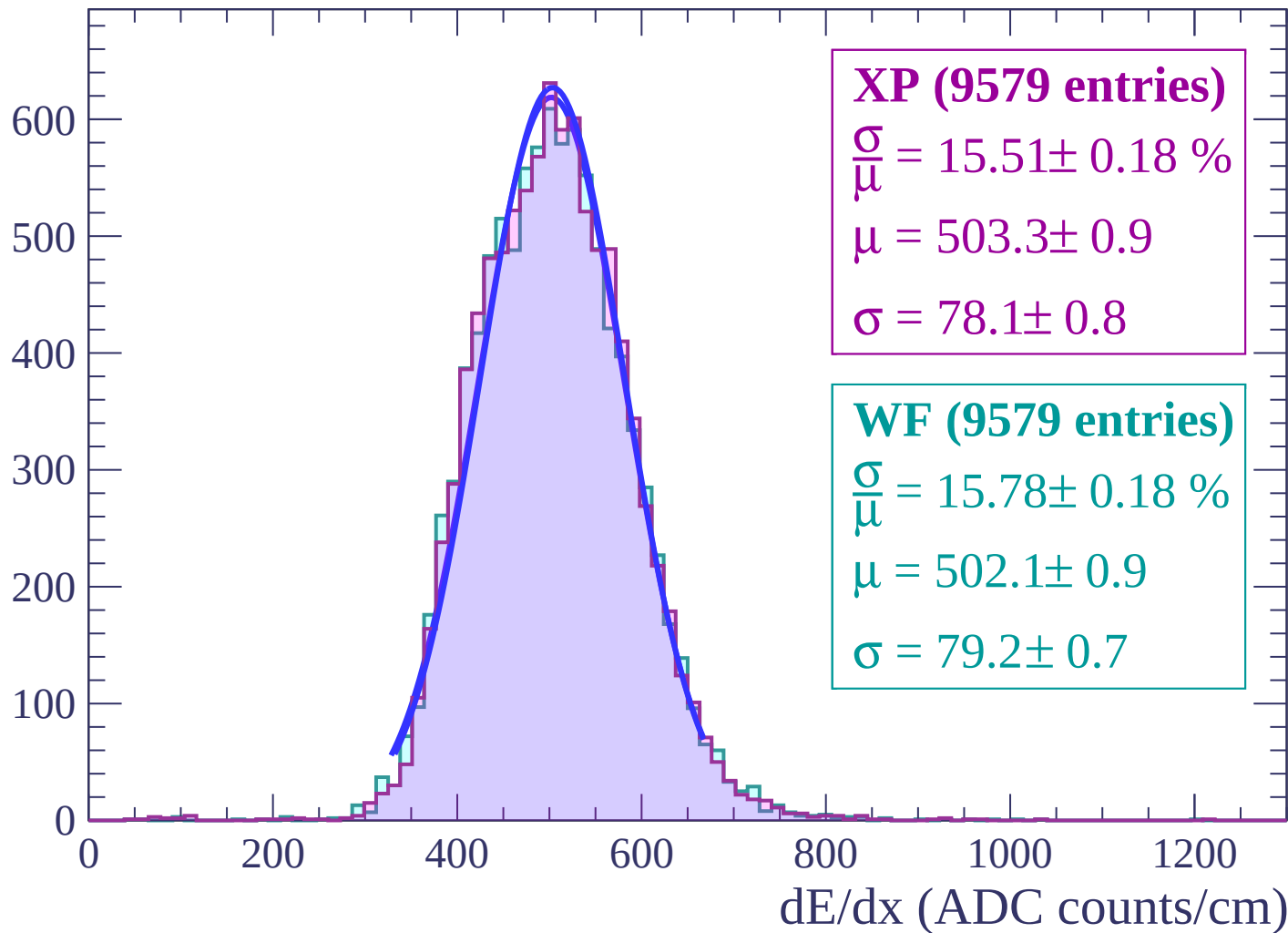
Energy loss | $30 < \theta < 32$

Count



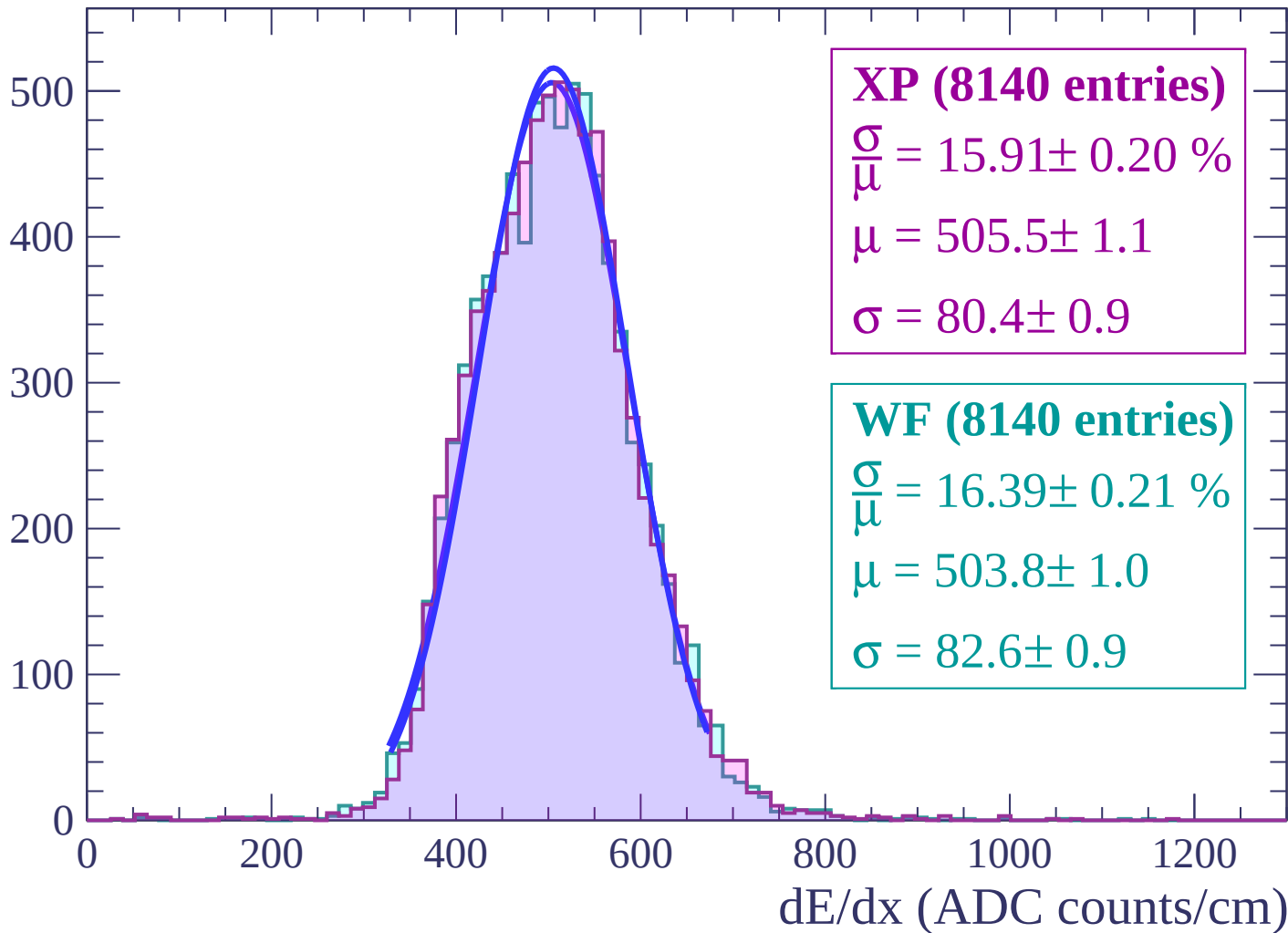
Energy loss | $32 < \theta < 34$

Count



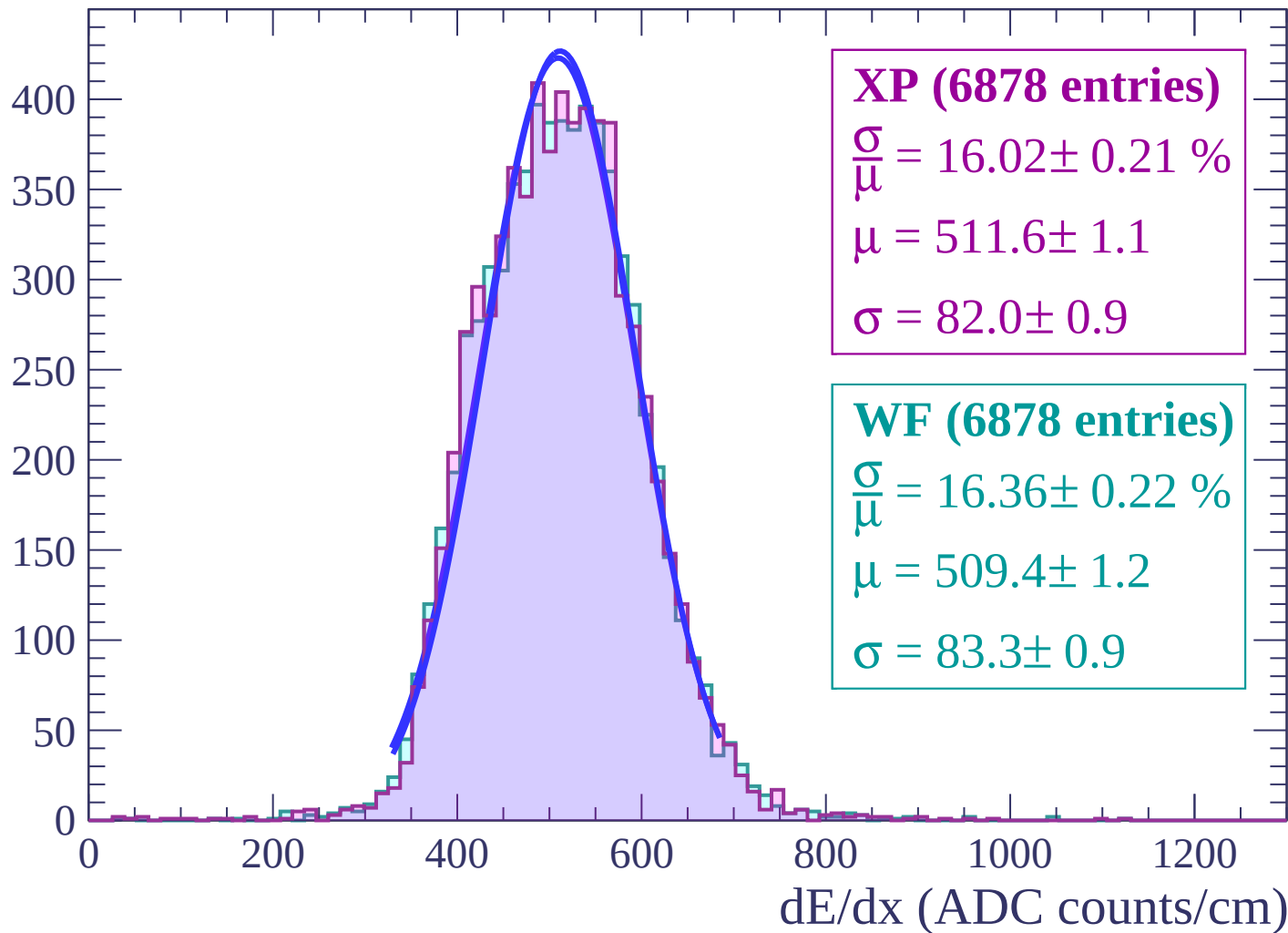
Energy loss | $34 < \theta < 36$

Count



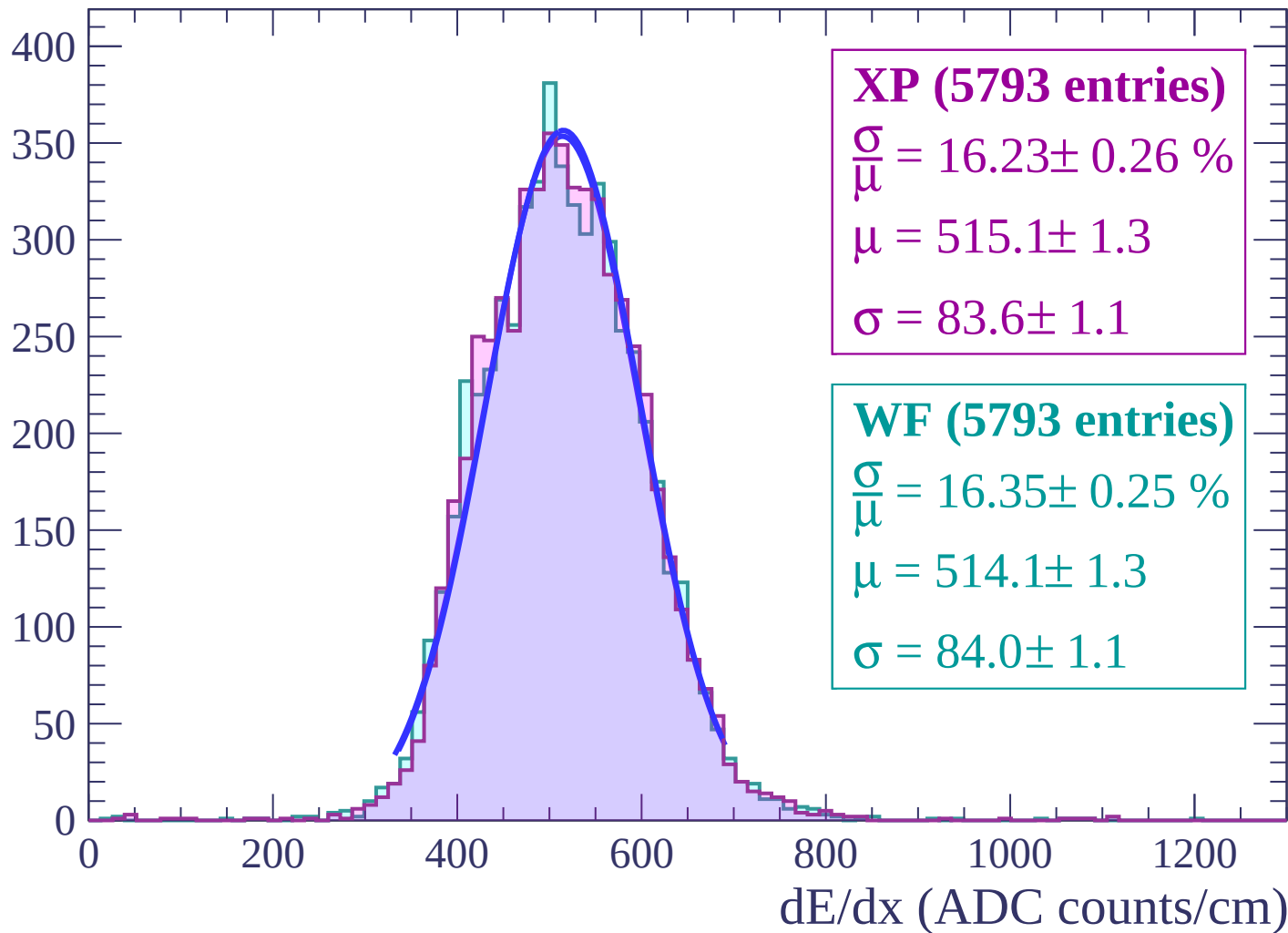
Energy loss | $36 < \theta < 38$

Count



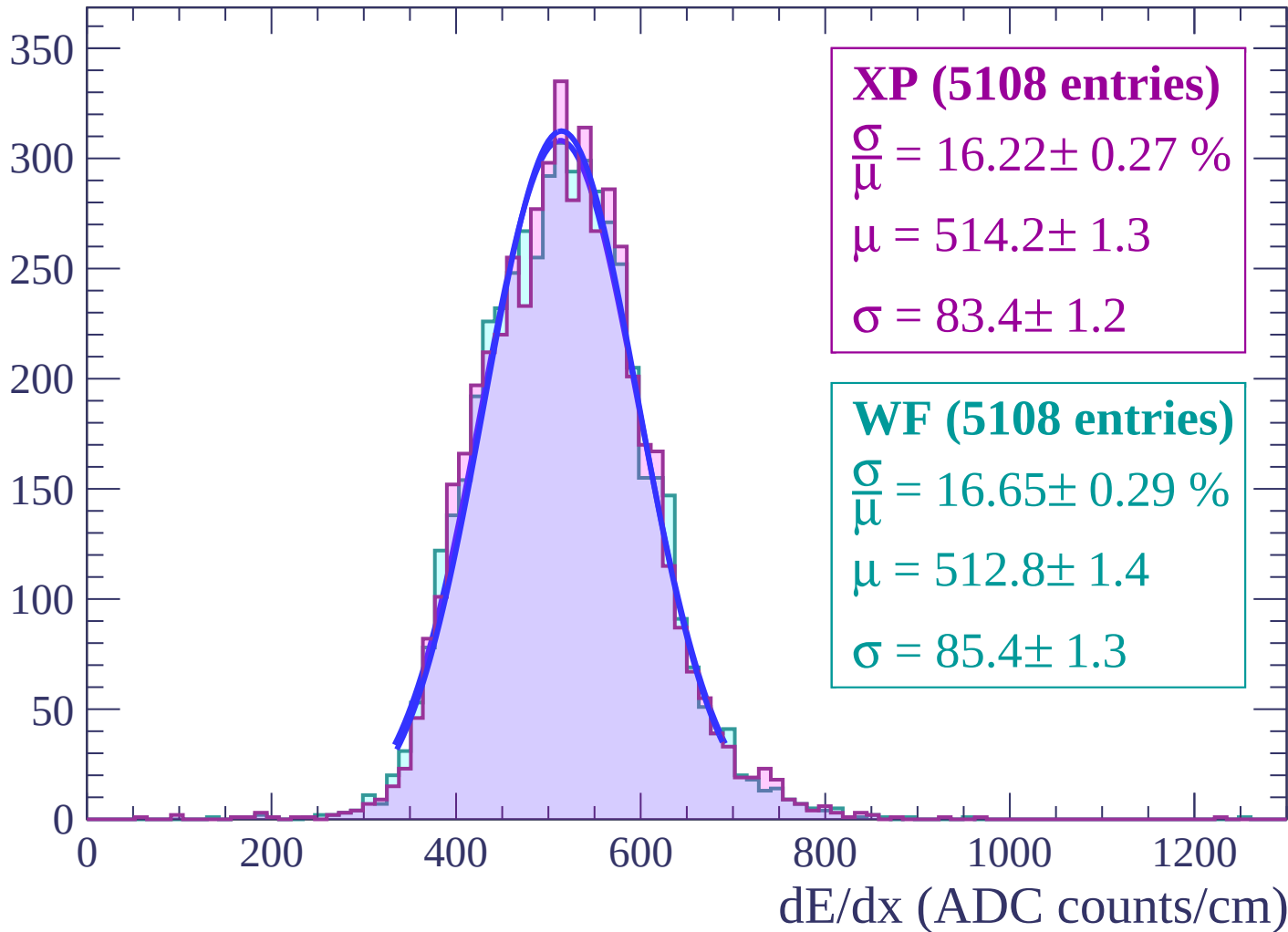
Energy loss | $38 < \theta < 40$

Count



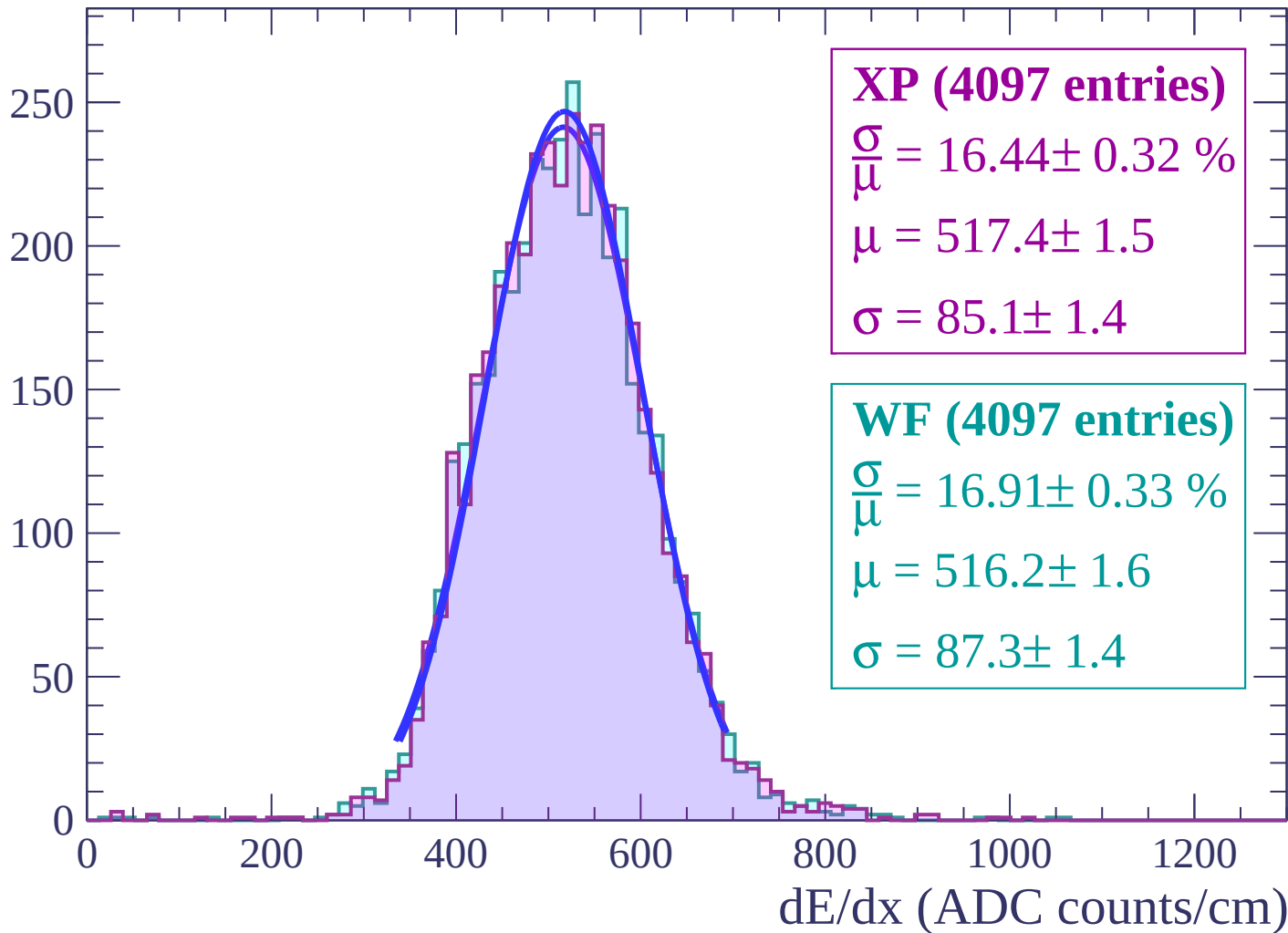
Energy loss | $40 < \theta < 42$

Count



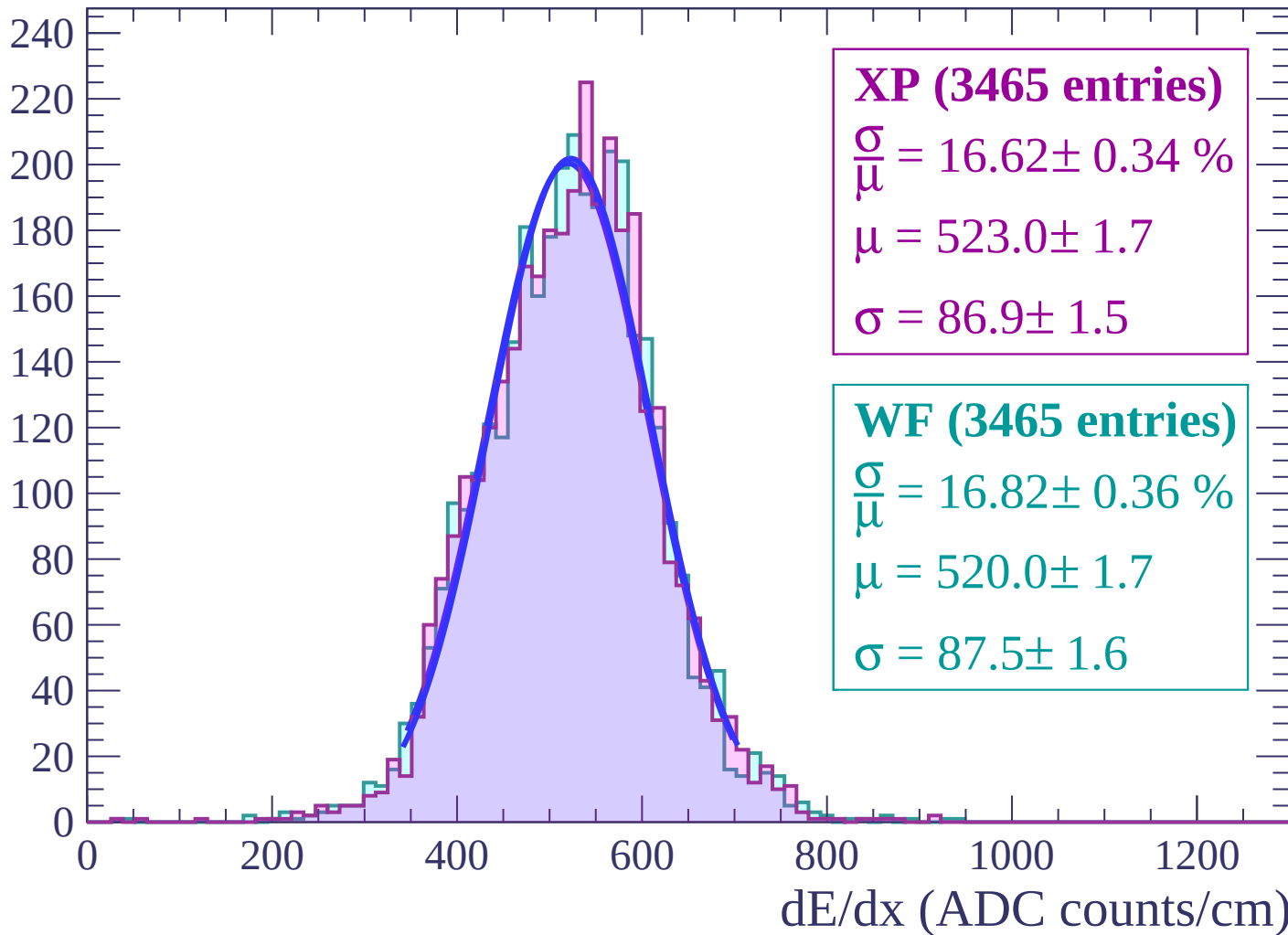
Energy loss | $42 < \theta < 44$

Count



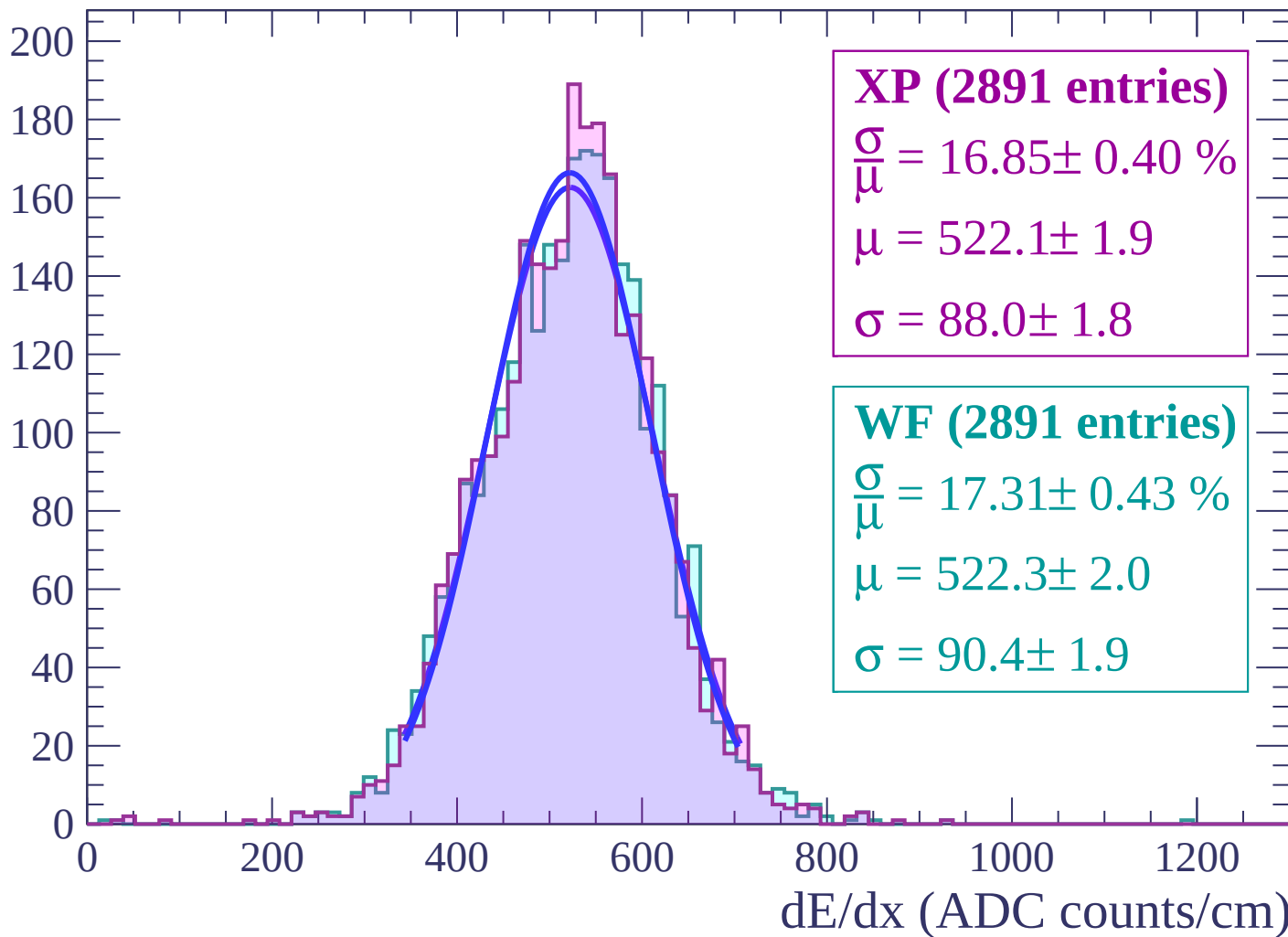
Energy loss | $44 < \theta < 46$

Count



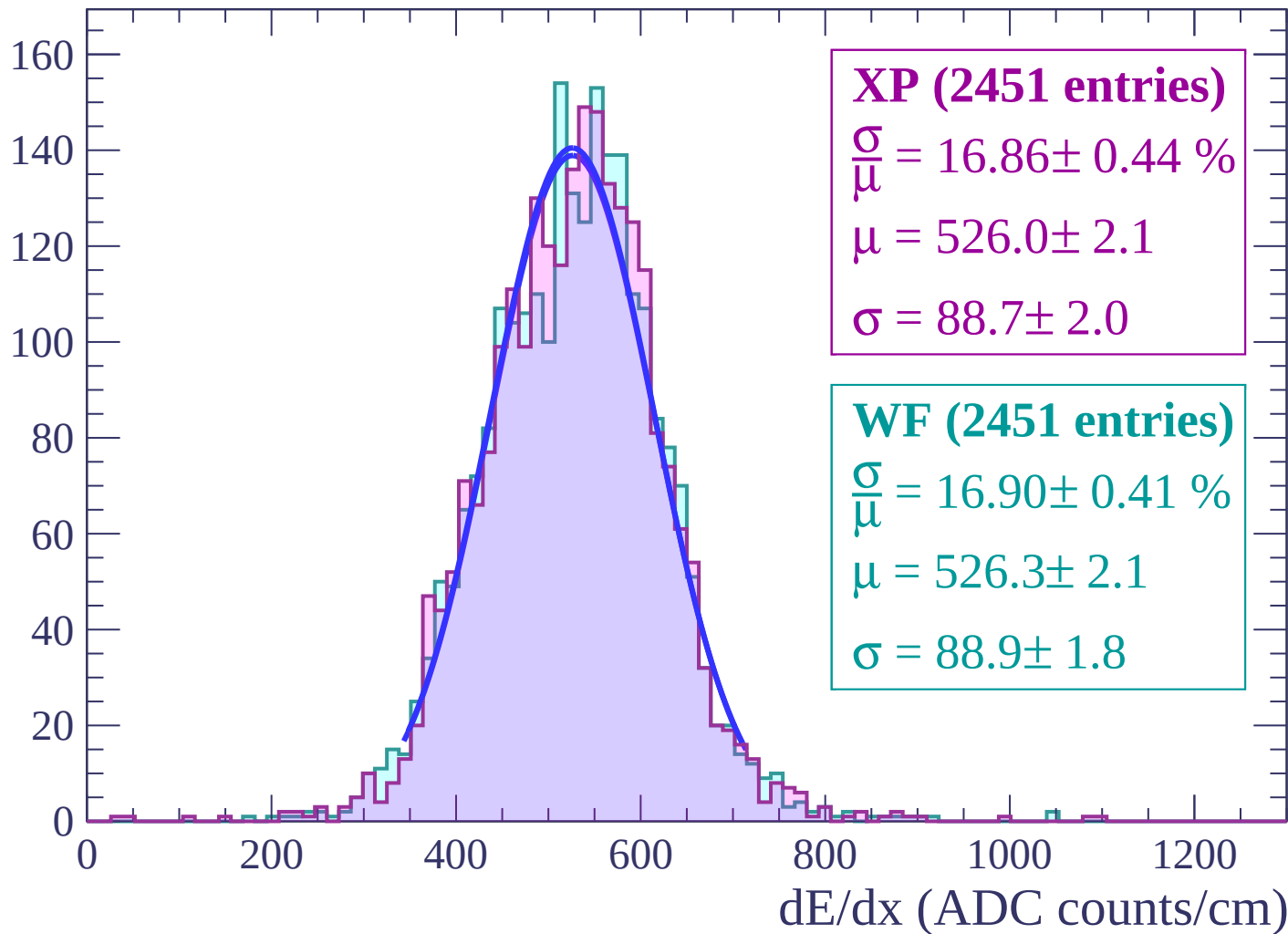
Energy loss | $46 < \theta < 48$

Count



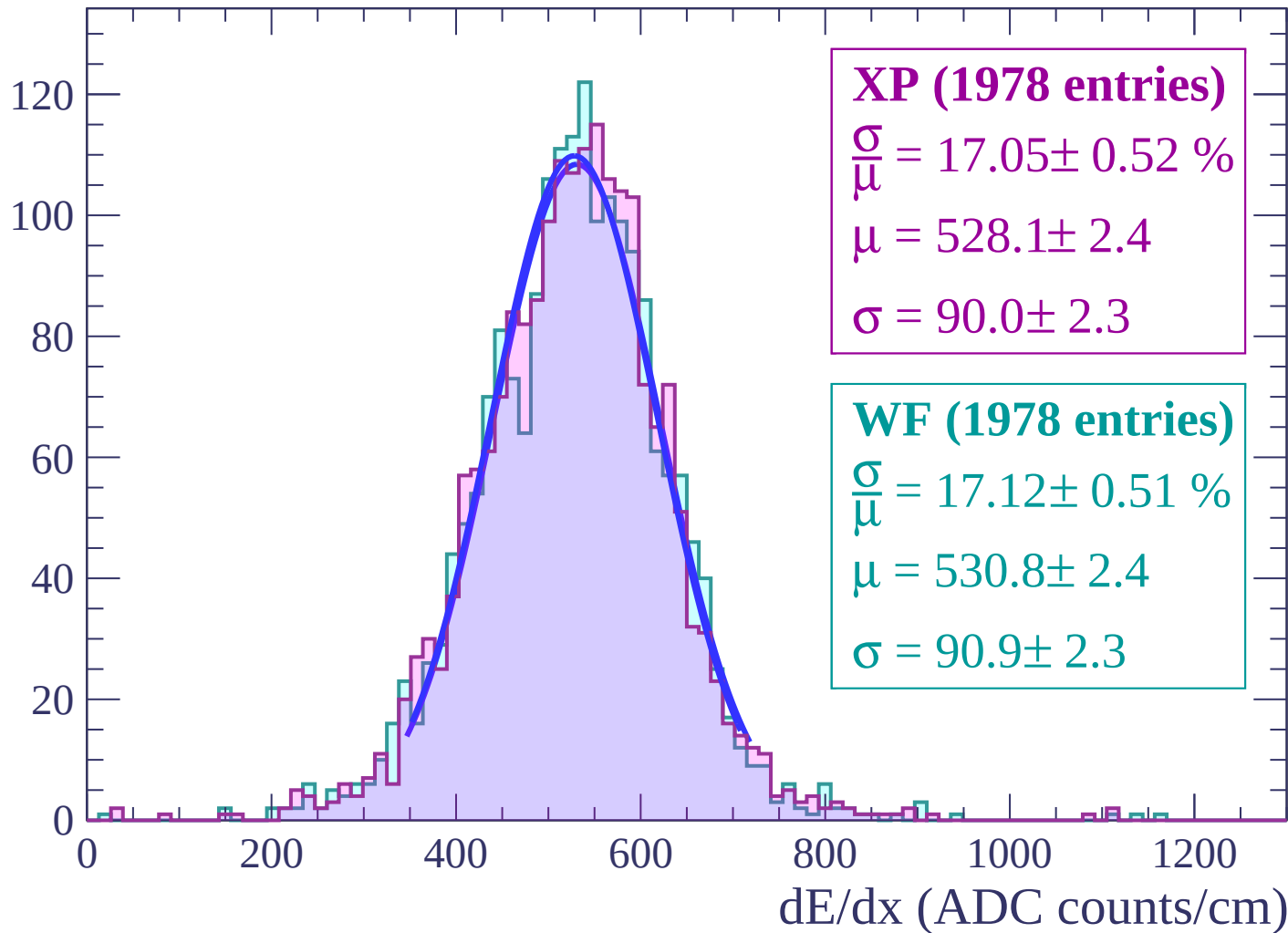
Energy loss | $48 < \theta < 50$

Count



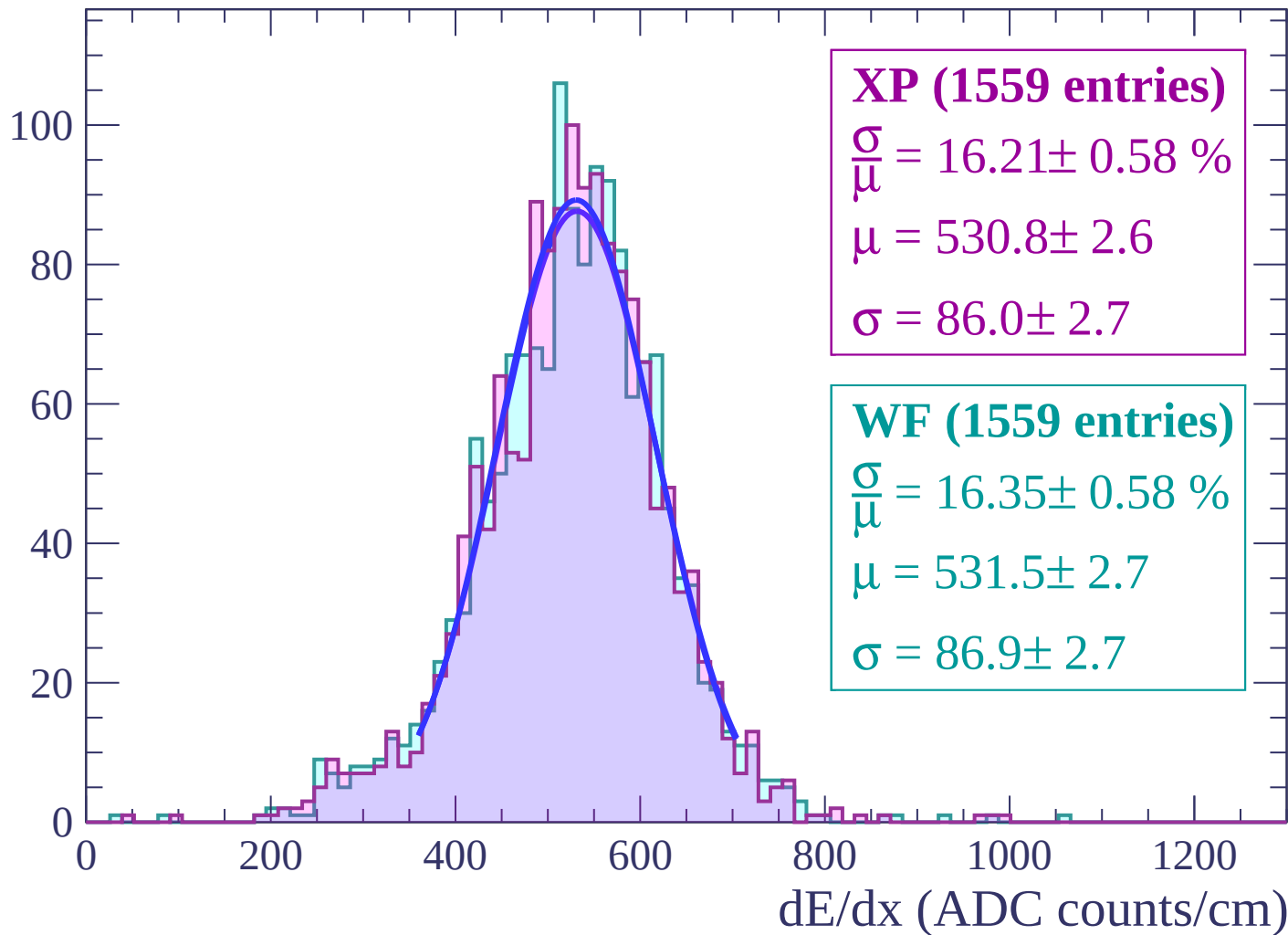
Energy loss | $50 < \theta < 52$

Count



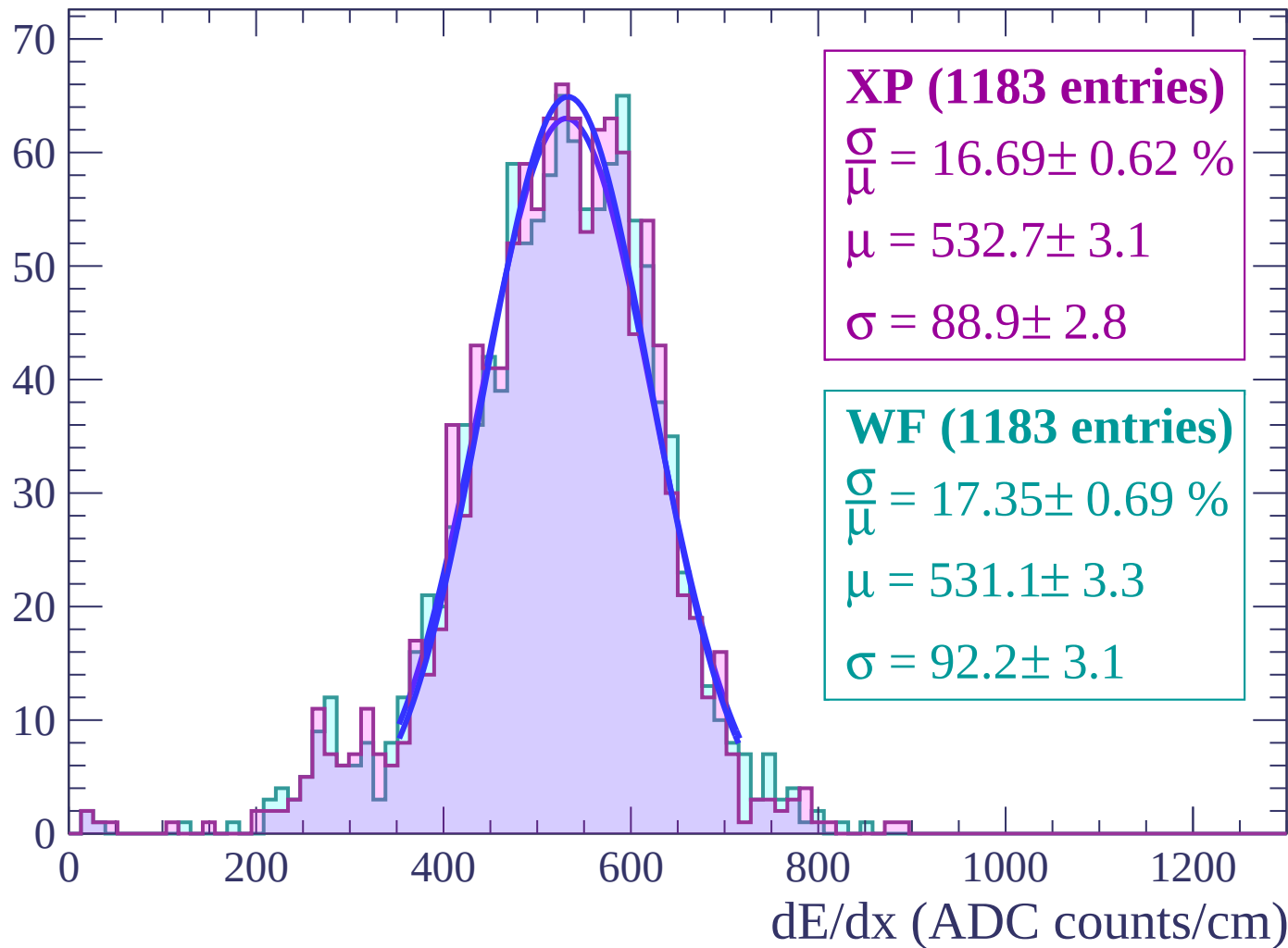
Energy loss | $52 < \theta < 54$

Count



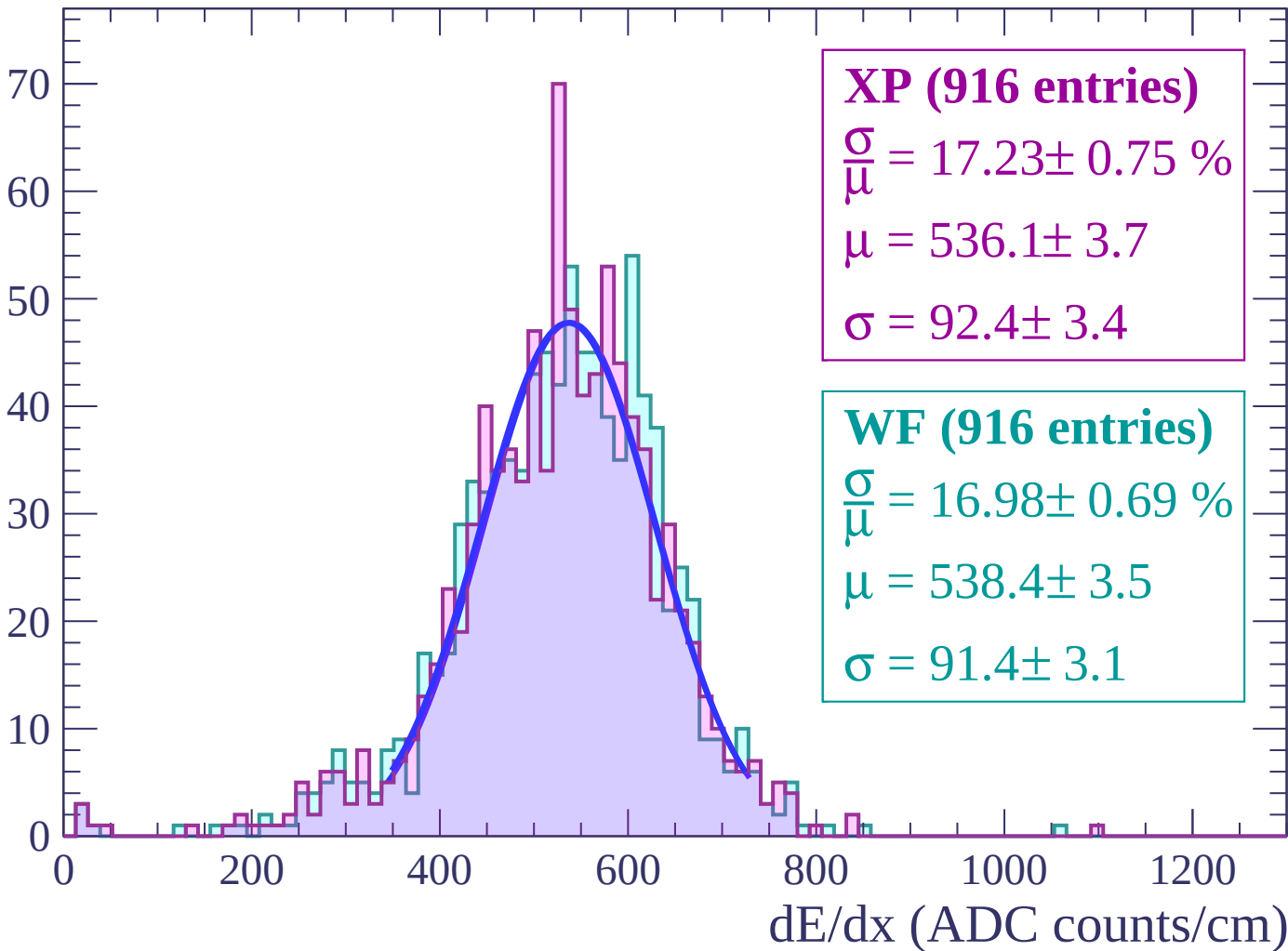
Energy loss | $54 < \theta < 56$

Count



Energy loss | $56 < \theta < 58$

Count



Energy loss | $58 < \theta < 60$

Count

XP (653 entries)

$$\frac{\sigma}{\mu} = 15.59 \pm 0.92 \%$$

$$\mu = 527.8 \pm 4.0$$

$$\sigma = 82.3 \pm 4.2$$

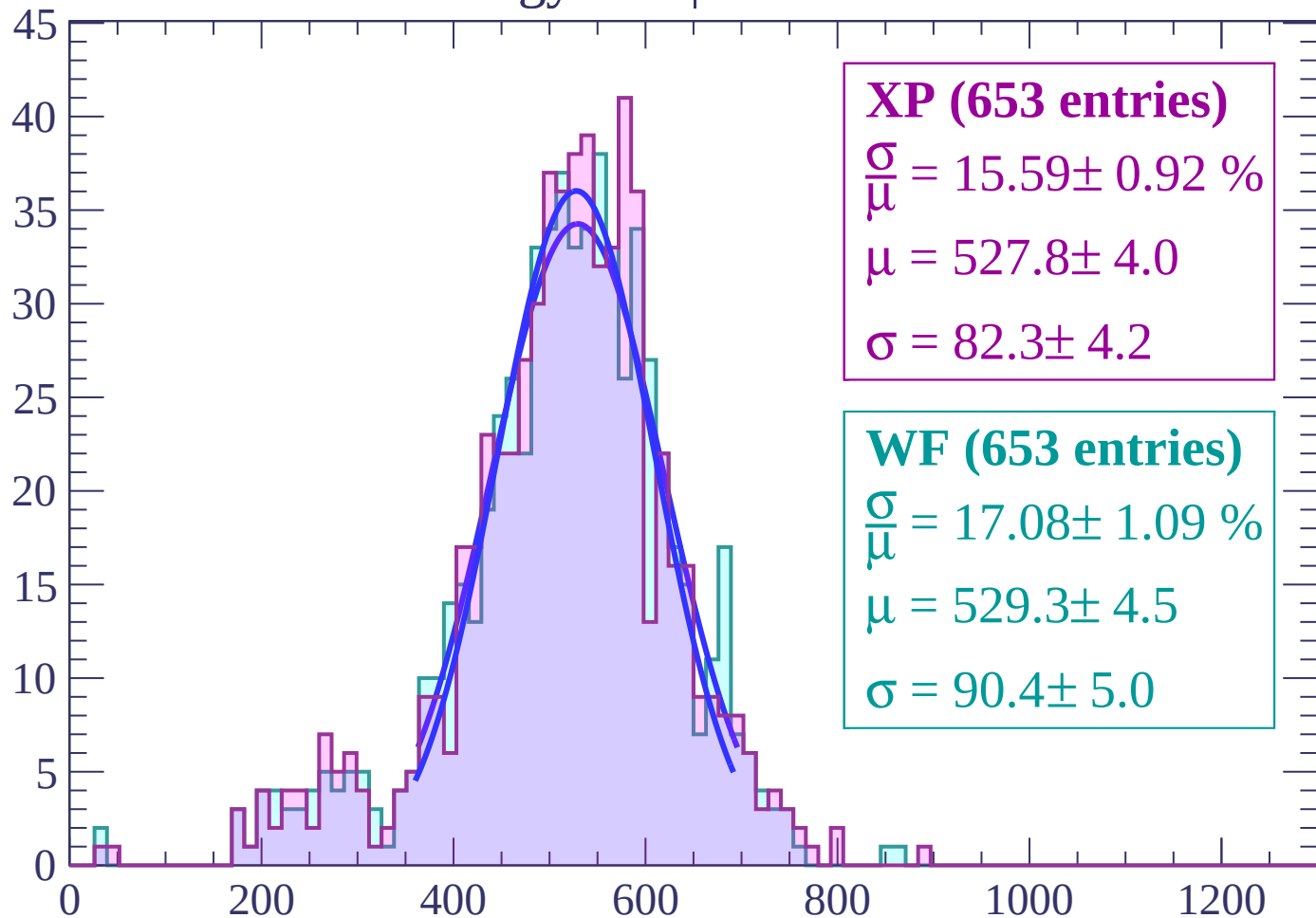
WF (653 entries)

$$\frac{\sigma}{\mu} = 17.08 \pm 1.09 \%$$

$$\mu = 529.3 \pm 4.5$$

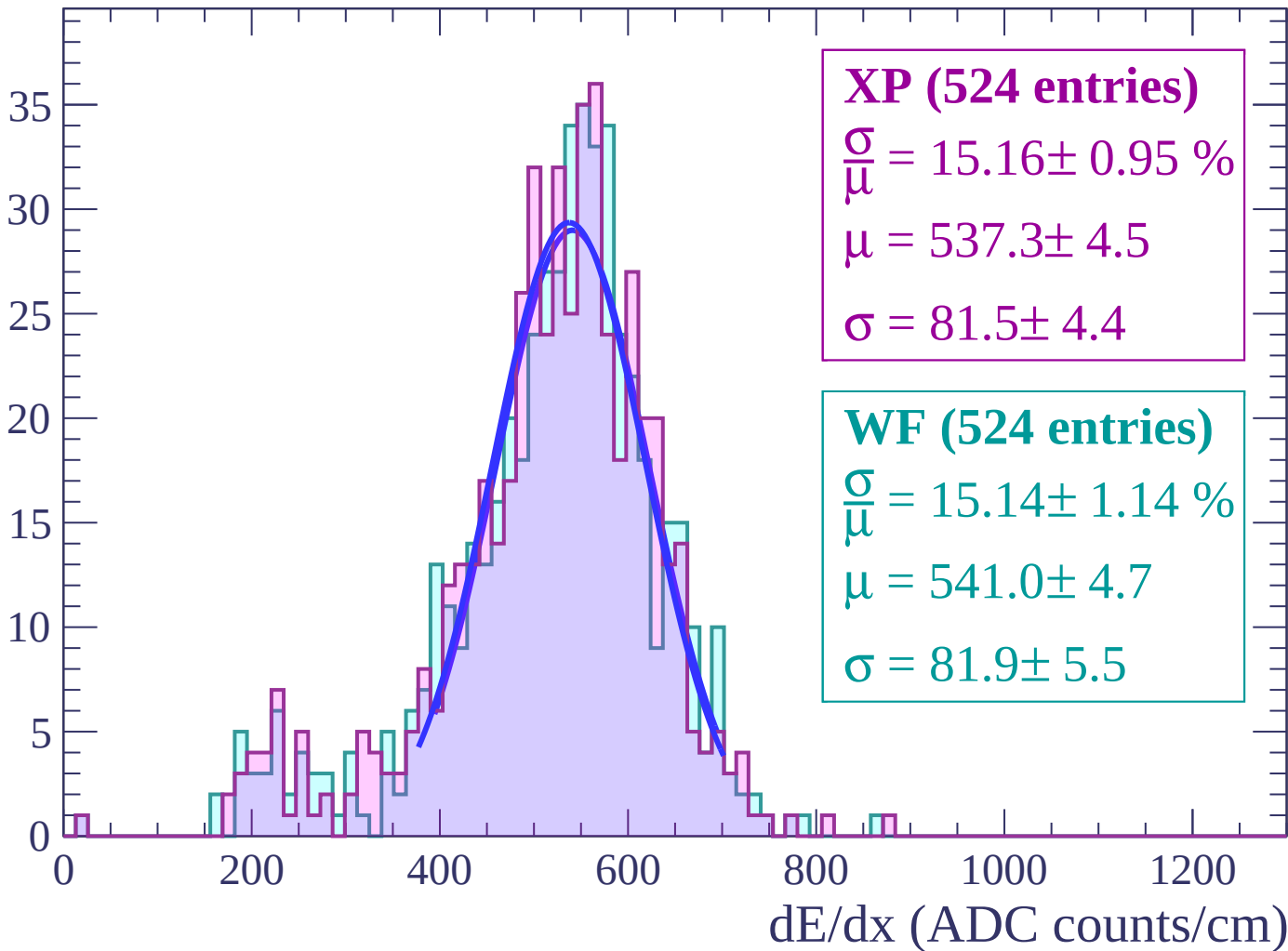
$$\sigma = 90.4 \pm 5.0$$

dE/dx (ADC counts/cm)



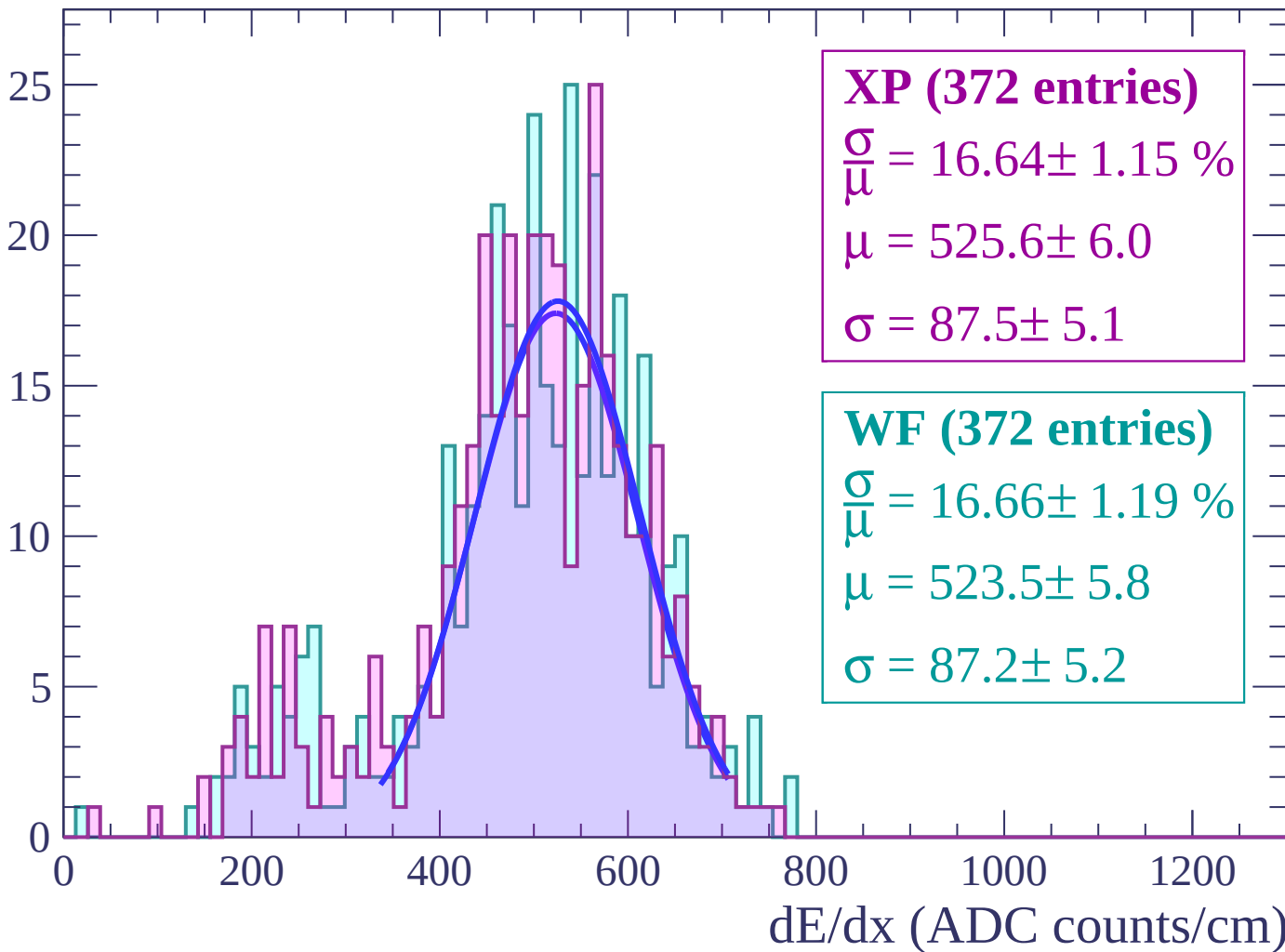
Energy loss | $60 < \theta < 62$

Count



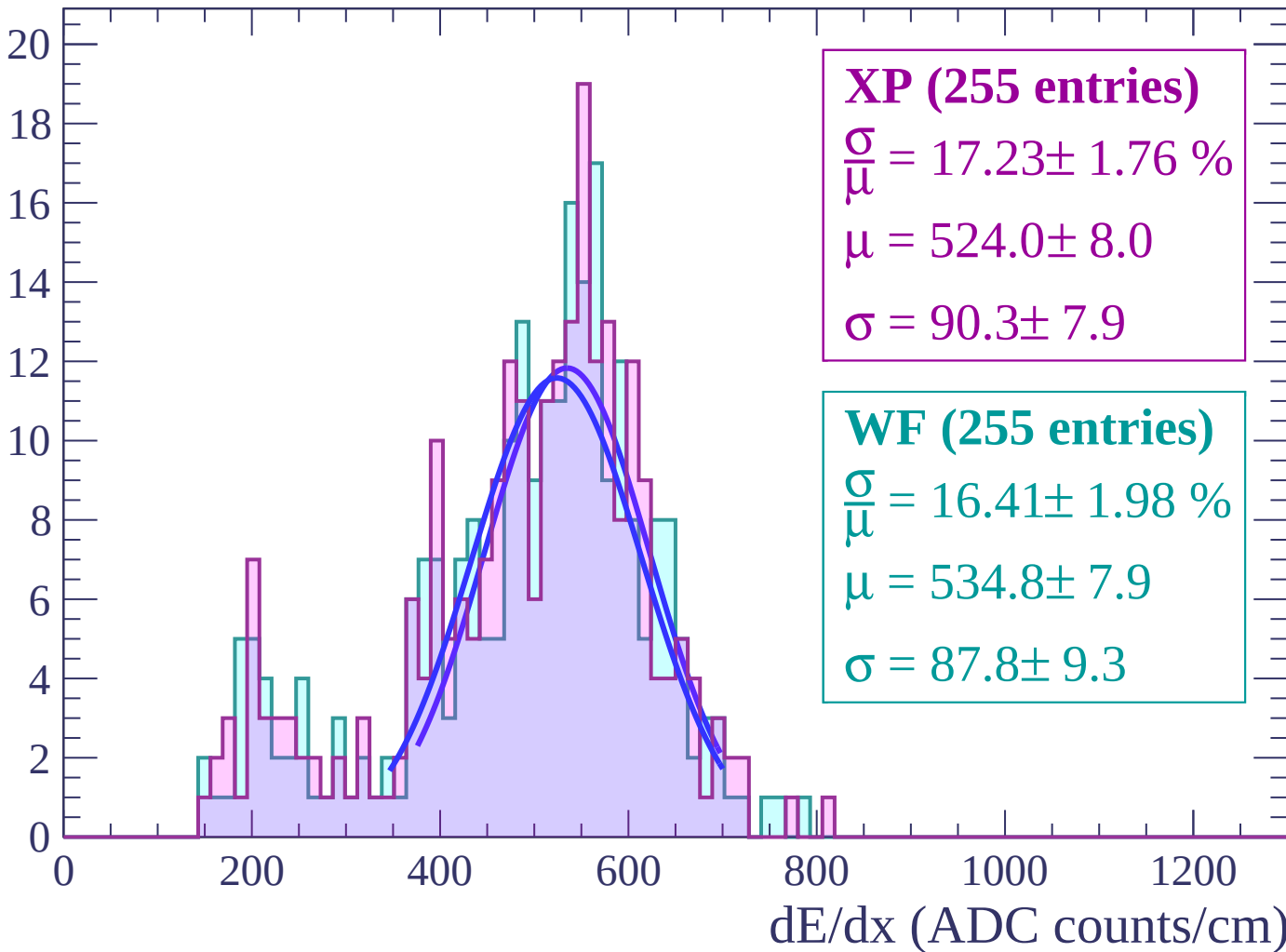
Energy loss | $62 < \theta < 64$

Count



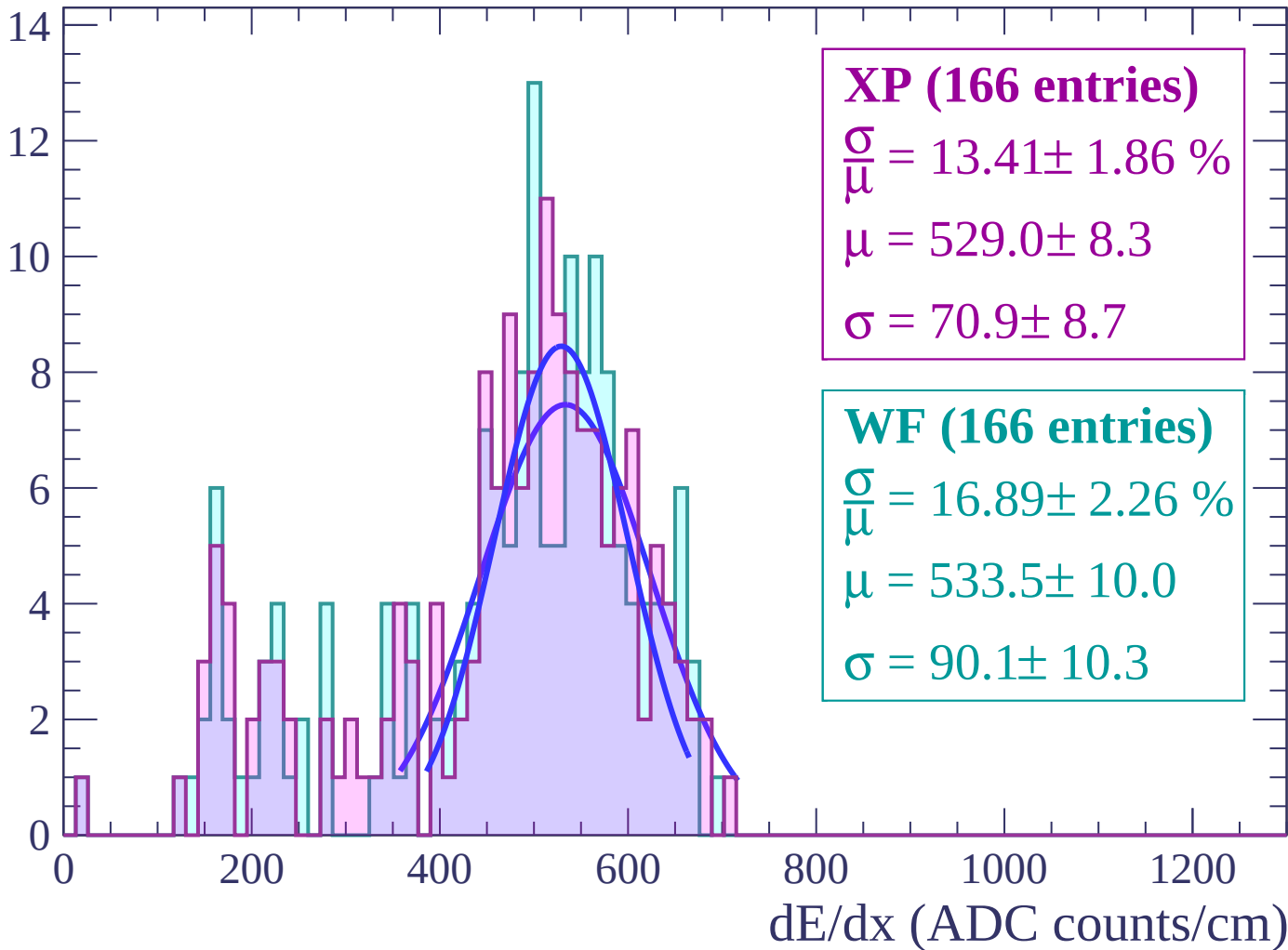
Energy loss | $64 < \theta < 66$

Count



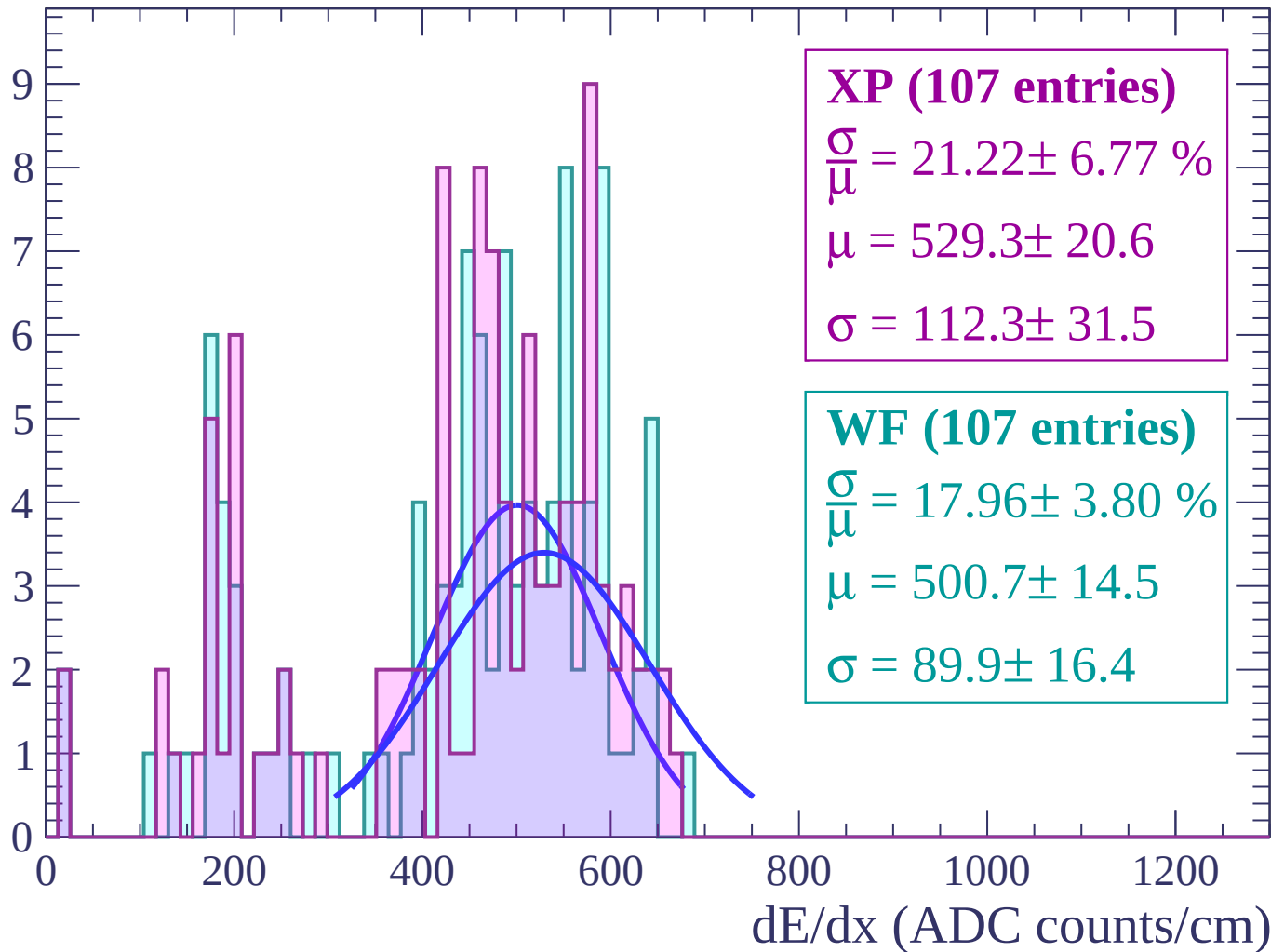
Energy loss | $66 < \theta < 68$

Count



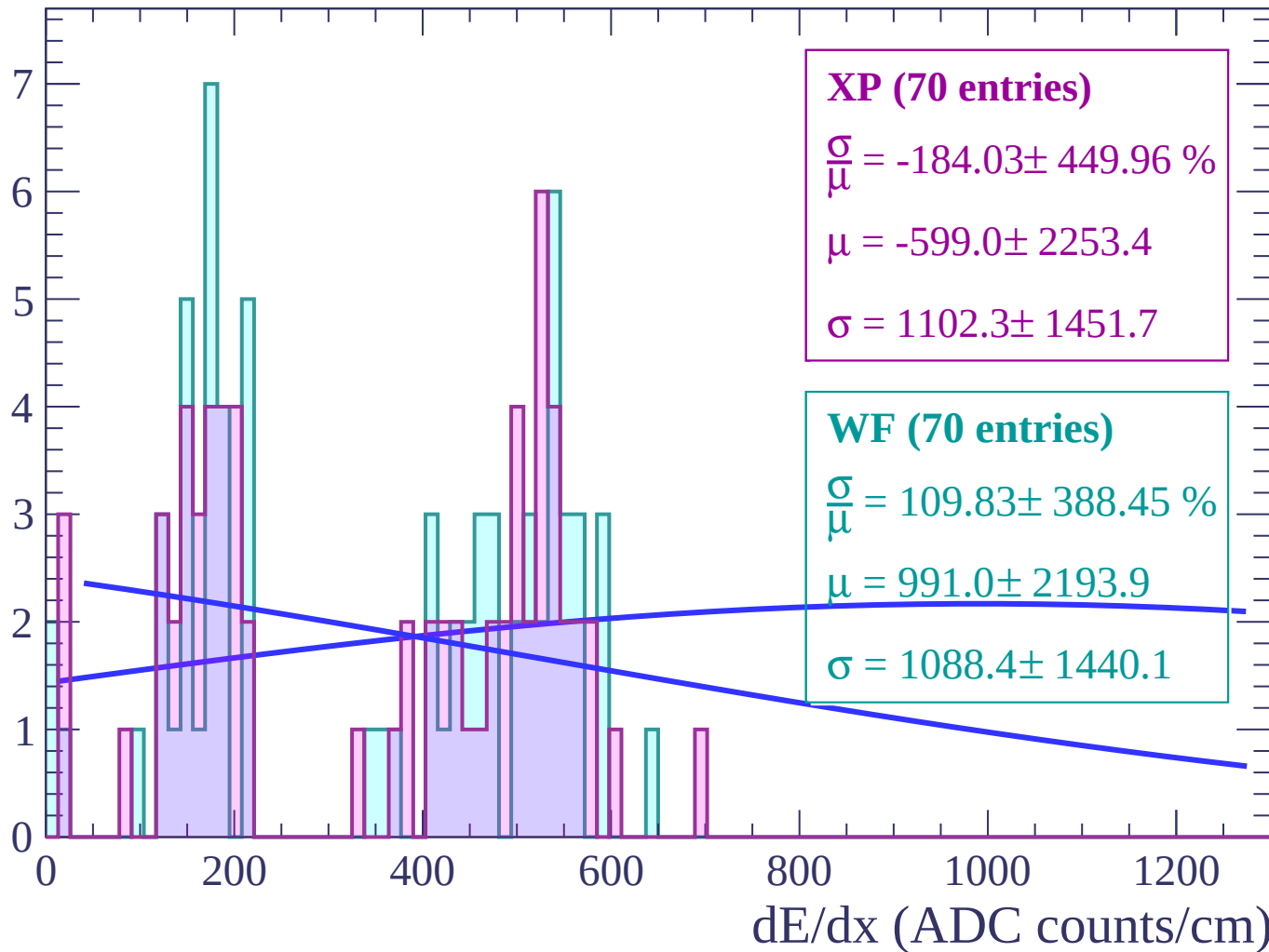
Energy loss | $68 < \theta < 70$

Count



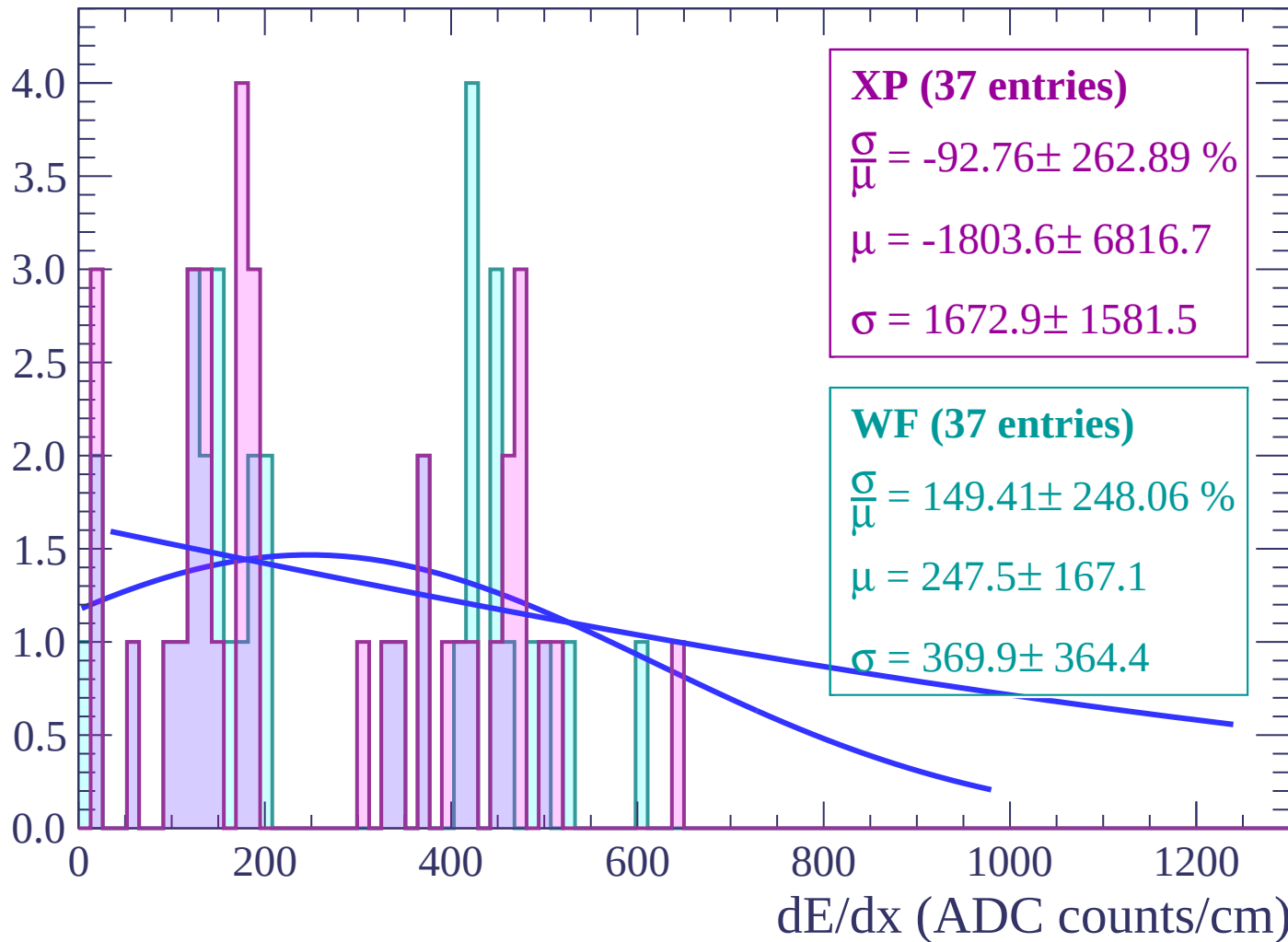
Energy loss | $70 < \theta < 72$

Count



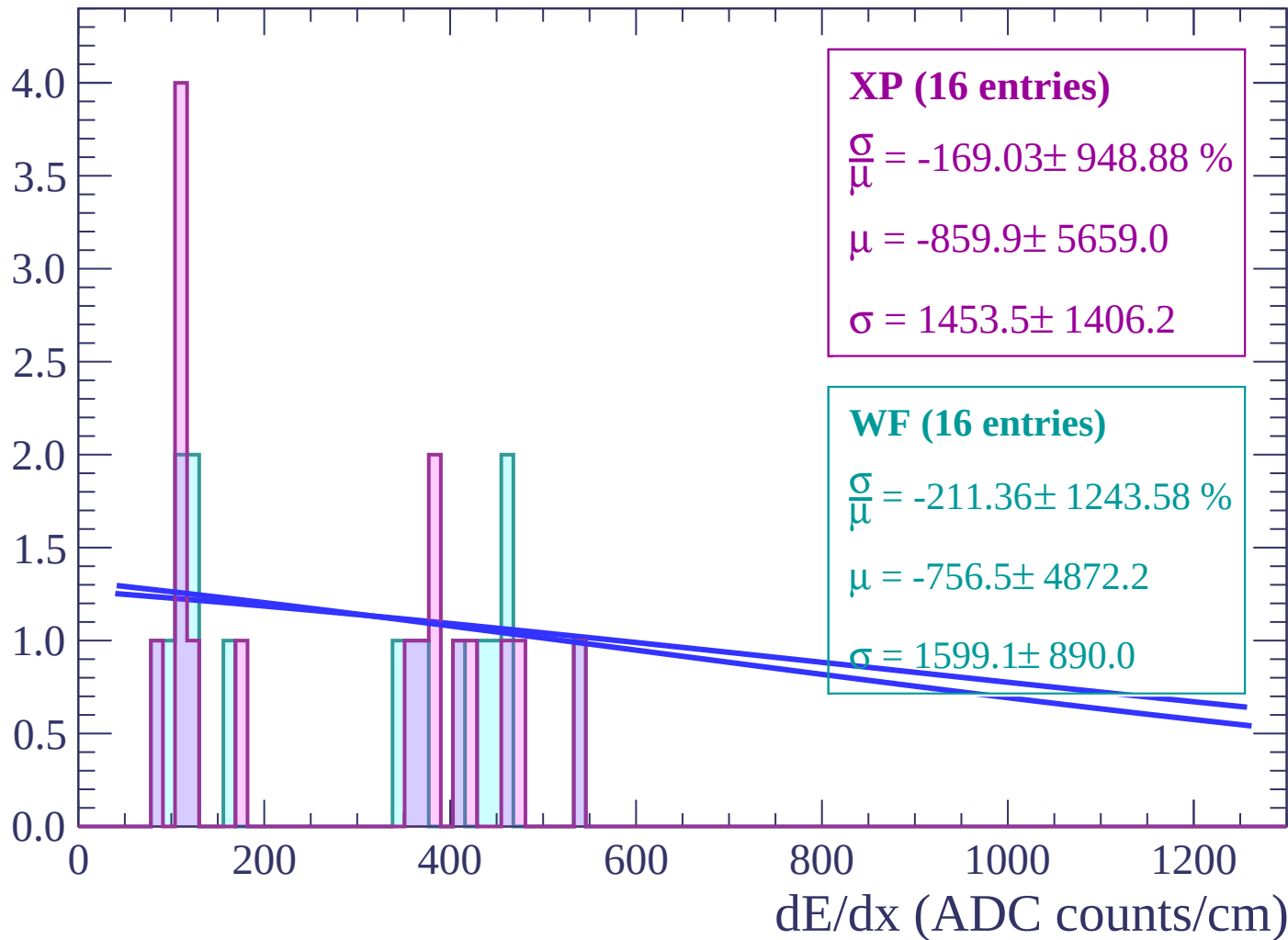
Energy loss | $72 < \theta < 74$

Count



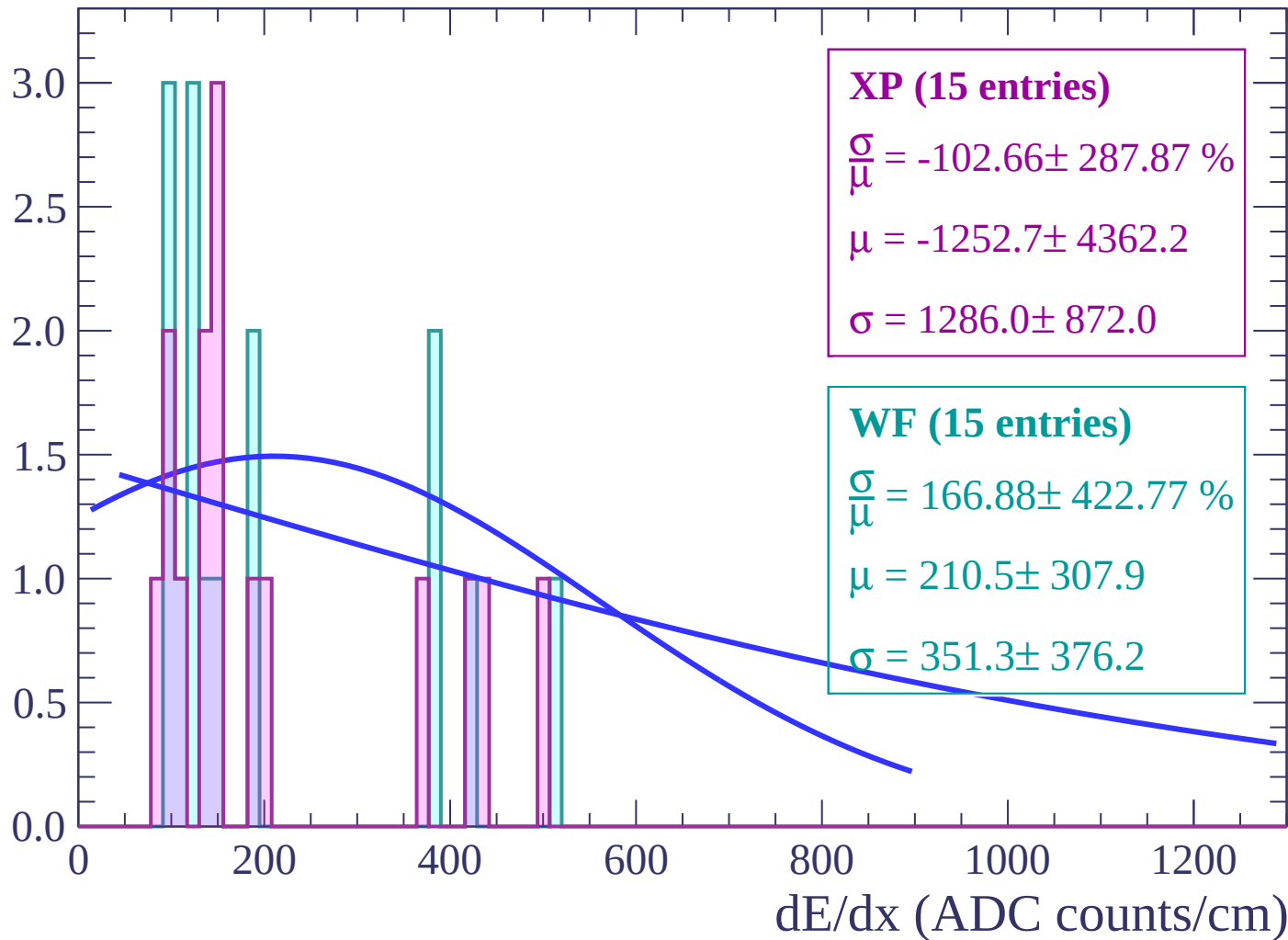
Energy loss | $74 < \theta < 76$

Count



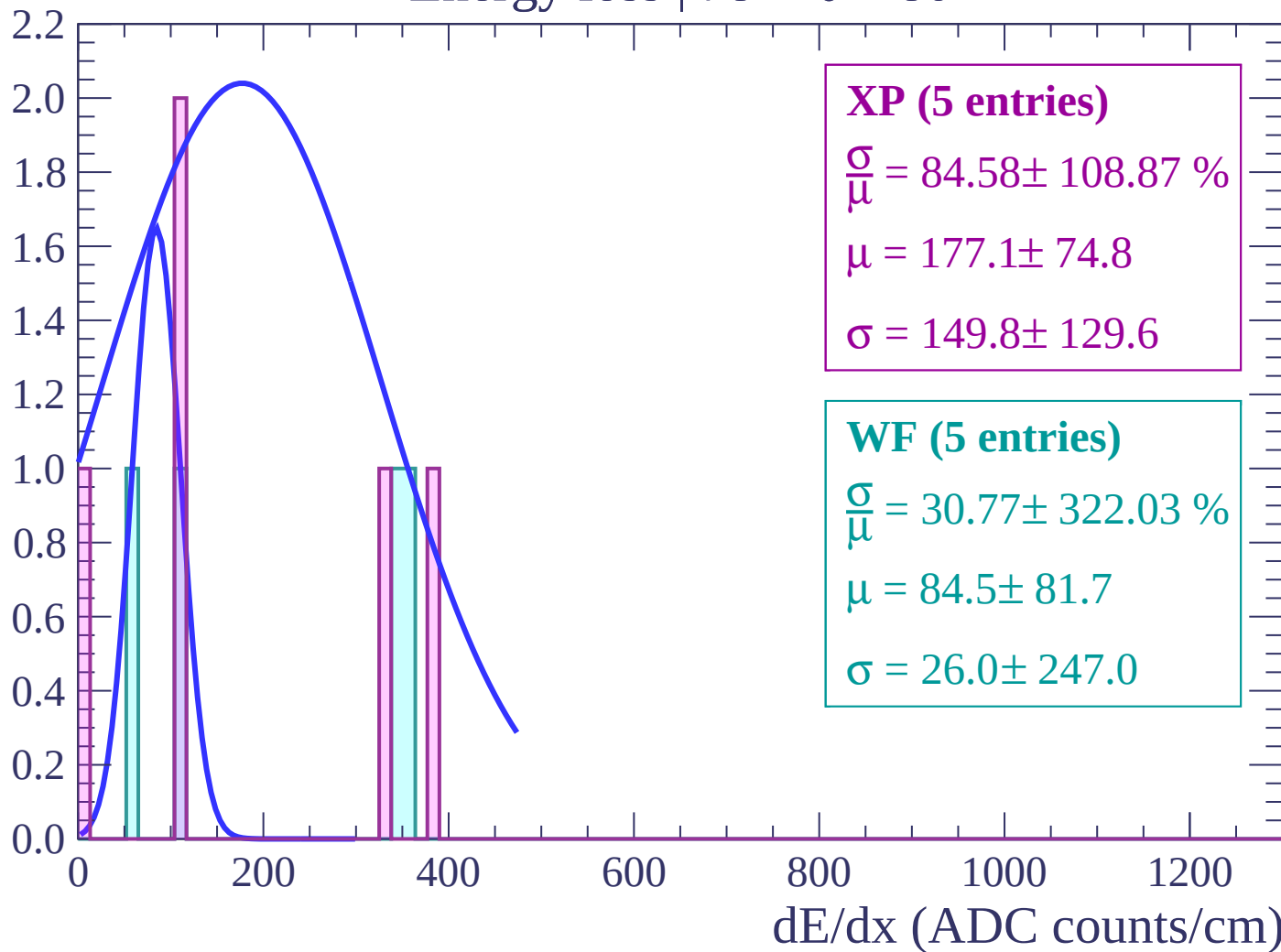
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count



Energy loss | $80 < \theta < 82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $82 < \theta < 84$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

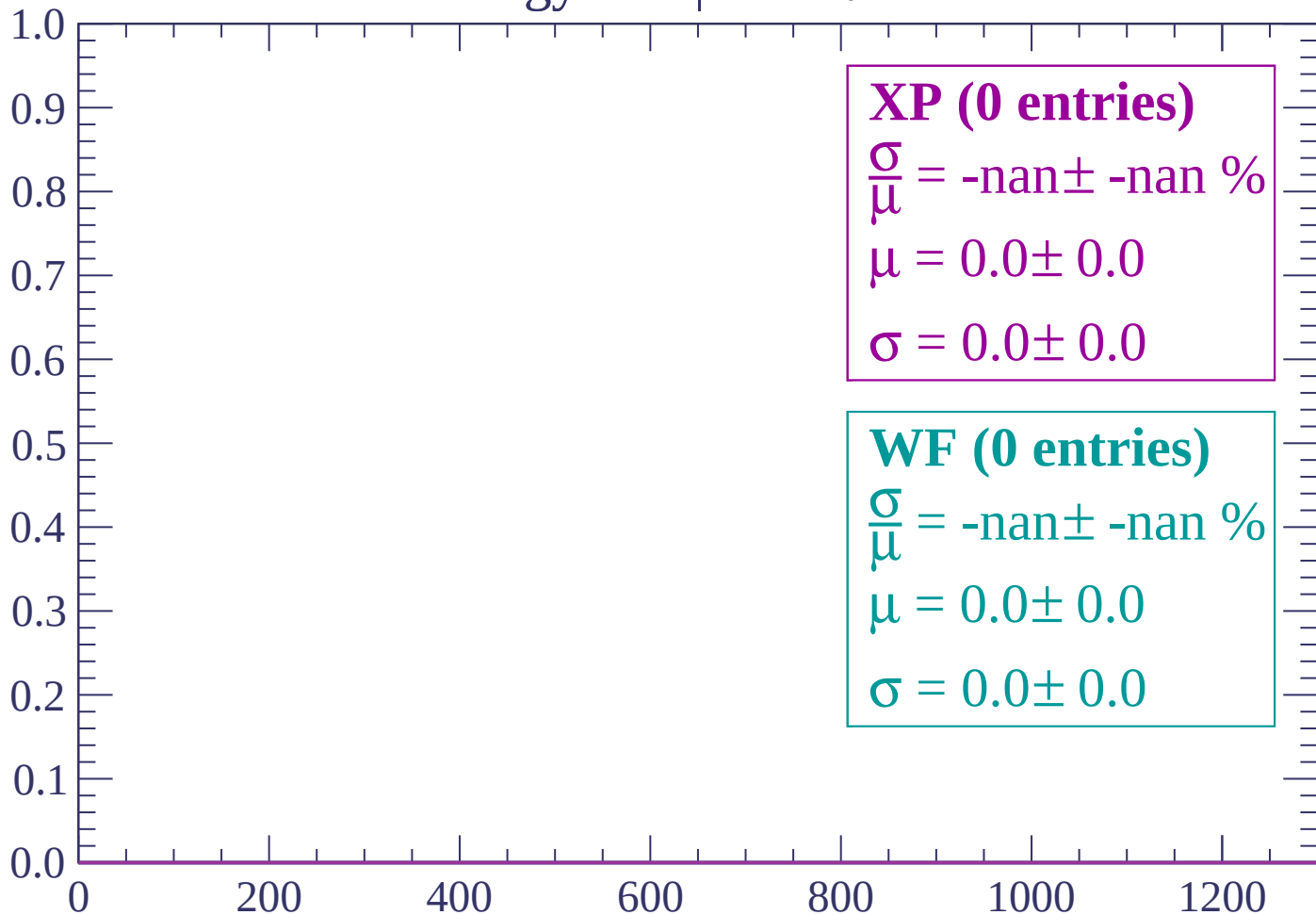
$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

